

OSTBAYERISCHE TECHNISCHE HOCHSCHULE AMBERG-WEIDEN

Department of Environmental Engineering

Modern Wind Turbine (HAWT) Design

Using QBlade Free Modules

Project with Seminar Report

M.Eng. International Energy Engineering

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July 7, 2026

Acknowledgement

I would like to express my deepest gratitude to my project advisor, **Prof. Dr. Stefan Beer**, for his invaluable guidance, patience, and insightful feedback throughout the course of this research. His expertise in wind turbine aerodynamics was instrumental in shaping the methodology and direction of this study.

I am also sincerely grateful to the developers and community of the open-source software QBlade. By providing comprehensive, freely accessible modules for aerodynamic, structural, and BEM analysis, they have made advanced wind turbine simulation accessible to independent researchers and students globally.

Additionally, I would like to extend my thanks to the **Department of Environmental Engineering at Ostbayerische Technische Hochschule Amberg-Weiden** for providing the academic environment and computing resources necessary to complete this project.

Finally, I am thankful to my family and peers for their continuous encouragement and support during the compilation of this report.

Abstract

As the global energy transition accelerates toward renewable sources, wind energy has emerged as one of the most prominent and rapidly growing industrial sectors [1]. Consequently, maximizing the power coefficient (C_p) of Horizontal Axis Wind Turbines (HAWT) remains critical to improving the net efficiency of fluid kinetic energy extraction [2], [3]. Approaching the theoretical Betz limit requires meticulous aerodynamic optimization, particularly regarding spanwise blade-level tailoring [4]. This study investigates methods to maximize C_p by evaluating a diverse selection of airfoil profiles including the FFA-W3, DU, NACA, SG, and Risø families—operating within a high-Reynolds-number regime ranging from 3.0×10^6 to 9.5×10^6 , resolved locally at each independent blade station. To maximize aerodynamic torque extraction, Schmitz’s planform optimization theory was applied to mathematically isolate and smoothly distribute the precise optimum local twist angle (θ) and chord length (c) based on the ideal angle of attack (α_{opt}) along the radius [5], [6].

Using the open-source QBlade wind turbine simulation software framework, a comprehensive decoupled design and validation pipeline was strictly executed using its foundational, freely accessible modules [7], [8]. The numerical methodology integrates viscous potential flow panel method computing (XFOIL) for primary polar generation alongside semi-empirical 360° Viterna-Corrigan extrapolation routines to characterize aerodynamic profile coefficients across a complete, un-bounded range of operating angles [9]. Utilizing classical Blade Element Momentum (BEM) theory and the built-in Blade Optimization scripts, custom multi-airfoil rotor geometries were generated using the NREL 5MW baseline configuration as an engineering reference template [4], [10]. Finally, the steady-state Turbine Performance and multi-body Structural Behavior modules were leveraged to evaluate aerodynamic torque, axial thrust states, cantilever elastic beam deflections, and basic structural aeroelastic modal responses under rated operational conditions.

The numerical simulation results revealed that while increasing total rotor solidity via a 4-blade layout yielded a peak C_p of 0.52, an exceptionally efficient C_p of 0.51 was achieved on a standard 3-blade architecture solely by optimizing spanwise structural stations using a targeted, strategically blended combination of DU and NACA airfoils. Beyond these explicit geometric performance findings, this research successfully benchmarks QBlade’s free ecosystem, demonstrating that high-fidelity aerodynamic optimization, advanced theoretical planning via Schmitz’s theory, and structural validation can be thoroughly achieved without relying on proprietary industrial software extensions.

Keywords: Wind Turbine Optimization, Power Coefficient (C_p), QBlade, Schmitz’s Theory, Reynolds Number, Airfoil Polars (DU, NACA).

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Nomenclature

Upper-Case Roman Letters

C_d	Drag coefficient (dimensionless profile resistance metric)	[—]
C_l	Lift coefficient (dimensionless profile aerodynamic force metric)	[—]
C_p	Power coefficient (rotor performance efficiency metric relative to the Betz limit)	[—]
E	Material Young's Modulus (measure of elastic axial stiffness)	[Pa]
F_x	Local out-of-plane aerodynamic force per unit span (Thrust force vector normal to the rotor plane)	[N/m]
F_y	Local in-plane aerodynamic force per unit span (Torque-generating force vector parallel to the rotor plane)	[N/m]
G	Shear modulus of the material (measure of cross-sectional structural resistance to torsional shear stress)	[Pa]
GJ	Torsional rigidity parameter (product of material shear modulus G and cross-sectional polar moment of inertia J)	[N · m ²]
I_x	Area moment of inertia about the local x-axis (governs structural resistance to chordwise edgewise bending)	[m ⁴]
I_y	Area moment of inertia about the local y-axis (governs structural resistance to out-of-plane flapwise bending)	[m ⁴]
M_x	Local in-plane edge bending moment vector acting along a spanwise station	[N · m]
M_y	Local out-of-plane flap bending moment vector acting along a spanwise station	[N · m]
R	Total tip radius of the wind turbine rotor	[m]
Re	Reynolds number (dimensionless ratio of fluid inertial forces to viscous forces)	[—]

Lower-Case Roman Letters

a	Axial induction factor (fractional decrease in wind velocity between freestream and rotor plane)	[—]
a'	Angular induction factor (fractional increase in angular velocity of the wind wake)	[—]
c	Local blade geometric chord length (straight line connecting the profile leading and trailing edges)	[m]
r	Local spanwise station radius (radial distance isolated from the center of the hub)	[m]
v	Oncoming freestream fluid velocity (ambient wind speed upstream of the rotor)	[m/s]
z	Spanwise coordinate directed axially along the longitudinal blade axis	[m]

Greek Letters

α	Angle of attack (geometric angle between the airfoil chord line and the relative oncoming fluid velocity vector)	[°]
α_{opt}	Optimal operational angle of attack (the angle maximizing the local lift-to-drag ratio)	[°]
ϵ	Aerodynamic lift-to-drag efficiency ratio (glide ratio representing cross-sectional profile efficiency)	[—]
θ	Local geometric structural twist angle (the spanwise variation angle relative to the plane of rotation)	[°]
ρ	Fluid density of the working medium (mass per unit volume of ambient air)	[kg/m ³]
Ω	Rotational velocity of the wind turbine rotor (angular speed of operation)	[rad/s]

Subscripts and Superscripts

flap	Structural metric or bending state oriented in the out-of-plane direction
opt	Mathematically optimized value computed under steady-state design conditions

Abbreviations and Acronyms

1P	Primary rotor rotational frequency excitation band (mechanical imbalances per rotation)
3P	Three-blade rotor passage frequency excitation band (aerodynamic tower-shadow shadow interaction pulse)
BEM	Blade Element Momentum theory (classical decoupled numerical method relating fluid forces to rotor induction)
CE	Community Edition (open-source software licensing classification framework)
CFD	Computational Fluid Dynamics (numerical fluid simulation modeling utilizing discretized transport equations)
CFRP	Carbon-Fiber Reinforced Polymer (high-strength, low-density composite material matrix)
DU	Delft University airfoil profile family (thick airfoils engineered for wind turbine structural root regions)
FEA	Finite Element Analysis (numerical stress-strain continuum mechanics structural solver)
FSI	Fluid-Structure Interaction (coupled multi-physics simulation charting aerodynamic-elastic feedback loops)
HAWT	Horizontal Axis Wind Turbine (dominant utility-scale rotor architecture aligned parallel to the wind vector)
NACA	National Advisory Committee for Aeronautics (standardized analytical airfoil shape classification geometry matrix)
PIV	Particle Image Velocimetry (optical laser fluid visualization method used to measure velocity fields)

1. Introduction

1.1 Background and Global Energy Context

Global energy and electricity demands are increasing on a daily basis. Given that fossil fuels continue to provide approximately 76% of the world's energy [1], environmental pollution and greenhouse gas accumulations are expanding at an alarming rate. Addressing this ecological crisis demands a substantial, coordinated global transition toward sustainable renewable energy infrastructure. According to the Global Energy Review [1], wind power has emerged as a cornerstone of this transition, with global utility-scale capacity increasing nearly threefold since 2015 [11].

Among the various clean energy technologies available, wind energy has proven to be one of the most commercially viable, scalable, and rapidly expanding sectors worldwide [2], [12]. Horizontal Axis Wind Turbines (HAWTs) represent the dominant mechanical architecture utilized for utility-scale electrical generation due to their superior aerodynamic efficiency and torque-to-weight ratios compared to vertical-axis alternatives [3]. However, the overarching viability of wind energy remains fundamentally contingent on regional meteorological conditions and local aerodynamic optimization. For instance, the wind energy sector experienced a notable drop in net energy output in 2024 due to less favorable atmospheric wind conditions when evaluated against the peak generation curves of 2023 [11]. This variability highlights the critical need for advanced aerodynamic and structural design methodologies capable of maximizing rotor power extraction even under suboptimal fluid velocities.

1.2 Theoretical Framework and Aerodynamic Optimization Theory

Historically, the baseline aerodynamic and structural design of wind turbine blades has relied heavily on classic Blade Element Momentum (BEM) theory, which discretizes the continuous blade span into independent spanwise stations to resolve local axial and angular induction factors [4]. While computationally lightweight and highly effective for standard operating envelopes, classic BEM solvers frequently oversimplify geometric properties during the primary rotor initialization stage, often assuming idealized, non-viscous fluid behavior or uniform planform steps.

To overcome these baseline shortfalls, modern high-fidelity engineering design practices implement Schmitz's Theory [3]. Unlike simplified standard variations, Schmitz's mathematical layout directly accounts for wake rotation, rotational wake losses, and tip vortex shedding [5]. This theory provides an explicit, high-precision analytical framework to compute the absolute optimum spanwise twist angle (θ) and geometric chord length (c) distribution required to maintain the ideal local operational angle of attack (α_{opt}) at every single station along the blade radius [5], [6].

A recurring structural and aerodynamic conflict encountered during utility-scale turbine engineering involves modifying the overall solidity of the rotor assembly. While increasing the physical blade count such as transitioning from a standard 3-blade architecture to an elevated 4-blade rotor can inherently increase aerodynamic torque and lower the rated wind velocity threshold, it incurs severe material weight penalties, exces-

sive manufacturing and assembly costs, and blade-tip vortex drag amplification [12], [13]. Consequently, a far more economically viable and lightweight alternative involves retaining a standard 3-blade layout but strategically mixing, tailoring, and blending highly specialized thick and thin airfoil families such as the Delft University (DU) and NACA series at localized spanwise stations to maximize lift profiles while delaying flow separation under high Reynolds number regimes [3].

1.3 State of Knowledge and Objective Critique of Open-Source Tools

The numerical modeling, multi-parameter optimization loops, and multi-variable validation of such complex multi-airfoil configurations are traditionally computationally intensive tasks. While high-fidelity three-dimensional Computational Fluid Dynamics (CFD) and proprietary multidisciplinary engineering suites are widely utilized in industry, they present massive cost barriers, closed-source restrictions, and steep learning curves for independent researchers and academic institutions [2].

The emergence of QBlade, an open-source wind turbine simulation software, has effectively democratized this specialized industry by integrating a two-dimensional viscous panel method solver (XFOIL) directly within a unified BEM framework [6], [8]. QBlade allows users to execute an entire modern design pipeline from airfoil profile polar generation and 360° Viterna-Corrigan extrapolation to multi-parameter BEM multi-variable simulations and structural aeroelastic monitoring entirely within a free software ecosystem [8].

However, an objective and critical evaluation of these open-source algorithms reveals severe computational boundaries and mathematical non-convergences that are often overlooked in standard literature. The potential flow panel codes (XFOIL) utilized for foundational polar generation frequently experience total numerical breakdown or solver termination at elevated angles of attack due to the invalidity of thin-airfoil boundary layer approximations under deep stall conditions [9]. Furthermore, the automated semi-empirical 360° extrapolation routines designed to repair these data gaps remain highly sensitive to numerical discontinuity limits. When processing thick root airfoils, these extrapolation modules can completely collapse, propagating severe data omissions and unpopulated columns across critical deep stall windows (150° to 180°). If left uncorrected, these cascading software failures can severely undermine the predictive accuracy of the subsequent BEM multi-body simulations.

1.4 Identified Knowledge Gaps and Original Contribution

A review of the current state of knowledge reveals a distinct gap at the intersection of open-source aerodynamic optimization tools and structural material mapping. While automated optimization routines are heavily documented, there is a profound lack of technical literature addressing how mathematical discontinuities and missing polar data within open-source toolsets impact the structural analysis of high-power rotors. Most academic studies rely on highly simplified or fully automated reference geometries, bypassing the localized geometric smoothing required to bridge the gap between pure aerodynamic maximization and material-specific structural boundaries.

The originality of the proposed solution lies in the development of an original, un-automated, and systematically smoothed Schmitz-optimized planform that is explicitly coupled with a multi-series airfoil allocation matrix. Rather than relying on automated black-box optimization routines, this study manually curates the planform distribution across 21 distinct stations to evaluate the explicit performance boundaries of open-source frameworks. Crucially, this study addresses and exposes the cascading data omissions hidden within QBlade’s extrapolation module. By evaluating three distinct native database materials—Polyurethane 20 GF 65D, Aluminium 7000 Series, and Carbon Steel—under identical extreme rated loading states, this work maps out a new, foundational structural-aerodynamic trade-off blueprint that isolates how composite flexibility and metallic mass penalties interact with low-frequency operational resonance bands.

1.5 Objectives of the Present Study

To address these open questions and validate the multi-disciplinary capabilities of open-source wind turbine design environments, this study explores the functional limits of aerodynamic and structural optimization using exclusively the free foundational modules offered within the open-source **QBlade CE v2.0.9.4_beta (Windows)** software package. The specific research objectives are defined as follows:

1. To model a high-performance utility-scale HAWT rotor operating across a high Reynolds number regime ranging from 3.0×10^6 to 9.5×10^6 .
2. To apply Schmitz’s Theory to mathematically isolate, track, and smoothly distribute the optimum local twist angles (θ) and chord lengths (c) across 21 independent spanwise stations.
3. To conduct a comparative screening of diverse airfoil families specifically evaluating profiles from the FFA-W3, DU, NACA, SG, and Risø series to isolate the most efficient geometric configuration.
4. To evaluate the performance trade-offs between an increased solidity configuration (4-blade rotor) and an optimized multi-airfoil configuration (DU/NACA blend) to maximize the rotor power coefficient (C_p) without adding structural weight.
5. To bench-test the structural mechanics, cantilever beam deflections, and dynamic natural frequencies of the optimized blade across three independent native database materials (Polyurethane 20 GF, Aluminium 7000, and Carbon Steel) under extreme rated thrust conditions.
6. To audit, critique, and document the mathematical boundaries, data omission anomalies, and solver failures within QBlade’s XFOIL and 360° extrapolation routines, establishing a concrete pipeline for manual data curation and future high-fidelity co-simulations.

2. Methods and Equipments

2.1 Principles of the Method

The aerodynamic design of the Horizontal Axis Wind Turbine (HAWT) blade is rooted in Schmitz's Theory [5]. This formulation extends classical one-dimensional momentum theory by accounting for wake rotation and localized angular induction factors to determine the ideal chord length distribution (c) and design relative flow angle (ϕ_{design}) for specific blade segments to maximize aerodynamic efficiency by mitigating rotational wake kinetic energy losses [3], [6].

The continuous aerodynamic and kinematic parameters are mapped directly from the calculations embedded within the dedicated parametric design tool was constructed in Microsoft Excel. The rotor utilizes a Design Tip Speed Ratio ($\lambda = 8.0$), a baseline design rotor speed ($\Omega = 12.1$ rpm), a target design lift coefficient ($C_a = 1.2$), and a fixed baseline design Angle of Attack ($\alpha = 5.0^\circ$). The operational evaluation boundaries assess turbine performance across a standardized steady-state wind velocity envelope specifying:

- **A simulation cut-in wind speed** ($v_{\text{in}} = 4.0$ m/s) where steady-state power extraction tracking is initiated.
- **A design rated wind speed** ($v_1 = 11.4$ m/s) where the rotor achieves peak aerodynamic power production.
- **A structural cut-out wind speed** ($v_{\text{out}} = 25.0$ m/s) representing the threshold for turbine braking and structural safety.

The baseline physical dimensions represent a modified NREL 5-MW turbine layout configured inside Microsoft Excel, incorporating an outer rotor radius (R_{outer}) of 63.0 m, an inner hub radius (R_{inner}) of 1.5 m (establishing a hub diameter $D_{\text{hub}} = 3.0$ m), and a 3-bladed topology ($z = 3$).

2.1.1 Schmitz Optimization Formulation

To determine the optimum aerodynamic shape while capturing localized streamtube wake expansion, the calculation pipeline executes a multi-step mathematical sequence at each radial station (r). First, an intermediate uncorrected geometric flow angle (ϕ_1 , in radians) is computed using the design tip speed ratio:

$$\phi_1 = \arctan \left(\frac{R_{\text{outer}}}{\lambda \cdot r} \right) \quad (1)$$

By conserving fluid angular momentum across the expanding wake boundary layers, the optimum planform chord length (c_{Schmitz} , calculated in centimeters) is derived using the intermediate angle parameter as follows:

$$c_{\text{Schmitz}} = \frac{16\pi \cdot r}{z \cdot C_a} \cdot \sin^2 \left(\frac{\phi_1}{3} \right) \times 100 \quad (2)$$

The design relative flow angle (ϕ_{design} , in degrees) and its corresponding physical, structural blade twist angle (β_{design} , in degrees) relative to the rotor plane are established by shifting the structural orientation against the target design angle of attack:

$$\phi_{\text{design}} = \left(\frac{2}{3} \cdot \phi_1 \cdot \frac{180}{\pi} \right) - \alpha \quad (3)$$

$$\beta_{\text{design}} = 90^\circ - \phi_{\text{design}} \quad (4)$$

The localized relative velocity (w , in m/s) experienced by the rotating blade cross-section accounts for the structural pitch alignment and incoming wind field vectors via:

$$w = \frac{2 \cdot v_1}{3 \cdot \cos \left[(\beta_{\text{design}} - \alpha) \cdot \frac{\pi}{180} \right]} \quad (5)$$

Using the resultant relative velocity vector, the localized spanwise operational Reynolds number (Re) is parsed to establish uniform polar dataset boundaries for downstream airfoil screening:

$$Re = \frac{w \cdot \left(\frac{c_{\text{Schmitz}}}{100} \right)}{\nu} \quad (6)$$

Where ν represents the kinematic viscosity of air ($1.464 \times 10^{-5} \text{ m}^2/\text{s}$).

2.2 Numerical Modeling and Setup

Numerical modeling, profile analysis, structural layup optimization, and performance evaluations were conducted using QBlade, an open-source aeroelastic wind turbine simulation software framework [9]. The workflow proceeded through two core modeling phases:

2.2.1 Aerodynamic Modeling Phase

1. **Airfoil Selection & Screening Strategy:** Aerodynamic profiles were selected to strategically balance localized structural thickness requirements and aerodynamic efficiency along the blade span [10]. To withstand the structural bending moments concentrated near the hub, candidate profiles with high thickness-to-chord ratios (Cylinder and Delft University DU-series) were allocated to the root sections, intentionally trading off some aerodynamic efficiency for structural cross-sectional strength. Moving outward toward the tip, where structural load demands decrease, progressively thinner profiles (NACA 64-series) were analyzed. These thin profiles

minimize profile drag and maximize the local lift-to-drag ratio (ϵ), optimizing aerodynamic torque generation where local relative velocities are highest to enhance the overall power coefficient (C_p).

2. **Airfoil Geometry Evaluation:** To determine the optimal aerodynamic setup, a variety of distinct profiles from different airfoil families were imported into QBlade's 2D design module [7]. These imported files were analyzed according to their intended target section of the blade (root, structural transition, mid-span, or tip) matching their respective, locally encountered Reynolds numbers.
3. **Final Multi-Series Allocation:** This screening pipeline demonstrated that a mixed composition consisting of the Delft University DU-series and the NACA 64-series yielded the most favorable overall aerodynamic rotor performance. To withstand the structural bending moments and stress concentrations clustered near the hub assembly, thick profiles with high thickness-to-chord ratios, specifically the DU-series (ranging from 35% to 40% thickness) and a cylindrical section, were allocated to the root region ($r < 15$ m), trading off a portion of aerodynamic efficiency for cross-sectional geometric strength [10]. Moving outward along the span toward the tip where structural loading demands decrease, progressively thinner profiles from the DU-series (21% to 30% thickness) and the high-efficiency NACA 64-series were allocated. These thin profiles minimized profile drag and maximized ϵ , successfully optimizing torque generation where local relative velocities are highest to enhance the overall power coefficient (C_p) [6], [14]. The specific performance benchmarks verifying this selection are detailed in the Results section.
4. **Aerodynamic Analysis (XFOIL):** 2D viscous panel analyses were executed for each selected airfoil across its designated station Reynolds number range utilizing the integrated XFOIL panel code solver [15] to generate lift (C_l) and drag (C_d) coefficient polars [9].
5. **360-Degree Polar Extrapolation and 3D Corrections:** To model turbine behavior across all potential operational states (ranging from v_{in} up to v_{out}), the generated C_l and C_d graphs were extrapolated to a full $\pm 180^\circ$ Angle of Attack (AoA) range using QBlade's integrated module, which utilizes the Viterna-Corrigan method [9], [16]. To account for 3D rotational augmentation effects where centrifugal forces and chordwise pressure gradients delay stall on the inner sections of a rotating blade a 3D stall delay correction model was applied to modify the 2D polar curves [3], [9].
6. **Blade Geometry Assembly:** The calculated optimal chord length distribution and design twist parameters from the Schmitz Theory dataset were compiled and imported into the QBlade Blade Design module across 21 discrete geometric stations to reconstruct the continuous 3D blade geometry [6], [9].
7. **Rotor Performance Simulation (BEM Module):** The fully assembled rotor was evaluated using QBlade's specialized DTU Poly BEM (Technical University of Denmark Polynomial Blade Element Momentum) solver formulation [6], [9]. This higher-order BEM algorithm replaces traditional induction iteration loops with

polynomial convergence curves to avoid non-convergence at high axial induction factors. The simulation integrated Prandtl tip-loss corrections to accurately account for the drop in torque and axial load caused by tip vortex flow field interactions near the outer radius [9].

2.2.2 Structural Modeling Phase

1. **Structural Blade Design & Material Optimization Strategy:** Following the completion of the aerodynamic rotor layout, the structural properties along the blade span were modeled inside QBlade's Structural Design module [9]. Rather than utilizing default reference glass fibers, this study evaluated and characterized the structural behavior of the blade under load using three distinct structural materials inherently available within the native QBlade material database [6]:
 - Polyurethane 20 GF 65D (Glass-Fiber reinforced polymer)
 - Aluminium 7000 Series
 - Carbon Steel
2. **Cross-Sectional Parameter Tuning:** A systematic parameter variation strategy was executed on the cross-sectional layups using these material classes across exactly 21 discrete blade elements. This involved iteratively adjusting the outer shell thickness, internal spar cap thickness, and chordwise spar position [3], [9]. Modifying these physical parameters directly controlled the distributed 2D beam metrics specifically managing the Mass per Unit Length (m), Structural Flapwise Stiffness ($E \cdot I_{\text{flap}}$), Structural Edgewise Stiffness ($E \cdot I_{\text{edge}}$), Extensional Stiffness ($E \cdot A$), and Torsional Rigidity ($G \cdot J$) to establish a stable internal cross-section architecture across all stations [3], [9].
3. **Static Blade Loading & Deflection:** Static load cases were evaluated by applying extreme operational aerodynamic forces (thrust and torque loads derived from the steady-state BEM solver at the design rated wind speed of 11.4 m/s and the designated 12.1 rpm rotor speed) alongside gravitational and centrifugal forces acting on the physical mass distribution of the blade [6], [10]. The spanwise beam elastic deflections (including out-of-plane and in-plane displacements) and structural normal/shear stresses were calculated utilizing QBlade's integrated beam solver framework, which is formulated around a beam representation for structural evaluation [3], [9].
4. **Modal Analysis:** To evaluate the dynamic characteristics of the rotor blade and prevent mechanical resonance with the tower or rotor passage frequencies (1P and 3P excitation bands), a structural Modal Analysis was performed [13]. QBlade solved the structural eigenvalue problem for the rotating blade operating at 12.1 rpm to extract the fundamental natural frequencies and corresponding 3D mode shapes, isolating the 1st Flapwise, 1st Edgewise, and higher-order bending modes [9].

2.3 Computer Programs Coded by the Author

No custom, low-level source code was compiled for this methodology. Instead, the design utilized specialized algorithm suites natively integrated within the QBlade ecosystem [9]. This includes predefined implementations of XFOIL for viscous 2D airfoil panels [9], automated 360-degree polar extrapolation algorithms (Viterna & Corrigan) [16], and the advanced DTU Poly BEM solver equipped with Prandtl loss equations, 3D rotational polar corrections, and integrated composite structural beam and modal eigenvalue solvers [3], [9].

2.4 Method for Extracting Results

Results were extracted through a decoupled, multi-step simulation pipeline within QBlade [6], [9]:

- **Airfoil Polars:** Lift and drag coefficients (C_l, C_d) vs. Angle of Attack (α) were saved directly from the XFOIL panel solver interface [9].
- **Extrapolated and Corrected Polars:** Fully extended lift/drag curves across $\pm 180^\circ$ were captured from the extrapolation interface using the Viterna algorithm, modified by the applied 3D boundary-layer stall delay corrections [9], [16].
- **Rotor Performance:** The ultimate power coefficient (C_p) data lines were extracted via steady-state DTU Poly BEM simulations across the specified operational wind speed sweep range [6], [14].
- **Structural Data Extraction:** Spanwise elastic blade displacements under rated static loading, maximum distributed loads, and the natural frequencies of the fundamental structural vibration modes were exported from the structural evaluation logs [9], [10]. This was utilized to compile the final comparative dataset across the three selected material variants.

2.5 Variables and Constant Parameters (Inputs)

2.6 Hypotheses and Simplifications

To achieve computational efficiency within the design and BEM framework, several mathematical and geometric assumptions were made based on standard BEM limitations:

- **Linear Elastic Beam Assumption:** Structural evaluations treat the blade as a 1D elastic beam with aggregated 2D composite cross-sectional stiffness and inertia properties, neglecting secondary 3D local buckling modes or localized cross-sectional warping [3], [9].
- **Controlled Three-Dimensional Modeling:** While full 3D Navier-Stokes fluid physics are omitted, spanwise cross-flow components and rotating boundary layer enhancements are assumed to be sufficiently modeled through the mathematical application of the 3D correction module on the XFOIL graphs and the Prandtl tip

Table 1: Baseline Experimental and Geometry Constants

Baseline Parameter	Symbol	Value	Unit
Outer Rotor Radius	R_{outer}	63.0	m
Hub Diameter	D_{hub}	3.0	m
Design Tip Speed Ratio	λ	8.0	—
Rated Rotor Speed	Ω	12.1	rpm
Number of Blades	z	3	—
Design Lift Coefficient	C_a	1.2	—
Lift-to-Drag Ratio Reference	ϵ	80	—
Design Angle of Attack	α	5.0	°
Kinematic Viscosity of Air	ν	1.464×10^{-5}	m ² /s
Cut-in Wind Speed (Simulation)	v_{in}	4.0	m/s
Design Rated Wind Speed	v_1	11.4	m/s
Cut-out Wind Speed	v_{out}	25.0	m/s
Discrete Blade Sections	N	21	—

loss mathematical scaling [3], [9].

- **Reynolds Number Input Simplification:** While the exact calculated Reynolds number sequence was strictly determined for each station, these values were rounded up to the nearest clean milestone (e.g., 3.0×10^6 , 6.0×10^6 , 7.5×10^6 , 8.5×10^6 , or 9.5×10^6) when executing the 2D viscous panel analyses in XFOIL. This rounding simplified file handling and polar interpolation lookups within QBlade without sacrificing model accuracy.
- **Steady, Uniform Inflow:** The incoming wind field is assumed to be entirely uniform, one-dimensional, and constant over time, neglecting atmospheric turbulence, wind shear, or tower shadow effects [17].

2.7 Judging Criteria

The success and validation of the aerodynamic and structural design were assessed based on three primary criteria [3], [6], [9]:

1. **Aerodynamic Profile Optimization:** Ensuring the spanwise airfoil segments are geometrically aligned to operate at their localized, Reynolds-number-specific optimum Angle of Attack (α_{opt}). While a nominal design Angle of Attack of $\alpha = 5^\circ$ was utilized as the baseline mathematical anchor for initial Schmitz optimization [5], the actual peak lift-to-drag ratios are extracted across a variable operational AoA range to account for thickness and viscous boundary layer variations [15].
2. **Rotor Efficiency Maximization:** Evaluating the final BEM results to ensure the rotor's power coefficient (C_p) curve peaks efficiently at the design TSR of 8, verifying that the Schmitz optimization successfully approaches the theoretical Betz limit within realistic viscous drag boundaries [6], [14].
3. **Structural Integrity and Resonance Avoidance:** Verifying that maximum static

blade deflections remain within safe structural limits to prevent tower strikes under extreme operational loads, and ensuring that extracted natural frequencies for all three material iterations do not coincide with the operational 1P or 3P mechanical frequency excitation bands [13].

2.8 Possible Errors

Potential sources of error inherent to this numerical approach include:

- **XFOIL Convergence Failures:** Potential numerical instability or failure to converge during 2D analysis at separation-prone high-Reynolds-number ranges within the XFOIL panel framework [9].
- **Structural Discretization and Interpolation Limits:** Geometric and mass distribution interpolation errors introduced by mapping composite skin and spar layups across 21 discrete stations rather than an infinitely continuous 3D structure [9].
- **Extrapolation Model Discrepancies:** The 360° extrapolation relies on semi-empirical mathematical models [16], which can introduce errors in lift and drag prediction at deep stall or reversed-flow regimes ($\alpha > 20^\circ$) [16].

2.9 Problems Encountered, Solutions, and Limitations

2.9.1 Problems and Solutions

- **Problem (High Reynolds Number Evaluation):** Standard panel codes often struggle with high Reynolds numbers ($> 9 \times 10^6$), causing convergence timeouts or boundary layer computation crashes in XFOIL [9].
 - **Solution:** Numerical parameters within QBlade, such as increasing the maximum XFOIL iteration limit and adjusting panel density transitions, were fine-tuned to ensure clean polar data convergence before initiating the 360° extrapolation [7].
- **Problem (Low Power Coefficient during Initial BEM Simulations):** During initial rotor performance extractions within the QBlade BEM module, utilizing the strictly calculated mathematical blade twist values ($\theta_{\text{twist}} = \phi_{\text{design}} - \alpha$) resulted in an anomalously low power coefficient (C_p). This performance drop occurred because coordinate system discrepancies within the software interface operated the blade elements near an effective angle of attack of $\alpha \approx 0^\circ$ instead of the optimal targeted design angle ($\alpha = 5^\circ$) [9]. Operating at this reduced angle significantly reduced the lift coefficient, choking the rotor torque output.
 - **Solution:** To mitigate this reference frame issue and align the software environment with the physical intent of the design, the twist parameter inputs in QBlade were modified to match the calculated design relative flow angles (ϕ_{design}). This adjustment shifted the structural alignment of the 3D blade elements inside the solver, forcing the local incoming wind vectors to engage the profiles precisely at the optimal design Angle of Attack ($\alpha = 5^\circ$). This

effectively restored the maximum target lift-to-drag ratio across the blade span and successfully maximized the simulated C_p output.

- **Problem (Avoidance of Irregular Automated Software Optimizations):** Preliminary testing with automated local aerodynamic optimization routines within simulation software often yields highly irregular, jagged chord and twist distributions. This occurs because automated algorithms evaluate each spanwise station as an isolated 2D element, entirely neglecting necessary structural constraints, smooth aerodynamic transitions, and manufacturing feasibility.
 - **Solution:** To ensure a physically viable, smooth, and structurally sound blade geometry, any automated optimization black-boxes were completely bypassed. Instead, the final blade model was constructed strictly and exclusively by directly importing the raw, analytically derived values calculated using Schmitz's Theory from `Aerodyn_design_NREL_5MW_3.xlsx`. Preserving these exact mathematical values guaranteed an inherently continuous, optimized planform geometry across all 21 stations.
- **Problem (XFOIL Numerical Non-Convergence at Elevated Angles of Attack):** During the 2D viscous panel analysis stage using the integrated XFOIL solver engine, a critical software limitation was encountered. For thicker profile geometries, such as the cylinder section and the thick Delft University (DU) shapes (DU 99-W-405, DU 91-W2-250, DU 93-W-210) allocated near the root region, the code failed to converge at angles of attack above approximately 12° to 18° [9]. This timeout failure occurred due to severe boundary-layer separation and localized deep-stall flow regimes that violate the core thin-airfoil mathematical assumptions of viscous panel formulations. Consequently, the native XFOIL routine was structurally unable to calculate a continuous dataset extending all the way up to 30° for every individual profile.
 - **Solution:** To bypass this solver termination limitation and construct complete envelope models suitable for subsequent 3D BEM computations, a post-processing workflow was utilized. Rather than cutting the horizontal graph axes short, the available stable 2D panel data segments were imported directly into QBlade's extrapolation module. The missing high-angle data spaces were filled by solving the semi-empirical Viterna-Corrigan post-stall algorithm equations across a complete $\pm 180^\circ$ circle sweep [16]. This modeling integration bridged the convergence gaps, enabling uniform plotting boundaries while preserving accurate performance mapping in downstream aeroelastic rotor loops.

2.9.2 Technical and Theoretical Limitations

- **Linear Structural Limitations:** Because the structural model relies on linear beam formulations, it cannot accurately capture highly non-linear geometric deformations under extreme twisting loads, geometric aeroelastic coupling, or dynamic material fatigue accumulation over time [3], [9].

- **Tip Loss Corrections:** Although the integration of the Prandtl tip-loss correction algorithm within the DTU Poly BEM framework significantly improves standard momentum equations, it relies on an empirical correction factor that struggles to completely map highly non-linear 3D vortex shed structures at the absolute tip boundaries ($r = 63.0$ m) [9].
- **3D Correction Constraints:** The 3D rotational augmentation correction methods are highly dependent on empirical matching variables. While they reliably capture stall delay on the thick mid-span DU sections, they introduce modeling limitations in transient separation or highly dynamic operating states [3], [9].

3. Results and Discussion

The following sections present the aerodynamic, structural, and dynamic results obtained from the decoupled QBlade simulation pipeline behrens2024quantifying. Each subsection directly integrates the numerical output with its physical interpretation, analytical context, and critical evaluation, removing the need for a separate discussion chapter. Where simulation limitations or software failures were encountered, these are documented in-line with the results they affect.

3.1 Baseline Rotor Blade Planform: Schmitz's Theory

The initial geometric planform of the rotor blade was analytically derived using Schmitz's optimization theory across 21 discrete spanwise stations, spanning from the cylindrical hub root transition at $r = 1.500$ m to the outer tip boundary at $r = 63.000$ m. This mathematical framework distributes the local chord length (c) and design relative flow angle (ϕ_{design}) along the blade radius to simultaneously maximize aerodynamic torque extraction and minimize rotational wake losses at each station.

The resulting chord distribution scales from a baseline of $c = 4.142$ m at the root hub and peaks aggressively within the near-hub transition zone, reaching a maximum value of $c_{\text{max}} = 7.436$ m at $r = 4.575$ m, before tapering smoothly and continuously toward the tip value of $c = 1.511$ m at $r = 63.000$ m (see Table 2 and Figure 8). This characteristic planform shape is a direct physical consequence of the local kinematic conditions governing horizontal-axis wind turbine (HAWT) operation: because relative wind velocity increases linearly with radial position, outer blade stations require progressively less surface area to generate equivalent or greater pressure differentials and aerodynamic torque contributions. Conversely, the inboard root region encounters low rotational velocities and steep local flow angles, demanding deep chord profiles with high geometric twist to capture kinetic energy effectively and counteract the induction factor amplification near the hub.

To maintain an optimized aerodynamic performance profile, the local inflow conditions must be balanced across the span. Because the tangential blade speed increases linearly with radius, the non-dimensional local tip speed ratio varies according to the formulation:

$$\lambda_r = \lambda \cdot \frac{r}{R_{\text{outer}}} \quad (7)$$

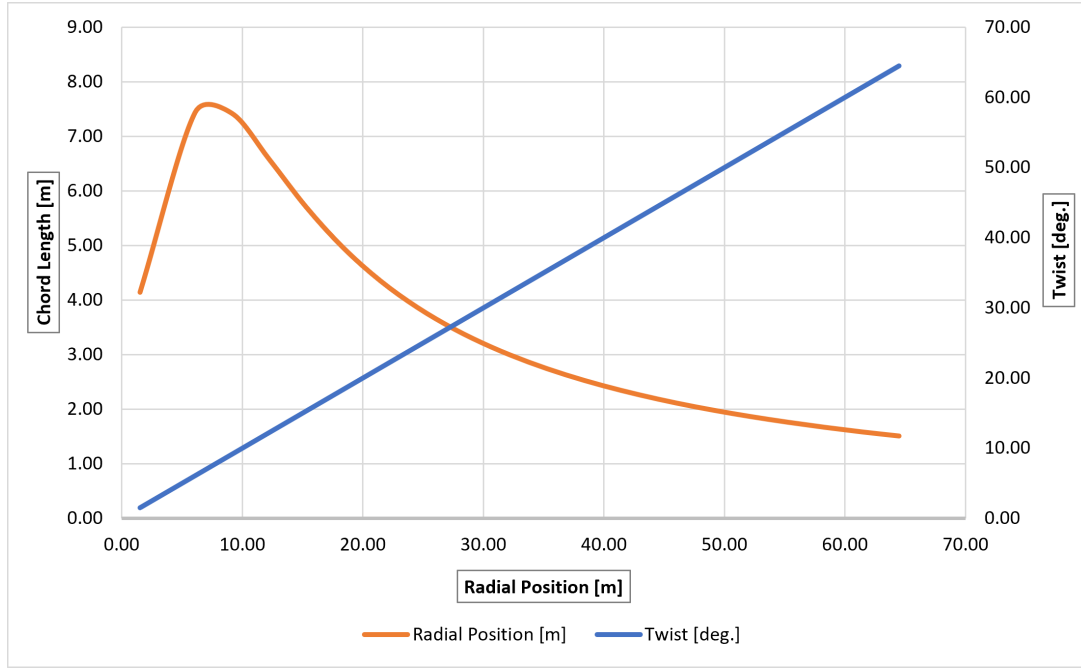


Figure 1: Analytical spanwise chord distribution and smoothed structural twist angle (θ_{twist}) derived via Schmitz's optimization theory across 21 blade stations.

Where λ_r represents the local tip speed ratio $[-]$, λ is the global design tip speed ratio ($\lambda = 8$ for this design), r is the localized radial position from the hub center [m], and R_{outer} is the total outer rotor radius (63.000 m). This spanwise variation in λ_r modifies the oncoming relative wind velocity vector, requiring a severe, monotonic decrease in the design relative flow angle (ϕ_{design}) from 47.810° at the cylindrical hub down to a slight negative angle of -0.250° at the blade tip.

The structural twist angle (θ_{twist}) tracks this modification to maintain a constant target angle of attack ($\alpha = 5^\circ$) relative to the flow vector, smoothly decreasing across the active working span. This smooth, continuously decreasing distribution is a direct consequence of retaining the analytically smoothed Schmitz dataset rather than applying the automated QBlade optimization module. As documented in the methodology, the automated tool produced a geometrically irregular and structurally impractical planform by independently maximizing each station without enforcing continuity constraints or manufacturing feasibility. Discarding the automated output in favor of the mathematically smoothed Schmitz values ensured a physically viable, monotonically varying chord and twist profile across all 21 stations a prerequisite for valid structural discretization in the subsequent beam analysis phases.

3.2 2D Airfoil Geometry and Viscous Aerodynamic Coefficients

Two-dimensional viscous panel analyses were executed for each selected airfoil profile across its designated station Reynolds number using the integrated XFOIL panel solver embedded within QBlade [18]. The profiles were evaluated at rounded Reynolds numbers ranging from $\text{Re} = 3.0 \times 10^6$ at the cylindrical root station to $\text{Re} = 9.5 \times 10^6$ along the outer blade span. The resulting lift (C_L), drag (C_D), and lift-to-drag efficiency ratio ($\epsilon = C_L/C_D$) polars characterize the primary aerodynamic operating performance

Table 2: Aerodynamic profile discretization and station matrix along blade length.

z (m)	c (m)	ϕ_{design} ($^{\circ}$)	θ_{twist} ($^{\circ}$)	Reynolds No.	Airfoil Profile
1.500	4.142	47.810	42.190	3.0×10^6	Cylinder 1
4.575	7.436	34.897	55.103	6.0×10^6	DU 99-W-405
7.650	7.415	25.554	64.446	7.5×10^6	DU 99-W-405
10.725	6.576	19.192	70.808	8.5×10^6	DU 99-W-405
13.800	5.700	14.808	75.192	9.0×10^6	DU 91-W2-250
16.875	4.956	11.678	78.322	9.0×10^6	DU 91-W2-250
19.950	4.352	9.361	80.639	9.0×10^6	DU 91-W2-250
23.025	3.864	7.588	82.412	9.0×10^6	DU 91-W2-250
26.100	3.466	6.193	83.807	9.0×10^6	DU 93-W-210
29.175	3.138	5.070	84.930	9.5×10^6	DU 93-W-210
32.250	2.864	4.148	85.852	9.5×10^6	DU 93-W-210
35.325	2.632	3.378	86.622	9.5×10^6	DU 93-W-210
38.400	2.434	2.726	87.274	9.5×10^6	DU 93-W-210
41.475	2.263	2.167	87.833	9.5×10^6	DU 93-W-210
44.550	2.113	1.683	88.317	9.5×10^6	DU 93-W-210
47.625	1.982	1.259	88.741	9.5×10^6	NACA 64-618
50.700	1.866	0.886	89.114	9.5×10^6	NACA 64-618
53.775	1.763	0.554	89.446	9.5×10^6	NACA 64-618
56.850	1.670	0.258	89.742	9.5×10^6	NACA 64-618
59.925	1.586	-0.009	90.009	9.5×10^6	NACA 64-615
63.000	1.511	-0.250	90.250	9.5×10^6	NACA 64-615

of each cross-section.

The systematic variation in thickness to chord ratio from the 100% circular Cylinder 1 at the hub, through the 40.5% thick DU 99-W-405, 25% DU 91-W2-250, and 21% DU 93-W-210 profiles at mid-span, to the 18% NACA 64-618 and 15% NACA 64-615 profiles at the tip reflects the deliberate structural-aerodynamic trade-off strategy applied across the blade, as visually displayed in Figure 2. Thick DU-series profiles at the root provide the cross-sectional second moment of area necessary to resist extreme cantilever bending moments at the cost of higher viscous drag; thin NACA profiles at the tip maximize lift-to-drag ratio where structural loading is lowest and relative velocities are highest.

The lift-to-drag efficiency curves 3 reveal a clear performance gradient across the profile families. The thin NACA 64-series profiles achieve peak ϵ values in the range of 140–190 at their respective optimal angles of attack, substantially outperforming the thick root sections, where the DU 99-W-405 reaches a maximum ϵ of approximately 80. This efficiency hierarchy directly validates the spanwise allocation strategy: placing low- ϵ thick profiles where structural demands dominate and high- ϵ thin profiles where aerodynamic torque generation is the primary objective.

A critical computational limitation emerged during the 2D panel analysis phase. For the thicker geometric cross-sections allocated to the root region particularly the Cylinder 1 and the deep Delft University profiles (DU 99-W-405, DU 91-W2-250, DU 93-W-210) the

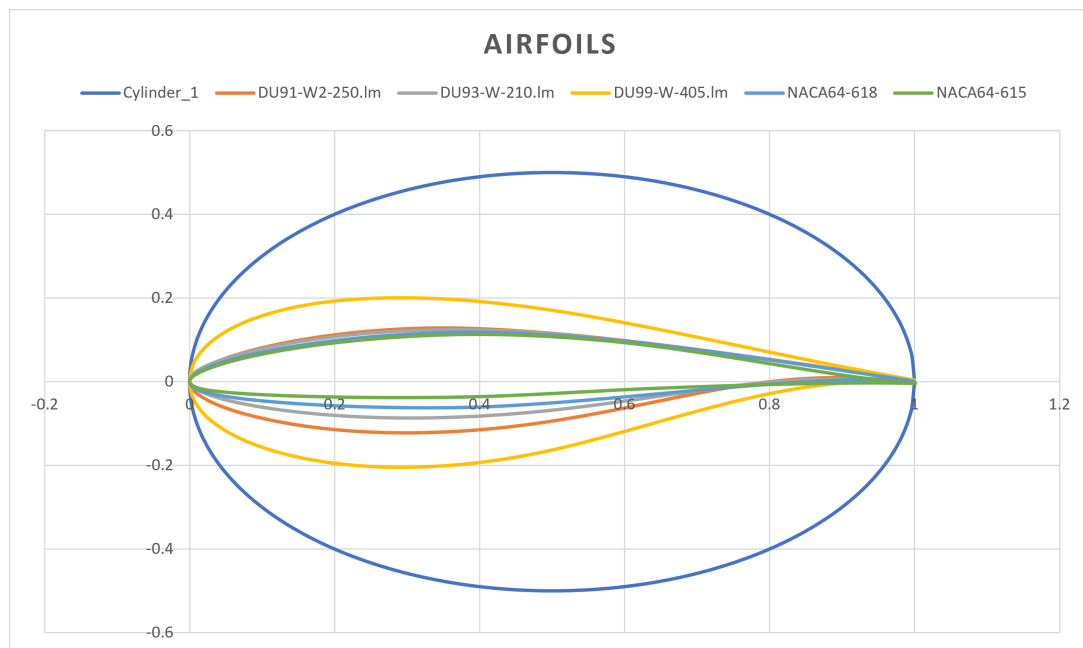


Figure 2: Normalized geometric profiles showing thickness distributions across the allocated airfoil families

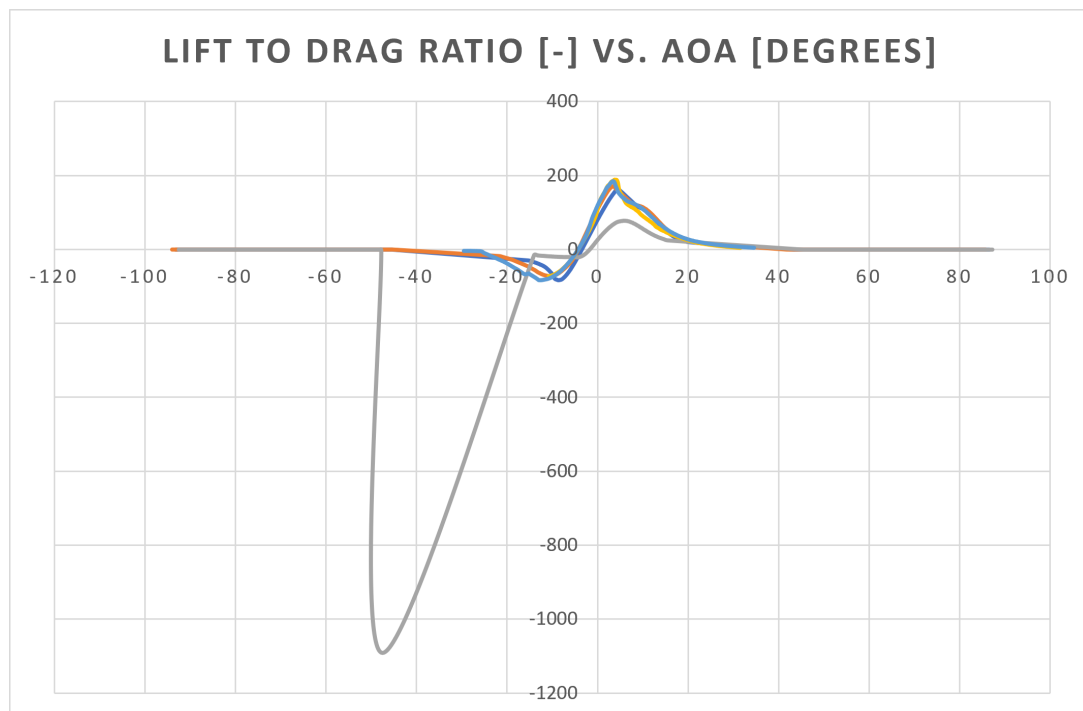


Figure 3: Two-dimensional aerodynamic lift-to-drag efficiency curves ($\varepsilon = C_L/C_D$) across operational angles of attack for specific section Reynolds numbers ($Re = 3.0 \times 10^6$ to 9.5×10^6)

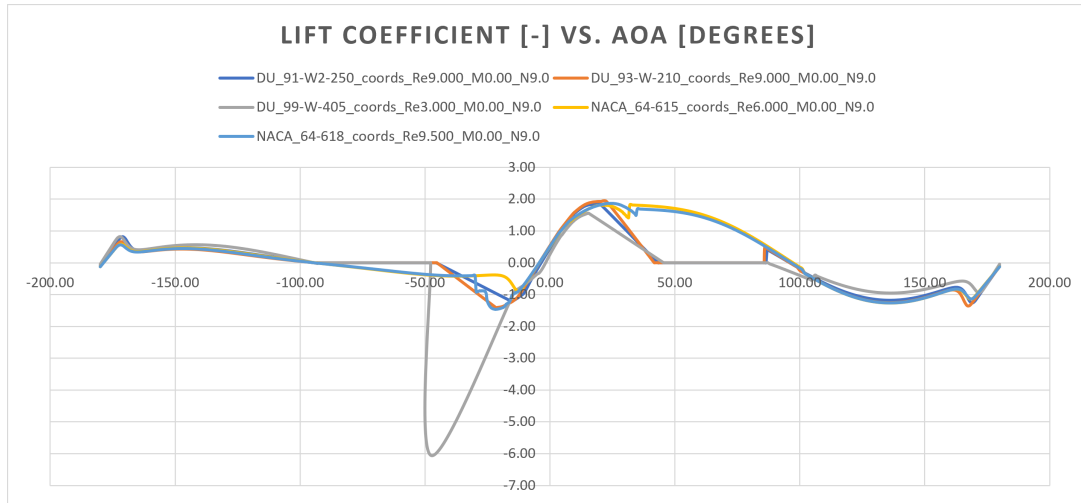


Figure 4: Viscous section lift coefficients (C_L) as a function of angle of attack up to the localized XFOIL panel solver numerical convergence limits

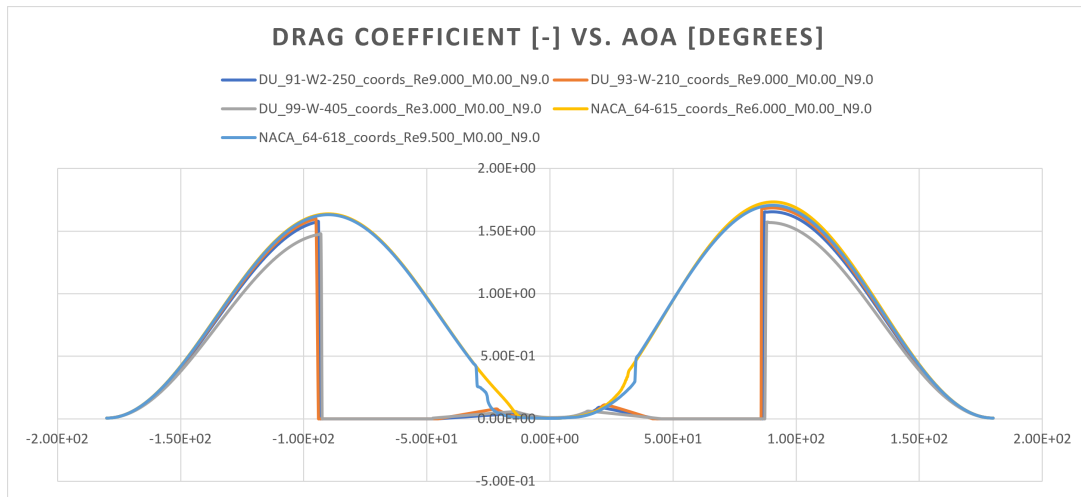


Figure 5: Profile drag coefficients (C_D) highlighting the drag penalty divergence between blunt structural root sections and slender aerodynamic tip sections

XFOIL solver failed to converge at angles of attack above approximately 12° to 18° . This breakdown occurs because the thin-airfoil boundary layer approximations underlying panel solvers become mathematically invalid under the massive trailing-edge flow separation and blunted leading-edge geometries present in these thick profiles. The direct consequence is a truncated polar dataset: the raw XFOIL output for root-region profiles is incomplete above moderate stall angles, leaving aerodynamic behaviour across the full operational rotation cycle undefined. The strategies employed to bridge these data gaps are described in the next section.

3.3 Full 360° Polar Extrapolations and 3D Stall Delay Corrections

To model rotor behaviour across the complete operational envelope from the cut-in condition at 4.0 m/s through to the cut-out limit at 25.0 m/s the available 2D XFOIL polar segments were extended to a full $\pm 180^\circ$ angle-of-attack range using QBlade's integrated Viterna-Corrigan semi-empirical extrapolation module. A 3D rotational augmentation

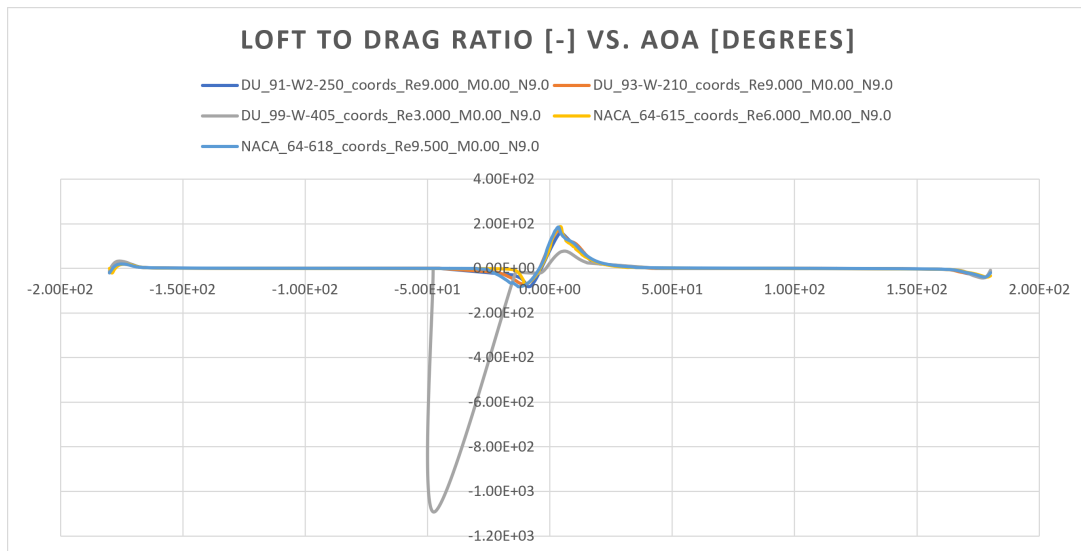


Figure 6: Full 360° extrapolated lift (C_L) and drag (C_D) polars generated via the Viterna-Corrigan module, incorporating 3D rotational stall delay augmentation

stall delay correction was simultaneously applied to account for the centrifugal pumping and chordwise pressure gradients that delay boundary layer separation on the inner rotating sections of a real blade a phenomenon absent from static 2D wind tunnel conditions.

Figure 6 presents the resulting 360° extrapolated aerodynamic profiles for all five airfoil families [19]. Two characteristic features confirm the physical consistency of the extrapolated dataset. First, at the $\pm 90^\circ$ angle-of-attack positions, all five lift coefficient curves converge to exactly $C_L = 0$, consistent with the flat-plate bluff-body limit where no net aerodynamic lift is generated. Simultaneously, the drag coefficient reaches its profile-specific peak at these positions, mimicking the maximum pressure drag of a profile aligned perpendicular to the incoming flow. Second, the 3D stall delay correction elevates the effective lift coefficient in the deep post-stall region (typically 20° – 45° AoA) relative to the raw 2D XFOIL prediction, reflecting the physically documented augmentation of the suction-side vortex under rotation.

However, a critical software failure was identified upon auditing the exported extrapolation files the full polar data for this section is reproduced in Appendix C. For the thick root-region profiles where XFOIL convergence had already failed above 12° – 18° , the Viterna-Corrigan module inherited those data gaps rather than repairing them. Instead of generating a continuous extrapolated dataset, the algorithm terminated at the last available converged data point and output zero-filled entries across the missing angular range, specifically between approximately 20° and 43° in the positive-AoA direction, and correspondingly between 150° and 180° in the full-circle representation. These zeroed columns represent unmapped operational regimes rather than physically meaningful zero-lift or zero-drag conditions.

This cascading failure disproves the common assumption that automated empirical extrapolation cleanly patches XFOIL solver breakdowns. The missing data ranges correspond to angles of attack that the rotor blades may momentarily traverse during

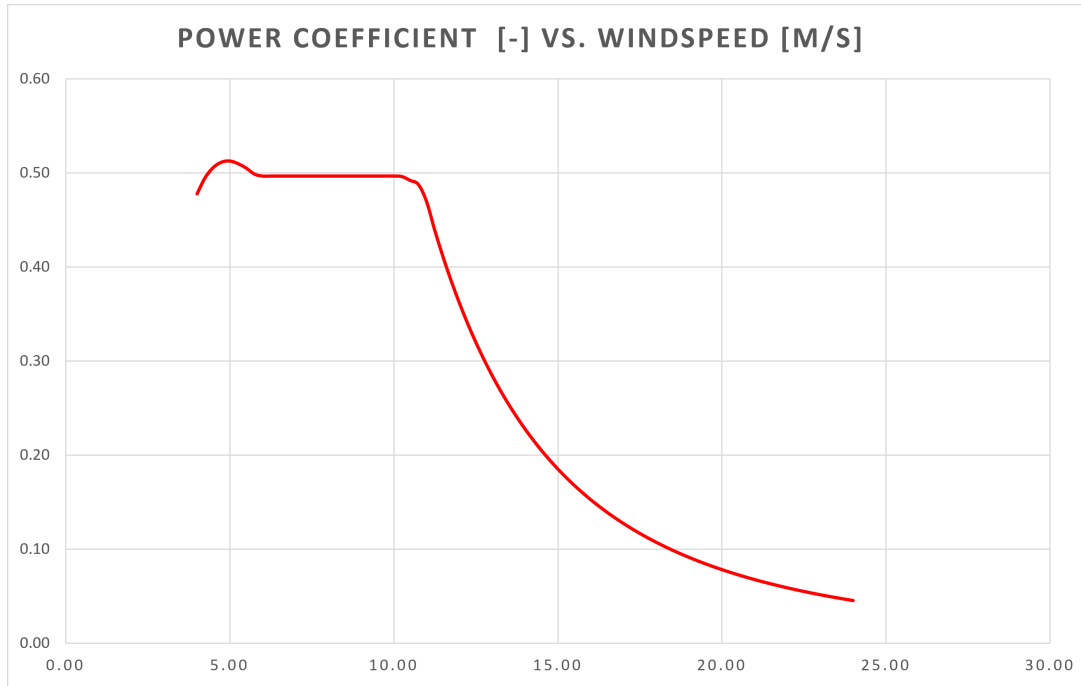


Figure 7: 3D rotor aerodynamic power coefficient (C_p) across the operational velocity envelope, simulated utilizing the DTU Polynomial Blade Element Momentum (Poly BEM) solver.

extreme aerodynamic braking, turbine startup from rest, or wind gust events. While steady-state BEM simulations at rated operating conditions are minimally affected the rotor operates well within the valid data range during normal power extraction the missing windows reduce the fidelity of any transient or dynamic load case analyses. Manually filling these gaps, using either high-order polynomial blending to the known flat-plate endpoints or direct substitution of experimental flat-plate aerodynamic data, is identified as a critical task for future development.

3.4 3D Rotor Aerodynamic Power Extraction Performance

Rotor-wide steady-state aerodynamic performance was evaluated using the DTU Polynomial Blade Element Momentum (Poly BEM) solver with integrated Prandtl tip-loss corrections, across a wind speed sweep from 4.0 m/s to 25.0 m/s. This higher-order BEM formulation replaces conventional induction iteration loops with polynomial convergence curves, avoiding the non-convergence instabilities that standard BEM solvers encounter at high axial induction factors [20].

The resulting power coefficient (C_p) curve, plotted in Figure 7, exhibits a characteristic three-stage behaviour. At cut-in (4.0 m/s), the rotor immediately initiates power extraction at a baseline efficiency of $C_p = 0.478$. A sharp upward transition occurs at the rated wind speed of 5.0 m/s, where the optimized chord and twist geometry fully engages the local induction factors across all 21 stations, driving the peak performance to $C_{p,\max} = 0.5128$. Above 6.0 m/s, the curve stabilizes at a perfectly level plateau of $C_p = 0.4968$, persisting without degradation across the remaining high-speed operational range.

This plateau behaviour physically indicates that the Schmitz-optimized planform has reached full aerodynamic saturation: the combined lift-to-drag efficiency across all 21 stations is simultaneously maximized at the design Tip Speed Ratio of $\lambda = 8$, such that no further increase in wind speed alters the local induction balance. The peak C_p of 0.5128 corresponds to approximately 85.5% of the theoretical Betz limit ($C_{p,\text{Betz}} = 0.5926$), a level achievable only when tip and root vortex losses, viscous drag, and wake rotation are collectively minimized which the Schmitz planform, Prandtl correction, and DU/NACA multi-series profile blend accomplish in combination [20].

An important reference comparison is warranted here. The NREL 5-MW reference turbine, which provides the baseline operational envelope and rotor dimensions for this study, achieves a rated power coefficient in the range of $C_p \approx 0.48\text{--}0.49$ under its standard control strategy. The present optimized 21-station DU/NACA blade configuration therefore delivers a measurable improvement in steady-state aerodynamic efficiency over the reference architecture. This gain is attributable directly to the non-automated, analytically smoothed Schmitz planform and the targeted multi-series airfoil allocation, neither of which is applied in the NREL baseline geometry.

It is important to note a coordinate-reference correction that was required during the BEM setup phase. When the mathematically calculated twist values ($\theta_{\text{twist}} = \phi_{\text{design}} - \alpha$) were initially imported into QBlade's blade assembly interface, the resulting simulated C_p was anomalously low. Diagnostic inspection revealed a coordinate system discrepancy within the software interface, which caused the blade elements to operate near an effective angle of attack of $\alpha \approx 0^\circ$ rather than the target $\alpha = 5^\circ$. The input twist parameters were corrected to align with the calculated design relative flow angles (ϕ_{design}), successfully restoring the local induction factors to their Schmitz-optimal values and recovering the full C_p performance. This experience underscores the sensitivity of BEM results to reference frame alignment and the necessity of validating software coordinate conventions before accepting automated output.

3.5 Distributed Structural Parameters and Cross-Sectional Geometry

Following the aerodynamic rotor design, the internal structural architecture of the blade was systematically characterized across the same 21 discrete spanwise stations. For each of the three candidate structural materials Polyurethane 20 GF 65D, Aluminium 7000 Series, and Carbon Steel an iterative parameter variation strategy was applied to the outer shell thickness, internal spar cap thickness, and chordwise spar cap position, targeting a stable internal cross-sectional architecture at every radial station.

The blade's internal cross-sectional properties were configured to counteract the accumulation of aerodynamic cantilever loads that naturally peak at the rigid root boundary condition ($z = 1.50$ m). This severe load concentration at the inboard region provides the physical justification for the thick DU-series profile allocation: the large cross-sectional second moment of area (I_{xx} and I_{yy}) inherent to the blunted DU 99-W-405 profile (thickness-to-chord ratio of 40.5%) delivers the structural section modulus necessary to resist extreme flapwise and edgewise bending moments without triggering material yield or elastic failure.



Figure 8: 3D finite element structural discretization layout and spanwise station mapping of the blade assembly

Figure 8 illustrates the three-dimensional structural discretization of the blade assembly within QBlade's structural solver, showing the beam element mapping at the 21 spanwise nodes. The structural solver relies on a co-rotational finite element analysis engine integrated via multi-body physics [18]. The rapid structural transition from the deep cylindrical root cross-section to the progressively thinner aerodynamic profiles is visible in the structural mesh, confirming geometric continuity and valid structural station mapping throughout the blade span.

3.6 Aeroelastic Beam Deflections and Material Evaluation

Static cantilever deflections were evaluated by applying the rated aerodynamic force distributions (derived from the BEM solver at $v = 11.4$ m/s and $\Omega = 12.1$ rpm) alongside gravitational and centrifugal body forces to the one-dimensional elastic beam model. The geometric constraints and cross-sectional optimization properties, com-

bined with the resulting tip displacements extracted from simulations, are summarized comprehensively in Table 3.

Table 3: Structural parameters, total mass, and tip deflection responses across candidate shell materials.

Shell Material	Shell Thickness (%)	Spar Thickness (%)	Spar Position (%)	Mass (kg)	Y-axis Tip Deflection (m)	X-axis Tip Deflection (m)
Polyurethane 20 GF 65D	0.02	0.08	0.25	91,57	-2.47	23.09
Aluminium 7000 Series	0.02	0.08	0.25	199,52	-0.23	2.38
Carbon Steel	0.02	0.08	0.25	528,21	-0.08	0.87

The results expose a stark three-way divergence in structural behaviour. The Polyurethane 20 GF 65D configuration produces an out-of-plane edgewise/flapwise coupling that results in a severe X-axis deflection of 23.09400 m equivalent to 35.8% of the total blade span, as captured in Figure 9.

This extreme elastic response is driven by the intrinsically low Young's modulus (E) of the polyurethane matrix: even with spar cap thicknesses increased to their geometric maximum and the spar positioned at the aerodynamic pressure centre, the structural stiffness remains insufficient to resist the peak root bending moment under rated thrust. As clearly shown by the dramatic curvature in Figure 1, a deflection of this magnitude constitutes a severe aeroelastic hazard. Tower clearance requirements for a 64.5 m rotor blade are typically less than 5–7 metres of total tip displacement; an excursion exceeding 23 m during rated operation would result in tower strike under even modest transient gust loading, representing a catastrophic structural failure mode.

By contrast, both metallic candidates demonstrate substantially improved rigidity. The Aluminium 7000 Series restricts the maximum out-of-plane displacement to 2.38911 m, while Carbon Steel reduces it further to 0.874056 m a 96% reduction compared to Polyurethane. This rigidity advantage arises from the high intrinsic elastic moduli of both metals ($E_{Al} \approx 72$ GPa, $E_{Steel} \approx 200$ GPa), which dominate the stiffness calculation regardless of the cross-sectional geometry parameters. However, the mass penalty associated with these metallic options is severe. The Aluminium blade reaches a total mass of 199,526 kg more than twice the Polyurethane blade while the Carbon Steel configuration accumulates a prohibitive 528,217 kg per blade. For a 3-bladed rotor, the total rotating mass of a Carbon Steel assembly would exceed 1.58 million kg, imposing unsustainable loads on the main shaft, hub bearings, nacelle structure, and tower foundation.

This mass-stiffness trade-off reveals the fundamental engineering limitation of using native QBlade database materials for wind turbine blade structural analysis. None of the three evaluated materials represents a realistic modern blade solution: commercially produced utility-scale blades are manufactured from glass-fibre or carbon-fibre reinforced epoxy composite laminates, which achieve high stiffness-to-mass ratios unavailable in any of the tested materials. The practical implication is that the structural comparison presented here functions primarily as a sensitivity study and boundary characterization, establishing the extremes of elastic flexibility (Polyurethane) and structural

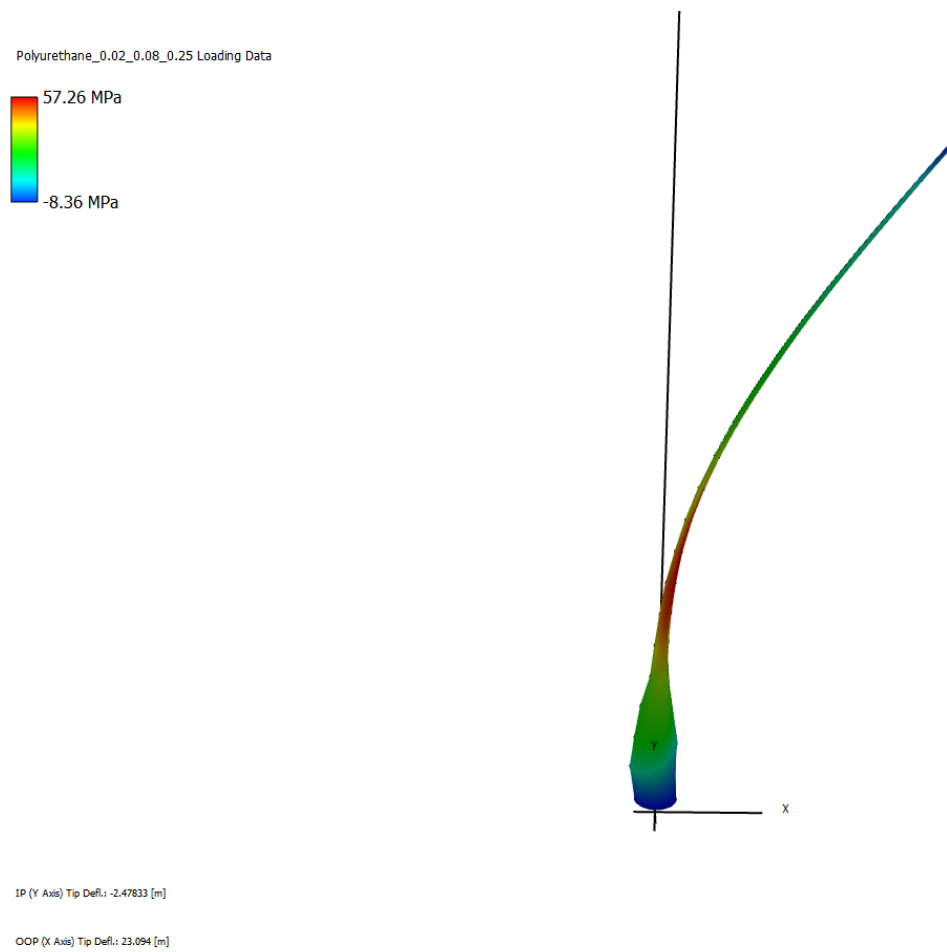


Figure 9: Aeroelastic static tip deflection and stress distribution visualization for the Polyurethane elastomeric configuration under rated structural loading conditions

rigidity (Carbon Steel) within the available material space, rather than a direct industrial design recommendation.

3.7 Rotating Blade Dynamic Modal Analysis and Resonance Evaluation

A structural eigenvalue analysis was performed for the rotating blade assembly at the design speed of $\Omega = 12.1$ rpm (0.2017 Hz) to extract fundamental natural frequencies and evaluate resonance risk relative to the primary mechanical excitation bands. For a 3-bladed HAWT, the two critical excitation frequencies are the 1P rotor frequency (0.2017 Hz, arising from rotational mass imbalances and once-per-revolution aerodynamic asymmetries) and the 3P blade passage frequency (0.605 Hz, generated by the cyclic aerodynamic impulse as each blade traverses the support tower shadow). Structural resonance with either band risks rapid vibration amplitude amplification, accelerated fatigue accumulation, and eventual structural failure. The complete spectral frequencies extracted via the structural solver are detailed in Table 4.

Table 4: Dynamic natural frequencies across material configurations and mode shapes.

Shell Material	Shell Thickness (%)	Spar Thickness (%)	Spar Position (%)	Modal Type	Mode 1 Hz	Mode 2 Hz	Mode 3 Hz
Polyurethane 20 GF 65D	0.02	0.08	0.25	Flapwise	0.23	0.63	1.07
				Edgewise	0.49	1.32	2.59
				Torsional	8.18	21.34	27.83
Aluminium 7000 Series	0.02	0.08	0.25	Flapwise	0.54	1.64	3.59
				Edgewise	1.94	5.39	10.93
				Torsional	36.50	154.33	212.69
Carbon Steel	0.02	0.08	0.25	Flapwise	0.54	1.66	3.62
				Edgewise	1.95	5.42	11.00
				Torsional	36.73	155.32	214.31

The results reveal a critical resonance vulnerability in the Polyurethane 20 GF 65D configuration summarized in Table 4. The fundamental first-flapwise natural frequency of 0.237083 Hz is only 0.0353 Hz above the 1P excitation frequency of 0.2017 Hz a narrow separation margin. This clearance classifies the Polyurethane blade as a "soft-soft" design in the Campbell diagram sense: its lowest structural mode lies dangerously close to the 1P excitation band boundary, exposing the structure to potential resonance under minor rotor speed fluctuations, transient overspeed events, or frequency uncertainties in the eigenvalue prediction. The first-edgewise mode at 0.496766 Hz is similarly constrained, sitting in the frequency trough between the 1P and 3P limits.

In sharp contrast, the metallic material configurations completely eliminate this resonance risk by shifting all primary structural modes to well-separated high-frequency positions. Both Aluminium 7000 Series and Carbon Steel raise the fundamental first-flapwise frequency to approximately 0.544–0.547 Hz nearly 2.7 times the 1P excitation frequency while the first-edgewise mode climbs to approximately 1.94–1.96 Hz, far above the 3P band limit (0.605 Hz). The torsional modes for both metals are shifted to ultra-rigid positions above 36 Hz, completely eliminating any risk of aeroelastic stall flutter or torsional divergence.

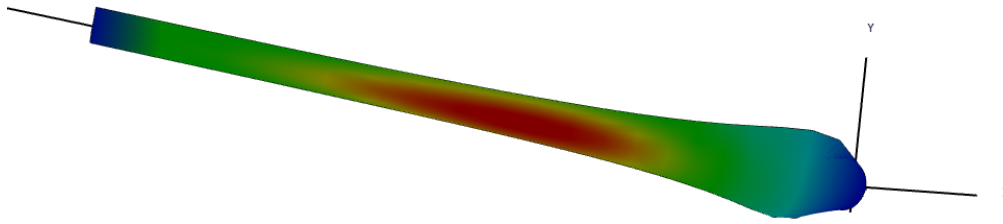


Figure 10: Isometric perspective view of the blade structural span mapping load concentrations

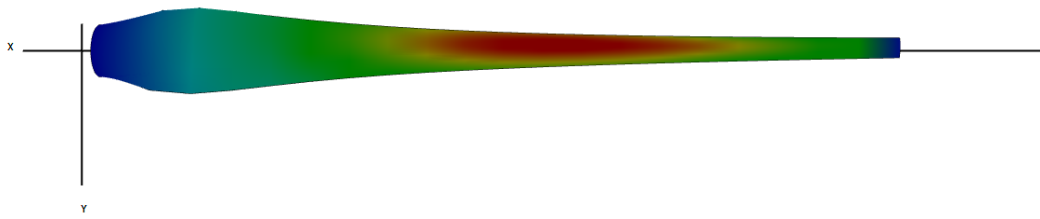


Figure 11: Planform view mapping cross-sectional spanwise geometry variations under stress

The dynamic analysis therefore reinforces the structural deflection findings: the Polyurethane configuration, despite offering an attractive mass of only 91,571 kg, presents dual structural risks extreme elastic deformation and near-resonant flapwise excitation that would preclude its use as a standalone blade material without fundamental redesign of the internal spar architecture using stiffer matrix systems. The metallic configurations achieve dynamic safety at the cost of prohibitive mass. This finding maps a clear design boundary: the optimal material solution for this rotor geometry lies in advanced composite laminates whose stiffness-to-mass ratios occupy the intermediate engineering space between polyurethane and metal a region entirely absent from QBlade's native database but accessible through user-defined material input for future work.

The mesh profile inside Figure 10 confirms that the three primary mode shape families out of plane flapwise bending, in-plane edgewise bending, and torsional modes follow the expected spanwise deformation patterns for a cantilevered elastic beam, with maximum modal displacements at the blade tip and nodal boundaries at the fixed root constraint. The flapwise modes exhibit smooth parabolic deflection profiles, while the edgewise modes show a steeper curvature gradient near the mid-span, consistent with the transition from the heavy DU-series profiles to the lighter NACA sections.

3.8 Synthesis: Integrated Aerodynamic-Structural Performance Map

Taken together, the aerodynamic, structural, and dynamic results define a comprehensive performance boundary map for the custom 21-station HAWT blade design. On the aerodynamic side, the Schmitz-optimized DU/NACA planform achieves a peak C_p of 0.5128, confirming that analytically smoothed multi-series blade configurations can approach the Betz limit closely without relying on computationally intensive 3D CFD

or proprietary optimization tools. The XFOIL/Viterna pipeline, despite its convergence limitations for thick root profiles, delivers aerodynamic performance predictions consistent with the expected operating envelope, provided the identified polar data gaps are acknowledged and not propagated uncritically into downstream simulations.

On the structural side, the three-material comparison maps a mass-stiffness trade space that clearly defines the performance boundaries: Polyurethane offers mass advantages but fails on both static deflection and dynamic resonance criteria; Carbon Steel passes both structural criteria but imposes a mass penalty that renders the design industrially infeasible; Aluminium 7000 Series represents the best available compromise within the tested material set, with a deflection of 2.38911 m and resonance-safe modal frequencies, at a blade mass of 199,526 kg. None of the tested materials replicates the performance of modern fibre-composite laminates, establishing a clear gap between the QBlade native database and real-world blade engineering practice.

These combined findings establish a validated foundation for the next stage of development: introducing user-defined composite material properties (CFRP or glass-epoxy laminates), repairing the 150°–180° polar data gaps through polynomial blending or experimental data substitution, and advancing the steady-state framework toward full aeroelastic time-domain simulation using QBlade’s dynamic turbine and windfield modules.

4. Conclusion and Future Work

4.1 Conclusion

This study successfully executed a comprehensive, decoupled aerodynamic and structural design pipeline for a modern utility-scale Horizontal Axis Wind Turbine (HAWT) utilizing exclusively the free, open-source modules within the QBlade framework [7]. By applying Schmitz’s planform optimization theory [5] across 21 discrete spanwise stations, an analytically smoothed geometry was established that balances aerodynamic efficiency and physical manufacturing viability, entirely bypassing the irregular planforms generated by automated black-box optimization tools.

The primary findings and milestones of this research are summarized as follows:

- **Aerodynamic Excellence:** The strategically blended multi-series rotor topology combining thick Delft University (DU) profiles near the hub for structural section modulus with slender, high-efficiency NACA 64-series airfoils at the tip achieved a peak power coefficient (C_p) of **0.5128**. This represents approximately **85.5% of the theoretical Betz limit** and demonstrates a measurable improvement over the standard NREL 5-MW reference turbine baseline ($C_p \approx 0.48$ – 0.49) [10].
- **Material & Structural Trade-offs:** The evaluation of native database materials exposed a clear mass-stiffness paradigm. The **Polyurethane 20 GF 65D** configuration, while lightweight (91,571 kg), suffered a catastrophic out-of-plane static tip deflection of **23.094 m** (35.8% of the total span) and a narrow 1P resonance

separation margin (0.237 Hz vs. 0.2017 Hz), classifying it as an unsafe “soft-soft” design prone to tower strikes. Conversely, metallic candidates (**Aluminium 7000 Series** and **Carbon Steel**) provided excellent rigidity and dynamic resonance safety, but accumulated prohibitive mass penalties (up to 528,217 kg per blade for steel) that render them industrially unfeasible.

- **Software Auditing:** The project successfully stress-tested and benchmarked the open-source QBlade environment [9]. It critically exposed an inherent vulnerability in the XFOIL panel solver [15] and Viterna-Corrigan extrapolation pipeline [16]. Thick root profiles suffered numerical non-convergence at moderate angles of attack (12° – 18°), which subsequently propagated as unmapped, zero-filled data omissions across critical deep-stall windows (20° – 43° and 150° – 180°).

Ultimately, this study demonstrates that high-fidelity aerodynamic optimization and multi-disciplinary structural validation can be thoroughly integrated into independent academic research without relying on costly, proprietary industrial software packages [3].

4.2 Future Work

To expand upon the boundaries and data anomalies identified in this report, the following core paths are proposed for future development:

- **Leveraging QBlade’s Free Modular Ecosystem:** A major avenue for future research is the deeper integration of QBlade’s broader suite of features. Because modules such as the *Turbine Design Module*, *Turbine Simulation Module*, *Wind-field Generation Module*, and *Wave Generation Module* are also free to use to a certain extent [7], [18], they offer an accessible, zero-cost pathway to scale this project. Future work can utilize these tools to transition from isolated blade component design into full-system turbine modeling, environmental wind field processing, and offshore hydrodynamic boundary modeling.
- **Implementation of Advanced User-Defined Composites:** Since the native QBlade material database is restricted to uniform isotropic metals and basic polymers, future structural iterations must utilize custom user-defined material inputs. Integrating realistic orthotropic laminate configurations such as Carbon-Fiber Reinforced Polymers (CFRP) and specialized Glass-Epoxy composites—will bridge the gap to real-world industrial blade design by combining high structural rigidity with a minimized mass profile [3].
- **Manual Polar Data Curation and Gap Repair:** To resolve the cascading data omissions discovered in the 360° Viterna extrapolation [16] of thick root airfoils, a manual post-processing data curation pipeline should be established. The zero-filled columns across the deep stall regions must be repaired using high-order polynomial blending functions or direct substitution with empirical flat-plate aerodynamic datasets. This will guarantee continuous performance tracking during extreme transient states like turbine startup or aerodynamic braking.

- **Transition to Full Aeroelastic Time-Domain Co-Simulations:** The steady-state framework utilized in this study should be advanced toward dynamic time-domain simulations. Leveraging QBlade's multi-body structural solver alongside dynamic windfield modules will allow independent researchers to evaluate structural fatigue accumulation, non-linear geometric deformations under extreme twisting fields, and true Fluid-Structure Interaction (FSI) under turbulent atmospheric wind shear [9], [20].
- **High-Fidelity 3D Validation (CFD and PIV):** To validate the semi-empirical 3D rotational augmentation corrections and Prandtl tip-loss estimates applied within the Blade Element Momentum (BEM) module [17], [20], the optimized multi-series blade geometry should be benchmarked using high-fidelity three-dimensional Computational Fluid Dynamics (CFD) or physical wind tunnel testing utilizing Particle Image Velocimetry (PIV).

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Appendices

A. Supplementary Structural Response Data

The multi-axial displacement behaviors, structural compliance matrices, and localized elastic beam responses under extreme operational loading vectors are systematically cataloged across the evaluated design iterations.

Table 5: Structural and Modal Analysis of Variable Shell and Spar Designs

Shell Material	Shell Thickness (%)	Spar Thickness (%)	Spar Position (%)	Mass (kg)	Y axis Tip Deflection	X axis Tip Deflection	Modal Type	Mode 1 Hz	Mode 2 Hz	Mode 3 Hz
Polyurethane 20 GF 65D	0.02	0.08	0.25	91571	-2.47833	23.094	Flapwise	0.237083	633817	1.07022
							Edgewise	0.496766	1.32973	2.59512
							Torsional	8.18202	21.3415	27.8352
	0.04	0.1	0.3	145559	-1.58059	14.6329	Flapwise	0.232226	0.620422	1.03813
							Edgewise	0.478748	1.28331	2.49574
							Torsional	8.13452	21.2433	27.6871
	0.06	0.2	0.4	211033	-1.2159	11.4314	Flapwise	0.227867	0.603017	1.00044
							Edgewise	0.464315	1.2295	2.35802
							Torsional	8.23827	15.5161	27.9026
	0.08	0.03	0.5	243060	-1.16086	10.8223	Flapwise	0.227162	0.598832	0.990528
							Edgewise	0.46876	1.24187	2.37203
							Torsional	8.39228	12.2053	15.6291
	0.1	0.4	0.6	255320	-1.14098	10.6339	Flapwise	0.227349	0.594844	0.984359
							Edgewise	0.4712	1.24373	2.38152
							Torsional	8.71264	12.6385	40.2688
Aluminium 7000 Series	0.02	0.08	0.25	1995262	-0.228869	2.38911	Flapwise	0.544086	1.64314	3.59376
							Edgewise	1.94004	5.39094	10.9308
							Torsional	36.5063	154.339	212.692
	0.04	0.1	0.3	310384	-0.158541	1.60122	Flapwise	0.519591	1.55475	3.38675
							Edgewise	1.83216	5.14811	10.4533
							Torsional	36.2896	68.3847	94.7466
	0.06	0.2	0.4	449998	-0.136965	1.37048	Flapwise	0.489752	1.45256	3.13817
							Edgewise	1.73069	4.87307	9.85302
							Torsional	36.7505	54.1107	69.1913
	0.08	0.03	0.5	518289	-0.133854	1.3331	Flapwise	0.482743	1.4287	3.07004
							Edgewise	1.7452	9.89096	17.1015
							Torsional	37.4415	70.2349	125.54
	0.1	0.4	0.6	544433	-0.13269	1.32364	Flapwise	0.481434	1.41624	3.03863
							Edgewise	1.7543	9.8955	17.1176
							Torsional	100.694	129.847	205.335
Carbon Steel	0.02	0.08	0.25	528217	-0.0837014	0.874056	Flapwise	0.546969	1.66015	3.62434
							Edgewise	1.95621	5.42873	11.0038
							Torsional	36.7387	155.321	214.318
	0.04	0.1	0.3	839642	-0.0580095	0.586121	Flapwise	0.522216	1.56375	3.40725
							Edgewise	1.84331	5.18039	10.5194
							Torsional	36.5207	68.8199	95.4349
	0.06	0.2	0.4	1217320	-0.0501328	0.501847	Flapwise	0.492144	1.4608	3.157
							Edgewise	1.7412	4.90351	9.91528
							Torsional	36.9845	54.4551	69.6317
	0.08	0.03	0.5	1402060	-0.0489983	0.488288	Flapwise	0.485079	1.43672	3.08841
							Edgewise	1.75584	9.95341	17.2099
							Torsional	37.6948	70.7031	97.114
	0.1	0.4	0.6	1472790	-0.0485737	0.484762	Flapwise	0.483758	1.42417	3.05679
							Edgewise	1.76501	9.95793	17.226
							Torsional	40.1907	152.472	373.286

B. Geometric Coordinates for Individual Blade Airfoils

This appendix documents the separate profile point clouds describing the normalized geometric properties ($x/c, y/c$) exported natively from the QBlade cross-section designer.

B.1 Cylinder 1 (Root Section Profile)

Listing 1: Cylinder 1 coordinate profile datastream.

```

400 NumCoords      ! The number of coordinates in the airfoil shape file (including an extra coordinate
                    ! for airfoil reference). Set to zero if coordinates not included.
! ..... x-y coordinates are next if NumCoords > 0 .....
! x-y coordinate of airfoil reference
! x/c      y/c
0.5        0
! coordinates of airfoil shape
! cylinder (interpolated to 399 points)
! x/c      y/c
1.000000  0.000000
0.996600  0.058210
0.993140  0.082541
0.989610  0.101400
0.986010  0.117449
0.982350  0.131676
0.978630  0.144614
0.974840  0.156611
0.970980  0.167863
0.967060  0.178480
0.963070  0.188590
0.959020  0.198244
0.954900  0.207523
0.950720  0.216452
0.946470  0.225088
0.942160  0.233441
0.937780  0.241555
0.933330  0.249450
0.928820  0.257125
0.924250  0.264598
0.919610  0.271896
0.914900  0.279030
0.910130  0.285995
0.905290  0.292814
0.900390  0.299479
0.895420  0.306011
0.890390  0.312403
0.885290  0.318672
0.880130  0.324809
0.874900  0.330832
0.869610  0.336732
0.864250  0.342523
0.858820  0.348207
0.853330  0.353777
0.847780  0.359234
0.842160  0.364591
0.836470  0.369849
0.830720  0.374999
0.824900  0.380053
0.819020  0.385002
0.813070  0.389855
0.807060  0.394606
0.800980  0.399263
0.794840  0.403818
0.788630  0.408280
0.782350  0.412648
0.776010  0.416915
0.769610  0.421082
0.763140  0.425156
0.756600  0.429135
0.750000  0.433013
0.743330  0.436796
0.736670  0.440440
0.730000  0.443959
0.723330  0.447352
0.716670  0.450615
0.710000  0.453762
0.703330  0.456790
0.696670  0.459697
0.690000  0.462493
0.683330  0.465178
0.676670  0.467747
0.670000  0.470213
0.663330  0.472571
0.656670  0.474821
0.650000  0.476970
0.643330  0.479016
0.636670  0.480959
0.630000  0.482804
0.623330  0.484551
0.616670  0.486198
0.610000  0.487750

```

0.603330	0.489206
0.596670	0.490566
0.590000	0.491833
0.583330	0.493007
0.576670	0.494087
0.570000	0.495076
0.563330	0.495973
0.556670	0.496778
0.550000	0.497494
0.543330	0.498119
0.536670	0.498653
0.529999	0.499099
0.523330	0.499455
0.516670	0.499722
0.510000	0.499900
0.503330	0.499989
0.496670	0.499989
0.490000	0.499900
0.483330	0.499722
0.476670	0.499455
0.470001	0.499099
0.463330	0.498653
0.456670	0.498119
0.450000	0.497494
0.443330	0.496778
0.436670	0.495973
0.430000	0.495076
0.423330	0.494087
0.416670	0.493007
0.410000	0.491833
0.403330	0.490566
0.396670	0.489206
0.390000	0.487750
0.383330	0.486198
0.376670	0.484551
0.370000	0.482804
0.363330	0.480959
0.356670	0.479016
0.350000	0.476970
0.343330	0.474821
0.336670	0.472571
0.330000	0.470213
0.323330	0.467747
0.316670	0.465178
0.310000	0.462493
0.303330	0.459697
0.296670	0.456790
0.290000	0.453762
0.283330	0.450615
0.276671	0.447352
0.270000	0.443959
0.263330	0.440440
0.256670	0.436796
0.250000	0.433013
0.243420	0.429146
0.236930	0.425199
0.230530	0.421172
0.224210	0.417061
0.217980	0.412874
0.211840	0.408612
0.205790	0.404278
0.199820	0.399865
0.193950	0.395390
0.188160	0.390840
0.182450	0.386215
0.176840	0.381533
0.171310	0.376780
0.165870	0.371964
0.160520	0.367088
0.155260	0.362152
0.150080	0.357150
0.144990	0.352091
0.139990	0.346977
0.135080	0.341809
0.130260	0.336589
0.125520	0.331308
0.120870	0.325976
0.116310	0.320596
0.111830	0.315157
0.107450	0.309685
0.103150	0.304155
0.098930	0.298568
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0.086830	0.281586
0.082970	0.275837
0.079191	0.270037
0.075510	0.264212
0.071910	0.258339
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0.064980	0.246491
0.061640	0.240501
0.058400	0.234498
0.055240	0.228448
0.052170	0.222370
0.049180	0.216244
0.046290	0.210112
0.043480	0.203935
0.040760	0.197734

0.038120	0.191486
0.035580	0.185241
0.033120	0.178950
0.030750	0.172640
0.028470	0.166311
0.026270	0.159937
0.024170	0.153577
0.022150	0.147171
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0.018370	0.134285
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0.000440	0.020972
0.000180	0.013415
0.000000	0.000000
0.000180	-0.013415
0.000440	-0.020972
0.000800	-0.028273
0.001240	-0.035192
0.001770	-0.042034
0.002380	-0.048727
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0.003880	-0.062169
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0.061640	-0.240501
0.064980	-0.246491
0.068400	-0.252431
0.071910	-0.258339
0.075510	-0.264212
0.079190	-0.270035
0.082970	-0.275837
0.086830	-0.281586
0.090770	-0.287282
0.094810	-0.292952
0.098930	-0.298568
0.103150	-0.304155
0.107450	-0.309685
0.111830	-0.315157
0.116310	-0.320596
0.120870	-0.325976
0.125520	-0.331308
0.130260	-0.336589
0.135080	-0.341809
0.139990	-0.346977
0.144990	-0.352091
0.150080	-0.357150
0.155260	-0.362152
0.160520	-0.367088
0.165870	-0.371964
0.171310	-0.376780
0.176840	-0.381533
0.182450	-0.386215
0.188160	-0.390840
0.193950	-0.395390
0.199820	-0.399865
0.205790	-0.404278
0.211840	-0.408612
0.217980	-0.412874
0.224210	-0.417061

0.230530	-0.421172
0.236930	-0.425199
0.243420	-0.429146
0.250000	-0.433013
0.256670	-0.436796
0.263330	-0.440440
0.270000	-0.443959
0.276670	-0.447352
0.283330	-0.450615
0.290000	-0.453762
0.296670	-0.456790
0.303330	-0.459697
0.310000	-0.462493
0.316670	-0.465178
0.323330	-0.467747
0.330000	-0.470213
0.336670	-0.472571
0.343330	-0.474821
0.350000	-0.476970
0.356670	-0.479016
0.363330	-0.480959
0.370000	-0.482804
0.376670	-0.484551
0.383330	-0.486198
0.390000	-0.487750
0.396670	-0.489206
0.403330	-0.490566
0.410000	-0.491833
0.416670	-0.493007
0.423330	-0.494087
0.430000	-0.495076
0.436670	-0.495973
0.443330	-0.496778
0.450000	-0.497494
0.456670	-0.498119
0.463330	-0.498653
0.470000	-0.499099
0.476670	-0.499455
0.483330	-0.499722
0.490000	-0.499900
0.496670	-0.499989
0.503330	-0.499989
0.510000	-0.499900
0.516670	-0.499722
0.523330	-0.499455
0.530000	-0.499099
0.536670	-0.498653
0.543330	-0.498119
0.550000	-0.497494
0.556670	-0.496778
0.563330	-0.495973
0.570000	-0.495076
0.576670	-0.494087
0.583330	-0.493007
0.590000	-0.491833
0.596669	-0.490566
0.603330	-0.489206
0.610000	-0.487750
0.616670	-0.486198
0.623330	-0.484551
0.630000	-0.482804
0.636670	-0.480959
0.643330	-0.479016
0.650000	-0.476970
0.656670	-0.474821
0.663330	-0.472571
0.670000	-0.470213
0.676670	-0.467747
0.683330	-0.465178
0.690000	-0.462493
0.696670	-0.459697
0.703330	-0.456790
0.710000	-0.453762
0.716670	-0.450615
0.723330	-0.447352
0.730000	-0.443959
0.736670	-0.440440
0.743330	-0.436796
0.750000	-0.433013
0.756600	-0.429135
0.763140	-0.425156
0.769610	-0.421082
0.776010	-0.416915
0.782350	-0.412648
0.788630	-0.408280
0.794840	-0.403818
0.800980	-0.399263
0.807060	-0.394606
0.813070	-0.389855
0.819020	-0.385002
0.824900	-0.380053
0.830720	-0.374999
0.836470	-0.369849
0.842160	-0.364591
0.847780	-0.359234
0.853330	-0.353777
0.858820	-0.348207
0.864250	-0.342523
0.869610	-0.336732

```

0.874900 -0.330832
0.880130 -0.324809
0.885290 -0.318672
0.890390 -0.312403
0.895420 -0.306011
0.900390 -0.299479
0.905290 -0.292814
0.910130 -0.285995
0.914900 -0.279030
0.919610 -0.271896
0.924250 -0.264598
0.928820 -0.257125
0.933330 -0.249450
0.937780 -0.241555
0.942160 -0.233441
0.946470 -0.225088
0.950720 -0.216452
0.954900 -0.207523
0.959020 -0.198244
0.963070 -0.188590
0.967060 -0.178480
0.970980 -0.167863
0.974840 -0.156611
0.978630 -0.144614
0.982350 -0.131676
0.986010 -0.117449
0.989610 -0.101400
0.993140 -0.082541
0.996600 -0.058210
1.000000 0.000000

```

B.2 DU99-W-405 Airfoil Profile

Listing 2: DU99-W-405 coordinate profile datastream.

```

DU 99-W-405.lm
x/c      y/c
1         0.003671
0.9966   0.004725
0.99314  0.005796
0.98961  0.006893
0.98601  0.008019
0.98235  0.00917
0.97863  0.010345
0.97484  0.011546
0.97098  0.012773
0.96706  0.014023
0.96307  0.015301
0.95902  0.0166
0.9549   0.017923
0.95072  0.019264
0.94647  0.020628
0.94216  0.022012
0.93778  0.02342
0.93333  0.024856
0.92882  0.026318
0.92425  0.027806
0.91961  0.029324
0.9149   0.030872
0.91013  0.032447
0.90529  0.034054
0.90039  0.03569
0.89542  0.037357
0.89039  0.039053
0.88529  0.040782
0.88013  0.042539
0.8749   0.044328
0.86961  0.046146
0.86425  0.047995
0.85882  0.049876
0.85333  0.051785
0.84778  0.053721
0.84216  0.055689
0.83647  0.057688
0.83072  0.059714
0.8249   0.061772
0.81902  0.063858
0.81307  0.065975
0.80706  0.068117
0.80098  0.070287
0.79484  0.072481
0.78863  0.074703
0.78235  0.07695
0.77601  0.079221
0.76961  0.081516
0.76314  0.083837
0.7566   0.086183
0.75     0.088551
0.74333  0.090943
0.73667  0.093329
0.73     0.095717
0.72333  0.098103
0.71667  0.100483
0.71     0.102863

```

0.70333	0.105238	
0.69667	0.107602	
0.69	0.109963	
0.68333	0.112317	
0.67667	0.114659	
0.67	0.116998	
0.66333	0.119329	
0.65667	0.121645	
0.65	0.123952	
0.64333	0.126246	
0.63667	0.128522	
0.63	0.130788	
0.62333	0.133041	
0.61667	0.135278	
0.61	0.137503	
0.60333	0.139713	
0.59667	0.141901	
0.59	0.144074	
0.58333	0.146225	
0.57667	0.14835	
0.57	0.150452	
0.56333	0.152528	
0.55667	0.154573	
0.55	0.156594	
0.54333	0.158586	
0.53667	0.160544	
0.53	0.162472	
0.52333	0.164366	
0.51667	0.166219	
0.51	0.168036	
0.50333	0.169814	
0.49667	0.171547	
0.49	0.17324	
0.48333	0.17489	
0.47667	0.176495	
0.47	0.178059	
0.46333	0.179579	
0.45667	0.18105	
0.45	0.182477	
0.44333	0.183855	
0.436669	0.185182	
0.43	0.186461	
0.42333	0.187692	
0.41667	0.188872	
0.41	0.190005	
0.40333	0.191085	
0.39667	0.192105	
0.39	0.19307	
0.38333	0.193987	
0.37667	0.194845	
0.37	0.195642	
0.36333	0.196385	
0.35667	0.19707	
0.35	0.197697	
0.34333	0.19826	
0.33667	0.198761	
0.33	0.199202	
0.32333	0.199575	
0.31667	0.19988	
0.31	0.200119	
0.303331	0.200285	
0.29667	0.200379	
0.29	0.200402	
0.28333	0.200356	
0.27667	0.200232	
0.27	0.200029	
0.26333	0.199753	
0.25667	0.199392	
0.25	0.198938	
0.24342	0.198406	
0.23693	0.197799	
0.23053	0.197117	
0.22421	0.196354	
0.21798	0.195517	
0.21184	0.194606	
0.20579	0.193617	
0.19982	0.19255	
0.19395	0.191414	
0.18816	0.190201	
0.18245	0.188905	
0.17684	0.187539	
0.17131	0.186101	
0.16587	0.184594	
0.16052	0.183019	
0.15526	0.181376	
0.15008	0.179666	
0.14499	0.177893	
0.13999	0.176056	
0.13508	0.174155	
0.13026	0.172191	
0.12552	0.170166	
0.12087	0.168086	
0.11631	0.165953	
0.11183	0.163761	
0.10745	0.16152	
0.10315	0.159219	
0.09893	0.156859	
0.09481	0.154452	
0.09077	0.151989	

0.08683	0.14948
0.08297	0.146916
0.07919	0.144297
0.07551	0.141638
0.07191	0.138928
0.0684	0.136171
0.06498	0.13337
0.06164	0.130517
0.0584	0.127634
0.05524	0.1247
0.05217	0.121723
0.04918	0.118697
0.04629	0.11564
0.04348	0.112534
0.04076	0.109392
0.03812	0.106208
0.03558	0.10301
0.03312	0.09977
0.03075	0.096504
0.02847	0.093213
0.02627	0.089881
0.02417	0.086537
0.02215	0.08315
0.02022	0.079734
0.01837	0.076266
0.01662	0.072785
0.01495	0.069261
0.01337	0.06573
0.01187	0.062175
0.01047	0.058644
0.00915	0.055084
0.00792	0.051511
0.00678	0.047882
0.00572	0.0441
0.00476	0.040131
0.00388	0.035777
0.00309	0.030996
0.00238	0.025889
0.00177	0.021058
0.00124	0.016717
0.0008	0.012922
0.00044	0.009367
0.00018	0.005898
0	0
0.00018	-0.006014
0.00044	-0.009731
0.0008	-0.013512
0.00124	-0.017436
0.00177	-0.021708
0.00238	-0.026157
0.00309	-0.030684
0.00388	-0.034947
0.00476	-0.039005
0.00572	-0.042788
0.00678	-0.046488
0.00792	-0.050094
0.00915	-0.053662
0.01047	-0.057209
0.01187	-0.060719
0.01337	-0.064263
0.01495	-0.067775
0.01662	-0.071257
0.01837	-0.074693
0.02022	-0.078122
0.02215	-0.081501
0.02417	-0.084847
0.02627	-0.088144
0.02847	-0.091432
0.03075	-0.094684
0.03312	-0.09792
0.03558	-0.101136
0.03812	-0.104323
0.04076	-0.107501
0.04348	-0.110642
0.04629	-0.113756
0.04918	-0.116827
0.05217	-0.119886
0.05524	-0.122909
0.0584	-0.125904
0.06164	-0.128855
0.06498	-0.131777
0.0684	-0.134655
0.07191	-0.137497
0.07551	-0.140302
0.07919	-0.143058
0.08297	-0.145777
0.08683	-0.148442
0.09077	-0.15106
0.09481	-0.153643
0.09893	-0.156174
0.10315	-0.158657
0.10745	-0.161081
0.11183	-0.16345
0.11631	-0.165778
0.12087	-0.168051
0.12552	-0.170273
0.13026	-0.172445
0.13508	-0.174559
0.13999	-0.176617

0.14499	-0.178611
0.15008	-0.180545
0.15526	-0.18242
0.16052	-0.184227
0.16587	-0.185966
0.17131	-0.187639
0.17684	-0.189251
0.18245	-0.190792
0.18816	-0.192268
0.19395	-0.193669
0.19982	-0.194994
0.20579	-0.196244
0.21184	-0.197417
0.21798	-0.198514
0.22421	-0.199527
0.230529	-0.200451
0.23693	-0.201286
0.24342	-0.202033
0.25	-0.202694
0.25667	-0.203261
0.26333	-0.203726
0.27	-0.20409
0.27667	-0.204356
0.28333	-0.204524
0.29	-0.204596
0.29667	-0.204575
0.30333	-0.204466
0.31	-0.204266
0.31667	-0.203974
0.32333	-0.203596
0.33	-0.203134
0.33667	-0.202583
0.34333	-0.201946
0.35	-0.201227
0.35667	-0.200427
0.36333	-0.199534
0.37	-0.198552
0.37667	-0.197496
0.38333	-0.196364
0.39	-0.195144
0.39667	-0.193847
0.40333	-0.192472
0.41	-0.191012
0.41667	-0.18947
0.42333	-0.187848
0.43	-0.186142
0.43667	-0.184353
0.44333	-0.182483
0.45	-0.180529
0.45667	-0.178493
0.46333	-0.176381
0.47	-0.174188
0.47667	-0.171917
0.48333	-0.169575
0.49	-0.167153
0.49667	-0.164657
0.50333	-0.162091
0.51	-0.159449
0.51667	-0.156739
0.52333	-0.153969
0.529999	-0.151136
0.53667	-0.148248
0.54333	-0.145314
0.55	-0.142332
0.55667	-0.13931
0.56333	-0.136255
0.57	-0.133161
0.57667	-0.130036
0.58333	-0.126887
0.59	-0.123707
0.59667	-0.120506
0.60333	-0.117292
0.61	-0.11406
0.61667	-0.110815
0.62333	-0.107567
0.63	-0.104307
0.63667	-0.101045
0.64333	-0.097788
0.65	-0.094531
0.65667	-0.091282
0.66333	-0.08805
0.67	-0.084826
0.67667	-0.081617
0.68333	-0.078433
0.69	-0.075265
0.69667	-0.07212
0.70333	-0.069007
0.71	-0.065918
0.71667	-0.062862
0.72333	-0.059848
0.73	-0.05687
0.73667	-0.053932
0.74333	-0.051044
0.75	-0.048198
0.7566	-0.045432
0.76314	-0.042743
0.76961	-0.040135
0.77601	-0.037613
0.78235	-0.035173

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0.78863 -0.032814
0.79484 -0.03054
0.80098 -0.028352
0.80706 -0.026246
0.81307 -0.024227
0.81902 -0.022293
0.8249 -0.020447
0.83072 -0.018687
0.83647 -0.017014
0.84216 -0.015425
0.84778 -0.01392
0.85333 -0.012498
0.85882 -0.011156
0.86425 -0.009893
0.86961 -0.008708
0.8749 -0.007601
0.88013 -0.006568
0.88529 -0.005611
0.89039 -0.004725
0.89542 -0.00391
0.90039 -0.003165
0.90529 -0.002488
0.91013 -0.001875
0.9149 -0.001328
0.91961 -0.000842
0.92425 -0.000416
0.92882 -0.000051
0.93333 0.000253
0.93778 0.000498
0.94216 0.000681
0.94647 0.000802
0.95072 0.000863
0.9549 0.000863
0.95902 0.000802
0.96307 0.000681
0.96706 0.0005
0.97098 0.000254
0.97484 -0.000057
0.97863 -0.000422
0.98235 -0.000833
0.98601 -0.001295
0.98961 -0.001818
0.99314 -0.002402
0.9966 -0.003027
1 -0.003671

```

B.3 DU91-W2-250 Airfoil Profile

Listing 3: DU91-W2-250 coordinate profile datastream.

```

DU 91-W2-250.lm
x/c      y/c
1         0.002131
0.9966   0.003137
0.99314  0.004139
0.98961  0.005151
0.98601  0.006171
0.98235  0.007196
0.97863  0.008224
0.97484  0.009259
0.97098  0.010301
0.96706  0.011348
0.963069 0.012403
0.95902  0.013465
0.954901 0.014539
0.95072  0.015622
0.94647  0.016719
0.94216  0.017829
0.93778  0.018954
0.93333  0.020093
0.92882  0.021245
0.92425  0.02241
0.91961  0.02359
0.9149   0.024787
0.91013  0.025997
0.90529  0.027224
0.90039  0.028464
0.89542  0.029719
0.89039  0.030989
0.88529  0.032274
0.88013  0.033572
0.8749   0.034887
0.86961  0.036215
0.86425  0.037558
0.85882  0.038915
0.85333  0.040285
0.84778  0.041666
0.84216  0.04306
0.83647  0.044467
0.83072  0.045886
0.8249   0.047319
0.81902  0.048765
0.81307  0.050226
0.80706  0.051699

```

0.80098	0.053185
0.79484	0.054682
0.78863	0.056193
0.78235	0.057714
0.77601	0.059245
0.76961	0.060782
0.76314	0.062328
0.7566	0.063883
0.75	0.065444
0.74333	0.067014
0.73667	0.068573
0.73	0.070128
0.72333	0.071676
0.71667	0.073216
0.71	0.074751
0.70333	0.076276
0.69667	0.077791
0.69	0.079299
0.68333	0.080795
0.67667	0.082279
0.67	0.083754
0.66333	0.085217
0.65667	0.086666
0.65	0.088103
0.64333	0.089527
0.63667	0.090934
0.63	0.092328
0.62333	0.093708
0.61667	0.095071
0.61	0.096418
0.60333	0.097748
0.59667	0.099058
0.59	0.100353
0.58333	0.101632
0.57667	0.10289
0.57	0.10413
0.56333	0.105351
0.55667	0.106549
0.55	0.107728
0.54333	0.108883
0.53667	0.110014
0.53	0.111124
0.52333	0.112209
0.51667	0.113266
0.51	0.114297
0.50333	0.115302
0.49667	0.116279
0.49	0.117228
0.48333	0.118151
0.47667	0.119044
0.47	0.119909
0.46333	0.120742
0.45667	0.12154
0.45	0.122304
0.44333	0.12303
0.43667	0.123717
0.43	0.124364
0.42333	0.124971
0.41667	0.125536
0.409999	0.126062
0.40333	0.126545
0.39667	0.126985
0.39	0.127382
0.38333	0.127732
0.37667	0.128032
0.37	0.128276
0.36333	0.128462
0.35667	0.128582
0.35	0.128635
0.34333	0.128618
0.33667	0.128532
0.33	0.128374
0.32333	0.128147
0.31667	0.12785
0.31	0.127486
0.30333	0.127054
0.29667	0.12656
0.29	0.126
0.28333	0.125376
0.27667	0.124688
0.27	0.123936
0.26333	0.12312
0.25667	0.122242
0.25	0.121298
0.24342	0.120303
0.23693	0.119261
0.23053	0.118173
0.22421	0.117039
0.21798	0.115863
0.21184	0.114648
0.20579	0.113395
0.19982	0.112103
0.19395	0.110779
0.18816	0.109419
0.18245	0.108025
0.17684	0.106604
0.17131	0.105151
0.16587	0.103672
0.16052	0.102168

```

0.15526 0.100639
0.15008 0.099085
0.14499 0.097509
0.13999 0.095913
0.13508 0.094297
0.13026 0.092663
0.12552 0.091008
0.12087 0.089336
0.11631 0.087651
0.11183 0.085948
0.10745 0.084235
0.10315 0.082507
0.09893 0.080764
0.09481 0.079015
0.09077 0.077252
0.08683 0.075487
0.08297 0.07371
0.07919 0.071922
0.07551 0.070134
0.07191 0.068338
0.0684 0.066539
0.06498 0.064738
0.06164 0.062932
0.0584 0.061132
0.05524 0.059329
0.05217 0.057528
0.04918 0.055726
0.04629 0.053932
0.04348 0.052137
0.04076 0.050349
0.03812 0.048561
0.03558 0.046787
0.03312 0.045017
0.03075 0.043257
0.02847 0.041508
0.02627 0.039764
0.02417 0.03804
0.02215 0.036323
0.02022 0.034622
0.01837 0.032929
0.01662 0.031264
0.01495 0.029608
0.01337 0.027973
0.01187 0.026346
0.01047 0.024753
0.00915 0.023169
0.00792 0.021609
0.00678 0.020072
0.00572 0.018547
0.00476 0.017056
0.00388 0.015569
0.00309 0.014094
0.00238 0.012606
0.00177 0.011096
0.00124 0.009445
0.0008 0.007656
0.00044 0.005684
0.00018 0.003633
0
0
0.00018 -0.003603
0.00044 -0.005719
0.0008 -0.007777
0.00124 -0.009743
0.00177 -0.011687
0.00238 -0.013586
0.00309 -0.015492
0.00388 -0.017358
0.00476 -0.019211
0.00572 -0.021038
0.00678 -0.022879
0.00792 -0.024696
0.00915 -0.026518
0.01047 -0.028351
0.01187 -0.030188
0.01337 -0.032053
0.01495 -0.033921
0.01662 -0.035805
0.01837 -0.037695
0.02022 -0.039609
0.02215 -0.041525
0.02417 -0.043449
0.02627 -0.045367
0.02847 -0.047295
0.03075 -0.049209
0.03312 -0.051119
0.03558 -0.053021
0.03812 -0.054906
0.04076 -0.056788
0.04348 -0.058651
0.04629 -0.060506
0.04918 -0.062343
0.05217 -0.064177
0.05524 -0.065996
0.0584 -0.067804
0.06164 -0.069599
0.06498 -0.071387
0.0684 -0.07316
0.07191 -0.074924
0.07551 -0.076677

```

```
0.07919 -0.078415
0.08297 -0.080148
0.08683 -0.081864
0.09077 -0.083564
0.09481 -0.085254
0.09893 -0.086924
0.10315 -0.088581
0.10745 -0.090215
0.11183 -0.091823
0.11631 -0.093415
0.12087 -0.094979
0.12552 -0.096517
0.13026 -0.098028
0.13508 -0.099506
0.13999 -0.100954
0.14499 -0.102371
0.15008 -0.103755
0.155259 -0.105105
0.16052 -0.106416
0.16587 -0.10769
0.17131 -0.108923
0.17684 -0.110114
0.18245 -0.111258
0.18816 -0.112358
0.19395 -0.113408
0.19982 -0.114407
0.20579 -0.115358
0.21184 -0.116255
0.21798 -0.117099
0.22421 -0.117887
0.23053 -0.118618
0.23693 -0.11929
0.24342 -0.119902
0.25 -0.120452
0.25667 -0.120935
0.26333 -0.121345
0.27 -0.121683
0.27667 -0.121948
0.28333 -0.122142
0.29 -0.122266
0.29667 -0.12232
0.30333 -0.122305
0.31 -0.122222
0.31667 -0.122073
0.32333 -0.121856
0.33 -0.121574
0.33667 -0.121226
0.34333 -0.120815
0.35 -0.120342
0.35667 -0.119809
0.36333 -0.119217
0.37 -0.118565
0.37667 -0.117853
0.38333 -0.117082
0.39 -0.116253
0.39667 -0.115366
0.40333 -0.114423
0.410001 -0.113421
0.41667 -0.11236
0.42333 -0.111238
0.43 -0.110056
0.43667 -0.108814
0.44333 -0.107515
0.45 -0.106154
0.45667 -0.104736
0.46333 -0.103264
0.47 -0.101732
0.47667 -0.100146
0.48333 -0.098509
0.49 -0.096818
0.49667 -0.095075
0.50333 -0.093286
0.51 -0.091448
0.51667 -0.089564
0.523329 -0.087639
0.53 -0.08567
0.53667 -0.08366
0.54333 -0.081615
0.55 -0.079532
0.55667 -0.077415
0.56333 -0.075269
0.57 -0.073089
0.57667 -0.070879
0.58333 -0.068647
0.589999 -0.066385
0.59667 -0.064101
0.60333 -0.061798
0.61 -0.059475
0.61667 -0.057133
0.62333 -0.054782
0.63 -0.052417
0.63667 -0.050045
0.64333 -0.047673
0.65 -0.0453
0.65667 -0.04293
0.66333 -0.040571
0.67 -0.038219
0.67667 -0.035883
0.68333 -0.033569
```

```

0.69      -0.031274
0.69667   -0.029003
0.70333   -0.026764
0.71       -0.024555
0.71667   -0.022384
0.72333   -0.020258
0.73       -0.018173
0.73667   -0.016137
0.74333   -0.014155
0.75       -0.012226
0.7566     -0.010378
0.76314   -0.008608
0.76961   -0.006923
0.77601   -0.005324
0.78235   -0.003807
0.78863   -0.002375
0.79484   -0.001027
0.80098    0.000232
0.80706    0.001409
0.81307    0.002501
0.819019   0.00351
0.8249     0.004437
0.83072    0.005286
0.83647    0.006056
0.84216    0.00675
0.84778    0.007368
0.85333    0.007912
0.85882    0.008387
0.86425    0.008792
0.86961    0.009128
0.8749     0.009399
0.88013    0.009606
0.88529    0.009749
0.89039    0.009831
0.89542    0.009853
0.90039    0.009816
0.90529    0.00972
0.91013    0.00957
0.9149     0.009364
0.91961    0.009104
0.92425    0.008796
0.92882    0.00844
0.93333    0.008039
0.93778    0.007596
0.94216    0.007115
0.94647    0.006602
0.95072    0.006059
0.9549     0.005491
0.95902    0.0049
0.96307    0.004291
0.96706    0.003666
0.97098    0.00303
0.97484    0.002384
0.97863    0.001734
0.98235    0.001083
0.98601    0.000432
0.98961    -0.000216
0.99314    -0.000857
0.9966     -0.001489
1          -0.002131

```

B.4 DU93-W-210 Airfoil Profile

Listing 4: DU93-W-210 coordinate profile datastream.

```

DU 93-W-210.1m
x/c      y/c
1         0.001939
0.9966   0.003036
0.99314  0.004112
0.98961  0.005156
0.98601  0.006177
0.98235  0.007196
0.97863  0.008227
0.97484  0.009271
0.97098  0.010325
0.96706  0.011386
0.96307  0.012458
0.95902  0.013536
0.9549   0.014626
0.95072  0.015727
0.94647  0.016844
0.94216  0.01797
0.93778  0.019107
0.93333  0.020259
0.92882  0.021423
0.92425  0.022599
0.91961  0.023788
0.9149   0.024994
0.91013  0.026213
0.90529  0.027444
0.90039  0.028688
0.89542  0.029946
0.89039  0.031215

```

0.88529	0.032498
0.88013	0.033793
0.8749	0.035103
0.86961	0.036426
0.86425	0.037763
0.85882	0.039117
0.85333	0.040484
0.84778	0.041863
0.84216	0.043257
0.83647	0.044664
0.83072	0.046083
0.8249	0.047516
0.81902	0.048959
0.81307	0.050414
0.80706	0.051879
0.80098	0.053355
0.79484	0.05484
0.78863	0.056336
0.78235	0.057843
0.77601	0.059358
0.76961	0.060882
0.76314	0.062415
0.7566	0.063956
0.75	0.065502
0.74333	0.067056
0.73667	0.068601
0.73	0.070141
0.72333	0.071674
0.71667	0.0732
0.71	0.074721
0.70333	0.076235
0.69667	0.077739
0.69	0.079235
0.68333	0.080721
0.67667	0.082195
0.67	0.083659
0.66333	0.085113
0.65667	0.086553
0.65	0.087983
0.64333	0.089402
0.63667	0.090804
0.63	0.092195
0.62333	0.093569
0.61667	0.094923
0.61	0.096264
0.60333	0.097585
0.59667	0.098886
0.59	0.100168
0.58333	0.10143
0.57667	0.10267
0.57	0.103893
0.56333	0.105091
0.55667	0.106266
0.55	0.107415
0.54333	0.108541
0.53667	0.109639
0.529999	0.11071
0.52333	0.111754
0.51667	0.112767
0.51	0.113751
0.50333	0.114704
0.49667	0.115618
0.49	0.116497
0.48333	0.117336
0.47667	0.118133
0.470001	0.118887
0.46333	0.119599
0.45667	0.120266
0.45	0.120889
0.44333	0.121467
0.43667	0.121999
0.43	0.122487
0.42333	0.122924
0.41667	0.123312
0.41	0.12365
0.40333	0.123938
0.39667	0.124175
0.39	0.12436
0.38333	0.12449
0.37667	0.124565
0.37	0.124586
0.36333	0.124551
0.35667	0.124458
0.35	0.12431
0.34333	0.124106
0.33667	0.123848
0.33	0.123534
0.32333	0.123168
0.31667	0.122749
0.31	0.122276
0.30333	0.121748
0.29667	0.121168
0.29	0.120531
0.28333	0.119839
0.276671	0.119095
0.27	0.118294
0.26333	0.11744
0.25667	0.116531
0.25	0.115566

0.24342	0.114559
0.23693	0.113512
0.23053	0.112428
0.22421	0.111304
0.21798	0.110147
0.21184	0.108956
0.20579	0.107734
0.19982	0.10648
0.19395	0.105199
0.18816	0.103889
0.18245	0.102549
0.17684	0.101187
0.17131	0.099799
0.16587	0.098389
0.16052	0.096958
0.15526	0.095504
0.15008	0.09403
0.14499	0.092538
0.13999	0.09103
0.13508	0.089503
0.13026	0.087963
0.12552	0.086405
0.12087	0.084833
0.11631	0.083249
0.11183	0.08165
0.10745	0.080044
0.10315	0.078425
0.09893	0.076794
0.09481	0.075159
0.09077	0.073511
0.08683	0.071863
0.08297	0.070207
0.079191	0.068541
0.07551	0.066874
0.07191	0.065202
0.0684	0.063529
0.06498	0.061853
0.06164	0.060173
0.0584	0.058501
0.05524	0.056824
0.05217	0.055151
0.04918	0.053476
0.04629	0.051811
0.04348	0.050145
0.04076	0.048487
0.03812	0.046832
0.03558	0.045193
0.03312	0.043559
0.03075	0.041938
0.02847	0.040333
0.02627	0.038736
0.02417	0.037163
0.02215	0.035596
0.02022	0.034039
0.01837	0.032485
0.01662	0.030943
0.01495	0.029395
0.01337	0.02785
0.01187	0.0263
0.01047	0.024763
0.00915	0.023224
0.00792	0.021692
0.00678	0.020167
0.00572	0.018638
0.00476	0.017133
0.00388	0.01562
0.00309	0.014102
0.00238	0.012534
0.00177	0.010942
0.00124	0.009227
0.0008	0.007396
0.00044	0.005366
0.00018	0.003327
0	0
0.00018	-0.002918
0.00044	-0.004434
0.0008	-0.005885
0.00124	-0.007266
0.00177	-0.008641
0.00238	-0.009983
0.00309	-0.011337
0.00388	-0.012657
0.00476	-0.01397
0.00572	-0.015262
0.00678	-0.016563
0.00792	-0.017849
0.00915	-0.019142
0.01047	-0.02044
0.01187	-0.021737
0.01337	-0.023057
0.01495	-0.024377
0.01662	-0.02571
0.01837	-0.027049
0.02022	-0.028405
0.02215	-0.029762
0.02417	-0.031124
0.02627	-0.03248
0.02847	-0.033844
0.03075	-0.035199

0.03312	-0.036549
0.03558	-0.037894
0.03812	-0.039227
0.04076	-0.040558
0.04348	-0.041878
0.04629	-0.043192
0.04918	-0.044494
0.05217	-0.045791
0.05524	-0.047078
0.0584	-0.048359
0.06164	-0.049627
0.06498	-0.050893
0.0684	-0.052148
0.07191	-0.053397
0.07551	-0.054638
0.07919	-0.055868
0.08297	-0.057094
0.08683	-0.058308
0.09077	-0.05951
0.09481	-0.060706
0.09893	-0.061888
0.10315	-0.063061
0.10745	-0.064218
0.11183	-0.065357
0.11631	-0.066483
0.12087	-0.067589
0.12552	-0.068676
0.13026	-0.069746
0.13508	-0.070793
0.13999	-0.071818
0.14499	-0.072822
0.15008	-0.073801
0.15526	-0.074756
0.16052	-0.075683
0.16587	-0.076583
0.17131	-0.077456
0.17684	-0.078299
0.18245	-0.07911
0.18816	-0.079888
0.19395	-0.08063
0.19982	-0.081337
0.20579	-0.082009
0.21184	-0.082644
0.21798	-0.083242
0.22421	-0.0838
0.23053	-0.084316
0.23693	-0.084792
0.24342	-0.085225
0.25	-0.085613
0.25667	-0.085953
0.26333	-0.086241
0.27	-0.08648
0.27667	-0.086667
0.28333	-0.086804
0.29	-0.086893
0.29667	-0.086932
0.30333	-0.086922
0.31	-0.086863
0.31667	-0.086756
0.32333	-0.086602
0.33	-0.0864
0.33667	-0.086153
0.34333	-0.085862
0.35	-0.085528
0.35667	-0.08515
0.36333	-0.08473
0.37	-0.084267
0.37667	-0.083762
0.38333	-0.083216
0.39	-0.08263
0.39667	-0.082002
0.40333	-0.081335
0.41	-0.080623
0.41667	-0.079871
0.42333	-0.079079
0.43	-0.078242
0.43667	-0.077361
0.44333	-0.076441
0.45	-0.075478
0.45667	-0.074475
0.46333	-0.07343
0.47	-0.072345
0.47667	-0.071221
0.48333	-0.070061
0.49	-0.068863
0.49667	-0.06763
0.50333	-0.066364
0.51	-0.065062
0.51667	-0.063728
0.52333	-0.062365
0.53	-0.06097
0.53667	-0.05955
0.54333	-0.058103
0.55	-0.05663
0.55667	-0.055131
0.56333	-0.053611
0.57	-0.052067
0.57667	-0.050504
0.58333	-0.048924

```

0.59      -0.047323
0.596669      -0.045706
0.60333      -0.044077
0.61      -0.042432
0.61667      -0.040776
0.62333      -0.039113
0.63      -0.037441
0.63667      -0.035763
0.64333      -0.034086
0.65      -0.032405
0.65667      -0.030727
0.66333      -0.029057
0.67      -0.027395
0.67667      -0.025743
0.68333      -0.024106
0.69      -0.022483
0.69667      -0.020878
0.70333      -0.019296
0.71      -0.017736
0.71667      -0.016201
0.72333      -0.014699
0.73      -0.013225
0.73667      -0.011788
0.74333      -0.010388
0.75      -0.009027
0.75667      -0.007721
0.76314      -0.006473
0.76961      -0.005284
0.77601      -0.004155
0.78235      -0.003084
0.78863      -0.002072
0.79484      -0.001122
0.80098      -0.000233
0.80706      0.000596
0.81307      0.001364
0.81902      0.002074
0.8249      0.002728
0.83072      0.003326
0.83647      0.003866
0.84216      0.004352
0.84778      0.004786
0.85333      0.005168
0.85882      0.005499
0.86425      0.005781
0.86961      0.006012
0.8749      0.006195
0.88013      0.006333
0.88529      0.006426
0.89039      0.006478
0.89542      0.006488
0.90039      0.006457
0.90529      0.006385
0.91013      0.006277
0.9149      0.006132
0.91961      0.005952
0.92425      0.005742
0.92882      0.005502
0.93333      0.005233
0.93778      0.004938
0.94216      0.00462
0.94647      0.004281
0.95072      0.003923
0.9549      0.003547
0.95902      0.003154
0.96307      0.002747
0.96706      0.002326
0.97098      0.001892
0.97484      0.001446
0.97863      0.00099
0.98235      0.000525
0.98601      0.000048
0.98961      -0.000044
0.99314      -0.000935
0.9966      -0.001433
1      -0.001939

```

B.5 NACA 64-618 Airfoil Profile

Listing 5: NACA 64-618 coordinate profile datastream.

```

NACA 64-618
x/c      y/c
1      0
0.99      0.003385
0.98      0.006126
0.97      0.008767
0.96      0.011357
0.95      0.013921
0.94      0.016478
0.93      0.019036
0.92      0.021598
0.91      0.024163
0.9      0.026733
0.875      0.033169

```

0.85	0.039603
0.825	0.046004
0.8	0.052337
0.775	0.058566
0.75	0.064656
0.725	0.070573
0.7	0.076285
0.675	0.081773
0.65	0.087012
0.625	0.091967
0.6	0.096599
0.575	0.100887
0.55	0.104799
0.525	0.108299
0.5	0.11136
0.475	0.113937
0.45	0.11599
0.44	0.11665
0.43	0.117211
0.42	0.117667
0.41	0.118003
0.4	0.118204
0.39	0.118258
0.38	0.118168
0.37	0.117946
0.36	0.117607
0.35	0.117159
0.34	0.116603
0.33	0.115942
0.32	0.115181
0.31	0.114319
0.3	0.113356
0.29	0.112292
0.28	0.111127
0.27	0.109863
0.26	0.108497
0.25	0.107027
0.24	0.105454
0.23	0.103774
0.22	0.101985
0.21	0.100084
0.2	0.098068
0.19	0.095934
0.18	0.093676
0.17	0.091291
0.16	0.088772
0.15	0.08611
0.14	0.083298
0.13	0.080329
0.12	0.077188
0.11	0.073862
0.1	0.070334
0.095	0.068489
0.09	0.066584
0.085	0.064617
0.08	0.062583
0.075	0.060478
0.07	0.058297
0.065	0.056032
0.06	0.053676
0.055	0.051218
0.05	0.048647
0.045	0.04595
0.04	0.04311
0.035	0.040102
0.03	0.036899
0.025	0.033454
0.02	0.029694
0.0175	0.027666
0.015	0.025511
0.0125	0.023197
0.01	0.020674
0.009	0.019587
0.008	0.018447
0.007	0.017241
0.006	0.015955
0.005	0.014565
0.004	0.013037
0.003	0.011296
0.002	0.009227
0.00175	0.008633
0.0015	0.007997
0.00125	0.007312
0.001	0.006555
0.00075	0.005695
0.0005	0.004672
0.0004	0.00419
0.0003	0.003636
0.0002	0.00297
0.0001	0.002104
0	0
0.0001	-0.002092
0.0002	-0.002954
0.0003	-0.003613
0.0004	-0.004169
0.0005	-0.004658
0.00075	-0.005698
0.001	-0.006572
0.00125	-0.00734

0.0015	-0.008027
0.00175	-0.008654
0.002	-0.009231
0.003	-0.011176
0.004	-0.012729
0.005	-0.01403
0.006	-0.01516
0.007	-0.016174
0.008	-0.017098
0.009	-0.017953
0.01	-0.01875
0.0125	-0.020537
0.015	-0.022107
0.0175	-0.023517
0.02	-0.024803
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0.03	-0.029102
0.035	-0.030904
0.04	-0.032546
0.045	-0.034061
0.05	-0.035472
0.055	-0.036792
0.06	-0.038035
0.065	-0.039209
0.07	-0.040322
0.075	-0.041381
0.08	-0.042389
0.085	-0.04335
0.09	-0.044268
0.095	-0.045145
0.1	-0.045987
0.11	-0.047576
0.12	-0.04905
0.13	-0.050417
0.14	-0.051688
0.15	-0.052868
0.16	-0.053967
0.17	-0.054988
0.18	-0.055934
0.19	-0.056811
0.2	-0.057621
0.21	-0.058365
0.22	-0.059045
0.23	-0.059664
0.24	-0.060224
0.25	-0.060723
0.26	-0.061163
0.27	-0.061545
0.28	-0.061871
0.29	-0.062137
0.3	-0.062343
0.31	-0.06249
0.32	-0.062577
0.33	-0.062602
0.34	-0.062563
0.35	-0.06246
0.36	-0.062287
0.37	-0.06204
0.38	-0.061716
0.39	-0.061301
0.4	-0.060778
0.41	-0.060138
0.42	-0.059388
0.43	-0.058544
0.44	-0.057622
0.45	-0.056632
0.475	-0.053897
0.5	-0.05085
0.525	-0.047539
0.55	-0.044014
0.575	-0.040316
0.6	-0.036486
0.625	-0.032565
0.65	-0.028585
0.675	-0.024603
0.7	-0.02066
0.725	-0.016795
0.75	-0.013048
0.775	-0.009472
0.8	-0.006122
0.825	-0.003065
0.85	-0.000376
0.875	0.001859
0.9	0.003536
0.91	0.004023
0.92	0.004387
0.93	0.00462
0.94	0.004702
0.95	0.004614
0.96	0.004338
0.97	0.003829
0.98	0.003042
0.99	0.00191
1	0

B.6 NACA 64-615 Airfoil Profile

Listing 6: NACA 64-615 coordinate profile datastream.

```

NACA64615a=0.6
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0.99723631 -0.00426969
0.99152154 -0.00352121
0.98406877 -0.00240480
0.97491466 -0.00086813
0.96410346 0.00114114
0.95168378 0.00365935
0.93770927 0.00671469
0.92223457 0.01031207
0.90532008 0.01445738
0.88702916 0.01914667
0.86742400 0.02435236
0.84657137 0.03004747
0.82453746 0.03618726
0.80138709 0.04270796
0.77718583 0.04953484
0.75199726 0.05657235
0.72588785 0.06372014
0.69892346 0.07085877
0.67117260 0.07785808
0.64270711 0.08457573
0.61359648 0.09082332
0.58390092 0.09630049
0.55376052 0.10107006
0.52330278 0.10514090
0.49264727 0.10846591
0.46191511 0.11098446
0.43123122 0.11263388
0.40072339 0.11328413
0.37053073 0.11284652
0.34079308 0.11139686
0.31163514 0.10911714
0.28317124 0.10613812
0.25551601 0.10250631
0.22878024 0.09826917
0.20306992 0.09348038
0.17848479 0.08820393
0.15512042 0.08250060
0.13306986 0.07642388
0.11242039 0.07003404
0.09325363 0.06339619
0.07564738 0.05657491
0.05968267 0.04962024
0.04544384 0.04257682
0.03300617 0.03551485
0.02240777 0.02857168
0.01370516 0.02185532
0.00707220 0.01537311
0.00270083 0.00925022
0.00052796 0.00392701
0.00000000 0.00000000
0.00000000 0.00000000
0.00066648 -0.00377350
0.00369883 -0.00847135
0.00893937 -0.01353367
0.01689951 -0.01770677
0.02708879 -0.02086096
0.03909989 -0.02350474
0.05277641 -0.02589307
0.06804877 -0.02802017
0.08484790 -0.02988953
0.10309909 -0.03155740
0.12273013 -0.03305566
0.14366886 -0.03438423
0.16583974 -0.03553793
0.18916367 -0.03650773
0.21355760 -0.03728980
0.23893536 -0.03786810
0.26520734 -0.03822060
0.29228051 -0.03833239
0.32005889 -0.03819228
0.34844386 -0.03778641
0.37733203 -0.03700878
0.40661209 -0.03570022
0.43617177 -0.03382962
0.46590851 -0.03152588
0.49572722 -0.02895264
0.52553084 -0.02620958
0.55522469 -0.02340521
0.58471611 -0.02065777
0.61391939 -0.01816533
0.64274121 -0.01605724
0.67106590 -0.01416309
0.69879045 -0.01242782
0.72581708 -0.01083575
0.75205149 -0.00938632
0.77740207 -0.00807624
0.80178087 -0.00691335
0.82510353 -0.00591173
0.84728854 -0.00507053
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0.90626351 -0.00349958
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0.93858521	-0.00317690
0.95246938	-0.00319875
0.96476865	-0.00332969
0.97543958	-0.00354762
0.98444463	-0.00382742
0.99175268	-0.00413356
0.99733879	-0.00443410
1.00118411	-0.00470033

C. 360-Degree Extrapolated Profile Polar Files

This appendix aggregates the fully resolved 360-degree aerodynamic polars generated via Viterna-Corrigan extrapolation loops to ensure high stability across extreme deep stall angles.

C.1 DU91-W2-250 Extrapolated Polars

Listing 7: Aerodynamic polar characteristics for DU91-W2-250.

-----QBlade Multi RE Polar File-----					
Generated with : QBlade CE v2.0.9.4_beta windows					
Archive Format: 310039					
Time : 19:42:31					
Date : 04.06.2026					
-----Object Names-----					
DU_91-W2-250_coords_Re9.000_M0.00_N9.0 360 M		POLARNAME	- the polar name		
../Airfoils/DU_91-W2-250_coords.afl		FOILNAME	- the airfoil name to which		
the polar(s) belong OR path to airfoil file					
-----Parameters-----					
25.0	THICKNESS	- the thickness of the airfoil in % of chord			
true	ISDECOMPOSED	- is the polar decomposed (imports Cl_Sep,			
Cl_att and f_st columns) [bool]					
REYNOLDS	9.0000E+06	- the list of Reynolds numbers for the imported polars			
-----Polar Data-----					
AOA	CL	CD	CM	CL_ATT	CL_SEP
	F_ST				
-180.000000	-0.099464	0.006000	-0.035817	0.000000	-0.099464
	0.000000				
-179.000000	0.012711	0.006493	0.004604	0.000000	0.012711
	0.000000				
-178.000000	0.124887	0.007968	0.044788	0.000000	0.124887
	0.000000				
-177.000000	0.237063	0.010425	0.084716	0.000000	0.237063
	0.000000				
-176.000000	0.349239	0.013857	0.124370	0.000000	0.349239
	0.000000				
-175.000000	0.461415	0.018261	0.163731	0.000000	0.461415
	0.000000				
-174.000000	0.573591	0.023628	0.202782	0.000000	0.573591
	0.000000				
-173.000000	0.685255	0.029952	0.241326	0.000000	0.685255
	0.000000				
-172.000000	0.783292	0.037224	0.274776	0.000000	0.783292
	0.000000				
-171.000000	0.822619	0.045432	0.287571	0.000000	0.822619
	0.000000				
-170.000000	0.763057	0.054568	0.266208	0.000000	0.763057
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	0.000000				
-168.000000	0.525490	0.075566	0.184230	0.000000	0.525490
	0.000000				
-167.000000	0.448095	0.087402	0.158218	0.000000	0.448095
	0.000000				
-166.000000	0.404182	0.100109	0.143963	0.000000	0.404182
	0.000000				
-165.000000	0.382571	0.113670	0.137524	0.000000	0.382571
	0.000000				
-164.000000	0.374484	0.128067	0.135851	0.000000	0.374484
	0.000000				
-163.000000	0.374349	0.143282	0.137023	0.000000	0.374349
	0.000000				
-162.000000	0.378810	0.159296	0.139897	0.000000	0.378810
	0.000000				
-161.000000	0.385852	0.176088	0.143790	0.000000	0.385852
	0.000000				
-160.000000	0.394242	0.193637	0.148292	0.000000	0.394242
	0.000000				
-159.000000	0.403211	0.211920	0.153153	0.000000	0.403211
	0.000000				
-158.000000	0.412266	0.230915	0.158215	0.000000	0.412266
	0.000000				
-157.000000	0.421086	0.250597	0.163381	0.000000	0.421086
	0.000000				
-156.000000	0.429458	0.270942	0.168587	0.000000	0.429458
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-153.000000	0.450675	0.335695	0.184133	0.000000	0.450675
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-151.000000	0.460960	0.381692	0.194312	0.000000	0.460960
	0.000000				

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-146.000000	0.472220	0.504904	0.219137	0.000000	0.472220
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-145.000000	0.472045	0.530713	0.224039	0.000000	0.472045
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-144.000000	0.471098	0.556838	0.228935	0.000000	0.471098
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	0.871851				
-4.750000	-0.159700	0.005820	-0.115100	-0.170758	-0.083926
	0.872647				
-4.500000	-0.126700	0.005770	-0.116300	-0.135703	-0.066667
	0.869584				
-4.250000	-0.093900	0.005740	-0.117400	-0.100649	-0.049436
	0.868226				
-4.000000	-0.061200	0.005710	-0.118500	-0.065594	-0.032218
	0.868358				

-3.750000	-0.028400	0.005680	-0.119600	-0.030539	-0.014987
	0.862480				
-3.500000	0.004200	0.005670	-0.120600	0.004516	0.002216
	0.862480				
-3.250000	0.036900	0.005640	-0.121700	0.039571	0.019434
	0.867351				
-3.000000	0.069400	0.005630	-0.122700	0.074626	0.036623
	0.862480				
-2.750000	0.102000	0.005610	-0.123700	0.109681	0.053827
	0.862480				
-2.500000	0.134400	0.005600	-0.124700	0.144736	0.071002
	0.859819				
-2.250000	0.166900	0.005590	-0.125700	0.179791	0.088192
	0.859267				
-2.000000	0.199300	0.005590	-0.126600	0.214846	0.105367
	0.858000				
-1.750000	0.231700	0.005580	-0.127600	0.249901	0.122543
	0.857089				
-1.500000	0.264000	0.005580	-0.128500	0.284956	0.139704
	0.855727				
-1.250000	0.296400	0.005570	-0.129500	0.320011	0.156880
	0.855265				
-1.000000	0.328600	0.005580	-0.130300	0.355066	0.174027
	0.853812				
-0.750000	0.361000	0.005570	-0.131300	0.390121	0.191203
	0.853605				
-0.500000	0.393200	0.005570	-0.132200	0.425176	0.208350
	0.852529				
-0.250000	0.425400	0.005570	-0.133100	0.460230	0.225498
	0.851617				
0.000000	0.457500	0.005580	-0.134000	0.495285	0.242631
	0.850446				
0.250000	0.489600	0.005590	-0.134800	0.530340	0.259764
	0.849431				
0.500000	0.521700	0.005590	-0.135700	0.565395	0.276897
	0.848542				
0.750000	0.553600	0.005620	-0.136500	0.600450	0.294002
	0.847119				
1.000000	0.585600	0.005630	-0.137300	0.635505	0.311121
	0.846154				
1.250000	0.617400	0.005650	-0.138100	0.670560	0.328212
	0.844719				
1.500000	0.649300	0.005670	-0.138900	0.705615	0.345317
	0.843699				
1.750000	0.681100	0.005690	-0.139600	0.740670	0.362407
	0.842517				
2.000000	0.712900	0.005700	-0.140400	0.775725	0.379498
	0.841442				
2.250000	0.744500	0.005740	-0.141100	0.810780	0.396560
	0.839989				
2.500000	0.776200	0.005760	-0.141900	0.845835	0.413636
	0.838883				
2.750000	0.807700	0.005800	-0.142600	0.880890	0.430683
	0.837431				
3.000000	0.839100	0.005840	-0.143200	0.915945	0.447717
	0.835882				
3.250000	0.870500	0.005880	-0.143900	0.951000	0.464750
	0.834448				
3.500000	0.901800	0.005930	-0.144600	0.986055	0.481769
	0.832923				
3.750000	0.933100	0.005970	-0.145200	1.021109	0.498787
	0.831503				
4.000000	0.964300	0.006020	-0.145800	1.056164	0.515791
	0.829998				
4.250000	0.995000	0.006100	-0.146400	1.091219	0.532724
	0.827717				
4.500000	1.025200	0.006240	-0.146900	1.126274	0.549583
	0.824734				
5.000000	1.081900	0.006810	-0.147200	1.196384	0.582765
	0.813428				
8.750000	1.462100	0.012550	-0.141900	1.722208	0.824373
	0.710294				
9.000000	1.485500	0.012880	-0.141200	1.757263	0.840133
	0.703681				
9.250000	1.508000	0.013220	-0.140400	1.792318	0.855742
	0.696428				
9.500000	1.529900	0.013580	-0.139400	1.827373	0.871244
	0.688878				
9.750000	1.550500	0.013950	-0.138300	1.862428	0.886524
	0.680370				
10.000000	1.569600	0.014310	-0.136800	1.897483	0.901543
	0.670781				
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	0.654066				
10.500000	1.596400	0.015130	-0.131900	1.967593	0.929595
	0.642395				
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	0.631808				
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	0.620853				
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	0.610150				
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	0.599605				
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	0.588886				
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	0.578724				

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	0.557816				
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	0.547698				
13.000000	1.740000	0.022890	-0.117000	2.318142	1.069685
	0.536915				
13.250000	1.752100	0.024070	-0.115900	2.353197	1.083112
	0.526727				
13.500000	1.763800	0.025340	-0.114900	2.388252	1.096428
	0.516612				
13.750000	1.773500	0.026780	-0.113800	2.423307	1.109311
	0.505473				
14.250000	1.791500	0.029910	-0.111700	2.493417	1.134662
	0.483412				
14.500000	1.799400	0.031630	-0.110700	2.528472	1.147041
	0.472234				
14.750000	1.805600	0.033500	-0.109700	2.563527	1.159010
	0.460365				
15.000000	1.811000	0.035490	-0.108800	2.598582	1.170747
	0.448408				
15.250000	1.817000	0.037470	-0.107900	2.633637	1.182550
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	0.426001				
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16.250000	1.832400	0.046540	-0.104800	2.773856	1.227183
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	0.379707				
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	0.357360				
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17.500000	1.840400	0.059640	-0.102000	2.949131	1.278540
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	0.325957				
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77.000000	0.000000	0.000000	0.000000	11.292207	0.000000
	0.000000				
77.250000	0.000000	0.000000	0.000000	11.327261	0.000000
	0.000000				
77.500000	0.000000	0.000000	0.000000	11.362316	0.000000
	0.000000				
77.750000	0.000000	0.000000	0.000000	11.397371	0.000000
	0.000000				
78.000000	0.000000	0.000000	0.000000	11.432426	0.000000
	0.000000				
78.250000	0.000000	0.000000	0.000000	11.467481	0.000000
	0.000000				
78.500000	0.000000	0.000000	0.000000	11.502536	0.000000
	0.000000				
78.750000	0.000000	0.000000	0.000000	11.537591	0.000000
	0.000000				
79.000000	0.000000	0.000000	0.000000	11.572646	0.000000
	0.000000				
79.250000	0.000000	0.000000	0.000000	11.607700	0.000000
	0.000000				
79.500000	0.000000	0.000000	0.000000	11.642756	0.000000
	0.000000				
79.750000	0.000000	0.000000	0.000000	11.677811	0.000000
	0.000000				
80.000000	0.000000	0.000000	0.000000	11.712866	0.000000
	0.000000				
80.250000	0.000000	0.000000	0.000000	11.747921	0.000000
	0.000000				
80.500000	0.000000	0.000000	0.000000	11.782975	0.000000
	0.000000				
80.750000	0.000000	0.000000	0.000000	11.818030	0.000000
	0.000000				
81.000000	0.000000	0.000000	0.000000	11.853086	0.000000
	0.000000				
81.250000	0.000000	0.000000	0.000000	11.888141	0.000000
	0.000000				
81.500000	0.000000	0.000000	0.000000	11.923195	0.000000
	0.000000				
81.750000	0.000000	0.000000	0.000000	11.958250	0.000000
	0.000000				
82.000000	0.000000	0.000000	0.000000	11.993305	0.000000
	0.000000				
82.250000	0.000000	0.000000	0.000000	12.028360	0.000000
	0.000000				
82.500000	0.000000	0.000000	0.000000	12.063415	0.000000
	0.000000				
82.750000	0.000000	0.000000	0.000000	12.098470	0.000000
	0.000000				
83.000000	0.000000	0.000000	0.000000	12.133525	0.000000
	0.000000				
83.250000	0.000000	0.000000	0.000000	12.168580	0.000000
	0.000000				
83.500000	0.000000	0.000000	0.000000	12.203634	0.000000
	0.000000				
83.750000	0.000000	0.000000	0.000000	12.238689	0.000000
	0.000000				
84.000000	0.000000	0.000000	0.000000	12.273745	0.000000
	0.000000				
84.250000	0.000000	0.000000	0.000000	12.308800	0.000000
	0.000000				
84.500000	0.000000	0.000000	0.000000	12.343855	0.000000
	0.000000				
84.750000	0.000000	0.000000	0.000000	12.378909	0.000000
	0.000000				
85.000000	0.000000	0.000000	0.000000	12.413964	0.000000
	0.000000				
85.250000	0.000000	0.000000	0.000000	12.449019	0.000000
	0.000000				
85.500000	0.000000	0.000000	0.000000	12.484075	0.000000
	0.000000				

85.750000	0.000000	0.000000	0.000000	12.519129	0.000000
	0.000000				
86.000000	0.000000	0.000000	0.000000	12.554184	0.000000
	0.000000				
86.250000	0.000000	0.000000	0.000000	12.589239	0.000000
	0.000000				
86.500000	0.000000	0.000000	0.000000	12.624294	0.000000
	0.000000				
87.000000	0.414878	1.649676	-0.014916	12.694404	0.414878
	0.000000				
88.000000	0.370039	1.652565	-0.044749	12.834623	0.370039
	0.000000				
89.000000	0.324810	1.654424	-0.074582	12.974843	0.324810
	0.000000				
90.000000	0.279250	1.655251	-0.104414	0.000000	0.279250
	0.000000				
91.000000	0.233417	1.655046	-0.134247	0.000000	0.233417
	0.000000				
92.000000	0.187371	1.653809	-0.164080	0.000000	0.187371
	0.000000				
93.000000	0.141174	1.651543	-0.193913	0.000000	0.141174
	0.000000				
94.000000	0.094887	1.648252	-0.223745	0.000000	0.094887
	0.000000				
95.000000	0.048573	1.643941	-0.253578	0.000000	0.048573
	0.000000				
96.000000	0.002295	1.638614	-0.283411	0.000000	0.002295
	0.000000				
97.000000	-0.043884	1.632281	-0.313243	0.000000	-0.043884
	0.000000				
98.000000	-0.089898	1.624948	-0.343076	0.000000	-0.089898
	0.000000				
99.000000	-0.135685	1.616627	-0.372909	0.000000	-0.135685
	0.000000				
100.000000	-0.181180	1.607327	-0.402742	0.000000	-0.181180
	0.000000				
101.000000	-0.226318	1.597061	-0.432574	0.000000	-0.226318
	0.000000				
102.000000	-0.316165	1.585842	-0.450922	0.000000	-0.316165
	0.000000				
103.000000	-0.359340	1.573686	-0.451915	0.000000	-0.359340
	0.000000				
104.000000	-0.401999	1.560607	-0.452917	0.000000	-0.401999
	0.000000				
105.000000	-0.444077	1.546622	-0.453924	0.000000	-0.444077
	0.000000				
106.000000	-0.485510	1.531749	-0.454931	0.000000	-0.485510
	0.000000				
107.000000	-0.526237	1.516008	-0.455936	0.000000	-0.526237
	0.000000				
108.000000	-0.566196	1.499418	-0.456930	0.000000	-0.566196
	0.000000				
109.000000	-0.605325	1.482001	-0.457909	0.000000	-0.605325
	0.000000				
110.000000	-0.643567	1.463777	-0.458866	0.000000	-0.643567
	0.000000				
111.000000	-0.680862	1.444772	-0.459792	0.000000	-0.680862
	0.000000				
112.000000	-0.717155	1.425008	-0.460680	0.000000	-0.717155
	0.000000				
113.000000	-0.752390	1.404510	-0.461522	0.000000	-0.752390
	0.000000				
114.000000	-0.786514	1.383305	-0.462307	0.000000	-0.786514
	0.000000				
115.000000	-0.819477	1.361419	-0.463025	0.000000	-0.819477
	0.000000				
116.000000	-0.851228	1.338880	-0.463668	0.000000	-0.851228
	0.000000				
117.000000	-0.881720	1.315716	-0.464224	0.000000	-0.881720
	0.000000				
118.000000	-0.910908	1.291956	-0.464682	0.000000	-0.910908
	0.000000				
119.000000	-0.938748	1.267629	-0.465030	0.000000	-0.938748
	0.000000				
120.000000	-0.965201	1.242767	-0.465257	0.000000	-0.965201
	0.000000				
121.000000	-0.990228	1.217400	-0.465351	0.000000	-0.990228
	0.000000				
122.000000	-1.013793	1.191561	-0.465300	0.000000	-1.013793
	0.000000				
123.000000	-1.035863	1.165281	-0.465091	0.000000	-1.035863
	0.000000				
124.000000	-1.056407	1.138593	-0.464713	0.000000	-1.056407
	0.000000				
125.000000	-1.075399	1.111530	-0.464153	0.000000	-1.075399
	0.000000				
126.000000	-1.092812	1.084127	-0.463399	0.000000	-1.092812
	0.000000				
127.000000	-1.108626	1.056417	-0.462439	0.000000	-1.108626
	0.000000				
128.000000	-1.122821	1.028434	-0.461262	0.000000	-1.122821
	0.000000				
129.000000	-1.135381	1.000214	-0.459855	0.000000	-1.135381
	0.000000				
130.000000	-1.146293	0.971791	-0.458207	0.000000	-1.146293
	0.000000				
131.000000	-1.155547	0.943200	-0.456308	0.000000	-1.155547
	0.000000				

132.000000	-1.163138	0.914478	-0.454147	0.000000	-1.163138
	0.000000				
133.000000	-1.169061	0.885658	-0.451716	0.000000	-1.169061
	0.000000				
134.000000	-1.173317	0.856778	-0.449003	0.000000	-1.173317
	0.000000				
135.000000	-1.175908	0.827873	-0.446001	0.000000	-1.175908
	0.000000				
136.000000	-1.176840	0.798977	-0.442703	0.000000	-1.176840
	0.000000				
137.000000	-1.176124	0.770128	-0.439101	0.000000	-1.176124
	0.000000				
138.000000	-1.173773	0.741359	-0.435189	0.000000	-1.173773
	0.000000				
139.000000	-1.169803	0.712707	-0.430962	0.000000	-1.169803
	0.000000				
140.000000	-1.164234	0.684207	-0.426416	0.000000	-1.164234
	0.000000				
141.000000	-1.157089	0.655894	-0.421547	0.000000	-1.157089
	0.000000				
142.000000	-1.148396	0.627801	-0.416355	0.000000	-1.148396
	0.000000				
143.000000	-1.138185	0.599964	-0.410838	0.000000	-1.138185
	0.000000				
144.000000	-1.126492	0.572417	-0.404997	0.000000	-1.126492
	0.000000				
145.000000	-1.113355	0.545193	-0.398834	0.000000	-1.113355
	0.000000				
146.000000	-1.098818	0.518324	-0.392353	0.000000	-1.098818
	0.000000				
147.000000	-1.082929	0.491845	-0.385560	0.000000	-1.082929
	0.000000				
148.000000	-1.065742	0.465786	-0.378463	0.000000	-1.065742
	0.000000				
149.000000	-1.047320	0.440180	-0.371071	0.000000	-1.047320
	0.000000				
150.000000	-1.027729	0.415057	-0.363398	0.000000	-1.027729
	0.000000				
151.000000	-1.007051	0.390448	-0.355460	0.000000	-1.007051
	0.000000				
152.000000	-0.985378	0.366383	-0.347281	0.000000	-0.985378
	0.000000				
153.000000	-0.962820	0.342889	-0.338887	0.000000	-0.962820
	0.000000				
154.000000	-0.939514	0.319996	-0.330315	0.000000	-0.939514
	0.000000				
155.000000	-0.915633	0.297731	-0.321615	0.000000	-0.915633
	0.000000				
156.000000	-0.891400	0.276120	-0.312854	0.000000	-0.891400
	0.000000				
157.000000	-0.867124	0.255190	-0.304127	0.000000	-0.867124
	0.000000				
158.000000	-0.843238	0.234964	-0.295574	0.000000	-0.843238
	0.000000				
159.000000	-0.820382	0.215467	-0.287402	0.000000	-0.820382
	0.000000				
160.000000	-0.799533	0.196723	-0.279938	0.000000	-0.799533
	0.000000				
161.000000	-0.782251	0.178753	-0.273703	0.000000	-0.782251
	0.000000				
162.000000	-0.771100	0.161577	-0.269567	0.000000	-0.771100
	0.000000				
163.000000	-0.770406	0.145218	-0.269007	0.000000	-0.770406
	0.000000				
164.000000	-0.787481	0.129692	-0.274538	0.000000	-0.787481
	0.000000				
165.000000	-0.833987	0.115019	-0.290206	0.000000	-0.833987
	0.000000				
166.000000	-0.924541	0.101215	-0.321164	0.000000	-0.924541
	0.000000				
167.000000	-1.062228	0.088297	-0.368767	0.000000	-1.062228
	0.000000				
168.000000	-1.203228	0.076278	-0.418282	0.000000	-1.203228
	0.000000				
169.000000	-1.261493	0.065173	-0.439967	0.000000	-1.261493
	0.000000				
170.000000	-1.211921	0.054993	-0.424500	0.000000	-1.211921
	0.000000				
171.000000	-1.108932	0.045751	-0.390194	0.000000	-1.108932
	0.000000				
172.000000	-0.996871	0.037456	-0.352321	0.000000	-0.996871
	0.000000				
173.000000	-0.884695	0.030118	-0.314011	0.000000	-0.884695
	0.000000				
174.000000	-0.772519	0.023744	-0.275320	0.000000	-0.772519
	0.000000				
175.000000	-0.660343	0.018340	-0.236266	0.000000	-0.660343
	0.000000				
176.000000	-0.548168	0.013913	-0.196868	0.000000	-0.548168
	0.000000				
177.000000	-0.435992	0.010465	-0.157142	0.000000	-0.435992
	0.000000				
178.000000	-0.323816	0.008002	-0.117106	0.000000	-0.323816
	0.000000				
179.000000	-0.211640	0.006523	-0.076779	0.000000	-0.211640
	0.000000				
180.000000	-0.099464	0.006000	-0.035817	0.000000	-0.099464
	0.000000				

C.2 DU93-W-210 Extrapolated Polars

Listing 8: Aerodynamic polar characteristics for DU93-W-210.

-----QBlade Multi RE Polar File-----					
Generated with : QBlade CE v2.0.9.4_beta windows					
Archive Format: 310039					
Time : 19:42:31					
Date : 04.06.2026					
-----Object Names-----					
DU_93-W-210_coords_Re9.000_M0.00_N9.0 360 M			POLARNAME	- the polar name	
../Airfoils/DU_93-W-210_coords.af1			FOILNAME	- the airfoil name to which	
the polar(s) belong OR path to airfoil file					
-----Parameters-----					
21.0		THICKNESS		- the thickness of the airfoil in % of chord	
true		ISDECOMPOSED		- is the polar decomposed (imports Cl_Sep,	
Cl_att and f_st columns) [bool]					
REYNOLDS		9.0000E+06		- the list of Reynolds numbers for the imported polars	
-----Polar Data-----					
AOA	CL	CD	CM	CL_ATT	CL_SEP
	F_ST				
-180.000000	-0.122012	0.006000	-0.042747	0.000000	-0.122012
	0.000000				
-179.000000	-0.018227	0.006501	-0.006329	0.000000	-0.018227
	0.000000				
-178.000000	0.085557	0.008004	0.029899	0.000000	0.085557
	0.000000				
-177.000000	0.189341	0.010504	0.065923	0.000000	0.189341
	0.000000				
-176.000000	0.293125	0.013999	0.101726	0.000000	0.293125
	0.000000				
-175.000000	0.396910	0.018481	0.137296	0.000000	0.396910
	0.000000				
-174.000000	0.500459	0.023945	0.172535	0.000000	0.500459
	0.000000				
-173.000000	0.596507	0.030382	0.204951	0.000000	0.596507
	0.000000				
-172.000000	0.656576	0.037783	0.224888	0.000000	0.656576
	0.000000				
-171.000000	0.645515	0.046138	0.220647	0.000000	0.645515
	0.000000				
-170.000000	0.570990	0.055436	0.195202	0.000000	0.570990
	0.000000				
-169.000000	0.482420	0.065664	0.165460	0.000000	0.482420
	0.000000				
-168.000000	0.414434	0.076808	0.143042	0.000000	0.414434
	0.000000				
-167.000000	0.372642	0.088854	0.129703	0.000000	0.372642
	0.000000				
-166.000000	0.350840	0.101786	0.123266	0.000000	0.350840
	0.000000				
-165.000000	0.342046	0.115587	0.121341	0.000000	0.342046
	0.000000				
-164.000000	0.341213	0.130238	0.122219	0.000000	0.341213
	0.000000				
-163.000000	0.345099	0.145722	0.124814	0.000000	0.345099
	0.000000				
-162.000000	0.351678	0.162018	0.128454	0.000000	0.351678
	0.000000				
-161.000000	0.359683	0.179105	0.132726	0.000000	0.359683
	0.000000				
-160.000000	0.368311	0.196962	0.137371	0.000000	0.368311
	0.000000				
-159.000000	0.377040	0.215567	0.142227	0.000000	0.377040
	0.000000				
-158.000000	0.385531	0.234894	0.147192	0.000000	0.385531
	0.000000				
-157.000000	0.393556	0.254921	0.152201	0.000000	0.393556
	0.000000				
-156.000000	0.400963	0.275621	0.157214	0.000000	0.400963
	0.000000				
-155.000000	0.407651	0.296970	0.162207	0.000000	0.407651
	0.000000				
-154.000000	0.413553	0.318940	0.167168	0.000000	0.413553
	0.000000				
-153.000000	0.418626	0.341503	0.172091	0.000000	0.418626
	0.000000				
-152.000000	0.422848	0.364633	0.176975	0.000000	0.422848
	0.000000				
-151.000000	0.426206	0.388299	0.181824	0.000000	0.426206
	0.000000				
-150.000000	0.428701	0.412473	0.186643	0.000000	0.428701
	0.000000				
-149.000000	0.430338	0.437125	0.191439	0.000000	0.430338
	0.000000				
-148.000000	0.431131	0.462225	0.196222	0.000000	0.431131
	0.000000				
-147.000000	0.431098	0.487741	0.200998	0.000000	0.431098
	0.000000				
-146.000000	0.430260	0.513641	0.205778	0.000000	0.430260
	0.000000				
-145.000000	0.428641	0.539895	0.210570	0.000000	0.428641
	0.000000				
-144.000000	0.426270	0.566470	0.215383	0.000000	0.426270
	0.000000				

-143.000000	0.423175	0.593334	0.220225	0.000000	0.423175
	0.000000				
-142.000000	0.419389	0.620453	0.225103	0.000000	0.419389
	0.000000				
-141.000000	0.414943	0.647794	0.230024	0.000000	0.414943
	0.000000				
-140.000000	0.409871	0.675325	0.234995	0.000000	0.409871
	0.000000				
-139.000000	0.404209	0.703011	0.240021	0.000000	0.404209
	0.000000				
-138.000000	0.397992	0.730819	0.245106	0.000000	0.397992
	0.000000				
-137.000000	0.391255	0.758716	0.250253	0.000000	0.391255
	0.000000				
-136.000000	0.384034	0.786666	0.255465	0.000000	0.384034
	0.000000				
-135.000000	0.376366	0.814637	0.260742	0.000000	0.376366
	0.000000				
-134.000000	0.368287	0.842595	0.266085	0.000000	0.368287
	0.000000				
-133.000000	0.359831	0.870505	0.271493	0.000000	0.359831
	0.000000				
-132.000000	0.351035	0.898335	0.276965	0.000000	0.351035
	0.000000				
-131.000000	0.341933	0.926049	0.282495	0.000000	0.341933
	0.000000				
-130.000000	0.332559	0.953616	0.288082	0.000000	0.332559
	0.000000				
-129.000000	0.322947	0.981002	0.293718	0.000000	0.322947
	0.000000				
-128.000000	0.313128	1.008173	0.299399	0.000000	0.313128
	0.000000				
-127.000000	0.303135	1.035098	0.305116	0.000000	0.303135
	0.000000				
-126.000000	0.292998	1.061744	0.310862	0.000000	0.292998
	0.000000				
-125.000000	0.282745	1.088078	0.316628	0.000000	0.282745
	0.000000				
-124.000000	0.272404	1.114070	0.322402	0.000000	0.272404
	0.000000				
-123.000000	0.262003	1.139688	0.328176	0.000000	0.262003
	0.000000				
-122.000000	0.251565	1.164901	0.333937	0.000000	0.251565
	0.000000				
-121.000000	0.241115	1.189680	0.339673	0.000000	0.241115
	0.000000				
-120.000000	0.230674	1.213995	0.345372	0.000000	0.230674
	0.000000				
-119.000000	0.220264	1.237816	0.351020	0.000000	0.220264
	0.000000				
-118.000000	0.209903	1.261115	0.356603	0.000000	0.209903
	0.000000				
-117.000000	0.199608	1.283865	0.362108	0.000000	0.199608
	0.000000				
-116.000000	0.189396	1.306038	0.367521	0.000000	0.189396
	0.000000				
-115.000000	0.179280	1.327608	0.372826	0.000000	0.179280
	0.000000				
-114.000000	0.169272	1.348549	0.378009	0.000000	0.169272
	0.000000				
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	0.879007				
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	0.939040				
0.500000	0.597800	0.004930	-0.136400	0.617646	0.306281
	0.936260				
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	0.934777				
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	0.931940				
1.500000	0.721300	0.005030	-0.139000	0.747377	0.370341
	0.930838				
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	0.923894				
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	0.913307				
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	0.901251				
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	0.796050				
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	0.784608				
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	0.778899				
9.000000	1.518800	0.012810	-0.136300	1.720354	0.832528
	0.772980				
9.250000	1.541400	0.013090	-0.135600	1.752787	0.847311
	0.766546				
9.500000	1.563300	0.013380	-0.134700	1.785219	0.861983
	0.759629				
9.750000	1.583600	0.013690	-0.133500	1.817652	0.876402
	0.751339				
10.000000	1.598400	0.013980	-0.131200	1.850084	0.889958
	0.737863				
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	0.725696				
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	0.706102				
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16.750000	1.902400	0.046280	-0.098300	2.725764	1.228640
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	0.386106				
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61.000000	0.000000	0.000000	0.000000	8.466331	0.000000
	0.000000				
61.250000	0.000000	0.000000	0.000000	8.498763	0.000000
	0.000000				
61.500000	0.000000	0.000000	0.000000	8.531196	0.000000
	0.000000				
61.750000	0.000000	0.000000	0.000000	8.563628	0.000000
	0.000000				
62.000000	0.000000	0.000000	0.000000	8.596061	0.000000
	0.000000				
62.250000	0.000000	0.000000	0.000000	8.628494	0.000000
	0.000000				
62.500000	0.000000	0.000000	0.000000	8.660927	0.000000
	0.000000				
62.750000	0.000000	0.000000	0.000000	8.693359	0.000000
	0.000000				
63.000000	0.000000	0.000000	0.000000	8.725792	0.000000
	0.000000				
63.250000	0.000000	0.000000	0.000000	8.758224	0.000000
	0.000000				
63.500000	0.000000	0.000000	0.000000	8.790657	0.000000
	0.000000				
63.750000	0.000000	0.000000	0.000000	8.823090	0.000000
	0.000000				
64.000000	0.000000	0.000000	0.000000	8.855522	0.000000
	0.000000				
64.250000	0.000000	0.000000	0.000000	8.887955	0.000000
	0.000000				
64.500000	0.000000	0.000000	0.000000	8.920387	0.000000
	0.000000				
64.750000	0.000000	0.000000	0.000000	8.952820	0.000000
	0.000000				
65.000000	0.000000	0.000000	0.000000	8.985252	0.000000
	0.000000				
65.250000	0.000000	0.000000	0.000000	9.017685	0.000000
	0.000000				
65.500000	0.000000	0.000000	0.000000	9.050117	0.000000
	0.000000				
65.750000	0.000000	0.000000	0.000000	9.082550	0.000000
	0.000000				
66.000000	0.000000	0.000000	0.000000	9.114983	0.000000
	0.000000				
66.250000	0.000000	0.000000	0.000000	9.147415	0.000000
	0.000000				
66.500000	0.000000	0.000000	0.000000	9.179848	0.000000
	0.000000				
66.750000	0.000000	0.000000	0.000000	9.212280	0.000000
	0.000000				
67.000000	0.000000	0.000000	0.000000	9.244713	0.000000
	0.000000				
67.250000	0.000000	0.000000	0.000000	9.277145	0.000000
	0.000000				
67.500000	0.000000	0.000000	0.000000	9.309578	0.000000
	0.000000				
67.750000	0.000000	0.000000	0.000000	9.342010	0.000000
	0.000000				
68.000000	0.000000	0.000000	0.000000	9.374443	0.000000
	0.000000				
68.250000	0.000000	0.000000	0.000000	9.406876	0.000000
	0.000000				
68.500000	0.000000	0.000000	0.000000	9.439308	0.000000
	0.000000				
68.750000	0.000000	0.000000	0.000000	9.471741	0.000000
	0.000000				
69.000000	0.000000	0.000000	0.000000	9.504173	0.000000
	0.000000				
69.250000	0.000000	0.000000	0.000000	9.536606	0.000000
	0.000000				
69.500000	0.000000	0.000000	0.000000	9.569038	0.000000
	0.000000				
69.750000	0.000000	0.000000	0.000000	9.601471	0.000000
	0.000000				
70.000000	0.000000	0.000000	0.000000	9.633904	0.000000
	0.000000				
70.250000	0.000000	0.000000	0.000000	9.666336	0.000000
	0.000000				
70.500000	0.000000	0.000000	0.000000	9.698769	0.000000
	0.000000				
70.750000	0.000000	0.000000	0.000000	9.731201	0.000000
	0.000000				
71.000000	0.000000	0.000000	0.000000	9.763634	0.000000
	0.000000				

71.250000	0.000000	0.000000	0.000000	9.796066	0.000000
	0.000000				
71.500000	0.000000	0.000000	0.000000	9.828499	0.000000
	0.000000				
71.750000	0.000000	0.000000	0.000000	9.860931	0.000000
	0.000000				
72.000000	0.000000	0.000000	0.000000	9.893365	0.000000
	0.000000				
72.250000	0.000000	0.000000	0.000000	9.925797	0.000000
	0.000000				
72.500000	0.000000	0.000000	0.000000	9.958230	0.000000
	0.000000				
72.750000	0.000000	0.000000	0.000000	9.990663	0.000000
	0.000000				
73.000000	0.000000	0.000000	0.000000	10.023095	0.000000
	0.000000				
73.250000	0.000000	0.000000	0.000000	10.055528	0.000000
	0.000000				
73.500000	0.000000	0.000000	0.000000	10.087960	0.000000
	0.000000				
73.750000	0.000000	0.000000	0.000000	10.120393	0.000000
	0.000000				
74.000000	0.000000	0.000000	0.000000	10.152825	0.000000
	0.000000				
74.250000	0.000000	0.000000	0.000000	10.185258	0.000000
	0.000000				
74.500000	0.000000	0.000000	0.000000	10.217690	0.000000
	0.000000				
74.750000	0.000000	0.000000	0.000000	10.250123	0.000000
	0.000000				
75.000000	0.000000	0.000000	0.000000	10.282556	0.000000
	0.000000				
75.250000	0.000000	0.000000	0.000000	10.314988	0.000000
	0.000000				
75.500000	0.000000	0.000000	0.000000	10.347421	0.000000
	0.000000				
75.750000	0.000000	0.000000	0.000000	10.379853	0.000000
	0.000000				
76.000000	0.000000	0.000000	0.000000	10.412286	0.000000
	0.000000				
76.250000	0.000000	0.000000	0.000000	10.444718	0.000000
	0.000000				
76.500000	0.000000	0.000000	0.000000	10.477151	0.000000
	0.000000				
76.750000	0.000000	0.000000	0.000000	10.509583	0.000000
	0.000000				
77.000000	0.000000	0.000000	0.000000	10.542016	0.000000
	0.000000				
77.250000	0.000000	0.000000	0.000000	10.574449	0.000000
	0.000000				
77.500000	0.000000	0.000000	0.000000	10.606881	0.000000
	0.000000				
77.750000	0.000000	0.000000	0.000000	10.639314	0.000000
	0.000000				
78.000000	0.000000	0.000000	0.000000	10.671746	0.000000
	0.000000				
78.250000	0.000000	0.000000	0.000000	10.704179	0.000000
	0.000000				
78.500000	0.000000	0.000000	0.000000	10.736611	0.000000
	0.000000				
78.750000	0.000000	0.000000	0.000000	10.769044	0.000000
	0.000000				
79.000000	0.000000	0.000000	0.000000	10.801476	0.000000
	0.000000				
79.250000	0.000000	0.000000	0.000000	10.833909	0.000000
	0.000000				
79.500000	0.000000	0.000000	0.000000	10.866342	0.000000
	0.000000				
79.750000	0.000000	0.000000	0.000000	10.898774	0.000000
	0.000000				
80.000000	0.000000	0.000000	0.000000	10.931207	0.000000
	0.000000				
80.250000	0.000000	0.000000	0.000000	10.963639	0.000000
	0.000000				
80.500000	0.000000	0.000000	0.000000	10.996072	0.000000
	0.000000				
80.750000	0.000000	0.000000	0.000000	11.028504	0.000000
	0.000000				
81.000000	0.000000	0.000000	0.000000	11.060937	0.000000
	0.000000				
81.250000	0.000000	0.000000	0.000000	11.093369	0.000000
	0.000000				
81.500000	0.000000	0.000000	0.000000	11.125802	0.000000
	0.000000				
81.750000	0.000000	0.000000	0.000000	11.158236	0.000000
	0.000000				
82.000000	0.000000	0.000000	0.000000	11.190668	0.000000
	0.000000				
82.250000	0.000000	0.000000	0.000000	11.223101	0.000000
	0.000000				
82.500000	0.000000	0.000000	0.000000	11.255533	0.000000
	0.000000				
82.750000	0.000000	0.000000	0.000000	11.287966	0.000000
	0.000000				
83.000000	0.000000	0.000000	0.000000	11.320398	0.000000
	0.000000				
83.250000	0.000000	0.000000	0.000000	11.352831	0.000000
	0.000000				

83.500000	0.000000	0.000000	0.000000	11.385263	0.000000
	0.000000				
83.750000	0.000000	0.000000	0.000000	11.417696	0.000000
	0.000000				
84.000000	0.000000	0.000000	0.000000	11.450129	0.000000
	0.000000				
84.250000	0.000000	0.000000	0.000000	11.482561	0.000000
	0.000000				
84.500000	0.000000	0.000000	0.000000	11.514994	0.000000
	0.000000				
84.750000	0.000000	0.000000	0.000000	11.547426	0.000000
	0.000000				
85.000000	0.000000	0.000000	0.000000	11.579859	0.000000
	0.000000				
85.250000	0.000000	0.000000	0.000000	11.612291	0.000000
	0.000000				
85.500000	0.000000	0.000000	0.000000	11.644724	0.000000
	0.000000				
85.750000	0.000000	0.000000	0.000000	11.677156	0.000000
	0.000000				
86.000000	0.491945	1.675973	-0.007624	11.709589	0.491945
	0.000000				
87.000000	0.444441	1.680004	-0.038120	11.839319	0.444441
	0.000000				
88.000000	0.396454	1.682985	-0.068616	11.969049	0.396454
	0.000000				
89.000000	0.348043	1.684915	-0.099112	12.098780	0.348043
	0.000000				
90.000000	0.299272	1.685792	-0.129608	0.000000	0.299272
	0.000000				
91.000000	0.250204	1.685614	-0.160104	0.000000	0.250204
	0.000000				
92.000000	0.200903	1.684385	-0.190600	0.000000	0.200903
	0.000000				
93.000000	0.151435	1.682105	-0.221096	0.000000	0.151435
	0.000000				
94.000000	0.101867	1.678778	-0.251592	0.000000	0.101867
	0.000000				
95.000000	0.052266	1.674410	-0.282088	0.000000	0.052266
	0.000000				
96.000000	0.002701	1.669007	-0.312584	0.000000	0.002701
	0.000000				
97.000000	-0.046761	1.662575	-0.343081	0.000000	-0.046761
	0.000000				
98.000000	-0.096050	1.655125	-0.373577	0.000000	-0.096050
	0.000000				
99.000000	-0.145097	1.646665	-0.404073	0.000000	-0.145097
	0.000000				
100.000000	-0.241907	1.637207	-0.434569	0.000000	-0.241907
	0.000000				
101.000000	-0.289123	1.626763	-0.458410	0.000000	-0.289123
	0.000000				
102.000000	-0.335924	1.615348	-0.459415	0.000000	-0.335924
	0.000000				
103.000000	-0.382239	1.602975	-0.460455	0.000000	-0.382239
	0.000000				
104.000000	-0.427999	1.589661	-0.461525	0.000000	-0.427999
	0.000000				
105.000000	-0.473135	1.575424	-0.462621	0.000000	-0.473135
	0.000000				
106.000000	-0.517578	1.560281	-0.463739	0.000000	-0.517578
	0.000000				
107.000000	-0.561262	1.544251	-0.464872	0.000000	-0.561262
	0.000000				
108.000000	-0.604121	1.527356	-0.466015	0.000000	-0.604121
	0.000000				
109.000000	-0.646088	1.509617	-0.467160	0.000000	-0.646088
	0.000000				
110.000000	-0.687102	1.491056	-0.468298	0.000000	-0.687102
	0.000000				
111.000000	-0.727098	1.471697	-0.469422	0.000000	-0.727098
	0.000000				
112.000000	-0.766017	1.451564	-0.470521	0.000000	-0.766017
	0.000000				
113.000000	-0.803799	1.430684	-0.471586	0.000000	-0.803799
	0.000000				
114.000000	-0.840388	1.409082	-0.472606	0.000000	-0.840388
	0.000000				
115.000000	-0.875727	1.386786	-0.473570	0.000000	-0.875727
	0.000000				
116.000000	-0.909765	1.363823	-0.474466	0.000000	-0.909765
	0.000000				
117.000000	-0.942450	1.340223	-0.475282	0.000000	-0.942450
	0.000000				
118.000000	-0.973734	1.316015	-0.476006	0.000000	-0.973734
	0.000000				
119.000000	-1.003569	1.291231	-0.476624	0.000000	-1.003569
	0.000000				
120.000000	-1.031914	1.265900	-0.477122	0.000000	-1.031914
	0.000000				
121.000000	-1.058726	1.240054	-0.477489	0.000000	-1.058726
	0.000000				
122.000000	-1.083967	1.213727	-0.477709	0.000000	-1.083967
	0.000000				
123.000000	-1.107602	1.186950	-0.477769	0.000000	-1.107602
	0.000000				
124.000000	-1.129598	1.159758	-0.477656	0.000000	-1.129598
	0.000000				

125.000000	-1.149926	1.132184	-0.477355	0.000000	-1.149926
	0.000000				
126.000000	-1.168560	1.104262	-0.476853	0.000000	-1.168560
	0.000000				
127.000000	-1.185475	1.076028	-0.476136	0.000000	-1.185475
	0.000000				
128.000000	-1.200651	1.047516	-0.475190	0.000000	-1.200651
	0.000000				
129.000000	-1.214073	1.018763	-0.474004	0.000000	-1.214073
	0.000000				
130.000000	-1.225726	0.989802	-0.472565	0.000000	-1.225726
	0.000000				
131.000000	-1.235599	0.960672	-0.470859	0.000000	-1.235599
	0.000000				
132.000000	-1.243688	0.931407	-0.468876	0.000000	-1.243688
	0.000000				
133.000000	-1.249987	0.902043	-0.466606	0.000000	-1.249987
	0.000000				
134.000000	-1.254498	0.872618	-0.464036	0.000000	-1.254498
	0.000000				
135.000000	-1.257224	0.843167	-0.461159	0.000000	-1.257224
	0.000000				
136.000000	-1.258173	0.813726	-0.457965	0.000000	-1.258173
	0.000000				
137.000000	-1.257356	0.784333	-0.454447	0.000000	-1.257356
	0.000000				
138.000000	-1.254787	0.755022	-0.450598	0.000000	-1.254787
	0.000000				
139.000000	-1.250485	0.725831	-0.446412	0.000000	-1.250485
	0.000000				
140.000000	-1.244473	0.696794	-0.441885	0.000000	-1.244473
	0.000000				
141.000000	-1.236776	0.667948	-0.437013	0.000000	-1.236776
	0.000000				
142.000000	-1.227424	0.639328	-0.431794	0.000000	-1.227424
	0.000000				
143.000000	-1.216452	0.610968	-0.426228	0.000000	-1.216452
	0.000000				
144.000000	-1.203899	0.582904	-0.420316	0.000000	-1.203899
	0.000000				
145.000000	-1.189808	0.555169	-0.414060	0.000000	-1.189808
	0.000000				
146.000000	-1.174228	0.527797	-0.407464	0.000000	-1.174228
	0.000000				
147.000000	-1.157214	0.500822	-0.400535	0.000000	-1.157214
	0.000000				
148.000000	-1.138826	0.474276	-0.393283	0.000000	-1.138826
	0.000000				
149.000000	-1.119135	0.448191	-0.385719	0.000000	-1.119135
	0.000000				
150.000000	-1.098222	0.422600	-0.377859	0.000000	-1.098222
	0.000000				
151.000000	-1.076181	0.397531	-0.369725	0.000000	-1.076181
	0.000000				
152.000000	-1.053125	0.373017	-0.361342	0.000000	-1.053125
	0.000000				
153.000000	-1.029194	0.349086	-0.352748	0.000000	-1.029194
	0.000000				
154.000000	-1.004568	0.325767	-0.343992	0.000000	-1.004568
	0.000000				
155.000000	-0.979483	0.303089	-0.335142	0.000000	-0.979483
	0.000000				
156.000000	-0.954271	0.281077	-0.326298	0.000000	-0.954271
	0.000000				
157.000000	-0.929409	0.259758	-0.317607	0.000000	-0.929409
	0.000000				
158.000000	-0.905624	0.239158	-0.309300	0.000000	-0.905624
	0.000000				
159.000000	-0.884067	0.219300	-0.301746	0.000000	-0.884067
	0.000000				
160.000000	-0.866647	0.200210	-0.295567	0.000000	-0.866647
	0.000000				
161.000000	-0.856648	0.181907	-0.291838	0.000000	-0.856648
	0.000000				
162.000000	-0.859866	0.164416	-0.292477	0.000000	-0.859866
	0.000000				
163.000000	-0.886456	0.147754	-0.300862	0.000000	-0.886456
	0.000000				
164.000000	-0.952499	0.131943	-0.322403	0.000000	-0.952499
	0.000000				
165.000000	-1.074200	0.117001	-0.362695	0.000000	-1.074200
	0.000000				
166.000000	-1.235870	0.102944	-0.416930	0.000000	-1.235870
	0.000000				
167.000000	-1.351591	0.089788	-0.456746	0.000000	-1.351591
	0.000000				
168.000000	-1.347773	0.077549	-0.457119	0.000000	-1.347773
	0.000000				
169.000000	-1.263099	0.066241	-0.430276	0.000000	-1.263099
	0.000000				
170.000000	-1.159854	0.055876	-0.396838	0.000000	-1.159854
	0.000000				
171.000000	-1.056070	0.046465	-0.362857	0.000000	-1.056070
	0.000000				
172.000000	-0.952286	0.038019	-0.328531	0.000000	-0.952286
	0.000000				
173.000000	-0.848501	0.030547	-0.293874	0.000000	-0.848501
	0.000000				

174.000000	-0.744717	0.024057	-0.258900	0.000000	-0.744717
	0.000000				
175.000000	-0.640933	0.018556	-0.223625	0.000000	-0.640933
	0.000000				
176.000000	-0.537149	0.014048	-0.188061	0.000000	-0.537149
	0.000000				
177.000000	-0.433364	0.010539	-0.152224	0.000000	-0.433364
	0.000000				
178.000000	-0.329580	0.008030	-0.116128	0.000000	-0.329580
	0.000000				
179.000000	-0.225796	0.006525	-0.079788	0.000000	-0.225796
	0.000000				
180.000000	-0.122012	0.006000	-0.042747	0.000000	-0.122012
	0.000000				

C.3 DU99-W-405 Extrapolated Polars

Listing 9: Aerodynamic polar characteristics for DU99-W-405.

-----QBlade Multi RE Polar File-----					
Generated with : QBlade CE v2.0.9.4_beta windows					
Archive Format: 310039					
Time : 19:42:31					
Date : 04.06.2026					
-----Object Names-----					
DU_99-W-405_coords_Re8.500.M0.00_N9.0 360 M			POLARNAME	- the polar name	
../Airfoils/DU_99-W-405_coords.afl			FOILNAME	- the airfoil name to which	
the polar(s) belong OR path to airfoil file					
-----Parameters-----					
40.5			THICKNESS	- the thickness of the airfoil in % of chord	
true			ISDECOMPOSED	- is the polar decomposed (imports Cl_Sep,	
Cl_att and f_st columns) [bool]					
REYNOLDS			8.5000E+06	- the list of Reynolds numbers for the imported polars	
-----Polar Data-----					
AOA	CL	CD	CM	CL_ATT	CL_SEP
	F_ST				
-180.000000	-0.054993	0.006000	-0.020888	0.000000	-0.054993
	0.000000				
-179.000000	0.072653	0.006458	0.027541	0.000000	0.072653
	0.000000				
-178.000000	0.200300	0.007831	0.075625	0.000000	0.200300
	0.000000				
-177.000000	0.327946	0.010116	0.123339	0.000000	0.327946
	0.000000				
-176.000000	0.455593	0.013310	0.170657	0.000000	0.455593
	0.000000				
-175.000000	0.583239	0.017407	0.217553	0.000000	0.583239
	0.000000				
-174.000000	0.710885	0.022402	0.264004	0.000000	0.710885
	0.000000				
-173.000000	0.838532	0.028288	0.309983	0.000000	0.838532
	0.000000				
-172.000000	0.965343	0.035056	0.355161	0.000000	0.965343
	0.000000				
-171.000000	1.068440	0.042697	0.391194	0.000000	1.068440
	0.000000				
-170.000000	1.072856	0.051202	0.391121	0.000000	1.072856
	0.000000				
-169.000000	0.934641	0.060558	0.339858	0.000000	0.934641
	0.000000				
-168.000000	0.740918	0.070754	0.269562	0.000000	0.740918
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58.500000	0.000000	0.000000	0.000000	9.668635	0.000000
	0.000000				
58.750000	0.000000	0.000000	0.000000	9.708525	0.000000
	0.000000				
59.000000	0.000000	0.000000	0.000000	9.748415	0.000000
	0.000000				
59.250000	0.000000	0.000000	0.000000	9.788304	0.000000
	0.000000				
59.500000	0.000000	0.000000	0.000000	9.828194	0.000000
	0.000000				
59.750000	0.000000	0.000000	0.000000	9.868083	0.000000
	0.000000				
60.000000	0.000000	0.000000	0.000000	9.907972	0.000000
	0.000000				
60.250000	0.000000	0.000000	0.000000	9.947862	0.000000
	0.000000				
60.500000	0.000000	0.000000	0.000000	9.987752	0.000000
	0.000000				
60.750000	0.000000	0.000000	0.000000	10.027641	0.000000
	0.000000				
61.000000	0.000000	0.000000	0.000000	10.067531	0.000000
	0.000000				
61.250000	0.000000	0.000000	0.000000	10.107420	0.000000
	0.000000				
61.500000	0.000000	0.000000	0.000000	10.147309	0.000000
	0.000000				
61.750000	0.000000	0.000000	0.000000	10.187199	0.000000
	0.000000				
62.000000	0.000000	0.000000	0.000000	10.227089	0.000000
	0.000000				
62.250000	0.000000	0.000000	0.000000	10.266978	0.000000
	0.000000				
62.500000	0.000000	0.000000	0.000000	10.306868	0.000000
	0.000000				
62.750000	0.000000	0.000000	0.000000	10.346757	0.000000
	0.000000				
63.000000	0.000000	0.000000	0.000000	10.386646	0.000000
	0.000000				
63.250000	0.000000	0.000000	0.000000	10.426536	0.000000
	0.000000				
63.500000	0.000000	0.000000	0.000000	10.466426	0.000000
	0.000000				
63.750000	0.000000	0.000000	0.000000	10.506315	0.000000
	0.000000				
64.000000	0.000000	0.000000	0.000000	10.546205	0.000000
	0.000000				
64.250000	0.000000	0.000000	0.000000	10.586094	0.000000
	0.000000				
64.500000	0.000000	0.000000	0.000000	10.625983	0.000000
	0.000000				
64.750000	0.000000	0.000000	0.000000	10.665874	0.000000
	0.000000				
65.000000	0.000000	0.000000	0.000000	10.705763	0.000000
	0.000000				
65.250000	0.000000	0.000000	0.000000	10.745652	0.000000
	0.000000				
65.500000	0.000000	0.000000	0.000000	10.785542	0.000000
	0.000000				
65.750000	0.000000	0.000000	0.000000	10.825431	0.000000
	0.000000				
66.000000	0.000000	0.000000	0.000000	10.865320	0.000000
	0.000000				
66.250000	0.000000	0.000000	0.000000	10.905210	0.000000
	0.000000				
66.500000	0.000000	0.000000	0.000000	10.945100	0.000000
	0.000000				
66.750000	0.000000	0.000000	0.000000	10.984989	0.000000
	0.000000				
67.000000	0.000000	0.000000	0.000000	11.024879	0.000000
	0.000000				
67.250000	0.000000	0.000000	0.000000	11.064768	0.000000
	0.000000				
67.500000	0.000000	0.000000	0.000000	11.104657	0.000000
	0.000000				

67.750000	0.000000	0.000000	0.000000	11.144547	0.000000
	0.000000				
68.000000	0.000000	0.000000	0.000000	11.184437	0.000000
	0.000000				
68.250000	0.000000	0.000000	0.000000	11.224326	0.000000
	0.000000				
68.500000	0.000000	0.000000	0.000000	11.264215	0.000000
	0.000000				
68.750000	0.000000	0.000000	0.000000	11.304105	0.000000
	0.000000				
69.000000	0.000000	0.000000	0.000000	11.343994	0.000000
	0.000000				
69.250000	0.000000	0.000000	0.000000	11.383884	0.000000
	0.000000				
69.500000	0.000000	0.000000	0.000000	11.423774	0.000000
	0.000000				
69.750000	0.000000	0.000000	0.000000	11.463663	0.000000
	0.000000				
70.000000	0.000000	0.000000	0.000000	11.503552	0.000000
	0.000000				
70.250000	0.000000	0.000000	0.000000	11.543442	0.000000
	0.000000				
70.500000	0.000000	0.000000	0.000000	11.583331	0.000000
	0.000000				
70.750000	0.000000	0.000000	0.000000	11.623221	0.000000
	0.000000				
71.000000	0.000000	0.000000	0.000000	11.663111	0.000000
	0.000000				
71.250000	0.000000	0.000000	0.000000	11.703000	0.000000
	0.000000				
71.500000	0.000000	0.000000	0.000000	11.742889	0.000000
	0.000000				
71.750000	0.000000	0.000000	0.000000	11.782779	0.000000
	0.000000				
72.000000	0.000000	0.000000	0.000000	11.822669	0.000000
	0.000000				
72.250000	0.000000	0.000000	0.000000	11.862558	0.000000
	0.000000				
72.500000	0.000000	0.000000	0.000000	11.902448	0.000000
	0.000000				
72.750000	0.000000	0.000000	0.000000	11.942337	0.000000
	0.000000				
73.000000	0.000000	0.000000	0.000000	11.982226	0.000000
	0.000000				
73.250000	0.000000	0.000000	0.000000	12.022116	0.000000
	0.000000				
73.500000	0.000000	0.000000	0.000000	12.062006	0.000000
	0.000000				
73.750000	0.000000	0.000000	0.000000	12.101895	0.000000
	0.000000				
74.000000	0.000000	0.000000	0.000000	12.141785	0.000000
	0.000000				
74.250000	0.000000	0.000000	0.000000	12.181674	0.000000
	0.000000				
74.500000	0.000000	0.000000	0.000000	12.221563	0.000000
	0.000000				
74.750000	0.000000	0.000000	0.000000	12.261453	0.000000
	0.000000				
75.000000	0.000000	0.000000	0.000000	12.301343	0.000000
	0.000000				
75.250000	0.000000	0.000000	0.000000	12.341232	0.000000
	0.000000				
75.500000	0.000000	0.000000	0.000000	12.381122	0.000000
	0.000000				
75.750000	0.000000	0.000000	0.000000	12.421011	0.000000
	0.000000				
76.000000	0.000000	0.000000	0.000000	12.460900	0.000000
	0.000000				
76.250000	0.000000	0.000000	0.000000	12.500790	0.000000
	0.000000				
76.500000	0.000000	0.000000	0.000000	12.540680	0.000000
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76.750000	0.000000	0.000000	0.000000	12.580569	0.000000
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77.000000	0.000000	0.000000	0.000000	12.620459	0.000000
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77.250000	0.000000	0.000000	0.000000	12.660348	0.000000
	0.000000				
77.500000	0.000000	0.000000	0.000000	12.700237	0.000000
	0.000000				
77.750000	0.000000	0.000000	0.000000	12.740127	0.000000
	0.000000				
78.000000	0.000000	0.000000	0.000000	12.780017	0.000000
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78.250000	0.000000	0.000000	0.000000	12.819906	0.000000
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78.500000	0.000000	0.000000	0.000000	12.859796	0.000000
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78.750000	0.000000	0.000000	0.000000	12.899685	0.000000
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79.000000	0.000000	0.000000	0.000000	12.939574	0.000000
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79.250000	0.000000	0.000000	0.000000	12.979465	0.000000
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79.500000	0.000000	0.000000	0.000000	13.019354	0.000000
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79.750000	0.000000	0.000000	0.000000	13.059243	0.000000
	0.000000				

80.000000	0.000000	0.000000	0.000000	13.099133	0.000000
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80.250000	0.000000	0.000000	0.000000	13.139022	0.000000
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80.500000	0.000000	0.000000	0.000000	13.178911	0.000000
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80.750000	0.000000	0.000000	0.000000	13.218801	0.000000
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81.000000	0.000000	0.000000	0.000000	13.258691	0.000000
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81.250000	0.000000	0.000000	0.000000	13.298580	0.000000
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81.500000	0.000000	0.000000	0.000000	13.338470	0.000000
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81.750000	0.000000	0.000000	0.000000	13.378359	0.000000
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82.000000	0.000000	0.000000	0.000000	13.418248	0.000000
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82.250000	0.000000	0.000000	0.000000	13.458138	0.000000
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82.500000	0.000000	0.000000	0.000000	13.498028	0.000000
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82.750000	0.000000	0.000000	0.000000	13.537917	0.000000
	0.000000				
83.000000	0.000000	0.000000	0.000000	13.577806	0.000000
	0.000000				
83.250000	0.000000	0.000000	0.000000	13.617696	0.000000
	0.000000				
83.500000	0.000000	0.000000	0.000000	13.657585	0.000000
	0.000000				
83.750000	0.000000	0.000000	0.000000	13.697475	0.000000
	0.000000				
84.000000	0.000000	0.000000	0.000000	13.737365	0.000000
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84.250000	0.000000	0.000000	0.000000	13.777254	0.000000
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84.500000	0.000000	0.000000	0.000000	13.817143	0.000000
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84.750000	0.000000	0.000000	0.000000	13.857033	0.000000
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85.000000	0.000000	0.000000	0.000000	13.896922	0.000000
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85.250000	0.000000	0.000000	0.000000	13.936812	0.000000
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85.500000	0.000000	0.000000	0.000000	13.976702	0.000000
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85.750000	0.000000	0.000000	0.000000	14.016591	0.000000
	0.000000				
86.000000	0.000000	0.000000	0.000000	14.056480	0.000000
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86.250000	0.000000	0.000000	0.000000	14.096370	0.000000
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86.500000	0.000000	0.000000	0.000000	14.136260	0.000000
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86.750000	0.000000	0.000000	0.000000	14.176149	0.000000
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87.000000	0.000000	0.000000	0.000000	14.216039	0.000000
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87.250000	0.000000	0.000000	0.000000	14.255928	0.000000
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88.000000	0.465588	1.517170	-0.020301	14.375597	0.465588
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89.000000	0.424623	1.519366	-0.047370	14.535154	0.424623
	0.000000				
90.000000	0.383475	1.520597	-0.074439	0.000000	0.383475
	0.000000				
91.000000	0.342192	1.520863	-0.101507	0.000000	0.342192
	0.000000				
92.000000	0.300818	1.520165	-0.128576	0.000000	0.300818
	0.000000				
93.000000	0.259401	1.518506	-0.155644	0.000000	0.259401
	0.000000				
94.000000	0.217989	1.515889	-0.182713	0.000000	0.217989
	0.000000				
95.000000	0.176630	1.512317	-0.209781	0.000000	0.176630
	0.000000				
96.000000	0.135373	1.507798	-0.236850	0.000000	0.135373
	0.000000				
97.000000	0.094269	1.502337	-0.263918	0.000000	0.094269
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98.000000	0.053367	1.495942	-0.290987	0.000000	0.053367
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99.000000	0.012718	1.488623	-0.318056	0.000000	0.012718
	0.000000				
100.000000	-0.027628	1.480389	-0.345124	0.000000	-0.027628
	0.000000				
101.000000	-0.067621	1.471253	-0.372193	0.000000	-0.067621
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102.000000	-0.107208	1.461226	-0.399261	0.000000	-0.107208
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103.000000	-0.302851	1.450321	-0.416286	0.000000	-0.302851
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104.000000	-0.338267	1.438555	-0.417216	0.000000	-0.338267
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105.000000	-0.373209	1.425941	-0.418110	0.000000	-0.373209
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106.000000	-0.407625	1.412497	-0.418966	0.000000	-0.407625
	0.000000				

107.000000	-0.441463	1.398240	-0.419780	0.000000	-0.441463
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108.000000	-0.474672	1.383189	-0.420550	0.000000	-0.474672
	0.000000				
109.000000	-0.507204	1.367363	-0.421270	0.000000	-0.507204
	0.000000				
110.000000	-0.539008	1.350784	-0.421936	0.000000	-0.539008
	0.000000				
111.000000	-0.570038	1.333472	-0.422543	0.000000	-0.570038
	0.000000				
112.000000	-0.600247	1.315449	-0.423086	0.000000	-0.600247
	0.000000				
113.000000	-0.629589	1.296739	-0.423558	0.000000	-0.629589
	0.000000				
114.000000	-0.658021	1.277367	-0.423953	0.000000	-0.658021
	0.000000				
115.000000	-0.685500	1.257355	-0.424265	0.000000	-0.685500
	0.000000				
116.000000	-0.711984	1.236731	-0.424486	0.000000	-0.711984
	0.000000				
117.000000	-0.737436	1.215521	-0.424608	0.000000	-0.737436
	0.000000				
118.000000	-0.761816	1.193750	-0.424625	0.000000	-0.761816
	0.000000				
119.000000	-0.785089	1.171448	-0.424527	0.000000	-0.785089
	0.000000				
120.000000	-0.807222	1.148642	-0.424307	0.000000	-0.807222
	0.000000				
121.000000	-0.828181	1.125360	-0.423957	0.000000	-0.828181
	0.000000				
122.000000	-0.847937	1.101634	-0.423467	0.000000	-0.847937
	0.000000				
123.000000	-0.866463	1.077492	-0.422829	0.000000	-0.866463
	0.000000				
124.000000	-0.883731	1.052964	-0.422033	0.000000	-0.883731
	0.000000				
125.000000	-0.899718	1.028083	-0.421073	0.000000	-0.899718
	0.000000				
126.000000	-0.914403	1.002878	-0.419937	0.000000	-0.914403
	0.000000				
127.000000	-0.927767	0.977381	-0.418618	0.000000	-0.927767
	0.000000				
128.000000	-0.939793	0.951625	-0.417108	0.000000	-0.939793
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129.000000	-0.950467	0.925642	-0.415397	0.000000	-0.950467
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130.000000	-0.959776	0.899464	-0.413478	0.000000	-0.959776
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131.000000	-0.967712	0.873124	-0.411342	0.000000	-0.967712
	0.000000				
132.000000	-0.974267	0.846654	-0.408982	0.000000	-0.974267
	0.000000				
133.000000	-0.979437	0.820088	-0.406390	0.000000	-0.979437
	0.000000				
134.000000	-0.983221	0.793459	-0.403561	0.000000	-0.983221
	0.000000				
135.000000	-0.985618	0.766799	-0.400487	0.000000	-0.985618
	0.000000				
136.000000	-0.986633	0.740142	-0.397162	0.000000	-0.986633
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137.000000	-0.986272	0.713521	-0.393582	0.000000	-0.986272
	0.000000				
138.000000	-0.984543	0.686968	-0.389742	0.000000	-0.984543
	0.000000				
139.000000	-0.981459	0.660517	-0.385638	0.000000	-0.981459
	0.000000				
140.000000	-0.977033	0.634201	-0.381266	0.000000	-0.977033
	0.000000				
141.000000	-0.971284	0.608051	-0.376624	0.000000	-0.971284
	0.000000				
142.000000	-0.964230	0.582100	-0.371711	0.000000	-0.964230
	0.000000				
143.000000	-0.955896	0.556379	-0.366526	0.000000	-0.955896
	0.000000				
144.000000	-0.946307	0.530921	-0.361070	0.000000	-0.946307
	0.000000				
145.000000	-0.935494	0.505757	-0.355344	0.000000	-0.935494
	0.000000				
146.000000	-0.923491	0.480917	-0.349352	0.000000	-0.923491
	0.000000				
147.000000	-0.910335	0.456432	-0.343096	0.000000	-0.910335
	0.000000				
148.000000	-0.896067	0.432332	-0.336585	0.000000	-0.896067
	0.000000				
149.000000	-0.880737	0.408646	-0.329824	0.000000	-0.880737
	0.000000				
150.000000	-0.864397	0.385403	-0.322825	0.000000	-0.864397
	0.000000				
151.000000	-0.847110	0.362631	-0.315602	0.000000	-0.847110
	0.000000				
152.000000	-0.828949	0.340358	-0.308170	0.000000	-0.828949
	0.000000				
153.000000	-0.809999	0.318611	-0.300552	0.000000	-0.809999
	0.000000				
154.000000	-0.790365	0.297416	-0.292777	0.000000	-0.790365
	0.000000				
155.000000	-0.770177	0.276799	-0.284881	0.000000	-0.770177
	0.000000				

156.000000	-0.749601	0.256784	-0.276917	0.000000	-0.749601
	0.000000				
157.000000	-0.728859	0.237397	-0.268954	0.000000	-0.728859
	0.000000				
158.000000	-0.708255	0.218659	-0.261094	0.000000	-0.708255
	0.000000				
159.000000	-0.688223	0.200594	-0.253482	0.000000	-0.688223
	0.000000				
160.000000	-0.669401	0.183223	-0.246339	0.000000	-0.669401
	0.000000				
161.000000	-0.652771	0.166566	-0.240008	0.000000	-0.652771
	0.000000				
162.000000	-0.639889	0.150645	-0.235039	0.000000	-0.639889
	0.000000				
163.000000	-0.633306	0.135476	-0.232345	0.000000	-0.633306
	0.000000				
164.000000	-0.637322	0.121079	-0.233471	0.000000	-0.637322
	0.000000				
165.000000	-0.659224	0.107470	-0.241058	0.000000	-0.659224
	0.000000				
166.000000	-0.710885	0.094665	-0.259442	0.000000	-0.710885
	0.000000				
167.000000	-0.808420	0.082680	-0.294590	0.000000	-0.808420
	0.000000				
168.000000	-0.960325	0.071527	-0.349888	0.000000	-0.960325
	0.000000				
169.000000	-1.130417	0.061220	-0.412644	0.000000	-1.130417
	0.000000				
170.000000	-1.222791	0.051771	-0.448052	0.000000	-1.222791
	0.000000				
171.000000	-1.186773	0.043191	-0.436943	0.000000	-1.186773
	0.000000				
172.000000	-1.075712	0.035488	-0.398053	0.000000	-1.075712
	0.000000				
173.000000	-0.948518	0.028673	-0.352715	0.000000	-0.948518
	0.000000				
174.000000	-0.820871	0.022751	-0.306691	0.000000	-0.820871
	0.000000				
175.000000	-0.693225	0.017730	-0.260173	0.000000	-0.693225
	0.000000				
176.000000	-0.565579	0.013615	-0.213185	0.000000	-0.565579
	0.000000				
177.000000	-0.437932	0.010410	-0.165752	0.000000	-0.437932
	0.000000				
178.000000	-0.310286	0.008117	-0.117899	0.000000	-0.310286
	0.000000				
179.000000	-0.182639	0.006740	-0.069651	0.000000	-0.182639
	0.000000				
180.000000	-0.054993	0.006000	-0.020888	0.000000	-0.054993
	0.000000				

C.4 NACA 64-615 Extrapolated Polars

Listing 10: Aerodynamic polar characteristics for NACA 64-615.

-----QBlade Multi RE Polar File-----					
Generated with : QBlade CE v2.0.9.4_beta windows					
Archive Format: 310039					
Time : 19:42:31					
Date : 04.06.2026					
-----Object Names-----					
NACA_64-615_coords_Re9.500_M0.00_N9.0 360 M			POLARNAME	- the polar name	
../Airfoils/NACA_64-615_coords.afl			FOILNAME	- the airfoil name to which	
the polar(s) belong OR path to airfoil file					
-----Parameters-----					
15.0	THICKNESS		- the thickness of the airfoil in % of chord		
true	ISDECOMPOSED		- is the polar decomposed (imports Cl_Sep,		
Cl_att and f_st columns) [bool]					
REYNOLDS	9.5000E+06	- the list of Reynolds numbers for the imported polars			
-----Polar Data-----					
AOA	CL	CD	CM	CL_ATT	CL_SEP
	F_ST				
-180.000000	-0.124021	0.006000	-0.043521	0.000000	-0.124021
	0.000000				
-179.000000	-0.029020	0.006515	-0.010115	0.000000	-0.029020
	0.000000				
-178.000000	0.065982	0.008057	0.023118	0.000000	0.065982
	0.000000				
-177.000000	0.160984	0.010622	0.056167	0.000000	0.160984
	0.000000				
-176.000000	0.255986	0.014206	0.089022	0.000000	0.255986
	0.000000				
-175.000000	0.350969	0.018803	0.121663	0.000000	0.350969
	0.000000				
-174.000000	0.443752	0.024404	0.153330	0.000000	0.443752
	0.000000				
-173.000000	0.520707	0.031002	0.179361	0.000000	0.520707
	0.000000				
-172.000000	0.555349	0.038587	0.190813	0.000000	0.555349
	0.000000				
-171.000000	0.533175	0.047147	0.182997	0.000000	0.533175
	0.000000				

-170.000000	0.476052	0.056671	0.163601	0.000000	0.476052
	0.000000				
-169.000000	0.418569	0.067145	0.144426	0.000000	0.418569
	0.000000				
-168.000000	0.377789	0.078555	0.131180	0.000000	0.377789
	0.000000				
-167.000000	0.354644	0.090884	0.124093	0.000000	0.354644
	0.000000				
-166.000000	0.344684	0.104118	0.121606	0.000000	0.344684
	0.000000				
-165.000000	0.343417	0.118237	0.122183	0.000000	0.343417
	0.000000				
-164.000000	0.347559	0.133224	0.124711	0.000000	0.347559
	0.000000				
-163.000000	0.354925	0.149058	0.128458	0.000000	0.354925
	0.000000				
-162.000000	0.364099	0.165719	0.132956	0.000000	0.364099
	0.000000				
-161.000000	0.374164	0.183185	0.137907	0.000000	0.374164
	0.000000				
-160.000000	0.384519	0.201433	0.143121	0.000000	0.384519
	0.000000				
-159.000000	0.394764	0.220441	0.148474	0.000000	0.394764
	0.000000				
-158.000000	0.404630	0.240183	0.153888	0.000000	0.404630
	0.000000				
-157.000000	0.413934	0.260635	0.159311	0.000000	0.413934
	0.000000				
-156.000000	0.422551	0.281770	0.164714	0.000000	0.422551
	0.000000				
-155.000000	0.430397	0.303562	0.170077	0.000000	0.430397
	0.000000				
-154.000000	0.437415	0.325983	0.175390	0.000000	0.437415
	0.000000				
-153.000000	0.443569	0.349005	0.180650	0.000000	0.443569
	0.000000				
-152.000000	0.448838	0.372600	0.185858	0.000000	0.448838
	0.000000				
-151.000000	0.453213	0.396737	0.191016	0.000000	0.453213
	0.000000				
-150.000000	0.456693	0.421386	0.196132	0.000000	0.456693
	0.000000				
-149.000000	0.459285	0.446518	0.201211	0.000000	0.459285
	0.000000				
-148.000000	0.461000	0.472100	0.206261	0.000000	0.461000
	0.000000				
-147.000000	0.461854	0.498101	0.211292	0.000000	0.461854
	0.000000				
-146.000000	0.461868	0.524489	0.216310	0.000000	0.461868
	0.000000				
-145.000000	0.461063	0.551231	0.221326	0.000000	0.461063
	0.000000				
-144.000000	0.459465	0.578295	0.226346	0.000000	0.459465
	0.000000				
-143.000000	0.457101	0.605647	0.231380	0.000000	0.457101
	0.000000				
-142.000000	0.454002	0.633254	0.236433	0.000000	0.454002
	0.000000				
-141.000000	0.450196	0.661082	0.241512	0.000000	0.450196
	0.000000				
-140.000000	0.445716	0.689097	0.246625	0.000000	0.445716
	0.000000				
-139.000000	0.440595	0.717265	0.251774	0.000000	0.440595
	0.000000				
-138.000000	0.434865	0.745552	0.256965	0.000000	0.434865
	0.000000				
-137.000000	0.428562	0.773923	0.262201	0.000000	0.428562
	0.000000				
-136.000000	0.421717	0.802344	0.267484	0.000000	0.421717
	0.000000				
-135.000000	0.414367	0.830781	0.272815	0.000000	0.414367
	0.000000				
-134.000000	0.406545	0.859199	0.278195	0.000000	0.406545
	0.000000				
-133.000000	0.398286	0.887564	0.283623	0.000000	0.398286
	0.000000				
-132.000000	0.389624	0.915842	0.289097	0.000000	0.389624
	0.000000				
-131.000000	0.380591	0.943999	0.294614	0.000000	0.380591
	0.000000				
-130.000000	0.371222	0.972001	0.300171	0.000000	0.371222
	0.000000				
-129.000000	0.361547	0.999815	0.305763	0.000000	0.361547
	0.000000				
-128.000000	0.351600	1.027406	0.311384	0.000000	0.351600
	0.000000				
-127.000000	0.341411	1.054743	0.317028	0.000000	0.341411
	0.000000				
-126.000000	0.331009	1.081792	0.322687	0.000000	0.331009
	0.000000				
-125.000000	0.320423	1.108522	0.328352	0.000000	0.320423
	0.000000				
-124.000000	0.309681	1.134900	0.334016	0.000000	0.309681
	0.000000				
-123.000000	0.298810	1.160895	0.339667	0.000000	0.298810
	0.000000				
-122.000000	0.287834	1.186475	0.345295	0.000000	0.287834
	0.000000				

-121.000000	0.276778	1.211612	0.350889	0.000000	0.276778
	0.000000				
-120.000000	0.265664	1.236274	0.356437	0.000000	0.265664
	0.000000				
-119.000000	0.254514	1.260433	0.361927	0.000000	0.254514
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-118.000000	0.243348	1.284059	0.367347	0.000000	0.243348
	0.000000				
-117.000000	0.232183	1.307126	0.372683	0.000000	0.232183
	0.000000				
-116.000000	0.221037	1.329605	0.377922	0.000000	0.221037
	0.000000				
-115.000000	0.209926	1.351470	0.383051	0.000000	0.209926
	0.000000				
-114.000000	0.198864	1.372696	0.388056	0.000000	0.198864
	0.000000				
-113.000000	0.187862	1.393257	0.392923	0.000000	0.187862
	0.000000				
-112.000000	0.176933	1.413128	0.397638	0.000000	0.176933
	0.000000				
-111.000000	0.166086	1.432288	0.402190	0.000000	0.166086
	0.000000				
-110.000000	0.155330	1.450712	0.406563	0.000000	0.155330
	0.000000				
-109.000000	0.144671	1.468380	0.410745	0.000000	0.144671
	0.000000				
-108.000000	0.134115	1.485270	0.414723	0.000000	0.134115
	0.000000				
-107.000000	0.123666	1.501364	0.418485	0.000000	0.123666
	0.000000				
-106.000000	0.113329	1.516640	0.422019	0.000000	0.113329
	0.000000				
-105.000000	0.102772	1.531083	0.425289	0.000000	0.102772
	0.000000				
-104.000000	0.092666	1.544675	0.428336	0.000000	0.092666
	0.000000				
-103.000000	0.082671	1.557400	0.431122	0.000000	0.082671
	0.000000				
-102.000000	0.072788	1.569242	0.433638	0.000000	0.072788
	0.000000				
-101.000000	0.063013	1.580189	0.435875	0.000000	0.063013
	0.000000				
-100.000000	0.053344	1.590228	0.437827	0.000000	0.053344
	0.000000				
-99.000000	0.043777	1.599346	0.439484	0.000000	0.043777
	0.000000				
-98.000000	0.034307	1.607533	0.440842	0.000000	0.034307
	0.000000				
-97.000000	0.024929	1.614779	0.441894	0.000000	0.024929
	0.000000				
-96.000000	0.015637	1.621076	0.442637	0.000000	0.015637
	0.000000				
-95.000000	0.006424	1.626417	0.443067	0.000000	0.006424
	0.000000				
-94.000000	-0.002715	1.630795	0.443180	0.000000	-0.002715
	0.000000				
-93.000000	-0.011788	1.634205	0.442977	0.000000	-0.011788
	0.000000				
-92.000000	-0.020802	1.636643	0.442455	0.000000	-0.020802
	0.000000				
-91.000000	-0.029764	1.638106	0.441616	0.000000	-0.029764
	0.000000				
-90.000000	-0.038683	1.638593	0.440460	0.000000	-0.038683
	0.000000				
-89.000000	-0.047564	1.638103	0.438989	-10.067963	-0.047564
	0.000000				
-88.000000	-0.056416	1.636637	0.437207	-9.949211	-0.056416
	0.000000				
-87.000000	-0.065245	1.634196	0.435117	-9.830459	-0.065245
	0.000000				
-86.000000	-0.074059	1.630783	0.432725	-9.711706	-0.074059
	0.000000				
-85.000000	-0.082863	1.626402	0.430036	-9.592954	-0.082863
	0.000000				
-84.000000	-0.091664	1.621058	0.427057	-9.474202	-0.091664
	0.000000				
-83.000000	-0.100466	1.614758	0.423795	-9.355450	-0.100466
	0.000000				
-82.000000	-0.109276	1.607508	0.420257	-9.236697	-0.109276
	0.000000				
-81.000000	-0.118098	1.599318	0.416454	-9.117946	-0.118098
	0.000000				
-80.000000	-0.126934	1.590197	0.412393	-8.999193	-0.126934
	0.000000				
-79.000000	-0.135788	1.580155	0.408086	-8.880441	-0.135788
	0.000000				
-78.000000	-0.144662	1.569204	0.403542	-8.761689	-0.144662
	0.000000				
-77.000000	-0.153558	1.557358	0.398773	-8.642937	-0.153558
	0.000000				
-76.000000	-0.162474	1.544629	0.393789	-8.524184	-0.162474
	0.000000				
-75.000000	-0.171412	1.531034	0.388604	-8.405432	-0.171412
	0.000000				
-74.000000	-0.180370	1.516587	0.383230	-8.286680	-0.180370
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-73.000000	-0.189344	1.501306	0.377678	-8.167928	-0.189344
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-72.000000	-0.198332	1.485208	0.371962	-8.049175	-0.198332
	0.000000				
-71.000000	-0.207328	1.468313	0.366094	-7.930423	-0.207328
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-70.000000	-0.216328	1.450641	0.360087	-7.811671	-0.216328
	0.000000				
-69.000000	-0.225324	1.432211	0.353955	-7.692919	-0.225324
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-68.000000	-0.234309	1.413046	0.347710	-7.574167	-0.234309
	0.000000				
-67.000000	-0.243274	1.393169	0.341366	-7.455414	-0.243274
	0.000000				
-66.000000	-0.252209	1.372602	0.334935	-7.336662	-0.252209
	0.000000				
-65.000000	-0.261104	1.351370	0.328430	-7.217910	-0.261104
	0.000000				
-64.000000	-0.269945	1.329498	0.321863	-7.099158	-0.269945
	0.000000				
-63.000000	-0.278721	1.307012	0.315247	-6.980406	-0.278721
	0.000000				
-62.000000	-0.287417	1.283938	0.308594	-6.861653	-0.287417
	0.000000				
-61.000000	-0.296018	1.260303	0.301915	-6.742901	-0.296018
	0.000000				
-60.000000	-0.304509	1.236136	0.295221	-6.624149	-0.304509
	0.000000				
-59.000000	-0.312873	1.211465	0.288523	-6.505397	-0.312873
	0.000000				
-58.000000	-0.321093	1.186319	0.281831	-6.386645	-0.321093
	0.000000				
-57.000000	-0.329149	1.160727	0.275155	-6.267892	-0.329149
	0.000000				
-56.000000	-0.337024	1.134721	0.268504	-6.149140	-0.337024
	0.000000				
-55.000000	-0.344697	1.108331	0.261887	-6.030388	-0.344697
	0.000000				
-54.000000	-0.352148	1.081589	0.255311	-5.911636	-0.352148
	0.000000				
-53.000000	-0.359356	1.054525	0.248786	-5.792884	-0.359356
	0.000000				
-52.000000	-0.366300	1.027173	0.242316	-5.674131	-0.366300
	0.000000				
-51.000000	-0.372959	0.999564	0.235909	-5.555379	-0.372959
	0.000000				
-50.000000	-0.379310	0.971733	0.229571	-5.436627	-0.379310
	0.000000				
-49.000000	-0.385331	0.943711	0.223306	-5.317875	-0.385331
	0.000000				
-48.000000	-0.391001	0.915532	0.217119	-5.199122	-0.391001
	0.000000				
-47.000000	-0.396296	0.887230	0.211013	-5.080370	-0.396296
	0.000000				
-46.000000	-0.401195	0.858838	0.204993	-4.961618	-0.401195
	0.000000				
-45.000000	-0.405676	0.830391	0.199060	-4.842866	-0.405676
	0.000000				
-44.000000	-0.409717	0.801922	0.193217	-4.724114	-0.409717
	0.000000				
-43.000000	-0.413299	0.773465	0.187465	-4.605361	-0.413299
	0.000000				
-42.000000	-0.416400	0.745054	0.181807	-4.486609	-0.416400
	0.000000				
-41.000000	-0.419003	0.716723	0.176241	-4.367857	-0.419003
	0.000000				
-40.000000	-0.421089	0.688505	0.170769	-4.249105	-0.421089
	0.000000				
-39.000000	-0.422641	0.660434	0.165390	-4.130353	-0.422641
	0.000000				
-38.000000	-0.423647	0.632542	0.160105	-4.011600	-0.423647
	0.000000				
-37.000000	-0.424095	0.604863	0.154914	-3.892848	-0.424095
	0.000000				
-36.000000	-0.423975	0.577428	0.149815	-3.774096	-0.423975
	0.000000				
-35.000000	-0.423284	0.550270	0.144808	-3.655344	-0.423284
	0.000000				
-34.000000	-0.422022	0.523418	0.139895	-3.536592	-0.422022
	0.000000				
-33.000000	-0.420195	0.496902	0.135076	-3.417840	-0.420195
	0.000000				
-32.000000	-0.417820	0.470751	0.130352	-3.299088	-0.417820
	0.000000				
-31.000000	-0.414921	0.444993	0.125728	-3.180335	-0.414921
	0.000000				
-30.000000	-0.411539	0.419652	0.121210	-3.061583	-0.411539
	0.000000				
-29.000000	-0.407737	0.394751	0.116806	-2.942831	-0.407737
	0.000000				
-28.000000	-0.403602	0.370308	0.112531	-2.824079	-0.403602
	0.000000				
-27.000000	-0.399262	0.346340	0.102700	-2.705326	-0.399262
	0.000000				
-26.000000	-0.394902	0.322854	0.089591	-2.586574	-0.394902
	0.000000				
-25.000000	-0.390788	0.299850	0.076482	-2.467822	-0.390788
	0.000000				
-24.000000	-0.387314	0.277314	0.063374	-2.349070	-0.387314
	0.000000				

-23.000000	-0.385058	0.255216	0.050265	-2.230318	-0.385058
	0.000000				
-22.000000	-0.384897	0.233493	0.037156	-2.111565	-0.384897
	0.000000				
-21.000000	-0.388172	0.212042	0.024047	-1.992813	-0.388172
	0.000000				
-20.000000	-0.396963	0.190693	0.010939	-1.874061	-0.396963
	0.000000				
-19.000000	-0.414515	0.169175	-0.002170	-1.755309	-0.413455
	0.000789				
-18.000000	-0.445793	0.147080	-0.015279	-1.636557	-0.443501
	0.001921				
-17.000000	-0.497865	0.123860	-0.028388	-1.517804	-0.475819
	0.021157				
-16.000000	-0.578646	0.099001	-0.041497	-1.399052	-0.505433
	0.081929				
-15.000000	-0.690005	0.072724	-0.054605	-1.280300	-0.524230
	0.219258				
-14.000000	-0.811301	0.047414	-0.067714	-1.161548	-0.523701
	0.450891				
-13.000000	-0.890353	0.027940	-0.080823	-1.042796	-0.499961
	0.719173				
-12.750000	-0.974200	0.011730	-0.084100	-1.013108	-0.501546
	0.923944				
-12.500000	-0.943300	0.012130	-0.084800	-0.983419	-0.486536
	0.919258				
-12.250000	-0.916800	0.011800	-0.084800	-0.953731	-0.472111
	0.923319				
-11.750000	-0.867800	0.010270	-0.084100	-0.894355	-0.443782
	0.941063				
-11.250000	-0.813100	0.009730	-0.084300	-0.834979	-0.414700
	0.947941				
-11.000000	-0.785300	0.009490	-0.084500	-0.805291	-0.400099
	0.950663				
-10.750000	-0.757400	0.009260	-0.084600	-0.775603	-0.385485
	0.953339				
-10.500000	-0.729400	0.009040	-0.084800	-0.745915	-0.370858
	0.955966				
-10.250000	-0.701200	0.008820	-0.085000	-0.716227	-0.356205
	0.958261				
-10.000000	-0.672900	0.008620	-0.085200	-0.686539	-0.341539
	0.960467				
-9.750000	-0.644500	0.008420	-0.085400	-0.656851	-0.326859
	0.962572				
-9.500000	-0.616000	0.008230	-0.085700	-0.627163	-0.312167
	0.964562				
-9.250000	-0.587400	0.008050	-0.085900	-0.597475	-0.297462
	0.966419				
-9.000000	-0.558700	0.007880	-0.086200	-0.567787	-0.282744
	0.968122				
-8.750000	-0.529900	0.007710	-0.086400	-0.538099	-0.268013
	0.969644				
-7.750000	-0.414300	0.007160	-0.087600	-0.419347	-0.209037
	0.976004				
-7.500000	-0.385200	0.007030	-0.087900	-0.389658	-0.194267
	0.977182				
-7.250000	-0.356100	0.006910	-0.088200	-0.359970	-0.179497
	0.978554				
-7.000000	-0.326900	0.006790	-0.088600	-0.330282	-0.164715
	0.979571				
-6.750000	-0.297600	0.006660	-0.088900	-0.300594	-0.149920
	0.980127				
-6.500000	-0.268300	0.006540	-0.089300	-0.270906	-0.135125
	0.980806				
-6.250000	-0.239000	0.006410	-0.089600	-0.241218	-0.120330
	0.981651				
-6.000000	-0.209800	0.006260	-0.090000	-0.211530	-0.105547
	0.983675				
-5.750000	-0.180900	0.006010	-0.090300	-0.181842	-0.090803
	0.989652				
-5.500000	-0.151400	0.005910	-0.090700	-0.152154	-0.075982
	0.990101				
-5.250000	-0.121900	0.005840	-0.091100	-0.122466	-0.061162
	0.990767				
-5.000000	-0.092400	0.005790	-0.091400	-0.092778	-0.046342
	0.991861				
-4.750000	-0.063000	0.005740	-0.091800	-0.063090	-0.031534
	0.997151				
-4.500000	-0.033400	0.005680	-0.092200	-0.033402	-0.016701
	0.999889				
-4.250000	-0.003700	0.005610	-0.092600	-0.003714	-0.001855
	0.992578				
-4.000000	0.025900	0.005550	-0.093100	0.025974	0.012978
	0.994286				
-3.750000	0.055500	0.005500	-0.093500	0.055662	0.027811
	0.994172				
-3.250000	0.114600	0.005450	-0.094300	0.115038	0.057464
	0.992385				
-3.000000	0.144100	0.005400	-0.094700	0.144726	0.072285
	0.991352				
-2.750000	0.173800	0.005330	-0.095100	0.174415	0.087130
	0.992960				
-2.500000	0.203400	0.005270	-0.095600	0.204103	0.101963
	0.993122				
-2.250000	0.233100	0.005240	-0.096000	0.233791	0.116809
	0.994096				
-2.000000	0.262600	0.005230	-0.096400	0.263479	0.131629
	0.993336				

-1.750000	0.292100	0.005200	-0.096800	0.293167	0.146450
	0.992730				
-1.500000	0.321700	0.005170	-0.097300	0.322855	0.161283
	0.992853				
-1.250000	0.351300	0.005120	-0.097700	0.352543	0.176116
	0.992956				
-0.750000	0.410600	0.004950	-0.098600	0.411919	0.205794
	0.993601				
-0.500000	0.440200	0.004890	-0.099100	0.441607	0.220627
	0.993633				
-0.250000	0.469700	0.004810	-0.099500	0.471295	0.235448
	0.993237				
0.000000	0.498900	0.004740	-0.100000	0.500983	0.250230
	0.991693				
0.250000	0.527800	0.004540	-0.100500	0.530671	0.264975
	0.989194				
0.500000	0.557000	0.004440	-0.100900	0.560359	0.279758
	0.988029				
0.750000	0.586200	0.004210	-0.101400	0.590047	0.294540
	0.986981				
1.000000	0.615400	0.004070	-0.101900	0.619735	0.309323
	0.986034				
1.250000	0.644700	0.004080	-0.102400	0.649423	0.324118
	0.985480				
1.500000	0.673800	0.004150	-0.102800	0.679111	0.338888
	0.984389				
1.750000	0.703100	0.004200	-0.103200	0.708799	0.353683
	0.983951				
2.000000	0.732700	0.004200	-0.103600	0.738487	0.368516
	0.984357				
2.250000	0.762200	0.004220	-0.104100	0.768176	0.383336
	0.984473				
2.500000	0.791100	0.004330	-0.104500	0.797864	0.398081
	0.983082				
2.750000	0.819700	0.004450	-0.104800	0.827552	0.412787
	0.981070				
3.000000	0.849200	0.004470	-0.105300	0.857240	0.427608
	0.981287				
3.250000	0.877300	0.004610	-0.105500	0.886928	0.442251
	0.978349				
3.500000	0.903700	0.004950	-0.105600	0.916616	0.456676
	0.971919				
3.750000	0.932300	0.005040	-0.105900	0.946304	0.471382
	0.970513				
4.250000	0.980600	0.006030	-0.105300	1.005680	0.499645
	0.950438				
4.500000	1.004600	0.006490	-0.104900	1.035368	0.513750
	0.941014				
4.750000	1.028900	0.006880	-0.104600	1.065056	0.527890
	0.932691				
5.000000	1.052900	0.007280	-0.104300	1.094744	0.541987
	0.924299				
5.250000	1.077300	0.007610	-0.104000	1.124432	0.556133
	0.917065				
5.500000	1.101600	0.007940	-0.103700	1.154120	0.570262
	0.910046				
5.750000	1.124900	0.008320	-0.103200	1.183808	0.584254
	0.901747				
6.000000	1.147300	0.008730	-0.102500	1.213496	0.598119
	0.892430				
6.250000	1.170600	0.009040	-0.102000	1.243184	0.612101
	0.884985				
6.500000	1.193100	0.009370	-0.101400	1.272872	0.625970
	0.876686				
6.750000	1.215100	0.009710	-0.100700	1.302560	0.639764
	0.868043				
7.000000	1.236700	0.010040	-0.099900	1.332248	0.653497
	0.859229				
7.250000	1.258900	0.010320	-0.099200	1.361937	0.667309
	0.851667				
7.500000	1.279600	0.010620	-0.098300	1.391625	0.680903
	0.842379				
7.750000	1.301200	0.010860	-0.097500	1.421313	0.694618
	0.834714				
8.000000	1.320700	0.011050	-0.096300	1.451001	0.708027
	0.824623				
8.250000	1.335100	0.011250	-0.094100	1.480689	0.720683
	0.808437				
8.500000	1.349600	0.011490	-0.092100	1.510377	0.733330
	0.793093				
8.750000	1.362000	0.011860	-0.089900	1.540065	0.745637
	0.775858				
9.500000	1.409700	0.012890	-0.085400	1.629129	0.784012
	0.740357				
9.750000	1.427200	0.013340	-0.084300	1.658817	0.797022
	0.731239				
10.000000	1.444100	0.013820	-0.083100	1.688505	0.809927
	0.721817				
10.250000	1.461200	0.014320	-0.082000	1.718193	0.822852
	0.712966				
10.500000	1.477600	0.014870	-0.080900	1.747881	0.835653
	0.703713				
10.750000	1.494600	0.015400	-0.079900	1.777569	0.848539
	0.695414				
11.000000	1.509500	0.016050	-0.078700	1.807257	0.861071
	0.685308				
11.250000	1.525100	0.016700	-0.077600	1.836945	0.873704
	0.676254				

11.500000	1.535100	0.017680	-0.076000	1.866633	0.885386
	0.662131				
11.750000	1.552300	0.018260	-0.075100	1.896321	0.898255
	0.655312				
12.000000	1.565400	0.019100	-0.073900	1.926010	0.910416
	0.644927				
12.250000	1.575800	0.020120	-0.072400	1.955698	0.922092
	0.632454				
12.500000	1.591300	0.020830	-0.071600	1.985386	0.934629
	0.624951				
13.000000	1.615500	0.022760	-0.069300	2.044762	0.958462
	0.604841				
13.250000	1.629500	0.023630	-0.068300	2.074450	0.970687
	0.596879				
13.500000	1.643900	0.024450	-0.067500	2.104138	0.982971
	0.589501				
14.000000	1.666200	0.026660	-0.065400	2.163514	1.006302
	0.570248				
14.250000	1.679700	0.027620	-0.064600	2.193202	1.018371
	0.562914				
14.500000	1.692900	0.028620	-0.063800	2.222890	1.030370
	0.555571				
14.750000	1.700600	0.030130	-0.062800	2.252578	1.041309
	0.544298				
15.000000	1.713000	0.031250	-0.062000	2.282266	1.053116
	0.536862				
15.250000	1.725700	0.032360	-0.061300	2.311954	1.064965
	0.529864				
15.500000	1.738500	0.033460	-0.060600	2.341642	1.076819
	0.523141				
15.750000	1.744300	0.035260	-0.059700	2.371330	1.087276
	0.511679				
16.000000	1.754800	0.036550	-0.059000	2.401018	1.098631
	0.503820				
16.250000	1.766700	0.037770	-0.058400	2.430706	1.110247
	0.497140				
16.500000	1.778000	0.039060	-0.057800	2.460394	1.121725
	0.490244				
16.750000	1.782500	0.041100	-0.057100	2.490083	1.131786
	0.479066				
17.000000	1.789300	0.042880	-0.056400	2.519771	1.142278
	0.469710				
17.250000	1.801800	0.044140	-0.056000	2.549459	1.153938
	0.464244				
17.500000	1.812600	0.045550	-0.055600	2.579147	1.165224
	0.457858				
17.750000	1.820200	0.047330	-0.055100	2.608835	1.175807
	0.449672				
18.000000	1.813600	0.050660	-0.054300	2.638523	1.183263
	0.433144				
18.250000	1.824900	0.052080	-0.054100	2.668211	1.194572
	0.427736				
18.500000	1.831000	0.054010	-0.053700	2.697899	1.204706
	0.419433				
18.750000	1.833300	0.056550	-0.053500	2.727587	1.213943
	0.409183				
19.000000	1.842900	0.058250	-0.053400	2.757275	1.224798
	0.403335				
19.250000	1.843200	0.060970	-0.053100	2.786963	1.233479
	0.392486				
19.500000	1.852200	0.062760	-0.053000	2.816651	1.244137
	0.386682				
19.750000	1.848000	0.066260	-0.053100	2.846339	1.251636
	0.373966				
20.000000	1.853900	0.068470	-0.053100	2.876027	1.853900
	0.000000				
20.250000	1.860500	0.070590	-0.053200	2.905715	1.860500
	0.000000				
20.500000	1.866300	0.072810	-0.053300	2.935403	1.866300
	0.000000				
20.750000	1.861000	0.076620	-0.053800	2.965091	1.861000
	0.000000				
21.000000	1.865700	0.079020	-0.054100	2.994779	1.865700
	0.000000				
21.250000	1.870900	0.081290	-0.054400	3.024467	1.870900
	0.000000				
21.500000	1.875400	0.083710	-0.054700	3.054155	1.875400
	0.000000				
21.750000	1.866800	0.088140	-0.055600	3.083843	1.866800
	0.000000				
22.000000	1.870700	0.090740	-0.056200	3.113532	1.870700
	0.000000				
22.250000	1.874800	0.093310	-0.056700	3.143219	1.874800
	0.000000				
22.500000	1.867300	0.097680	-0.057900	3.172908	1.867300
	0.000000				
22.750000	1.866800	0.101000	-0.058800	3.202596	1.866800
	0.000000				
23.000000	1.870200	0.103680	-0.059600	3.232284	1.870200
	0.000000				
23.250000	1.872600	0.106510	-0.060400	3.261972	1.872600
	0.000000				
23.500000	1.860700	0.111680	-0.062200	3.291660	1.860700
	0.000000				
23.750000	1.861800	0.114730	-0.063300	3.321348	1.861800
	0.000000				
24.000000	1.863800	0.117610	-0.064400	3.351036	1.863800
	0.000000				

24.250000	1.864000	0.120680	-0.065600	3.380724	1.864000
	0.000000				
24.500000	1.849600	0.126220	-0.068000	3.410412	1.849600
	0.000000				
24.750000	1.850300	0.129310	-0.069400	3.440100	1.850300
	0.000000				
25.000000	1.850200	0.132550	-0.070900	3.469788	1.850200
	0.000000				
25.250000	1.835300	0.138410	-0.073800	3.499476	1.835300
	0.000000				
25.500000	1.834500	0.141770	-0.075500	3.529164	1.834500
	0.000000				
25.750000	1.832700	0.145320	-0.077400	3.558852	1.832700
	0.000000				
26.000000	1.817400	0.151330	-0.080800	3.588540	1.817400
	0.000000				
26.250000	1.815400	0.154910	-0.082900	3.618228	1.815400
	0.000000				
26.500000	1.798400	0.161280	-0.086700	3.647916	1.798400
	0.000000				
26.750000	1.795500	0.165050	-0.089100	3.677604	1.795500
	0.000000				
27.000000	1.778300	0.171510	-0.093300	3.707293	1.778300
	0.000000				
27.250000	1.773300	0.175680	-0.096100	3.736981	1.773300
	0.000000				
27.500000	1.767900	0.179920	-0.099100	3.766669	1.767900
	0.000000				
27.750000	1.749100	0.186850	-0.103900	3.796357	1.749100
	0.000000				
28.000000	1.742000	0.191260	-0.107300	3.826045	1.742000
	0.000000				
28.250000	1.723300	0.197350	-0.112600	3.855733	1.723300
	0.000000				
28.500000	1.716200	0.201470	-0.116300	3.885421	1.716200
	0.000000				
28.750000	1.694300	0.208780	-0.122300	3.915109	1.694300
	0.000000				
29.000000	1.672500	0.216530	-0.128500	3.944797	1.672500
	0.000000				
29.250000	1.662700	0.221890	-0.132800	3.974485	1.662700
	0.000000				
29.500000	1.639500	0.230640	-0.139400	4.004173	1.639500
	0.000000				
29.750000	1.629300	0.236490	-0.143900	4.033861	1.629300
	0.000000				
30.500000	1.563800	0.264890	-0.164000	4.122925	1.563800
	0.000000				
30.750000	1.531400	0.278580	-0.172900	4.152613	1.531400
	0.000000				
31.000000	1.841997	0.350500	-0.176696	4.182302	1.841997
	0.000000				
32.000000	1.837608	0.378421	-0.191881	4.301054	1.837608
	0.000000				
33.000000	1.832976	0.407057	-0.207066	4.419806	1.832976
	0.000000				
34.000000	1.828149	0.436346	-0.222251	4.538558	1.828149
	0.000000				
35.000000	1.823148	0.466225	-0.237436	4.657310	1.823148
	0.000000				
36.000000	1.817979	0.496634	-0.252620	4.776062	1.817979
	0.000000				
37.000000	1.812630	0.527516	-0.267805	4.894814	1.812630
	0.000000				
38.000000	1.807078	0.558814	-0.282990	5.013567	1.807078
	0.000000				
39.000000	1.801289	0.590473	-0.298175	5.132319	1.801289
	0.000000				
40.000000	1.795221	0.622440	-0.313360	5.251071	1.795221
	0.000000				
41.000000	1.788827	0.654663	-0.328545	5.369823	1.788827
	0.000000				
42.000000	1.782054	0.687093	-0.343729	5.488575	1.782054
	0.000000				
43.000000	1.774849	0.719680	-0.358914	5.607328	1.774849
	0.000000				
44.000000	1.767152	0.752377	-0.374099	5.726080	1.767152
	0.000000				
45.000000	1.758909	0.785137	-0.389284	5.844832	1.758909
	0.000000				
46.000000	1.750060	0.817915	-0.403582	5.963584	1.750060
	0.000000				
47.000000	1.740549	0.850665	-0.406500	6.082336	1.740549
	0.000000				
48.000000	1.730320	0.883346	-0.409415	6.201089	1.730320
	0.000000				
49.000000	1.719322	0.915913	-0.412314	6.319841	1.719322
	0.000000				
50.000000	1.707501	0.948325	-0.415184	6.438593	1.707501
	0.000000				
51.000000	1.694810	0.980541	-0.418016	6.557345	1.694810
	0.000000				
52.000000	1.681202	1.012521	-0.420797	6.676097	1.681202
	0.000000				
53.000000	1.666635	1.044225	-0.423519	6.794850	1.666635
	0.000000				
54.000000	1.651069	1.075613	-0.426173	6.913602	1.651069
	0.000000				

55.000000	1.634467	1.106649	-0.428751	7.032354	1.634467
	0.000000				
56.000000	1.616797	1.137293	-0.431244	7.151106	1.616797
	0.000000				
57.000000	1.598028	1.167511	-0.433647	7.269858	1.598028
	0.000000				
58.000000	1.578135	1.197265	-0.435954	7.388611	1.578135
	0.000000				
59.000000	1.557095	1.226520	-0.438161	7.507363	1.557095
	0.000000				
60.000000	1.534888	1.255242	-0.440263	7.626115	1.534888
	0.000000				
61.000000	1.511499	1.283397	-0.442257	7.744867	1.511499
	0.000000				
62.000000	1.486915	1.310952	-0.444142	7.863620	1.486915
	0.000000				
63.000000	1.461128	1.337874	-0.445915	7.982372	1.461128
	0.000000				
64.000000	1.434133	1.364133	-0.447576	8.101124	1.434133
	0.000000				
65.000000	1.405926	1.389697	-0.449125	8.219876	1.405926
	0.000000				
66.000000	1.376511	1.414537	-0.450563	8.338629	1.376511
	0.000000				
67.000000	1.345890	1.438625	-0.451891	8.457380	1.345890
	0.000000				
68.000000	1.314073	1.461931	-0.453111	8.576133	1.314073
	0.000000				
69.000000	1.281071	1.484431	-0.454226	8.694885	1.281071
	0.000000				
70.000000	1.246898	1.506097	-0.455240	8.813637	1.246898
	0.000000				
71.000000	1.211571	1.526905	-0.456157	8.932389	1.211571
	0.000000				
72.000000	1.175111	1.546832	-0.456979	9.051142	1.175111
	0.000000				
73.000000	1.137541	1.565853	-0.457714	9.169894	1.137541
	0.000000				
74.000000	1.098888	1.583948	-0.458365	9.288646	1.098888
	0.000000				
75.000000	1.059182	1.601097	-0.458939	9.407398	1.059182
	0.000000				
76.000000	1.018453	1.617279	-0.459441	9.526151	1.018453
	0.000000				
77.000000	0.976737	1.632477	-0.459878	9.644902	0.976737
	0.000000				
78.000000	0.934071	1.646673	-0.460256	9.763655	0.934071
	0.000000				
79.000000	0.890495	1.659852	-0.460582	9.882407	0.890495
	0.000000				
80.000000	0.846051	1.671999	-0.460862	10.001160	0.846051
	0.000000				
81.000000	0.800783	1.683102	-0.461104	10.119911	0.800783
	0.000000				
82.000000	0.754740	1.693147	-0.461314	10.238664	0.754740
	0.000000				
83.000000	0.707969	1.702124	-0.461499	10.357416	0.707969
	0.000000				
84.000000	0.660522	1.710024	-0.461666	10.476168	0.660522
	0.000000				
85.000000	0.612452	1.716839	-0.461822	10.594920	0.612452
	0.000000				
86.000000	0.563814	1.722562	-0.461972	10.713673	0.563814
	0.000000				
87.000000	0.514664	1.727186	-0.462124	10.832425	0.514664
	0.000000				
88.000000	0.465062	1.730709	-0.462283	10.951177	0.465062
	0.000000				
89.000000	0.415066	1.733128	-0.462455	11.069929	0.415066
	0.000000				
90.000000	0.364739	1.734440	-0.462645	0.000000	0.364739
	0.000000				
91.000000	0.314143	1.734646	-0.462858	0.000000	0.314143
	0.000000				
92.000000	0.263341	1.733747	-0.463099	0.000000	0.263341
	0.000000				
93.000000	0.212399	1.731746	-0.463370	0.000000	0.212399
	0.000000				
94.000000	0.161383	1.728647	-0.463676	0.000000	0.161383
	0.000000				
95.000000	0.110358	1.724455	-0.464020	0.000000	0.110358
	0.000000				
96.000000	0.059393	1.719177	-0.464402	0.000000	0.059393
	0.000000				
97.000000	0.008555	1.712820	-0.464826	0.000000	0.008555
	0.000000				
98.000000	-0.042088	1.705395	-0.465292	0.000000	-0.042088
	0.000000				
99.000000	-0.092466	1.696911	-0.465800	0.000000	-0.092466
	0.000000				
100.000000	-0.142512	1.687381	-0.466350	0.000000	-0.142512
	0.000000				
101.000000	-0.192156	1.676818	-0.466940	0.000000	-0.192156
	0.000000				
102.000000	-0.338117	1.665236	-0.473172	0.000000	-0.338117
	0.000000				
103.000000	-0.384711	1.652651	-0.474188	0.000000	-0.384711
	0.000000				

104.000000	-0.430750	1.639080	-0.475221	0.000000	-0.430750
	0.000000				
105.000000	-0.476166	1.624541	-0.476267	0.000000	-0.476166
	0.000000				
106.000000	-0.520890	1.609054	-0.477321	0.000000	-0.520890
	0.000000				
107.000000	-0.564854	1.592639	-0.478378	0.000000	-0.564854
	0.000000				
108.000000	-0.607992	1.575318	-0.479432	0.000000	-0.607992
	0.000000				
109.000000	-0.650239	1.557113	-0.480476	0.000000	-0.650239
	0.000000				
110.000000	-0.691532	1.538048	-0.481503	0.000000	-0.691532
	0.000000				
111.000000	-0.731807	1.518149	-0.482504	0.000000	-0.731807
	0.000000				
112.000000	-0.771004	1.497440	-0.483471	0.000000	-0.771004
	0.000000				
113.000000	-0.809065	1.475949	-0.484394	0.000000	-0.809065
	0.000000				
114.000000	-0.845930	1.453704	-0.485263	0.000000	-0.845930
	0.000000				
115.000000	-0.881546	1.430732	-0.486067	0.000000	-0.881546
	0.000000				
116.000000	-0.915859	1.407065	-0.486796	0.000000	-0.915859
	0.000000				
117.000000	-0.948817	1.382732	-0.487437	0.000000	-0.948817
	0.000000				
118.000000	-0.980373	1.357764	-0.487979	0.000000	-0.980373
	0.000000				
119.000000	-1.010479	1.332193	-0.488410	0.000000	-1.010479
	0.000000				
120.000000	-1.039092	1.306053	-0.488717	0.000000	-1.039092
	0.000000				
121.000000	-1.066170	1.279375	-0.488886	0.000000	-1.066170
	0.000000				
122.000000	-1.091676	1.252195	-0.488906	0.000000	-1.091676
	0.000000				
123.000000	-1.115573	1.224547	-0.488762	0.000000	-1.115573
	0.000000				
124.000000	-1.137829	1.196465	-0.488442	0.000000	-1.137829
	0.000000				
125.000000	-1.158413	1.167985	-0.487932	0.000000	-1.158413
	0.000000				
126.000000	-1.177299	1.139143	-0.487220	0.000000	-1.177299
	0.000000				
127.000000	-1.194464	1.109977	-0.486292	0.000000	-1.194464
	0.000000				
128.000000	-1.209886	1.080521	-0.485136	0.000000	-1.209886
	0.000000				
129.000000	-1.223550	1.050814	-0.483739	0.000000	-1.223550
	0.000000				
130.000000	-1.235440	1.020892	-0.482089	0.000000	-1.235440
	0.000000				
131.000000	-1.245547	0.990793	-0.480176	0.000000	-1.245547
	0.000000				
132.000000	-1.253863	0.960556	-0.477987	0.000000	-1.253863
	0.000000				
133.000000	-1.260386	0.930216	-0.475512	0.000000	-1.260386
	0.000000				
134.000000	-1.265114	0.899813	-0.472742	0.000000	-1.265114
	0.000000				
135.000000	-1.268051	0.869385	-0.469667	0.000000	-1.268051
	0.000000				
136.000000	-1.269205	0.838968	-0.466279	0.000000	-1.269205
	0.000000				
137.000000	-1.268585	0.808601	-0.462571	0.000000	-1.268585
	0.000000				
138.000000	-1.266207	0.778322	-0.458536	0.000000	-1.266207
	0.000000				
139.000000	-1.262088	0.748167	-0.454169	0.000000	-1.262088
	0.000000				
140.000000	-1.256250	0.718174	-0.449465	0.000000	-1.256250
	0.000000				
141.000000	-1.248719	0.688381	-0.444421	0.000000	-1.248719
	0.000000				
142.000000	-1.239523	0.658823	-0.439036	0.000000	-1.239523
	0.000000				
143.000000	-1.228698	0.629537	-0.433308	0.000000	-1.228698
	0.000000				
144.000000	-1.216279	0.600558	-0.427238	0.000000	-1.216279
	0.000000				
145.000000	-1.202310	0.571923	-0.420830	0.000000	-1.202310
	0.000000				
146.000000	-1.186838	0.543666	-0.414086	0.000000	-1.186838
	0.000000				
147.000000	-1.169915	0.515822	-0.407013	0.000000	-1.169915
	0.000000				
148.000000	-1.151600	0.488424	-0.399620	0.000000	-1.151600
	0.000000				
149.000000	-1.131957	0.461506	-0.391917	0.000000	-1.131957
	0.000000				
150.000000	-1.111063	0.435099	-0.383917	0.000000	-1.111063
	0.000000				
151.000000	-1.089000	0.409237	-0.375640	0.000000	-1.089000
	0.000000				
152.000000	-1.065868	0.383950	-0.367107	0.000000	-1.065868
	0.000000				

153.000000	-1.041784	0.359268	-0.358346	0.000000	-1.041784
	0.000000				
154.000000	-1.016890	0.335220	-0.349397	0.000000	-1.016890
	0.000000				
155.000000	-0.991364	0.311837	-0.340309	0.000000	-0.991364
	0.000000				
156.000000	-0.965437	0.289145	-0.331148	0.000000	-0.965437
	0.000000				
157.000000	-0.939419	0.267171	-0.322010	0.000000	-0.939419
	0.000000				
158.000000	-0.913742	0.245941	-0.313029	0.000000	-0.913742
	0.000000				
159.000000	-0.889027	0.225480	-0.304404	0.000000	-0.889027
	0.000000				
160.000000	-0.866208	0.205812	-0.296438	0.000000	-0.866208
	0.000000				
161.000000	-0.846724	0.186961	-0.289605	0.000000	-0.846724
	0.000000				
162.000000	-0.832855	0.168947	-0.284661	0.000000	-0.832855
	0.000000				
163.000000	-0.828241	0.151791	-0.282826	0.000000	-0.828241
	0.000000				
164.000000	-0.838545	0.135514	-0.286012	0.000000	-0.838545
	0.000000				
165.000000	-0.871616	0.120134	-0.296902	0.000000	-0.871616
	0.000000				
166.000000	-0.934537	0.105669	-0.317992	0.000000	-0.934537
	0.000000				
167.000000	-1.021946	0.092133	-0.347683	0.000000	-1.021946
	0.000000				
168.000000	-1.098956	0.079543	-0.374404	0.000000	-1.098956
	0.000000				
169.000000	-1.116462	0.067913	-0.381467	0.000000	-1.116462
	0.000000				
170.000000	-1.065562	0.057255	-0.365443	0.000000	-1.065562
	0.000000				
171.000000	-0.978768	0.047580	-0.337019	0.000000	-0.978768
	0.000000				
172.000000	-0.884035	0.038899	-0.305598	0.000000	-0.884035
	0.000000				
173.000000	-0.789034	0.031221	-0.273794	0.000000	-0.789034
	0.000000				
174.000000	-0.694032	0.024552	-0.241711	0.000000	-0.694032
	0.000000				
175.000000	-0.599030	0.018901	-0.209362	0.000000	-0.599030
	0.000000				
176.000000	-0.504028	0.014272	-0.176756	0.000000	-0.504028
	0.000000				
177.000000	-0.409027	0.010668	-0.143906	0.000000	-0.409027
	0.000000				
178.000000	-0.314025	0.008093	-0.110822	0.000000	-0.314025
	0.000000				
179.000000	-0.219023	0.006548	-0.077515	0.000000	-0.219023
	0.000000				
180.000000	-0.124021	0.006000	-0.043521	0.000000	-0.124021
	0.000000				

C.5 NACA 64-618 Extrapolated Polars

Listing 11: Aerodynamic polar characteristics for NACA 64-618.

-----QBlade Multi RE Polar File-----					
Generated with : QBlade CE v2.0.9.4_beta windows					
Archive Format: 310039					
Time : 19:42:31					
Date : 04.06.2026					
-----Object Names-----					
NACA_64-618_coords_Re9.500_M0.00_N9.0 360 M		POLARNAME	- the polar name		
../Airfoils/NACA_64-618_coords.afl		FOILNAME	- the airfoil name to which		
the polar(s) belong OR path to airfoil file					
-----Parameters-----					
18.0		THICKNESS	- the thickness of the airfoil in % of chord		
true		ISDECOMPOSED	- is the polar decomposed (imports Cl_Sep,		
Cl_att and f_st columns) [bool]					
REYNOLDS	9.5000E+06		- the list of Reynolds numbers for the imported polars		
-----Polar Data-----					
AOA	CL	CD	CM	CL_ATT	CL_SEP
	F_ST				
-180.000000	-0.124808	0.006000	-0.043659	0.000000	-0.124808
	0.000000				
-179.000000	-0.028857	0.006508	-0.010027	0.000000	-0.028857
	0.000000				
-178.000000	0.067094	0.008030	0.023433	0.000000	0.067094
	0.000000				
-177.000000	0.163045	0.010564	0.056709	0.000000	0.163045
	0.000000				
-176.000000	0.258997	0.014104	0.089790	0.000000	0.258997
	0.000000				
-175.000000	0.354937	0.018646	0.122660	0.000000	0.354937
	0.000000				
-174.000000	0.448944	0.024181	0.154647	0.000000	0.448944
	0.000000				

-173.000000	0.527647	0.030702	0.181191	0.000000	0.527647
	0.000000				
-172.000000	0.563669	0.038200	0.193066	0.000000	0.563669
	0.000000				
-171.000000	0.540692	0.046664	0.184985	0.000000	0.540692
	0.000000				
-170.000000	0.480520	0.056082	0.164603	0.000000	0.480520
	0.000000				
-169.000000	0.419544	0.066443	0.144302	0.000000	0.419544
	0.000000				
-168.000000	0.375998	0.077731	0.130166	0.000000	0.375998
	0.000000				
-167.000000	0.350927	0.089931	0.122456	0.000000	0.350927
	0.000000				
-166.000000	0.339633	0.103029	0.119537	0.000000	0.339633
	0.000000				
-165.000000	0.337397	0.117006	0.119795	0.000000	0.337397
	0.000000				
-164.000000	0.340782	0.131845	0.122070	0.000000	0.340782
	0.000000				
-163.000000	0.347515	0.147526	0.125604	0.000000	0.347515
	0.000000				
-162.000000	0.356132	0.164029	0.129913	0.000000	0.356132
	0.000000				
-161.000000	0.365686	0.181332	0.134690	0.000000	0.365686
	0.000000				
-160.000000	0.375559	0.199415	0.139740	0.000000	0.375559
	0.000000				
-159.000000	0.385341	0.218253	0.144937	0.000000	0.385341
	0.000000				
-158.000000	0.394759	0.237823	0.150201	0.000000	0.394759
	0.000000				
-157.000000	0.403626	0.258101	0.155481	0.000000	0.403626
	0.000000				
-156.000000	0.411815	0.279060	0.160744	0.000000	0.411815
	0.000000				
-155.000000	0.419241	0.300674	0.165973	0.000000	0.419241
	0.000000				
-154.000000	0.425848	0.322917	0.171156	0.000000	0.425848
	0.000000				
-153.000000	0.431599	0.345760	0.176292	0.000000	0.431599
	0.000000				
-152.000000	0.436475	0.369176	0.181380	0.000000	0.436475
	0.000000				
-151.000000	0.440465	0.393134	0.186425	0.000000	0.440465
	0.000000				
-150.000000	0.443571	0.417605	0.191432	0.000000	0.443571
	0.000000				
-149.000000	0.445800	0.442559	0.196408	0.000000	0.445800
	0.000000				
-148.000000	0.447163	0.467966	0.201363	0.000000	0.447163
	0.000000				
-147.000000	0.447677	0.493793	0.206304	0.000000	0.447677
	0.000000				
-146.000000	0.447365	0.520009	0.211239	0.000000	0.447365
	0.000000				
-145.000000	0.446248	0.546582	0.216179	0.000000	0.446248
	0.000000				
-144.000000	0.444354	0.573478	0.221131	0.000000	0.444354
	0.000000				
-143.000000	0.441711	0.600666	0.226103	0.000000	0.441711
	0.000000				
-142.000000	0.438349	0.628112	0.231102	0.000000	0.438349
	0.000000				
-141.000000	0.434299	0.655782	0.236136	0.000000	0.434299
	0.000000				
-140.000000	0.429594	0.683643	0.241210	0.000000	0.429594
	0.000000				
-139.000000	0.424268	0.711661	0.246330	0.000000	0.424268
	0.000000				
-138.000000	0.418355	0.739801	0.251499	0.000000	0.418355
	0.000000				
-137.000000	0.411889	0.768029	0.256721	0.000000	0.411889
	0.000000				
-136.000000	0.404906	0.796312	0.261998	0.000000	0.404906
	0.000000				
-135.000000	0.397440	0.824614	0.267332	0.000000	0.397440
	0.000000				
-134.000000	0.389527	0.852903	0.272722	0.000000	0.389527
	0.000000				
-133.000000	0.381202	0.881142	0.278168	0.000000	0.381202
	0.000000				
-132.000000	0.372498	0.909300	0.283669	0.000000	0.372498
	0.000000				
-131.000000	0.363452	0.937340	0.289220	0.000000	0.363452
	0.000000				
-130.000000	0.354095	0.965230	0.294819	0.000000	0.354095
	0.000000				
-129.000000	0.344461	0.992936	0.300459	0.000000	0.344461
	0.000000				
-128.000000	0.334582	1.020425	0.306136	0.000000	0.334582
	0.000000				
-127.000000	0.324489	1.047664	0.311843	0.000000	0.324489
	0.000000				
-126.000000	0.314212	1.074619	0.317571	0.000000	0.314212
	0.000000				
-125.000000	0.303780	1.101260	0.323312	0.000000	0.303780
	0.000000				

-124.000000	0.293221	1.127553	0.329056	0.000000	0.293221
	0.000000				
-123.000000	0.282561	1.153467	0.334793	0.000000	0.282561
	0.000000				
-122.000000	0.271827	1.178972	0.340513	0.000000	0.271827
	0.000000				
-121.000000	0.261041	1.204036	0.346203	0.000000	0.261041
	0.000000				
-120.000000	0.250226	1.228630	0.351851	0.000000	0.250226
	0.000000				
-119.000000	0.239404	1.252725	0.357445	0.000000	0.239404
	0.000000				
-118.000000	0.228594	1.276292	0.362971	0.000000	0.228594
	0.000000				
-117.000000	0.217813	1.299302	0.368416	0.000000	0.217813
	0.000000				
-116.000000	0.207080	1.321729	0.373766	0.000000	0.207080
	0.000000				
-115.000000	0.196407	1.343545	0.379007	0.000000	0.196407
	0.000000				
-114.000000	0.185810	1.364725	0.384126	0.000000	0.185810
	0.000000				
-113.000000	0.175300	1.385244	0.389107	0.000000	0.175300
	0.000000				
-112.000000	0.164887	1.405076	0.393937	0.000000	0.164887
	0.000000				
-111.000000	0.154580	1.424199	0.398602	0.000000	0.154580
	0.000000				
-110.000000	0.144388	1.442590	0.403088	0.000000	0.144388
	0.000000				
-109.000000	0.134314	1.460227	0.407381	0.000000	0.134314
	0.000000				
-108.000000	0.124366	1.477089	0.411468	0.000000	0.124366
	0.000000				
-107.000000	0.114544	1.493156	0.415337	0.000000	0.114544
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-106.000000	0.104853	1.508410	0.418974	0.000000	0.104853
	0.000000				
-105.000000	0.094965	1.522831	0.422345	0.000000	0.094965
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-104.000000	0.085538	1.536403	0.425488	0.000000	0.085538
	0.000000				
-103.000000	0.076239	1.549111	0.428366	0.000000	0.076239
	0.000000				
-102.000000	0.067064	1.560938	0.430970	0.000000	0.067064
	0.000000				
-101.000000	0.058010	1.571871	0.433291	0.000000	0.058010
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-100.000000	0.049073	1.581897	0.435320	0.000000	0.049073
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-99.000000	0.040247	1.591004	0.437050	0.000000	0.040247
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-98.000000	0.031527	1.599181	0.438474	0.000000	0.031527
	0.000000				
-97.000000	0.022904	1.606419	0.439587	0.000000	0.022904
	0.000000				
-96.000000	0.014372	1.612709	0.440385	0.000000	0.014372
	0.000000				
-95.000000	0.005923	1.618044	0.440863	0.000000	0.005923
	0.000000				
-94.000000	-0.002451	1.622417	0.441020	0.000000	-0.002451
	0.000000				
-93.000000	-0.010759	1.625824	0.440852	0.000000	-0.010759
	0.000000				
-92.000000	-0.019010	1.628260	0.440361	0.000000	-0.019010
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-91.000000	-0.027212	1.629722	0.439545	0.000000	-0.027212
	0.000000				
-90.000000	-0.035375	1.630208	0.438406	0.000000	-0.035375
	0.000000				
-89.000000	-0.043507	1.629719	0.436947	-10.158627	-0.043507
	0.000000				
-88.000000	-0.051617	1.628254	0.435170	-10.038688	-0.051617
	0.000000				
-87.000000	-0.059714	1.625815	0.433080	-9.918749	-0.059714
	0.000000				
-86.000000	-0.067805	1.622406	0.430682	-9.798810	-0.067805
	0.000000				
-85.000000	-0.075900	1.618030	0.427982	-9.678871	-0.075900
	0.000000				
-84.000000	-0.084004	1.612692	0.424986	-9.558932	-0.084004
	0.000000				
-83.000000	-0.092125	1.606398	0.421702	-9.438993	-0.092125
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4.750000	1.039800	0.006860	-0.124300	1.085657	0.536908
	0.916434				
5.250000	1.086100	0.007530	-0.122800	1.145626	0.565069
	0.897467				
5.500000	1.109300	0.007840	-0.122100	1.175611	0.579149
	0.888826				
5.750000	1.132400	0.008140	-0.121400	1.205596	0.593210
	0.880475				
6.000000	1.154800	0.008470	-0.120600	1.235581	0.607169
	0.871453				
6.250000	1.176400	0.008800	-0.119600	1.265565	0.621011
	0.861664				
6.500000	1.198300	0.009110	-0.118600	1.295550	0.634889
	0.852799				
6.750000	1.220600	0.009390	-0.117700	1.325535	0.648817
	0.844936				
7.000000	1.242500	0.009660	-0.116800	1.355520	0.662683
	0.836874				
7.250000	1.264200	0.009910	-0.115800	1.385504	0.676515
	0.828905				
7.500000	1.285500	0.010150	-0.114800	1.415489	0.690284
	0.820756				
7.750000	1.305600	0.010420	-0.113500	1.445474	0.703872
	0.811390				
8.000000	1.321200	0.010640	-0.111300	1.475459	0.716790
	0.796672				
8.250000	1.335800	0.010890	-0.108900	1.505443	0.729537
	0.781361				
8.500000	1.351900	0.011160	-0.106900	1.535428	0.742488
	0.768547				
8.750000	1.367500	0.011440	-0.104800	1.565413	0.755343
	0.755684				
9.000000	1.382800	0.011730	-0.102700	1.595398	0.768134
	0.743011				
9.250000	1.397300	0.012070	-0.100600	1.625382	0.780780
	0.729953				
9.500000	1.412300	0.012400	-0.098600	1.655367	0.793484
	0.717981				
9.750000	1.427700	0.012740	-0.096700	1.685352	0.806233
	0.706920				
10.000000	1.441500	0.013150	-0.094600	1.715337	0.818706
	0.694594				
10.250000	1.455000	0.013590	-0.092600	1.745321	0.831108
	0.682436				
10.500000	1.469500	0.014020	-0.090700	1.775306	0.843653
	0.671759				
10.750000	1.481900	0.014540	-0.088700	1.805291	0.855828
	0.659396				

11.000000	1.493900	0.015130	-0.086700	1.835276	0.867910
	0.647108				
11.250000	1.507800	0.015670	-0.085000	1.865260	0.880292
	0.637084				
11.500000	1.519600	0.016320	-0.083000	1.895245	0.892290
	0.625462				
11.750000	1.532200	0.016980	-0.081300	1.925230	0.904404
	0.614988				
12.000000	1.544400	0.017670	-0.079600	1.955215	0.916425
	0.604526				
12.250000	1.558400	0.018320	-0.078200	1.985199	0.928746
	0.596007				
12.500000	1.569000	0.019150	-0.076500	2.015184	0.940435
	0.584848				
12.750000	1.581300	0.019950	-0.075000	2.045169	0.952408
	0.575508				
13.000000	1.590700	0.020920	-0.073300	2.075154	0.963820
	0.564079				
13.250000	1.604000	0.021710	-0.072100	2.105139	0.975935
	0.556202				
13.500000	1.614500	0.022690	-0.070700	2.135123	0.987502
	0.546346				
13.750000	1.625400	0.023680	-0.069300	2.165108	0.999120
	0.537124				
14.000000	1.637000	0.024660	-0.068100	2.195093	1.010850
	0.528735				
14.250000	1.644700	0.025920	-0.066700	2.225077	1.021793
	0.517672				
14.500000	1.656200	0.026970	-0.065600	2.255062	1.033455
	0.509775				
14.750000	1.663600	0.028350	-0.064300	2.285047	1.044273
	0.499146				
15.000000	1.675100	0.029470	-0.063400	2.315032	1.055888
	0.491772				
15.250000	1.685600	0.030670	-0.062400	2.345016	1.067279
	0.483919				
15.500000	1.693800	0.032100	-0.061500	2.375001	1.078171
	0.474718				
16.000000	1.706100	0.034900	-0.058800	2.434971	1.098987
	0.454431				
16.250000	1.715300	0.036360	-0.058100	2.464956	1.109990
	0.446734				
16.500000	1.725600	0.037740	-0.057400	2.494940	1.121207
	0.439964				
16.750000	1.731000	0.039660	-0.056700	2.524925	1.131327
	0.430306				
17.000000	1.740600	0.041170	-0.056100	2.554910	1.142338
	0.423527				
17.250000	1.748900	0.042840	-0.055600	2.584895	1.153034
	0.416148				
17.750000	1.761900	0.046670	-0.054700	2.644864	1.173519
	0.399893				
19.000000	1.792600	0.057060	-0.053600	2.794788	1.223679
	0.362114				
19.250000	1.800300	0.059020	-0.053500	2.824773	1.233988
	0.355995				
19.500000	1.806600	0.061200	-0.053500	2.854757	1.243923
	0.349308				
19.750000	1.807700	0.064070	-0.053700	2.884742	1.252517
	0.340139				
20.000000	1.814700	0.066190	-0.053800	2.914727	1.814700
	0.000000				
20.250000	1.820600	0.068470	-0.053900	2.944712	1.820600
	0.000000				
20.500000	1.820300	0.071670	-0.054400	2.974696	1.820300
	0.000000				
20.750000	1.827700	0.073830	-0.054700	3.004681	1.827700
	0.000000				
21.000000	1.834800	0.076010	-0.055000	3.034666	1.834800
	0.000000				
21.250000	1.834300	0.079340	-0.055600	3.064651	1.834300
	0.000000				
21.500000	1.839500	0.081840	-0.056100	3.094635	1.839500
	0.000000				
21.750000	1.846300	0.084110	-0.056600	3.124620	1.846300
	0.000000				
22.000000	1.852700	0.086450	-0.057100	3.154605	1.852700
	0.000000				
22.250000	1.850400	0.090130	-0.058100	3.184590	1.850400
	0.000000				
22.500000	1.855500	0.092670	-0.058800	3.214574	1.855500
	0.000000				
22.750000	1.861500	0.095080	-0.059400	3.244559	1.861500
	0.000000				
23.000000	1.865700	0.097750	-0.060200	3.274544	1.865700
	0.000000				
23.250000	1.864100	0.101340	-0.061400	3.304529	1.864100
	0.000000				
23.500000	1.863400	0.104780	-0.062600	3.334513	1.863400
	0.000000				
24.000000	1.862600	0.111540	-0.065200	3.394483	1.862600
	0.000000				
24.250000	1.864400	0.114570	-0.066400	3.424468	1.864400
	0.000000				
24.500000	1.864000	0.117920	-0.067800	3.454453	1.864000
	0.000000				
24.750000	1.863400	0.121280	-0.069300	3.484437	1.863400
	0.000000				

25.000000	1.863700	0.124490	-0.070900	3.514422	1.863700
	0.000000				
25.250000	1.863800	0.127690	-0.072400	3.544407	1.863800
	0.000000				
25.500000	1.862000	0.131150	-0.074200	3.574392	1.862000
	0.000000				
25.750000	1.860100	0.134630	-0.076100	3.604376	1.860100
	0.000000				
26.000000	1.857800	0.138130	-0.078000	3.634361	1.857800
	0.000000				
26.250000	1.855200	0.141670	-0.080100	3.664346	1.855200
	0.000000				
26.500000	1.852000	0.145250	-0.082300	3.694330	1.852000
	0.000000				
26.750000	1.848600	0.148860	-0.084500	3.724315	1.848600
	0.000000				
27.000000	1.844200	0.152570	-0.087000	3.754300	1.844200
	0.000000				
27.250000	1.840500	0.156180	-0.089400	3.784285	1.840500
	0.000000				
27.500000	1.835000	0.160030	-0.092100	3.814270	1.835000
	0.000000				
27.750000	1.829100	0.163910	-0.094900	3.844254	1.829100
	0.000000				
28.000000	1.822500	0.167910	-0.097800	3.874239	1.822500
	0.000000				
28.250000	1.814300	0.172130	-0.101000	3.904224	1.814300
	0.000000				
28.500000	1.808000	0.176030	-0.104100	3.934209	1.808000
	0.000000				
28.750000	1.799200	0.180320	-0.107600	3.964193	1.799200
	0.000000				
29.000000	1.789700	0.184690	-0.111200	3.994178	1.789700
	0.000000				
29.250000	1.780000	0.189090	-0.114900	4.024163	1.780000
	0.000000				
29.500000	1.771000	0.193350	-0.118600	4.054148	1.771000
	0.000000				
29.750000	1.763000	0.197430	-0.122200	4.084132	1.763000
	0.000000				
30.000000	1.753700	0.201720	-0.126000	4.114117	1.753700
	0.000000				
30.250000	1.742700	0.206290	-0.130200	4.144102	1.742700
	0.000000				
30.500000	1.734200	0.210410	-0.134000	4.174087	1.734200
	0.000000				
30.750000	1.722200	0.215140	-0.138500	4.204072	1.722200
	0.000000				
31.000000	1.710400	0.219830	-0.142900	4.234056	1.710400
	0.000000				
31.250000	1.697100	0.224780	-0.147600	4.264041	1.697100
	0.000000				
31.500000	1.684700	0.229560	-0.152200	4.294026	1.684700
	0.000000				
31.750000	1.670000	0.234800	-0.157300	4.324010	1.670000
	0.000000				
32.000000	1.655800	0.239960	-0.162400	4.353995	1.655800
	0.000000				
32.250000	1.648700	0.243790	-0.166300	4.383980	1.648700
	0.000000				
32.500000	1.635000	0.248820	-0.171300	4.413965	1.635000
	0.000000				
32.750000	1.622300	0.253730	-0.176200	4.443950	1.622300
	0.000000				
33.000000	1.604600	0.259620	-0.181900	4.473934	1.604600
	0.000000				
33.250000	1.587900	0.265370	-0.187600	4.503919	1.587900
	0.000000				
33.500000	1.572300	0.271010	-0.193100	4.533904	1.572300
	0.000000				
33.750000	1.552600	0.277530	-0.199300	4.563889	1.552600
	0.000000				
34.000000	1.535600	0.283620	-0.205100	4.593873	1.535600
	0.000000				
34.250000	1.510500	0.291660	-0.212300	4.623858	1.510500
	0.000000				
34.500000	1.489000	0.299140	-0.218800	4.653843	1.489000
	0.000000				
35.000000	1.690188	0.479798	-0.224794	4.713812	1.690188
	0.000000				
36.000000	1.686562	0.509740	-0.236781	4.833751	1.686562
	0.000000				
37.000000	1.682970	0.540077	-0.248769	4.953691	1.682970
	0.000000				
38.000000	1.679350	0.570760	-0.260756	5.073629	1.679350
	0.000000				
39.000000	1.675631	0.601740	-0.272744	5.193568	1.675631
	0.000000				
40.000000	1.671742	0.632970	-0.284731	5.313508	1.671742
	0.000000				
41.000000	1.667607	0.664405	-0.296719	5.433446	1.667607
	0.000000				
42.000000	1.663152	0.695999	-0.308706	5.553385	1.663152
	0.000000				
43.000000	1.658302	0.727710	-0.320694	5.673325	1.658302
	0.000000				
44.000000	1.652983	0.759494	-0.332681	5.793263	1.652983
	0.000000				

45.000000	1.647123	0.791309	-0.344669	5.913203	1.647123
	0.000000				
46.000000	1.640654	0.823116	-0.356656	6.033142	1.640654
	0.000000				
47.000000	1.633509	0.854872	-0.368643	6.153080	1.633509
	0.000000				
48.000000	1.625624	0.886540	-0.380631	6.273020	1.625624
	0.000000				
49.000000	1.616941	0.918079	-0.392618	6.392959	1.616941
	0.000000				
50.000000	1.607402	0.949451	-0.401867	6.512897	1.607402
	0.000000				
51.000000	1.596956	0.980618	-0.405065	6.632837	1.596956
	0.000000				
52.000000	1.585554	1.011543	-0.408195	6.752776	1.585554
	0.000000				
53.000000	1.573151	1.042188	-0.411247	6.872715	1.573151
	0.000000				
54.000000	1.559705	1.072518	-0.414212	6.992654	1.559705
	0.000000				
55.000000	1.545180	1.102496	-0.417084	7.112593	1.545180
	0.000000				
56.000000	1.529542	1.132088	-0.419854	7.232532	1.529542
	0.000000				
57.000000	1.512762	1.161258	-0.422518	7.352471	1.512762
	0.000000				
58.000000	1.494815	1.189972	-0.425069	7.472410	1.494815
	0.000000				
59.000000	1.475677	1.218198	-0.427505	7.592349	1.475677
	0.000000				
60.000000	1.455331	1.245901	-0.429821	7.712288	1.455331
	0.000000				
61.000000	1.433762	1.273051	-0.432015	7.832227	1.433762
	0.000000				
62.000000	1.410960	1.299615	-0.434086	7.952166	1.410960
	0.000000				
63.000000	1.386916	1.325562	-0.436033	8.072105	1.386916
	0.000000				
64.000000	1.361626	1.350863	-0.437855	8.192044	1.361626
	0.000000				
65.000000	1.335091	1.375489	-0.439555	8.311983	1.335091
	0.000000				
66.000000	1.307313	1.399409	-0.441133	8.431922	1.307313
	0.000000				
67.000000	1.278297	1.422598	-0.442591	8.551861	1.278297
	0.000000				
68.000000	1.248055	1.445029	-0.443933	8.671801	1.248055
	0.000000				
69.000000	1.216596	1.466674	-0.445161	8.791739	1.216596
	0.000000				
70.000000	1.183939	1.487511	-0.446280	8.911678	1.183939
	0.000000				
71.000000	1.150101	1.507513	-0.447295	9.031617	1.150101
	0.000000				
72.000000	1.115104	1.526660	-0.448210	9.151556	1.115104
	0.000000				
73.000000	1.078972	1.544928	-0.449030	9.271495	1.078972
	0.000000				
74.000000	1.041734	1.562296	-0.449763	9.391435	1.041734
	0.000000				
75.000000	1.003419	1.578745	-0.450413	9.511374	1.003419
	0.000000				
76.000000	0.964061	1.594257	-0.450988	9.631312	0.964061
	0.000000				
77.000000	0.923694	1.608812	-0.451494	9.751251	0.923694
	0.000000				
78.000000	0.882357	1.622396	-0.451938	9.871190	0.882357
	0.000000				
79.000000	0.840091	1.634992	-0.452327	9.991130	0.840091
	0.000000				
80.000000	0.796938	1.646586	-0.452669	10.111069	0.796938
	0.000000				
81.000000	0.752944	1.657166	-0.452971	10.231008	0.752944
	0.000000				
82.000000	0.708156	1.666720	-0.453240	10.350946	0.708156
	0.000000				
83.000000	0.662624	1.675237	-0.453483	10.470885	0.662624
	0.000000				
84.000000	0.616399	1.682708	-0.453708	10.590824	0.616399
	0.000000				
85.000000	0.569534	1.689126	-0.453921	10.710764	0.569534
	0.000000				
86.000000	0.522085	1.694482	-0.454129	10.830703	0.522085
	0.000000				
87.000000	0.474109	1.698773	-0.454339	10.950642	0.474109
	0.000000				
88.000000	0.425664	1.701994	-0.454557	11.070580	0.425664
	0.000000				
89.000000	0.376811	1.704142	-0.454789	11.190519	0.376811
	0.000000				
90.000000	0.327610	1.705217	-0.455040	0.000000	0.327610
	0.000000				
91.000000	0.278125	1.705216	-0.455315	0.000000	0.278125
	0.000000				
92.000000	0.228419	1.704142	-0.455619	0.000000	0.228419
	0.000000				
93.000000	0.178556	1.701998	-0.455957	0.000000	0.178556
	0.000000				

94.000000	0.128604	1.698787	-0.456330	0.000000	0.128604
	0.000000				
95.000000	0.078628	1.694513	-0.456744	0.000000	0.078628
	0.000000				
96.000000	0.028695	1.689185	-0.457198	0.000000	0.028695
	0.000000				
97.000000	-0.021127	1.682808	-0.457696	0.000000	-0.021127
	0.000000				
98.000000	-0.070770	1.675392	-0.458239	0.000000	-0.070770
	0.000000				
99.000000	-0.120165	1.666948	-0.458825	0.000000	-0.120165
	0.000000				
100.000000	-0.169244	1.657486	-0.459456	0.000000	-0.169244
	0.000000				
101.000000	-0.217939	1.647020	-0.460130	0.000000	-0.217939
	0.000000				
102.000000	-0.335052	1.635563	-0.464831	0.000000	-0.335052
	0.000000				
103.000000	-0.381281	1.623130	-0.465831	0.000000	-0.381281
	0.000000				
104.000000	-0.426960	1.609739	-0.466855	0.000000	-0.426960
	0.000000				
105.000000	-0.472022	1.595406	-0.467898	0.000000	-0.472022
	0.000000				
106.000000	-0.516399	1.580150	-0.468957	0.000000	-0.516399
	0.000000				
107.000000	-0.560025	1.563991	-0.470024	0.000000	-0.560025
	0.000000				
108.000000	-0.602833	1.546950	-0.471095	0.000000	-0.602833
	0.000000				
109.000000	-0.644759	1.529049	-0.472162	0.000000	-0.644759
	0.000000				
110.000000	-0.685740	1.510310	-0.473217	0.000000	-0.685740
	0.000000				
111.000000	-0.725713	1.490759	-0.474253	0.000000	-0.725713
	0.000000				
112.000000	-0.764619	1.470419	-0.475260	0.000000	-0.764619
	0.000000				
113.000000	-0.802399	1.449317	-0.476229	0.000000	-0.802399
	0.000000				
114.000000	-0.838996	1.427479	-0.477149	0.000000	-0.838996
	0.000000				
115.000000	-0.874354	1.404934	-0.478010	0.000000	-0.874354
	0.000000				
116.000000	-0.908421	1.381710	-0.478800	0.000000	-0.908421
	0.000000				
117.000000	-0.941147	1.357837	-0.479508	0.000000	-0.941147
	0.000000				
118.000000	-0.972483	1.333345	-0.480121	0.000000	-0.972483
	0.000000				
119.000000	-1.002382	1.308264	-0.480628	0.000000	-1.002382
	0.000000				
120.000000	-1.030801	1.282627	-0.481014	0.000000	-1.030801
	0.000000				
121.000000	-1.057699	1.256465	-0.481268	0.000000	-1.057699
	0.000000				
122.000000	-1.083038	1.229812	-0.481376	0.000000	-1.083038
	0.000000				
123.000000	-1.106782	1.202701	-0.481324	0.000000	-1.106782
	0.000000				
124.000000	-1.128899	1.175166	-0.481099	0.000000	-1.128899
	0.000000				
125.000000	-1.149358	1.147242	-0.480688	0.000000	-1.149358
	0.000000				
126.000000	-1.168133	1.118963	-0.480078	0.000000	-1.168133
	0.000000				
127.000000	-1.185201	1.090366	-0.479255	0.000000	-1.185201
	0.000000				
128.000000	-1.200541	1.061485	-0.478207	0.000000	-1.200541
	0.000000				
129.000000	-1.214136	1.032357	-0.476921	0.000000	-1.214136
	0.000000				
130.000000	-1.225971	1.003018	-0.475385	0.000000	-1.225971
	0.000000				
131.000000	-1.236037	0.973505	-0.473587	0.000000	-1.236037
	0.000000				
132.000000	-1.244326	0.943854	-0.471515	0.000000	-1.244326
	0.000000				
133.000000	-1.250834	0.914102	-0.469160	0.000000	-1.250834
	0.000000				
134.000000	-1.255562	0.884286	-0.466511	0.000000	-1.255562
	0.000000				
135.000000	-1.258512	0.854442	-0.463559	0.000000	-1.258512
	0.000000				
136.000000	-1.259692	0.824609	-0.460296	0.000000	-1.259692
	0.000000				
137.000000	-1.259111	0.794822	-0.456713	0.000000	-1.259111
	0.000000				
138.000000	-1.256784	0.765118	-0.452805	0.000000	-1.256784
	0.000000				
139.000000	-1.252729	0.735534	-0.448566	0.000000	-1.252729
	0.000000				
140.000000	-1.246966	0.706107	-0.443991	0.000000	-1.246966
	0.000000				
141.000000	-1.239522	0.676872	-0.439077	0.000000	-1.239522
	0.000000				
142.000000	-1.230425	0.647866	-0.433822	0.000000	-1.230425
	0.000000				

143.000000	-1.219710	0.619123	-0.428225	0.000000	-1.219710
	0.000000				
144.000000	-1.207412	0.590679	-0.422287	0.000000	-1.207412
	0.000000				
145.000000	-1.193575	0.562569	-0.416010	0.000000	-1.193575
	0.000000				
146.000000	-1.178245	0.534827	-0.409398	0.000000	-1.178245
	0.000000				
147.000000	-1.161474	0.507486	-0.402457	0.000000	-1.161474
	0.000000				
148.000000	-1.143321	0.480580	-0.395196	0.000000	-1.143321
	0.000000				
149.000000	-1.123852	0.454142	-0.387626	0.000000	-1.123852
	0.000000				
150.000000	-1.103139	0.428204	-0.379759	0.000000	-1.103139
	0.000000				
151.000000	-1.081270	0.402796	-0.371615	0.000000	-1.081270
	0.000000				
152.000000	-1.058341	0.377950	-0.363215	0.000000	-1.058341
	0.000000				
153.000000	-1.034472	0.353695	-0.354590	0.000000	-1.034472
	0.000000				
154.000000	-1.009806	0.330061	-0.345777	0.000000	-1.009806
	0.000000				
155.000000	-0.984524	0.307075	-0.336826	0.000000	-0.984524
	0.000000				
156.000000	-0.958861	0.284766	-0.327807	0.000000	-0.958861
	0.000000				
157.000000	-0.933133	0.263159	-0.318817	0.000000	-0.933133
	0.000000				
158.000000	-0.907785	0.242281	-0.309992	0.000000	-0.907785
	0.000000				
159.000000	-0.883460	0.222156	-0.301539	0.000000	-0.883460
	0.000000				
160.000000	-0.861125	0.202808	-0.293773	0.000000	-0.861125
	0.000000				
161.000000	-0.842289	0.184259	-0.287186	0.000000	-0.842289
	0.000000				
162.000000	-0.829349	0.166533	-0.282574	0.000000	-0.829349
	0.000000				
163.000000	-0.826167	0.149648	-0.281226	0.000000	-0.826167
	0.000000				
164.000000	-0.838782	0.133625	-0.285177	0.000000	-0.838782
	0.000000				
165.000000	-0.875592	0.118482	-0.297286	0.000000	-0.875592
	0.000000				
166.000000	-0.944010	0.104237	-0.320162	0.000000	-0.944010
	0.000000				
167.000000	-1.037436	0.090906	-0.351810	0.000000	-1.037436
	0.000000				
168.000000	-1.117190	0.078504	-0.379421	0.000000	-1.117190
	0.000000				
169.000000	-1.132235	0.067045	-0.385680	0.000000	-1.132235
	0.000000				
170.000000	-1.077345	0.056542	-0.368384	0.000000	-1.077345
	0.000000				
171.000000	-0.988218	0.047006	-0.339267	0.000000	-0.988218
	0.000000				
172.000000	-0.892418	0.038449	-0.307585	0.000000	-0.892418
	0.000000				
173.000000	-0.796467	0.030878	-0.275558	0.000000	-0.796467
	0.000000				
174.000000	-0.700516	0.024302	-0.243249	0.000000	-0.700516
	0.000000				
175.000000	-0.604564	0.018728	-0.210670	0.000000	-0.604564
	0.000000				
176.000000	-0.508613	0.014161	-0.177833	0.000000	-0.508613
	0.000000				
177.000000	-0.412662	0.010606	-0.144750	0.000000	-0.412662
	0.000000				
178.000000	-0.316711	0.008064	-0.111433	0.000000	-0.316711
	0.000000				
179.000000	-0.220759	0.006539	-0.077892	0.000000	-0.220759
	0.000000				
180.000000	-0.124808	0.006000	-0.043659	0.000000	-0.124808
	0.000000				

D. 3D Blade Profile Nodal Mesh File

This section stores the complete, structural multi-body finite element layout tracing out the 3D grid coordinate system for downstream structural and modal calculations.

Listing 12: Multi-Body Solver Discretization Coordinates Map (New_Blade.txt).

New Blade	x	y	z
1.51371e+00	1.41374e+00	1.50000e+00	
1.78214e+00	1.05483e+00	1.50000e+00	
1.91818e+00	7.80470e-01	1.50000e+00	
2.01090e+00	4.94940e-01	1.50000e+00	
2.06118e+00	2.00989e-01	1.50000e+00	
2.06872e+00	-9.61954e-02	1.50000e+00	
2.03374e+00	-3.90965e-01	1.50000e+00	
1.95702e+00	-6.77403e-01	1.50000e+00	
1.84029e+00	-9.49861e-01	1.50000e+00	
1.68589e+00	-1.20276e+00	1.50000e+00	
1.49703e+00	-1.43103e+00	1.50000e+00	
1.27754e+00	-1.62998e+00	1.50000e+00	
1.03188e+00	-1.79557e+00	1.50000e+00	
7.65076e-01	-1.92446e+00	1.50000e+00	
4.82496e-01	-2.01395e+00	1.50000e+00	
1.89931e-01	-2.06226e+00	1.50000e+00	
-1.06838e-01	-2.06822e+00	1.50000e+00	
-4.01838e-01	-2.03163e+00	1.50000e+00	
-6.89433e-01	-1.95282e+00	1.50000e+00	
-9.64391e-01	-1.83272e+00	1.50000e+00	
-1.22411e+00	-1.67047e+00	1.50000e+00	
-1.53445e+00	-1.39087e+00	1.50000e+00	
-1.78232e+00	-1.05463e+00	1.50000e+00	
-1.91836e+00	-7.80276e-01	1.50000e+00	
-2.01095e+00	-4.94879e-01	1.50000e+00	
-2.06122e+00	-2.00948e-01	1.50000e+00	
-2.06874e+00	9.62170e-02	1.50000e+00	
-2.03375e+00	3.90977e-01	1.50000e+00	
-1.95702e+00	6.77403e-01	1.50000e+00	
-1.84029e+00	9.49861e-01	1.50000e+00	
-1.68589e+00	1.20276e+00	1.50000e+00	
-1.49703e+00	1.43103e+00	1.50000e+00	
-1.27754e+00	1.62998e+00	1.50000e+00	
-1.03188e+00	1.79557e+00	1.50000e+00	
-7.65076e-01	1.92446e+00	1.50000e+00	
-4.82496e-01	2.01395e+00	1.50000e+00	
-1.89920e-01	2.06225e+00	1.50000e+00	
1.06858e-01	2.06820e+00	1.50000e+00	
4.01874e-01	2.03159e+00	1.50000e+00	
6.89489e-01	1.95276e+00	1.50000e+00	
9.64566e-01	1.83253e+00	1.50000e+00	
1.22429e+00	1.67027e+00	1.50000e+00	
3.79141e-01	2.66544e+00	1.50000e+00	
1.58698e+00	1.49760e+00	1.81658e+00	
1.80589e+00	1.18655e+00	1.81658e+00	
1.92928e+00	9.20796e-01	1.81658e+00	
2.01085e+00	6.39681e-01	1.81658e+00	
2.05162e+00	3.46021e-01	1.81658e+00	
2.05161e+00	4.49241e-02	1.81658e+00	
2.01119e+00	-2.57882e-01	1.81658e+00	
1.93117e+00	-5.56238e-01	1.81658e+00	
1.81312e+00	-8.44109e-01	1.81658e+00	
1.65910e+00	-1.11537e+00	1.81658e+00	
1.47183e+00	-1.36425e+00	1.81658e+00	
1.25431e+00	-1.58508e+00	1.81658e+00	
1.01038e+00	-1.77298e+00	1.81658e+00	
7.44756e-01	-1.92395e+00	1.81658e+00	
4.62607e-01	-2.03472e+00	1.81658e+00	
1.69599e-01	-2.10301e+00	1.81658e+00	
-1.28632e-01	-2.12712e+00	1.81658e+00	
-4.26019e-01	-2.10654e+00	1.81658e+00	
-7.16760e-01	-2.04137e+00	1.81658e+00	
-9.95448e-01	-1.93235e+00	1.81658e+00	
-1.25918e+00	-1.77863e+00	1.81658e+00	
-1.57546e+00	-1.50564e+00	1.81658e+00	
-1.82699e+00	-1.17176e+00	1.81658e+00	
-1.96358e+00	-8.96803e-01	1.81658e+00	
-2.05508e+00	-6.08808e-01	1.81658e+00	
-2.10257e+00	-3.10467e-01	1.81658e+00	
-2.10540e+00	-7.39160e-03	1.81658e+00	
-2.06372e+00	2.94529e-01	1.81658e+00	
-1.97819e+00	5.89040e-01	1.81658e+00	
-1.85039e+00	8.70106e-01	1.81658e+00	
-1.68267e+00	1.13181e+00	1.81658e+00	
-1.47828e+00	1.36874e+00	1.81658e+00	
-1.24134e+00	1.57604e+00	1.81658e+00	
-9.77948e-01	1.75036e+00	1.81658e+00	
-6.95312e-01	1.88946e+00	1.81658e+00	
-4.00495e-01	1.99140e+00	1.81658e+00	
-1.00397e-01	2.05473e+00	1.81658e+00	
1.98650e-01	2.07827e+00	1.81658e+00	
4.90915e-01	2.06126e+00	1.81658e+00	
7.71542e-01	2.00314e+00	1.81658e+00	
1.03622e+00	1.90384e+00	1.81658e+00	

1.28298e+00	1.76196e+00	1.81658e+00
4.94280e-01	2.69607e+00	1.81658e+00
1.57701e+00	1.50456e+00	1.81658e+00
1.63951e+00	1.60433e+00	2.13317e+00
1.82965e+00	1.31827e+00	2.13317e+00
1.94037e+00	1.06112e+00	2.13317e+00
2.01080e+00	7.84421e-01	2.13317e+00
2.04206e+00	4.91054e-01	2.13317e+00
2.03449e+00	1.86044e-01	2.13317e+00
1.98864e+00	-1.24799e-01	2.13317e+00
1.90531e+00	-4.35074e-01	2.13317e+00
1.78595e+00	-7.38356e-01	2.13317e+00
1.63232e+00	-1.02799e+00	2.13317e+00
1.44663e+00	-1.29747e+00	2.13317e+00
1.23108e+00	-1.54019e+00	2.13317e+00
9.88877e-01	-1.75038e+00	2.13317e+00
7.24437e-01	-1.92344e+00	2.13317e+00
4.42719e-01	-2.05549e+00	2.13317e+00
1.49267e-01	-2.14375e+00	2.13317e+00
-1.50427e-01	-2.18602e+00	2.13317e+00
-4.50201e-01	-2.18145e+00	2.13317e+00
-7.44087e-01	-2.12992e+00	2.13317e+00
-1.02651e+00	-2.03197e+00	2.13317e+00
-1.29425e+00	-1.88679e+00	2.13317e+00
-1.61647e+00	-1.62041e+00	2.13317e+00
-1.87167e+00	-1.28889e+00	2.13317e+00
-2.00880e+00	-1.01333e+00	2.13317e+00
-2.09921e+00	-7.22738e-01	2.13317e+00
-2.14392e+00	-4.19986e-01	2.13317e+00
-2.14206e+00	-1.11000e-01	2.13317e+00
-2.09370e+00	1.98081e-01	2.13317e+00
-1.99936e+00	5.00677e-01	2.13317e+00
-1.86049e+00	7.90350e-01	2.13317e+00
-1.67945e+00	1.06086e+00	2.13317e+00
-1.45953e+00	1.30646e+00	2.13317e+00
-1.20514e+00	1.52209e+00	2.13317e+00
-9.24018e-01	1.70514e+00	2.13317e+00
-6.25549e-01	1.85446e+00	2.13317e+00
-3.18494e-01	1.96884e+00	2.13317e+00
-1.08735e-02	2.04722e+00	2.13317e+00
2.90443e-01	2.08835e+00	2.13317e+00
5.79956e-01	2.09093e+00	2.13317e+00
8.53595e-01	2.05352e+00	2.13317e+00
1.10788e+00	1.97515e+00	2.13317e+00
1.34167e+00	1.85364e+00	2.13317e+00
6.09420e-01	2.72670e+00	2.13317e+00
1.61957e+00	1.61825e+00	2.13317e+00
1.69204e+00	1.71107e+00	2.44975e+00
1.85340e+00	1.44999e+00	2.44975e+00
1.95146e+00	1.20145e+00	2.44975e+00
2.01076e+00	9.29162e-01	2.44975e+00
2.03249e+00	6.36087e-01	2.44975e+00
2.01737e+00	3.27163e-01	2.44975e+00
1.96610e+00	8.28372e-03	2.44975e+00
1.87946e+00	-3.13909e-01	2.44975e+00
1.75879e+00	-6.32604e-01	2.44975e+00
1.60554e+00	-9.40607e-01	2.44975e+00
1.42144e+00	-1.23069e+00	2.44975e+00
1.20785e+00	-1.49529e+00	2.44975e+00
9.67377e-01	-1.72779e+00	2.44975e+00
7.04118e-01	-1.92292e+00	2.44975e+00
4.22830e-01	-2.07627e+00	2.44975e+00
1.28935e-01	-2.18450e+00	2.44975e+00
-1.72221e-01	-2.24492e+00	2.44975e+00
-4.74383e-01	-2.25636e+00	2.44975e+00
-7.71414e-01	-2.21847e+00	2.44975e+00
-1.05756e+00	-2.13160e+00	2.44975e+00
-1.32932e+00	-1.99495e+00	2.44975e+00
-1.65748e+00	-1.73518e+00	2.44975e+00
-1.91634e+00	-1.40602e+00	2.44975e+00
-2.05402e+00	-1.12986e+00	2.44975e+00
-2.14334e+00	-8.36667e-01	2.44975e+00
-2.18528e+00	-5.29505e-01	2.44975e+00
-2.17872e+00	-2.14609e-01	2.44975e+00
-2.12367e+00	1.01634e-01	2.44975e+00
-2.02054e+00	4.12314e-01	2.44975e+00
-1.87060e+00	7.10595e-01	2.44975e+00
-1.67623e+00	9.89912e-01	2.44975e+00
-1.44078e+00	1.24418e+00	2.44975e+00
-1.16894e+00	1.46815e+00	2.44975e+00
-8.70087e-01	1.65993e+00	2.44975e+00
-5.55785e-01	1.81946e+00	2.44975e+00
-2.36493e-01	1.94629e+00	2.44975e+00
7.86499e-02	2.03970e+00	2.44975e+00
3.82235e-01	2.09842e+00	2.44975e+00
6.68997e-01	2.12060e+00	2.44975e+00
9.35648e-01	2.10389e+00	2.44975e+00
1.17954e+00	2.04646e+00	2.44975e+00
1.40037e+00	1.94533e+00	2.44975e+00
7.24559e-01	2.75732e+00	2.44975e+00
1.66212e+00	1.73194e+00	2.44975e+00
1.74457e+00	1.81780e+00	2.76633e+00
1.87715e+00	1.58170e+00	2.76633e+00
1.96256e+00	1.34178e+00	2.76633e+00
2.01071e+00	1.07390e+00	2.76633e+00
2.02293e+00	7.81120e-01	2.76633e+00

2.00025e+00	4.68283e-01	2.76633e+00
1.94355e+00	1.41366e-01	2.76633e+00
1.85361e+00	-1.92744e-01	2.76633e+00
1.73162e+00	-5.26851e-01	2.76633e+00
1.57876e+00	-8.53223e-01	2.76633e+00
1.39624e+00	-1.16391e+00	2.76633e+00
1.18462e+00	-1.45039e+00	2.76633e+00
9.45876e-01	-1.70520e+00	2.76633e+00
6.83798e-01	-1.92241e+00	2.76633e+00
4.02942e-01	-2.09704e+00	2.76633e+00
1.08602e-01	-2.22524e+00	2.76633e+00
-1.94015e-01	-2.30382e+00	2.76633e+00
-4.98565e-01	-2.33127e+00	2.76633e+00
-7.98741e-01	-2.30702e+00	2.76633e+00
-1.08862e+00	-2.23122e+00	2.76633e+00
-1.36439e+00	-2.10311e+00	2.76633e+00
-1.69849e+00	-1.84995e+00	2.76633e+00
-1.96101e+00	-1.52316e+00	2.76633e+00
-2.09924e+00	-1.24638e+00	2.76633e+00
-2.18746e+00	-9.50596e-01	2.76633e+00
-2.22663e+00	-6.39024e-01	2.76633e+00
-2.21538e+00	-3.18217e-01	2.76633e+00
-2.15364e+00	5.18589e-03	2.76633e+00
-2.04171e+00	3.23951e-01	2.76633e+00
-1.88070e+00	6.30839e-01	2.76633e+00
-1.67301e+00	9.18963e-01	2.76633e+00
-1.42202e+00	1.18190e+00	2.76633e+00
-1.13274e+00	1.41421e+00	2.76633e+00
-8.16157e-01	1.61472e+00	2.76633e+00
-4.86022e-01	1.78446e+00	2.76633e+00
-1.54493e-01	1.92374e+00	2.76633e+00
1.68173e-01	2.03218e+00	2.76633e+00
4.74028e-01	2.10850e+00	2.76633e+00
7.58038e-01	2.15027e+00	2.76633e+00
1.01770e+00	2.15427e+00	2.76633e+00
1.25120e+00	2.11777e+00	2.76633e+00
1.45906e+00	2.03701e+00	2.76633e+00
8.39699e-01	2.78795e+00	2.76633e+00
1.70468e+00	1.84562e+00	2.76633e+00
1.79710e+00	1.92454e+00	3.08291e+00
1.90091e+00	1.71342e+00	3.08291e+00
1.97365e+00	1.48210e+00	3.08291e+00
2.01066e+00	1.21864e+00	3.08291e+00
2.01337e+00	9.26153e-01	3.08291e+00
1.98314e+00	6.09402e-01	3.08291e+00
1.92100e+00	2.74449e-01	3.08291e+00
1.82775e+00	-7.15794e-02	3.08291e+00
1.70446e+00	-4.21099e-01	3.08291e+00
1.55198e+00	-7.65839e-01	3.08291e+00
1.37104e+00	-1.09713e+00	3.08291e+00
1.16139e+00	-1.40550e+00	3.08291e+00
9.24376e-01	-1.68261e+00	3.08291e+00
6.63479e-01	-1.92190e+00	3.08291e+00
3.83054e-01	-2.11781e+00	3.08291e+00
8.82702e-02	-2.26599e+00	3.08291e+00
-2.15809e-01	-2.36272e+00	3.08291e+00
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-8.26068e-01	-2.39556e+00	3.08291e+00
-1.11968e+00	-2.33085e+00	3.08291e+00
-1.39946e+00	-2.21127e+00	3.08291e+00
-1.73949e+00	-1.96472e+00	3.08291e+00
-2.00569e+00	-1.64029e+00	3.08291e+00
-2.14446e+00	-1.36291e+00	3.08291e+00
-2.23159e+00	-1.06453e+00	3.08291e+00
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-2.06288e+00	2.35588e-01	3.08291e+00
-1.89081e+00	5.51084e-01	3.08291e+00
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2.57697e-01	2.02466e+00	3.08291e+00
5.65821e-01	2.11857e+00	3.08291e+00
8.47079e-01	2.17994e+00	3.08291e+00
1.09975e+00	2.20464e+00	3.08291e+00
1.32286e+00	2.18908e+00	3.08291e+00
1.51775e+00	2.12870e+00	3.08291e+00
9.54838e-01	2.81858e+00	3.08291e+00
1.74724e+00	1.95931e+00	3.08291e+00
1.84963e+00	2.03127e+00	3.39950e+00
1.92466e+00	1.84514e+00	3.39950e+00
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1.96602e+00	7.50522e-01	3.39950e+00
1.89845e+00	4.07532e-01	3.39950e+00
1.80190e+00	4.95854e-02	3.39950e+00
1.67729e+00	-3.15346e-01	3.39950e+00
1.52520e+00	-6.78455e-01	3.39950e+00
1.34584e+00	-1.03035e+00	3.39950e+00
1.13816e+00	-1.36060e+00	3.39950e+00
9.02875e-01	-1.66002e+00	3.39950e+00
6.43160e-01	-1.92139e+00	3.39950e+00

3.63165e-01	-2.13858e+00	3.39950e+00
6.79379e-02	-2.30674e+00	3.39950e+00
-2.37604e-01	-2.42162e+00	3.39950e+00
-5.46929e-01	-2.48110e+00	3.39950e+00
-8.53395e-01	-2.48411e+00	3.39950e+00
-1.15073e+00	-2.43047e+00	3.39950e+00
-1.43453e+00	-2.31942e+00	3.39950e+00
-1.78050e+00	-2.07949e+00	3.39950e+00
-2.05036e+00	-1.75742e+00	3.39950e+00
-2.18968e+00	-1.47944e+00	3.39950e+00
-2.27572e+00	-1.17846e+00	3.39950e+00
-2.30933e+00	-8.58062e-01	3.39950e+00
-2.28870e+00	-5.25435e-01	3.39950e+00
-2.21359e+00	-1.87709e-01	3.39950e+00
-2.08405e+00	1.47224e-01	3.39950e+00
-1.90091e+00	4.71328e-01	3.39950e+00
-1.66657e+00	7.77065e-01	3.39950e+00
-1.38452e+00	1.05734e+00	3.39950e+00
-1.06034e+00	1.30632e+00	3.39950e+00
-7.08297e-01	1.52429e+00	3.39950e+00
-3.46495e-01	1.71445e+00	3.39950e+00
9.50869e-03	1.87863e+00	3.39950e+00
3.47220e-01	2.01715e+00	3.39950e+00
6.57613e-01	2.12864e+00	3.39950e+00
9.36120e-01	2.20962e+00	3.39950e+00
1.18181e+00	2.25502e+00	3.39950e+00
1.39451e+00	2.26039e+00	3.39950e+00
1.57645e+00	2.22039e+00	3.39950e+00
1.06998e+00	2.84921e+00	3.39950e+00
1.78980e+00	2.07300e+00	3.39950e+00
1.90216e+00	2.13801e+00	3.71608e+00
1.94842e+00	1.97686e+00	3.71608e+00
1.99584e+00	1.76275e+00	3.71608e+00
2.01057e+00	1.50812e+00	3.71608e+00
1.99424e+00	1.21622e+00	3.71608e+00
1.94890e+00	8.91641e-01	3.71608e+00
1.87590e+00	5.40615e-01	3.71608e+00
1.77605e+00	1.70750e-01	3.71608e+00
1.65013e+00	-2.09594e-01	3.71608e+00
1.49842e+00	-5.91071e-01	3.71608e+00
1.32064e+00	-9.63576e-01	3.71608e+00
1.11493e+00	-1.31571e+00	3.71608e+00
8.81375e-01	-1.63743e+00	3.71608e+00
6.22840e-01	-1.92088e+00	3.71608e+00
3.43277e-01	-2.15935e+00	3.71608e+00
4.76057e-02	-2.34748e+00	3.71608e+00
-2.59398e-01	-2.48052e+00	3.71608e+00
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-1.18179e+00	-2.53010e+00	3.71608e+00
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-1.82151e+00	-2.19426e+00	3.71608e+00
-2.09504e+00	-1.87455e+00	3.71608e+00
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-2.31984e+00	-1.29238e+00	3.71608e+00
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-2.32536e+00	-6.29043e-01	3.71608e+00
-2.24356e+00	-2.84157e-01	3.71608e+00
-2.10523e+00	5.88613e-02	3.71608e+00
-1.91102e+00	3.91573e-01	3.71608e+00
-1.66335e+00	7.06116e-01	3.71608e+00
-1.36576e+00	9.95053e-01	3.71608e+00
-1.02414e+00	1.25238e+00	3.71608e+00
-6.54367e-01	1.47908e+00	3.71608e+00
-2.76732e-01	1.67945e+00	3.71608e+00
9.15094e-02	1.85608e+00	3.71608e+00
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7.49406e-01	2.13872e+00	3.71608e+00
1.02516e+00	2.23929e+00	3.71608e+00
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1.46617e+00	2.33170e+00	3.71608e+00
1.63514e+00	2.31207e+00	3.71608e+00
1.18512e+00	2.87984e+00	3.71608e+00
1.83236e+00	2.18669e+00	3.71608e+00
1.95469e+00	2.24474e+00	4.03266e+00
1.97217e+00	2.10858e+00	4.03266e+00
2.00693e+00	1.90308e+00	4.03266e+00
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1.85336e+00	6.73698e-01	4.03266e+00
1.75020e+00	2.91915e-01	4.03266e+00
1.62296e+00	-1.03841e-01	4.03266e+00
1.47164e+00	-5.03687e-01	4.03266e+00
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3.23388e-01	-2.18013e+00	4.03266e+00
2.72735e-02	-2.38823e+00	4.03266e+00
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5.26267e-01	2.00211e+00	4.03266e+00
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1.11420e+00	2.26896e+00	4.03266e+00
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1.53783e+00	2.40301e+00	4.03266e+00
1.69383e+00	2.40376e+00	4.03266e+00
1.30026e+00	2.91047e+00	4.03266e+00
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2.00721e+00	2.35148e+00	4.34925e+00
1.99593e+00	2.24029e+00	4.34925e+00
2.01803e+00	2.04341e+00	4.34925e+00
2.01048e+00	1.79760e+00	4.34925e+00
1.97512e+00	1.50628e+00	4.34925e+00
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1.83081e+00	8.06780e-01	4.34925e+00
1.72434e+00	4.13080e-01	4.34925e+00
1.59580e+00	1.91130e-03	4.34925e+00
1.44486e+00	-4.16303e-01	4.34925e+00
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1.06847e+00	-1.22591e+00	4.34925e+00
8.38374e-01	-1.59224e+00	4.34925e+00
5.82202e-01	-1.91985e+00	4.34925e+00
3.03500e-01	-2.20090e+00	4.34925e+00
6.94122e-03	-2.42897e+00	4.34925e+00
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-2.30351e+00	-4.77052e-01	4.34925e+00
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-1.65692e+00	5.64219e-01	4.34925e+00
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-9.51746e-01	1.14450e+00	4.34925e+00
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2.55511e-01	1.81097e+00	4.34925e+00
6.15791e-01	1.99459e+00	4.34925e+00
9.32991e-01	2.15887e+00	4.34925e+00
1.20324e+00	2.29863e+00	4.34925e+00
1.42797e+00	2.40615e+00	4.34925e+00
1.60949e+00	2.47432e+00	4.34925e+00
1.75253e+00	2.49544e+00	4.34925e+00
1.41540e+00	2.94109e+00	4.34925e+00
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2.01968e+00	2.37201e+00	4.66583e+00
2.02912e+00	2.18373e+00	4.66583e+00
2.01043e+00	1.94234e+00	4.66583e+00
1.96556e+00	1.65132e+00	4.66583e+00
1.89755e+00	1.31500e+00	4.66583e+00
1.80826e+00	9.39863e-01	4.66583e+00
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5.61882e-01	-1.91934e+00	4.66583e+00
2.83612e-01	-2.22167e+00	4.66583e+00
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-2.22906e+00	-2.22594e+00	4.66583e+00
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-2.43534e+00	-9.39869e-01	4.66583e+00
-2.33348e+00	-5.73500e-01	4.66583e+00
-2.16874e+00	-2.06228e-01	4.66583e+00
-1.94133e+00	1.52306e-01	4.66583e+00
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1.29228e+00	2.32830e+00	4.66583e+00
1.51002e+00	2.45652e+00	4.66583e+00
1.68114e+00	2.54562e+00	4.66583e+00
1.81122e+00	2.58713e+00	4.66583e+00
1.53054e+00	2.97172e+00	4.66583e+00
1.96003e+00	2.52776e+00	4.66583e+00
2.11227e+00	2.56495e+00	4.98241e+00
2.04344e+00	2.50373e+00	4.98241e+00
2.04022e+00	2.32406e+00	4.98241e+00
2.01038e+00	2.08709e+00	4.98241e+00
1.95600e+00	1.79635e+00	4.98241e+00
1.88043e+00	1.45612e+00	4.98241e+00
1.78571e+00	1.07295e+00	4.98241e+00
1.67264e+00	6.55409e-01	4.98241e+00
1.54147e+00	2.13416e-01	4.98241e+00
1.39130e+00	-2.41535e-01	4.98241e+00
1.21984e+00	-6.96463e-01	4.98241e+00
1.02201e+00	-1.13612e+00	4.98241e+00
7.95373e-01	-1.54706e+00	4.98241e+00
5.41563e-01	-1.91883e+00	4.98241e+00
2.63723e-01	-2.24244e+00	4.98241e+00
-3.37233e-02	-2.51046e+00	4.98241e+00
-3.46575e-01	-2.71612e+00	4.98241e+00
-6.67839e-01	-2.85565e+00	4.98241e+00
-9.90030e-01	-2.92685e+00	4.98241e+00
-1.30602e+00	-2.92860e+00	4.98241e+00
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-1.98555e+00	-2.65334e+00	4.98241e+00
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-2.41579e+00	-2.06207e+00	4.98241e+00
-2.49635e+00	-1.74810e+00	4.98241e+00
-2.51610e+00	-1.40566e+00	4.98241e+00
-2.47200e+00	-1.04348e+00	4.98241e+00
-2.36346e+00	-6.69948e-01	4.98241e+00
-2.18991e+00	-2.94591e-01	4.98241e+00
-1.95144e+00	7.25505e-02	4.98241e+00
-1.65048e+00	4.22321e-01	4.98241e+00
-1.29075e+00	7.45927e-01	4.98241e+00
-8.79346e-01	1.03661e+00	4.98241e+00
-4.38646e-01	1.29823e+00	4.98241e+00
2.32160e-03	1.53945e+00	4.98241e+00
4.19512e-01	1.76586e+00	4.98241e+00
7.94838e-01	1.97956e+00	4.98241e+00
1.11658e+00	2.17901e+00	4.98241e+00
1.38132e+00	2.35797e+00	4.98241e+00
1.59207e+00	2.50690e+00	4.98241e+00
1.75280e+00	2.61693e+00	4.98241e+00
1.86991e+00	2.67882e+00	4.98241e+00
1.64568e+00	3.00235e+00	4.98241e+00
2.00259e+00	2.64145e+00	4.98241e+00
2.16480e+00	2.67168e+00	5.29899e+00
2.06719e+00	2.63545e+00	5.29899e+00
2.05131e+00	2.46439e+00	5.29899e+00
2.01034e+00	2.23183e+00	5.29899e+00
1.94643e+00	1.94138e+00	5.29899e+00
1.86331e+00	1.59724e+00	5.29899e+00
1.76316e+00	1.20603e+00	5.29899e+00
1.64678e+00	7.76574e-01	5.29899e+00
1.51430e+00	3.19169e-01	5.29899e+00
1.36452e+00	-1.54151e-01	5.29899e+00
1.19464e+00	-6.29684e-01	5.29899e+00
9.98776e-01	-1.09122e+00	5.29899e+00
7.73872e-01	-1.52447e+00	5.29899e+00
5.21244e-01	-1.91831e+00	5.29899e+00
2.43835e-01	-2.26321e+00	5.29899e+00
-5.40555e-02	-2.55121e+00	5.29899e+00
-3.68369e-01	-2.77502e+00	5.29899e+00
-6.92021e-01	-2.93056e+00	5.29899e+00
-1.01736e+00	-3.01540e+00	5.29899e+00
-1.33708e+00	-3.02822e+00	5.29899e+00
-1.64495e+00	-2.96838e+00	5.29899e+00
-2.02656e+00	-2.76811e+00	5.29899e+00
-2.31841e+00	-2.46020e+00	5.29899e+00
-2.46101e+00	-2.17859e+00	5.29899e+00
-2.54048e+00	-1.86203e+00	5.29899e+00
-2.55745e+00	-1.51518e+00	5.29899e+00
-2.50866e+00	-1.14709e+00	5.29899e+00
-2.39343e+00	-7.66396e-01	5.29899e+00
-2.21109e+00	-3.82954e-01	5.29899e+00
-1.96154e+00	-7.20497e-03	5.29899e+00
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8.84361e-01	1.97204e+00	5.29899e+00
1.20837e+00	2.18909e+00	5.29899e+00
1.47037e+00	2.38764e+00	5.29899e+00
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1.82446e+00	2.68824e+00	5.29899e+00

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1.84620e+00	1.73836e+00	5.61558e+00
1.74061e+00	1.33911e+00	5.61558e+00
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1.48714e+00	4.24921e-01	5.61558e+00
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1.16944e+00	-5.62906e-01	5.61558e+00
9.75546e-01	-1.04633e+00	5.61558e+00
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2.23947e-01	-2.28399e+00	5.61558e+00
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-7.16203e-01	-3.00547e+00	5.61558e+00
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-1.36813e+00	-3.12785e+00	5.61558e+00
-1.68002e+00	-3.07654e+00	5.61558e+00
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-2.36308e+00	-2.57733e+00	5.61558e+00
-2.50623e+00	-2.29512e+00	5.61558e+00
-2.58460e+00	-1.97596e+00	5.61558e+00
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-2.54532e+00	-1.25070e+00	5.61558e+00
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-2.23226e+00	-4.71317e-01	5.61558e+00
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-1.64404e+00	2.80423e-01	5.61558e+00
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9.73885e-01	1.96452e+00	5.61558e+00
1.30016e+00	2.19916e+00	5.61558e+00
1.55941e+00	2.41731e+00	5.61558e+00
1.75618e+00	2.60765e+00	5.61558e+00
1.89612e+00	2.75955e+00	5.61558e+00
1.98730e+00	2.86219e+00	5.61558e+00
1.87595e+00	3.06361e+00	5.61558e+00
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2.26986e+00	2.88515e+00	5.93216e+00
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2.07350e+00	2.74504e+00	5.93216e+00
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1.92731e+00	2.23145e+00	5.93216e+00
1.82908e+00	1.87948e+00	5.93216e+00
1.71807e+00	1.47219e+00	5.93216e+00
1.59508e+00	1.01890e+00	5.93216e+00
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1.14424e+00	-4.96127e-01	5.93216e+00
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4.80605e-01	-1.91729e+00	5.93216e+00
2.04058e-01	-2.30476e+00	5.93216e+00
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-1.39919e+00	-3.22747e+00	5.93216e+00
-1.71509e+00	-3.18470e+00	5.93216e+00
-2.10858e+00	-2.99765e+00	5.93216e+00
-2.40776e+00	-2.69447e+00	5.93216e+00
-2.55145e+00	-2.41165e+00	5.93216e+00
-2.62873e+00	-2.08989e+00	5.93216e+00
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-2.58198e+00	-1.35430e+00	5.93216e+00
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-1.98175e+00	-1.66716e-01	5.93216e+00
-1.64082e+00	2.09474e-01	5.93216e+00
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-2.76856e-01	1.16259e+00	5.93216e+00
2.11612e-01	1.43445e+00	5.93216e+00
6.65514e-01	1.69820e+00	5.93216e+00
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2.10757e+00	3.06119e+00	6.24874e+00
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1.29715e+00	7.18697e-02	6.24874e+00
1.13611e+00	-4.55213e-01	6.24874e+00
9.49954e-01	-9.71671e-01	6.24874e+00
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2.11977e+00	3.05073e+00	6.24874e+00
2.21189e+00	2.98504e+00	6.56533e+00
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2.00069e+00	2.85482e+00	6.56533e+00
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9.64751e-01	-9.54352e-01	6.56533e+00
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-2.58659e+00	-2.20418e+00	6.56533e+00
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-1.97600e+00	-2.53456e-01	6.56533e+00
-1.63975e+00	1.34690e-01	6.56533e+00
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2.05845e-01	1.42130e+00	6.56533e+00
6.57140e-01	1.70184e+00	6.56533e+00
1.05055e+00	1.97711e+00	6.56533e+00
1.37255e+00	2.24512e+00	6.56533e+00
1.62079e+00	2.49766e+00	6.56533e+00
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1.92108e+00	2.90734e+00	6.56533e+00
1.98985e+00	3.04107e+00	6.56533e+00
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2.15916e+00	3.01845e+00	6.88191e+00
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1.89187e+00	2.66111e+00	6.88191e+00
1.81732e+00	2.36890e+00	6.88191e+00
1.73114e+00	2.01234e+00	6.88191e+00
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9.79547e-01	-9.37033e-01	6.88191e+00
7.80540e-01	-1.43227e+00	6.88191e+00
5.50889e-01	-1.88896e+00	6.88191e+00

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-2.41391e+00	-1.10338e+00	6.88191e+00
-2.22798e+00	-6.93156e-01	6.88191e+00
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-2.99980e-01	1.12665e+00	6.88191e+00
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1.33329e+00	2.26535e+00	6.88191e+00
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1.28771e+00	1.36516e-01	7.19849e+00
1.15379e+00	-3.96258e-01	7.19849e+00
9.94344e-01	-9.19714e-01	7.19849e+00
8.03834e-01	-1.41782e+00	7.19849e+00
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3.30815e-01	-2.28902e+00	7.19849e+00
5.35196e-02	-2.64168e+00	7.19849e+00
-2.46644e-01	-2.92722e+00	7.19849e+00
-5.62694e-01	-3.14036e+00	7.19849e+00
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-1.21136e+00	-3.33682e+00	7.19849e+00
-1.52951e+00	-3.31691e+00	7.19849e+00
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-2.25118e+00	-2.87481e+00	7.19849e+00
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-2.50646e+00	-2.28540e+00	7.19849e+00
-2.53772e+00	-1.93043e+00	7.19849e+00
-2.50041e+00	-1.54653e+00	7.19849e+00
-2.39312e+00	-1.14284e+00	7.19849e+00
-2.21425e+00	-7.29882e-01	7.19849e+00
-1.96267e+00	-3.18997e-01	7.19849e+00
-1.64025e+00	7.94578e-02	7.19849e+00
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-8.00292e-01	8.00836e-01	7.19849e+00
-3.18599e-01	1.12056e+00	7.19849e+00
1.57753e-01	1.42471e+00	7.19849e+00
5.98550e-01	1.71968e+00	7.19849e+00
9.81776e-01	2.00744e+00	7.19849e+00
1.29402e+00	2.28558e+00	7.19849e+00
1.53319e+00	2.54582e+00	7.19849e+00
1.70570e+00	2.77595e+00	7.19849e+00
1.81904e+00	2.96455e+00	7.19849e+00
1.88316e+00	3.10026e+00	7.19849e+00
1.84732e+00	3.21062e+00	7.19849e+00
1.95742e+00	3.14316e+00	7.19849e+00
2.05369e+00	3.08526e+00	7.51508e+00
1.88516e+00	3.09721e+00	7.51508e+00
1.84996e+00	2.94482e+00	7.51508e+00
1.79901e+00	2.71957e+00	7.51508e+00
1.73440e+00	2.42541e+00	7.51508e+00
1.66034e+00	2.06663e+00	7.51508e+00
1.57889e+00	1.65020e+00	7.51508e+00
1.49012e+00	1.18555e+00	7.51508e+00
1.39298e+00	6.83797e-01	7.51508e+00
1.28456e+00	1.58065e-01	7.51508e+00
1.15968e+00	-3.76607e-01	7.51508e+00
1.00914e+00	-9.02395e-01	7.51508e+00
8.27127e-01	-1.40337e+00	7.51508e+00
6.13233e-01	-1.86691e+00	7.51508e+00
3.69062e-01	-2.28194e+00	7.51508e+00
9.78762e-02	-2.63903e+00	7.51508e+00
-1.97290e-01	-2.92942e+00	7.51508e+00
-5.09549e-01	-3.14773e+00	7.51508e+00
-8.31124e-01	-3.29035e+00	7.51508e+00
-1.15448e+00	-3.35496e+00	7.51508e+00
-1.47275e+00	-3.34041e+00	7.51508e+00
-1.87990e+00	-3.18777e+00	7.51508e+00
-2.20137e+00	-2.91073e+00	7.51508e+00

-2.36570e+00	-2.64068e+00	7.51508e+00
-2.46639e+00	-2.32600e+00	7.51508e+00
-2.50360e+00	-1.97180e+00	7.51508e+00
-2.47277e+00	-1.58752e+00	7.51508e+00
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1.48940e+00	2.56989e+00	7.51508e+00
1.65792e+00	2.80277e+00	7.51508e+00
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1.82982e+00	3.12985e+00	7.51508e+00
1.79217e+00	3.23952e+00	7.51508e+00
1.90330e+00	3.17397e+00	7.51508e+00
2.00096e+00	3.11866e+00	7.83166e+00
1.83234e+00	3.12775e+00	7.83166e+00
1.79972e+00	2.97482e+00	7.83166e+00
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1.62495e+00	2.09377e+00	7.83166e+00
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1.38095e+00	7.06970e-01	7.83166e+00
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1.02394e+00	-8.85077e-01	7.83166e+00
8.50420e-01	-1.38892e+00	7.83166e+00
6.44405e-01	-1.85588e+00	7.83166e+00
4.07308e-01	-2.27487e+00	7.83166e+00
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-1.41600e+00	-3.36392e+00	7.83166e+00
-1.82556e+00	-3.21819e+00	7.83166e+00
-2.15157e+00	-2.94666e+00	7.83166e+00
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-2.42632e+00	-2.36661e+00	7.83166e+00
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-2.44512e+00	-1.62852e+00	7.83166e+00
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-1.94935e+00	-3.84538e-01	7.83166e+00
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1.21549e+00	2.32604e+00	7.83166e+00
1.44560e+00	2.59397e+00	7.83166e+00
1.61014e+00	2.82960e+00	7.83166e+00
1.71699e+00	3.02175e+00	7.83166e+00
1.77647e+00	3.15944e+00	7.83166e+00
1.73702e+00	3.26842e+00	7.83166e+00
1.84919e+00	3.20478e+00	7.83166e+00
1.94823e+00	3.15206e+00	8.14824e+00
1.77952e+00	3.15828e+00	8.14824e+00
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1.70614e+00	2.77803e+00	8.14824e+00
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1.52217e+00	1.70207e+00	8.14824e+00
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1.27826e+00	2.01163e-01	8.14824e+00
1.17147e+00	-3.37304e-01	8.14824e+00
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4.45555e-01	-2.26779e+00	8.14824e+00
1.86590e-01	-2.63373e+00	8.14824e+00
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-1.77122e+00	-3.24860e+00	8.14824e+00
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-2.27506e+00	-2.71827e+00	8.14824e+00
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-2.17305e+00	-8.40059e-01	8.14824e+00
-1.94269e+00	-4.17308e-01	8.14824e+00
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5.10665e-01	1.74644e+00	8.14824e+00
8.78622e-01	2.05295e+00	8.14824e+00
1.17623e+00	2.34627e+00	8.14824e+00
1.40180e+00	2.61805e+00	8.14824e+00
1.56236e+00	2.85642e+00	8.14824e+00
1.66597e+00	3.05036e+00	8.14824e+00
1.72312e+00	3.18903e+00	8.14824e+00
1.68187e+00	3.29732e+00	8.14824e+00
1.79507e+00	3.23559e+00	8.14824e+00
1.89550e+00	3.18547e+00	8.46482e+00
1.72670e+00	3.18881e+00	8.46482e+00
1.69924e+00	3.03482e+00	8.46482e+00
1.65971e+00	2.80726e+00	8.46482e+00
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1.17737e+00	-3.17653e-01	8.46482e+00
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8.97006e-01	-1.36003e+00	8.46482e+00
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4.83802e-01	-2.26072e+00	8.46482e+00
2.30946e-01	-2.63108e+00	8.46482e+00
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3.75224e-02	1.43325e+00	8.78141e+00
4.52075e-01	1.76427e+00	8.78141e+00
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1.09770e+00	2.38673e+00	8.78141e+00
1.31421e+00	2.66620e+00	8.78141e+00
1.46680e+00	2.91007e+00	8.78141e+00
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1.61643e+00	3.24821e+00	8.78141e+00
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1.79003e+00	3.25228e+00	9.09799e+00
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1.41902e+00	2.93689e+00	9.09799e+00
1.51290e+00	3.13617e+00	9.09799e+00
1.56308e+00	3.27780e+00	9.09799e+00
1.51642e+00	3.38403e+00	9.09799e+00
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1.73612e+00	3.23935e+00	9.41457e+00
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1.44381e+00	2.19678e+00	9.41457e+00
1.40273e+00	1.77803e+00	9.41457e+00
1.35897e+00	1.31098e+00	9.41457e+00
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1.25308e+00	2.77324e-01	9.41457e+00
1.18031e+00	-2.62251e-01	9.41457e+00
1.08135e+00	-7.95481e-01	9.41457e+00
9.48985e-01	-1.30719e+00	9.41457e+00
7.81617e-01	-1.78495e+00	9.41457e+00
5.79782e-01	-2.21770e+00	9.41457e+00
3.45808e-01	-2.59579e+00	9.41457e+00
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-8.22814e-01	-3.42075e+00	9.41457e+00
-1.13836e+00	-3.43645e+00	9.41457e+00
-1.55467e+00	-3.32422e+00	9.41457e+00
-1.89817e+00	-3.08110e+00	9.41457e+00
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-2.21498e+00	-2.52893e+00	9.41457e+00
-2.28516e+00	-2.18280e+00	9.41457e+00
-2.29101e+00	-1.80056e+00	9.41457e+00
-2.23013e+00	-1.39107e+00	9.41457e+00
-2.09978e+00	-9.64518e-01	9.41457e+00
-1.89760e+00	-5.31934e-01	9.41457e+00
-1.62426e+00	-1.03732e-01	9.41457e+00
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3.98697e-01	1.76008e+00	9.41457e+00
7.44199e-01	2.08601e+00	9.41457e+00
1.02086e+00	2.39458e+00	9.41457e+00
1.22754e+00	2.67742e+00	9.41457e+00
1.37190e+00	2.92322e+00	9.41457e+00
1.46260e+00	3.12153e+00	9.41457e+00
1.51070e+00	3.26231e+00	9.41457e+00
1.46415e+00	3.36665e+00	9.41457e+00
1.57908e+00	3.31280e+00	9.41457e+00
1.67732e+00	3.21908e+00	9.73116e+00
1.51579e+00	3.21188e+00	9.73116e+00
1.49728e+00	3.05981e+00	9.73116e+00
1.47095e+00	2.83513e+00	9.73116e+00
1.43845e+00	2.54192e+00	9.73116e+00

1.40344e+00	2.18478e+00	9.73116e+00
1.36718e+00	1.77074e+00	9.73116e+00
1.32879e+00	1.30895e+00	9.73116e+00
1.28609e+00	8.10175e-01	9.73116e+00
1.23487e+00	2.86867e-01	9.73116e+00
1.16856e+00	-2.46850e-01	9.73116e+00
1.07632e+00	-7.74570e-01	9.73116e+00
9.50860e-01	-1.28138e+00	9.73116e+00
7.90475e-01	-1.75501e+00	9.73116e+00
5.95581e-01	-2.18454e+00	9.73116e+00
3.68377e-01	-2.56042e+00	9.73116e+00
1.10916e-01	-2.87382e+00	9.73116e+00
-1.70654e-01	-3.11877e+00	9.73116e+00
-4.69128e-01	-3.29096e+00	9.73116e+00
-7.77411e-01	-3.38735e+00	9.73116e+00
-1.08893e+00	-3.40611e+00	9.73116e+00
-1.50130e+00	-3.29955e+00	9.73116e+00
-1.84310e+00	-3.06296e+00	9.73116e+00
-2.03110e+00	-2.81692e+00	9.73116e+00
-2.16173e+00	-2.52082e+00	9.73116e+00
-2.23463e+00	-2.17966e+00	9.73116e+00
-2.24435e+00	-1.80217e+00	9.73116e+00
-2.18845e+00	-1.39706e+00	9.73116e+00
-2.06410e+00	-9.74386e-01	9.73116e+00
-1.86886e+00	-5.45014e-01	9.73116e+00
-1.60329e+00	-1.19234e-01	9.73116e+00
-1.26936e+00	2.92715e-01	9.73116e+00
-8.73527e-01	6.82300e-01	9.73116e+00
-4.43773e-01	1.05065e+00	9.73116e+00
-1.73440e-02	1.40324e+00	9.73116e+00
3.75638e-01	1.74264e+00	9.73116e+00
7.13542e-01	2.06814e+00	9.73116e+00
9.83633e-01	2.37580e+00	9.73116e+00
1.18486e+00	2.65730e+00	9.73116e+00
1.32491e+00	2.90158e+00	9.73116e+00
1.41245e+00	3.09840e+00	9.73116e+00
1.45851e+00	3.23795e+00	9.73116e+00
1.41245e+00	3.34017e+00	9.73116e+00
1.52553e+00	3.28853e+00	9.73116e+00
1.62063e+00	3.19800e+00	1.00477e+01
1.46317e+00	3.18845e+00	1.00477e+01
1.44650e+00	3.03805e+00	1.00477e+01
1.42287e+00	2.81585e+00	1.00477e+01
1.39387e+00	2.52590e+00	1.00477e+01
1.36307e+00	2.17278e+00	1.00477e+01
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1.29861e+00	1.30693e+00	1.00477e+01
1.26171e+00	8.13828e-01	1.00477e+01
1.21666e+00	2.96410e-01	1.00477e+01
1.15681e+00	-2.31449e-01	1.00477e+01
1.07129e+00	-7.53660e-01	1.00477e+01
9.52734e-01	-1.25557e+00	1.00477e+01
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6.11380e-01	-2.15139e+00	1.00477e+01
3.90947e-01	-2.52505e+00	1.00477e+01
1.39966e-01	-2.83732e+00	1.00477e+01
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-1.44793e+00	-3.27489e+00	1.00477e+01
-1.78803e+00	-3.04482e+00	1.00477e+01
-1.97632e+00	-2.80379e+00	1.00477e+01
-2.10849e+00	-2.51272e+00	1.00477e+01
-2.18410e+00	-2.17652e+00	1.00477e+01
-2.19770e+00	-1.80377e+00	1.00477e+01
-2.14676e+00	-1.40305e+00	1.00477e+01
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-4.49438e-01	1.03304e+00	1.00477e+01
-3.19812e-02	1.38581e+00	1.00477e+01
3.52579e-01	1.72520e+00	1.00477e+01
6.82885e-01	2.05028e+00	1.00477e+01
9.46401e-01	2.35701e+00	1.00477e+01
1.14218e+00	2.63718e+00	1.00477e+01
1.27792e+00	2.87994e+00	1.00477e+01
1.36230e+00	3.07527e+00	1.00477e+01
1.40632e+00	3.21359e+00	1.00477e+01
1.36076e+00	3.31370e+00	1.00477e+01
1.47197e+00	3.26426e+00	1.00477e+01
1.56395e+00	3.17693e+00	1.03643e+01
1.41055e+00	3.16502e+00	1.03643e+01
1.39571e+00	3.01629e+00	1.03643e+01
1.37479e+00	2.79657e+00	1.03643e+01
1.34929e+00	2.50988e+00	1.03643e+01
1.32270e+00	2.16078e+00	1.03643e+01
1.29610e+00	1.75615e+00	1.03643e+01
1.26842e+00	1.30490e+00	1.03643e+01
1.23733e+00	8.17482e-01	1.03643e+01
1.19844e+00	3.05953e-01	1.03643e+01
1.14506e+00	-2.16049e-01	1.03643e+01
1.06626e+00	-7.32750e-01	1.03643e+01
9.54608e-01	-1.22975e+00	1.03643e+01
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6.27179e-01	-2.11824e+00	1.03643e+01
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-1.00447e-01	-3.04572e+00	1.03643e+01
-3.87940e-01	-3.21999e+00	1.03643e+01
-6.86605e-01	-3.32054e+00	1.03643e+01
-9.90067e-01	-3.34543e+00	1.03643e+01
-1.39456e+00	-3.25022e+00	1.03643e+01
-1.73296e+00	-3.02668e+00	1.03643e+01
-1.92155e+00	-2.79067e+00	1.03643e+01
-2.05524e+00	-2.50461e+00	1.03643e+01
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-2.15104e+00	-1.80537e+00	1.03643e+01
-2.10508e+00	-1.40904e+00	1.03643e+01
-1.99274e+00	-9.94124e-01	1.03643e+01
-1.81139e+00	-5.71173e-01	1.03643e+01
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-8.66496e-01	6.47040e-01	1.03643e+01
-4.55103e-01	1.01543e+00	1.03643e+01
-4.66185e-02	1.36838e+00	1.03643e+01
3.29520e-01	1.70775e+00	1.03643e+01
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1.09949e+00	2.61706e+00	1.03643e+01
1.23093e+00	2.85830e+00	1.03643e+01
1.31215e+00	3.05214e+00	1.03643e+01
1.35412e+00	3.18923e+00	1.03643e+01
1.30906e+00	3.28722e+00	1.03643e+01
1.41842e+00	3.23998e+00	1.03643e+01
1.50727e+00	3.15586e+00	1.06809e+01
1.35793e+00	3.14158e+00	1.06809e+01
1.34493e+00	2.99452e+00	1.06809e+01
1.32670e+00	2.77730e+00	1.06809e+01
1.30471e+00	2.49386e+00	1.06809e+01
1.28233e+00	2.14878e+00	1.06809e+01
1.26055e+00	1.74885e+00	1.06809e+01
1.23824e+00	1.30288e+00	1.06809e+01
1.21295e+00	8.21135e-01	1.06809e+01
1.18023e+00	3.15496e-01	1.06809e+01
1.13331e+00	-2.00648e-01	1.06809e+01
1.06122e+00	-7.11839e-01	1.06809e+01
9.56483e-01	-1.20394e+00	1.06809e+01
8.17049e-01	-1.66521e+00	1.06809e+01
6.42978e-01	-2.08509e+00	1.06809e+01
4.36085e-01	-2.45432e+00	1.06809e+01
1.98066e-01	-2.76433e+00	1.06809e+01
-6.53441e-02	-3.00919e+00	1.06809e+01
-3.47346e-01	-3.18451e+00	1.06809e+01
-6.41203e-01	-3.28713e+00	1.06809e+01
-9.40637e-01	-3.31509e+00	1.06809e+01
-1.34119e+00	-3.22556e+00	1.06809e+01
-1.67790e+00	-3.00854e+00	1.06809e+01
-1.86677e+00	-2.77754e+00	1.06809e+01
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-2.10438e+00	-1.80697e+00	1.06809e+01
-2.06340e+00	-1.41503e+00	1.06809e+01
-1.95706e+00	-1.00399e+00	1.06809e+01
-1.78265e+00	-5.84253e-01	1.06809e+01
-1.54039e+00	-1.65741e-01	1.06809e+01
-1.23185e+00	2.41622e-01	1.06809e+01
-8.62981e-01	6.29409e-01	1.06809e+01
-4.60768e-01	9.97824e-01	1.06809e+01
-6.12557e-02	1.35095e+00	1.06809e+01
3.06461e-01	1.69031e+00	1.06809e+01
6.21572e-01	2.01454e+00	1.06809e+01
8.71936e-01	2.31943e+00	1.06809e+01
1.05681e+00	2.59694e+00	1.06809e+01
1.18394e+00	2.83666e+00	1.06809e+01
1.26200e+00	3.02900e+00	1.06809e+01
1.30193e+00	3.16487e+00	1.06809e+01
1.25736e+00	3.26074e+00	1.06809e+01
1.36487e+00	3.21571e+00	1.06809e+01
1.45059e+00	3.13478e+00	1.09975e+01
1.30531e+00	3.11815e+00	1.09975e+01
1.29414e+00	2.97276e+00	1.09975e+01
1.27862e+00	2.75802e+00	1.09975e+01
1.26013e+00	2.47785e+00	1.09975e+01
1.24196e+00	2.13679e+00	1.09975e+01
1.22501e+00	1.74156e+00	1.09975e+01
1.20805e+00	1.30085e+00	1.09975e+01
1.18857e+00	8.24788e-01	1.09975e+01
1.16202e+00	3.25039e-01	1.09975e+01
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9.58357e-01	-1.17813e+00	1.09975e+01
8.25907e-01	-1.63527e+00	1.09975e+01
6.58777e-01	-2.05194e+00	1.09975e+01
4.58655e-01	-2.41895e+00	1.09975e+01
2.27117e-01	-2.72784e+00	1.09975e+01
-3.02409e-02	-2.97267e+00	1.09975e+01
-3.06753e-01	-3.14903e+00	1.09975e+01
-5.95800e-01	-3.25373e+00	1.09975e+01
-8.91207e-01	-3.28475e+00	1.09975e+01
-1.28782e+00	-3.20089e+00	1.09975e+01
-1.62283e+00	-2.99040e+00	1.09975e+01

-1.81200e+00	-2.76441e+00	1.09975e+01
-1.94875e+00	-2.48840e+00	1.09975e+01
-2.03250e+00	-2.16710e+00	1.09975e+01
-2.05773e+00	-1.80858e+00	1.09975e+01
-2.02172e+00	-1.42102e+00	1.09975e+01
-1.92138e+00	-1.01386e+00	1.09975e+01
-1.75391e+00	-5.97332e-01	1.09975e+01
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2.83402e-01	1.67287e+00	1.09975e+01
5.90915e-01	1.99668e+00	1.09975e+01
8.34704e-01	2.30064e+00	1.09975e+01
1.01412e+00	2.57682e+00	1.09975e+01
1.13695e+00	2.81502e+00	1.09975e+01
1.21185e+00	3.00587e+00	1.09975e+01
1.24974e+00	3.14051e+00	1.09975e+01
1.20566e+00	3.23426e+00	1.09975e+01
1.31132e+00	3.19144e+00	1.09975e+01
1.39390e+00	3.11371e+00	1.13141e+01
1.25269e+00	3.09471e+00	1.13141e+01
1.24336e+00	2.95100e+00	1.13141e+01
1.23054e+00	2.73874e+00	1.13141e+01
1.21555e+00	2.46183e+00	1.13141e+01
1.20159e+00	2.12479e+00	1.13141e+01
1.18947e+00	1.73427e+00	1.13141e+01
1.17787e+00	1.29883e+00	1.13141e+01
1.16419e+00	8.28441e-01	1.13141e+01
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1.05116e+00	-6.70018e-01	1.13141e+01
9.60231e-01	-1.15232e+00	1.13141e+01
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6.74576e-01	-2.01878e+00	1.13141e+01
4.81224e-01	-2.38358e+00	1.13141e+01
2.56167e-01	-2.69135e+00	1.13141e+01
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2.60343e-01	1.65543e+00	1.13141e+01
5.60258e-01	1.97881e+00	1.13141e+01
7.97472e-01	2.28185e+00	1.13141e+01
9.71437e-01	2.55670e+00	1.13141e+01
1.08996e+00	2.79338e+00	1.13141e+01
1.16170e+00	2.98274e+00	1.13141e+01
1.19755e+00	3.11615e+00	1.13141e+01
1.15396e+00	3.20779e+00	1.13141e+01
1.25777e+00	3.16717e+00	1.13141e+01
1.33722e+00	3.09264e+00	1.16307e+01
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1.19258e+00	2.92924e+00	1.16307e+01
1.18246e+00	2.71946e+00	1.16307e+01
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1.16122e+00	2.11279e+00	1.16307e+01
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1.14769e+00	1.29680e+00	1.16307e+01
1.13981e+00	8.32095e-01	1.16307e+01
1.12559e+00	3.44124e-01	1.16307e+01
1.09806e+00	-1.54446e-01	1.16307e+01
1.04613e+00	-6.49108e-01	1.16307e+01
9.62106e-01	-1.12651e+00	1.16307e+01
8.43623e-01	-1.57540e+00	1.16307e+01
6.90375e-01	-1.98563e+00	1.16307e+01
5.03793e-01	-2.34822e+00	1.16307e+01
2.85217e-01	-2.65485e+00	1.16307e+01
3.99654e-02	-2.89962e+00	1.16307e+01
-2.25565e-01	-3.07807e+00	1.16307e+01
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-7.92347e-01	-3.22408e+00	1.16307e+01
-1.18108e+00	-3.15157e+00	1.16307e+01
-1.51269e+00	-2.95412e+00	1.16307e+01
-1.70245e+00	-2.73816e+00	1.16307e+01
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-1.93144e+00	-2.16082e+00	1.16307e+01
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2.37284e-01	1.63798e+00	1.16307e+01
5.29601e-01	1.96094e+00	1.16307e+01
7.60240e-01	2.26306e+00	1.16307e+01
9.28753e-01	2.53658e+00	1.16307e+01
1.04297e+00	2.77174e+00	1.16307e+01
1.11155e+00	2.95961e+00	1.16307e+01
1.14535e+00	3.09179e+00	1.16307e+01
1.10226e+00	3.18131e+00	1.16307e+01
1.20422e+00	3.14289e+00	1.16307e+01
1.28054e+00	3.07156e+00	1.19472e+01
1.14745e+00	3.04785e+00	1.19472e+01
1.14179e+00	2.90748e+00	1.19472e+01
1.13437e+00	2.70018e+00	1.19472e+01
1.12639e+00	2.42979e+00	1.19472e+01
1.12085e+00	2.10079e+00	1.19472e+01
1.11838e+00	1.71968e+00	1.19472e+01
1.11750e+00	1.29478e+00	1.19472e+01
1.11543e+00	8.35748e-01	1.19472e+01
1.10738e+00	3.53667e-01	1.19472e+01
1.08631e+00	-1.39045e-01	1.19472e+01
1.04110e+00	-6.28198e-01	1.19472e+01
9.63980e-01	-1.10070e+00	1.19472e+01
8.52481e-01	-1.54547e+00	1.19472e+01
7.06174e-01	-1.95248e+00	1.19472e+01
5.26363e-01	-2.31285e+00	1.19472e+01
3.14268e-01	-2.61836e+00	1.19472e+01
7.50686e-02	-2.86309e+00	1.19472e+01
-1.84971e-01	-3.04258e+00	1.19472e+01
-4.59591e-01	-3.15352e+00	1.19472e+01
-7.42917e-01	-3.19374e+00	1.19472e+01
-1.12771e+00	-3.12690e+00	1.19472e+01
-1.45763e+00	-2.93598e+00	1.19472e+01
-1.64768e+00	-2.72504e+00	1.19472e+01
-1.78901e+00	-2.46409e+00	1.19472e+01
-1.88091e+00	-2.15768e+00	1.19472e+01
-1.91776e+00	-1.81338e+00	1.19472e+01
-1.89667e+00	-1.43899e+00	1.19472e+01
-1.81434e+00	-1.04347e+00	1.19472e+01
-1.66770e+00	-6.36571e-01	1.19472e+01
-1.45652e+00	-2.27749e-01	1.19472e+01
-1.18184e+00	1.73497e-01	1.19472e+01
-8.48920e-01	5.58889e-01	1.19472e+01
-4.83427e-01	9.27387e-01	1.19472e+01
-1.19805e-01	1.28123e+00	1.19472e+01
2.14225e-01	1.62054e+00	1.19472e+01
4.98945e-01	1.94308e+00	1.19472e+01
7.23008e-01	2.24427e+00	1.19472e+01
8.86068e-01	2.51646e+00	1.19472e+01
9.95976e-01	2.75010e+00	1.19472e+01
1.06141e+00	2.93647e+00	1.19472e+01
1.09316e+00	3.06743e+00	1.19472e+01
1.05056e+00	3.15483e+00	1.19472e+01
1.15066e+00	3.11862e+00	1.19472e+01
1.24503e+00	3.04230e+00	1.22638e+01
1.09674e+00	3.02305e+00	1.22638e+01
1.09276e+00	2.88441e+00	1.22638e+01
1.08777e+00	2.67967e+00	1.22638e+01
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1.08271e+00	1.71157e+00	1.22638e+01
1.08638e+00	1.29215e+00	1.22638e+01
1.08919e+00	8.39030e-01	1.22638e+01
1.08634e+00	3.63096e-01	1.22638e+01
1.07074e+00	-1.23487e-01	1.22638e+01
1.03133e+00	-6.06859e-01	1.22638e+01
9.60322e-01	-1.07419e+00	1.22638e+01
8.55090e-01	-1.51457e+00	1.22638e+01
7.15060e-01	-1.91808e+00	1.22638e+01
5.41540e-01	-2.27597e+00	1.22638e+01
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1.02763e-01	-2.82471e+00	1.22638e+01
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-4.20108e-01	-3.11823e+00	1.22638e+01
-6.98028e-01	-3.16166e+00	1.22638e+01
-1.07643e+00	-3.10084e+00	1.22638e+01
-1.40247e+00	-2.91677e+00	1.22638e+01
-1.59146e+00	-2.71104e+00	1.22638e+01
-1.73317e+00	-2.45531e+00	1.22638e+01
-1.82681e+00	-2.15404e+00	1.22638e+01
-1.86678e+00	-1.81465e+00	1.22638e+01
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-1.77337e+00	-1.05333e+00	1.22638e+01
-1.63345e+00	-6.49813e-01	1.22638e+01
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-1.16391e+00	1.55960e-01	1.22638e+01
-8.40340e-01	5.40591e-01	1.22638e+01
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-1.30284e-01	1.26281e+00	1.22638e+01
1.94914e-01	1.60196e+00	1.22638e+01
4.71620e-01	1.92397e+00	1.22638e+01
6.88702e-01	2.22417e+00	1.22638e+01
8.45967e-01	2.49500e+00	1.22638e+01
9.51298e-01	2.72710e+00	1.22638e+01
1.01342e+00	2.91198e+00	1.22638e+01

1.04314e+00	3.04168e+00	1.22638e+01
1.00161e+00	3.12688e+00	1.22638e+01
1.09909e+00	3.09297e+00	1.22638e+01
1.18048e+00	3.01662e+00	1.25804e+01
1.05971e+00	2.98852e+00	1.25804e+01
1.05628e+00	2.85190e+00	1.25804e+01
1.05175e+00	2.65026e+00	1.25804e+01
1.04717e+00	2.38737e+00	1.25804e+01
1.04514e+00	2.06769e+00	1.25804e+01
1.04609e+00	1.69761e+00	1.25804e+01
1.04852e+00	1.28516e+00	1.25804e+01
1.04970e+00	8.39653e-01	1.25804e+01
1.04505e+00	3.71712e-01	1.25804e+01
1.02783e+00	-1.06797e-01	1.25804e+01
9.87677e-01	-5.82453e-01	1.25804e+01
9.17079e-01	-1.04273e+00	1.25804e+01
8.12982e-01	-1.47677e+00	1.25804e+01
6.74483e-01	-1.87475e+00	1.25804e+01
5.03827e-01	-2.22829e+00	1.25804e+01
3.03020e-01	-2.52962e+00	1.25804e+01
7.74432e-02	-2.77301e+00	1.25804e+01
-1.66655e-01	-2.95406e+00	1.25804e+01
-4.22977e-01	-3.06949e+00	1.25804e+01
-6.85627e-01	-3.11716e+00	1.25804e+01
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-1.34670e+00	-2.88984e+00	1.25804e+01
-1.52494e+00	-2.69081e+00	1.25804e+01
-1.65876e+00	-2.44166e+00	1.25804e+01
-1.74718e+00	-2.14683e+00	1.25804e+01
-1.78488e+00	-1.81354e+00	1.25804e+01
-1.76854e+00	-1.44940e+00	1.25804e+01
-1.69455e+00	-1.06314e+00	1.25804e+01
-1.55972e+00	-6.64214e-01	1.25804e+01
-1.36361e+00	-2.61825e-01	1.25804e+01
-1.10720e+00	1.34805e-01	1.25804e+01
-7.95514e-01	5.17522e-01	1.25804e+01
-4.52653e-01	8.84619e-01	1.25804e+01
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2.02420e-01	1.57526e+00	1.25804e+01
4.68142e-01	1.89595e+00	1.25804e+01
6.75336e-01	2.19471e+00	1.25804e+01
8.24345e-01	2.46394e+00	1.25804e+01
9.23164e-01	2.69443e+00	1.25804e+01
9.80961e-01	2.87773e+00	1.25804e+01
1.00863e+00	3.00601e+00	1.25804e+01
9.72288e-01	3.08836e+00	1.25804e+01
1.06166e+00	3.05744e+00	1.25804e+01
1.13564e+00	2.98414e+00	1.28970e+01
1.02268e+00	2.95398e+00	1.28970e+01
1.01980e+00	2.81940e+00	1.28970e+01
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1.00925e+00	2.04759e+00	1.28970e+01
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9.84913e-01	-9.01071e-02	1.28970e+01
9.44021e-01	-5.58048e-01	1.28970e+01
8.73837e-01	-1.01127e+00	1.28970e+01
7.70874e-01	-1.43898e+00	1.28970e+01
6.33906e-01	-1.83143e+00	1.28970e+01
4.66114e-01	-2.18061e+00	1.28970e+01
2.70296e-01	-2.47910e+00	1.28970e+01
5.21236e-02	-2.72132e+00	1.28970e+01
-1.82066e-01	-2.90293e+00	1.28970e+01
-4.25847e-01	-3.02074e+00	1.28970e+01
-6.73226e-01	-3.07265e+00	1.28970e+01
-1.00385e+00	-3.02879e+00	1.28970e+01
-1.29093e+00	-2.86292e+00	1.28970e+01
-1.45841e+00	-2.67058e+00	1.28970e+01
-1.58436e+00	-2.42800e+00	1.28970e+01
-1.66755e+00	-2.13962e+00	1.28970e+01
-1.70299e+00	-1.81242e+00	1.28970e+01
-1.68698e+00	-1.45399e+00	1.28970e+01
-1.61573e+00	-1.07295e+00	1.28970e+01
-1.48599e+00	-6.78615e-01	1.28970e+01
-1.29722e+00	-2.80063e-01	1.28970e+01
-1.05049e+00	1.13649e-01	1.28970e+01
-7.50688e-01	4.94452e-01	1.28970e+01
-4.20813e-01	8.60285e-01	1.28970e+01
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2.09926e-01	1.54855e+00	1.28970e+01
4.64663e-01	1.86793e+00	1.28970e+01
6.61970e-01	2.16525e+00	1.28970e+01
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9.74118e-01	2.97035e+00	1.28970e+01
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1.09081e+00	2.95167e+00	1.32136e+01
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9.73358e-01	2.02749e+00	1.32136e+01
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9.70712e-01	8.40900e-01	1.32136e+01
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9.42000e-01	-7.34174e-02	1.32136e+01
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7.28767e-01	-1.40119e+00	1.32136e+01
5.93329e-01	-1.78810e+00	1.32136e+01
4.28401e-01	-2.13293e+00	1.32136e+01
2.37573e-01	-2.42857e+00	1.32136e+01
2.68040e-02	-2.66962e+00	1.32136e+01
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-1.60542e+00	-1.45858e+00	1.32136e+01
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-1.41227e+00	-6.93016e-01	1.32136e+01
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2.17432e-01	1.52184e+00	1.32136e+01
4.61185e-01	1.83991e+00	1.32136e+01
6.48603e-01	2.13579e+00	1.32136e+01
7.81101e-01	2.40182e+00	1.32136e+01
8.66896e-01	2.62909e+00	1.32136e+01
9.16035e-01	2.80922e+00	1.32136e+01
9.39609e-01	2.93468e+00	1.32136e+01
9.13648e-01	3.01132e+00	1.32136e+01
9.86809e-01	2.98639e+00	1.32136e+01
1.04598e+00	2.91920e+00	1.35302e+01
9.48628e-01	2.88491e+00	1.35302e+01
9.46847e-01	2.75440e+00	1.35302e+01
9.43682e-01	2.56206e+00	1.35302e+01
9.39968e-01	2.31155e+00	1.35302e+01
9.37468e-01	2.00739e+00	1.35302e+01
9.36223e-01	1.65575e+00	1.35302e+01
9.34929e-01	1.26422e+00	1.35302e+01
9.31219e-01	8.41523e-01	1.35302e+01
9.21177e-01	3.97558e-01	1.35302e+01
8.99087e-01	-5.67276e-02	1.35302e+01
8.56708e-01	-5.09237e-01	1.35302e+01
7.87351e-01	-9.48345e-01	1.35302e+01
6.86659e-01	-1.36340e+00	1.35302e+01
5.52752e-01	-1.74477e+00	1.35302e+01
3.90688e-01	-2.08524e+00	1.35302e+01
2.04850e-01	-2.37805e+00	1.35302e+01
1.48434e-03	-2.61793e+00	1.35302e+01
-2.12887e-01	-2.80068e+00	1.35302e+01
-4.31586e-01	-2.92326e+00	1.35302e+01
-6.48424e-01	-2.98363e+00	1.35302e+01
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-1.17938e+00	-2.80907e+00	1.35302e+01
-1.32536e+00	-2.63013e+00	1.35302e+01
-1.43555e+00	-2.40070e+00	1.35302e+01
-1.50829e+00	-2.12520e+00	1.35302e+01
-1.53919e+00	-1.81019e+00	1.35302e+01
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-1.45809e+00	-1.09257e+00	1.35302e+01
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2.24939e-01	1.49514e+00	1.35302e+01
4.57706e-01	1.81190e+00	1.35302e+01
6.35237e-01	2.10633e+00	1.35302e+01
7.59479e-01	2.37076e+00	1.35302e+01
8.38761e-01	2.59642e+00	1.35302e+01
8.83572e-01	2.77497e+00	1.35302e+01
9.05099e-01	2.89901e+00	1.35302e+01
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9.49382e-01	2.95086e+00	1.35302e+01
1.00114e+00	2.88672e+00	1.38467e+01
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9.01578e-01	1.98729e+00	1.38467e+01
8.99601e-01	1.64180e+00	1.38467e+01
8.97067e-01	1.25724e+00	1.38467e+01
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8.79887e-01	4.06174e-01	1.38467e+01
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7.44108e-01	-9.16884e-01	1.38467e+01
6.44551e-01	-1.32560e+00	1.38467e+01

5.12174e-01	-1.70144e+00	1.38467e+01
3.52975e-01	-2.03756e+00	1.38467e+01
1.72126e-01	-2.32753e+00	1.38467e+01
-2.38353e-02	-2.56623e+00	1.38467e+01
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-1.25883e+00	-2.60990e+00	1.38467e+01
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-1.42866e+00	-2.11799e+00	1.38467e+01
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-1.37928e+00	-1.10238e+00	1.38467e+01
-1.26481e+00	-7.21818e-01	1.38467e+01
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2.32445e-01	1.46843e+00	1.38467e+01
4.54228e-01	1.78388e+00	1.38467e+01
6.21871e-01	2.07687e+00	1.38467e+01
7.37858e-01	2.33971e+00	1.38467e+01
8.10627e-01	2.56375e+00	1.38467e+01
8.51110e-01	2.74072e+00	1.38467e+01
8.70590e-01	2.86335e+00	1.38467e+01
8.55008e-01	2.93428e+00	1.38467e+01
9.11955e-01	2.91533e+00	1.38467e+01
9.56312e-01	2.85425e+00	1.41633e+01
8.74574e-01	2.81584e+00	1.41633e+01
8.73890e-01	2.68940e+00	1.41633e+01
8.71638e-01	2.50326e+00	1.41633e+01
8.68502e-01	2.26101e+00	1.41633e+01
8.65687e-01	1.96719e+00	1.41633e+01
8.62980e-01	1.62785e+00	1.41633e+01
8.59205e-01	1.25026e+00	1.41633e+01
8.52233e-01	8.42770e-01	1.41633e+01
8.38597e-01	4.14789e-01	1.41633e+01
8.13261e-01	-2.33480e-02	1.41633e+01
7.69396e-01	-4.60426e-01	1.41633e+01
7.00866e-01	-8.85422e-01	1.41633e+01
6.02443e-01	-1.28781e+00	1.41633e+01
4.71597e-01	-1.65811e+00	1.41633e+01
3.15262e-01	-1.98988e+00	1.41633e+01
1.39403e-01	-2.27700e+00	1.41633e+01
-4.91549e-02	-2.51453e+00	1.41633e+01
-2.43708e-01	-2.69843e+00	1.41633e+01
-4.37325e-01	-2.82577e+00	1.41633e+01
-6.23623e-01	-2.89461e+00	1.41633e+01
-8.58699e-01	-2.88469e+00	1.41633e+01
-1.06783e+00	-2.75522e+00	1.41633e+01
-1.19231e+00	-2.58967e+00	1.41633e+01
-1.28674e+00	-2.37340e+00	1.41633e+01
-1.34903e+00	-2.11079e+00	1.41633e+01
-1.37539e+00	-1.80795e+00	1.41633e+01
-1.36073e+00	-1.47234e+00	1.41633e+01
-1.30046e+00	-1.11219e+00	1.41633e+01
-1.19108e+00	-7.36218e-01	1.41633e+01
-1.03169e+00	-3.53014e-01	1.41633e+01
-8.23631e-01	2.90252e-02	1.41633e+01
-5.71384e-01	4.02173e-01	1.41633e+01
-2.93453e-01	7.62949e-01	1.41633e+01
-1.46272e-02	1.10990e+00	1.41633e+01
2.39951e-01	1.44172e+00	1.41633e+01
4.50749e-01	1.75586e+00	1.41633e+01
6.08505e-01	2.04741e+00	1.41633e+01
7.16236e-01	2.30865e+00	1.41633e+01
7.82493e-01	2.53107e+00	1.41633e+01
8.18647e-01	2.70647e+00	1.41633e+01
8.36080e-01	2.82768e+00	1.41633e+01
8.25688e-01	2.89576e+00	1.41633e+01
8.74529e-01	2.87980e+00	1.41633e+01
9.11480e-01	2.82177e+00	1.44799e+01
8.37547e-01	2.78130e+00	1.44799e+01
8.37411e-01	2.65689e+00	1.44799e+01
8.35616e-01	2.47385e+00	1.44799e+01
8.32770e-01	2.23573e+00	1.44799e+01
8.29797e-01	1.94709e+00	1.44799e+01
8.26358e-01	1.61390e+00	1.44799e+01
8.21343e-01	1.24327e+00	1.44799e+01
8.12740e-01	8.43394e-01	1.44799e+01
7.97306e-01	4.23405e-01	1.44799e+01
7.70348e-01	-6.65827e-03	1.44799e+01
7.25740e-01	-4.36021e-01	1.44799e+01
6.57623e-01	-8.53960e-01	1.44799e+01
5.60336e-01	-1.25002e+00	1.44799e+01
4.31020e-01	-1.61479e+00	1.44799e+01
2.77549e-01	-1.94219e+00	1.44799e+01
1.06680e-01	-2.22648e+00	1.44799e+01
-7.44745e-02	-2.46284e+00	1.44799e+01
-2.59119e-01	-2.64731e+00	1.44799e+01
-4.40195e-01	-2.77703e+00	1.44799e+01
-6.11222e-01	-2.85010e+00	1.44799e+01
-8.22410e-01	-2.84867e+00	1.44799e+01
-1.01206e+00	-2.72829e+00	1.44799e+01

-1.12578e+00	-2.56944e+00	1.44799e+01
-1.21234e+00	-2.35975e+00	1.44799e+01
-1.26940e+00	-2.10358e+00	1.44799e+01
-1.29349e+00	-1.80684e+00	1.44799e+01
-1.27917e+00	-1.47693e+00	1.44799e+01
-1.22164e+00	-1.12200e+00	1.44799e+01
-1.11735e+00	-7.50619e-01	1.44799e+01
-9.65303e-01	-3.71251e-01	1.44799e+01
-7.66917e-01	7.86938e-03	1.44799e+01
-5.26558e-01	3.79104e-01	1.44799e+01
-2.61613e-01	7.38615e-01	1.44799e+01
4.64879e-03	1.08441e+00	1.44799e+01
2.47457e-01	1.41502e+00	1.44799e+01
4.47271e-01	1.72785e+00	1.44799e+01
5.95138e-01	2.01795e+00	1.44799e+01
6.94614e-01	2.27759e+00	1.44799e+01
7.54359e-01	2.49840e+00	1.44799e+01
7.86184e-01	2.67222e+00	1.44799e+01
8.01571e-01	2.79202e+00	1.44799e+01
7.96367e-01	2.85724e+00	1.44799e+01
8.37102e-01	2.84427e+00	1.44799e+01
8.66647e-01	2.78930e+00	1.47965e+01
8.00519e-01	2.74677e+00	1.47965e+01
8.00932e-01	2.62439e+00	1.47965e+01
7.99594e-01	2.44445e+00	1.47965e+01
7.97037e-01	2.21046e+00	1.47965e+01
7.93907e-01	1.92699e+00	1.47965e+01
7.89737e-01	1.59994e+00	1.47965e+01
7.83481e-01	1.23629e+00	1.47965e+01
7.73246e-01	8.44017e-01	1.47965e+01
7.56016e-01	4.32020e-01	1.47965e+01
7.27435e-01	1.00315e-02	1.47965e+01
6.82084e-01	-4.11615e-01	1.47965e+01
6.14380e-01	-8.22498e-01	1.47965e+01
5.18228e-01	-1.21222e+00	1.47965e+01
3.90443e-01	-1.57146e+00	1.47965e+01
2.39836e-01	-1.89451e+00	1.47965e+01
7.39562e-02	-2.17596e+00	1.47965e+01
-9.97941e-02	-2.41114e+00	1.47965e+01
-2.74530e-01	-2.59618e+00	1.47965e+01
-4.43064e-01	-2.72829e+00	1.47965e+01
-5.98821e-01	-2.80559e+00	1.47965e+01
-7.86121e-01	-2.81264e+00	1.47965e+01
-9.56286e-01	-2.70137e+00	1.47965e+01
-1.05926e+00	-2.54921e+00	1.47965e+01
-1.13793e+00	-2.34610e+00	1.47965e+01
-1.18977e+00	-2.09637e+00	1.47965e+01
-1.21160e+00	-1.80572e+00	1.47965e+01
-1.19761e+00	-1.48152e+00	1.47965e+01
-1.14282e+00	-1.13181e+00	1.47965e+01
-1.04363e+00	-7.65020e-01	1.47965e+01
-8.98920e-01	-3.89489e-01	1.47965e+01
-7.10203e-01	-1.32865e-02	1.47965e+01
-4.81732e-01	3.56034e-01	1.47965e+01
-2.29773e-01	7.14281e-01	1.47965e+01
2.39248e-02	1.05892e+00	1.47965e+01
2.54964e-01	1.38831e+00	1.47965e+01
4.43792e-01	1.69983e+00	1.47965e+01
5.81772e-01	1.98849e+00	1.47965e+01
6.72992e-01	2.24653e+00	1.47965e+01
7.26225e-01	2.46573e+00	1.47965e+01
7.53721e-01	2.63797e+00	1.47965e+01
7.67061e-01	2.75635e+00	1.47965e+01
7.67047e-01	2.81872e+00	1.47965e+01
7.99675e-01	2.80875e+00	1.47965e+01
8.21814e-01	2.75682e+00	1.51131e+01
7.63492e-01	2.71223e+00	1.51131e+01
7.64454e-01	2.59189e+00	1.51131e+01
7.63572e-01	2.41505e+00	1.51131e+01
7.61304e-01	2.18519e+00	1.51131e+01
7.58017e-01	1.90689e+00	1.51131e+01
7.53116e-01	1.58599e+00	1.51131e+01
7.45620e-01	1.22931e+00	1.51131e+01
7.33753e-01	8.44640e-01	1.51131e+01
7.14726e-01	4.40636e-01	1.51131e+01
6.84523e-01	2.67213e-02	1.51131e+01
6.38427e-01	-3.87210e-01	1.51131e+01
5.71137e-01	-7.91037e-01	1.51131e+01
4.76120e-01	-1.17443e+00	1.51131e+01
3.49866e-01	-1.52813e+00	1.51131e+01
2.02123e-01	-1.84683e+00	1.51131e+01
4.12328e-02	-2.12543e+00	1.51131e+01
-1.25114e-01	-2.35945e+00	1.51131e+01
-2.89940e-01	-2.54506e+00	1.51131e+01
-4.45934e-01	-2.67954e+00	1.51131e+01
-5.86420e-01	-2.76109e+00	1.51131e+01
-7.49833e-01	-2.77662e+00	1.51131e+01
-9.00513e-01	-2.67444e+00	1.51131e+01
-9.92730e-01	-2.52899e+00	1.51131e+01
-1.06353e+00	-2.33245e+00	1.51131e+01
-1.11014e+00	-2.08916e+00	1.51131e+01
-1.12970e+00	-1.80460e+00	1.51131e+01
-1.11604e+00	-1.48611e+00	1.51131e+01
-1.06400e+00	-1.14162e+00	1.51131e+01
-9.69899e-01	-7.79421e-01	1.51131e+01
-8.32536e-01	-4.07727e-01	1.51131e+01
-6.53490e-01	-3.44423e-02	1.51131e+01

-4.36906e-01	3.32964e-01	1.51131e+01
-1.97933e-01	6.89948e-01	1.51131e+01
4.32009e-02	1.03344e+00	1.51131e+01
2.62470e-01	1.36160e+00	1.51131e+01
4.40314e-01	1.67181e+00	1.51131e+01
5.68406e-01	1.95903e+00	1.51131e+01
6.51370e-01	2.21547e+00	1.51131e+01
6.98091e-01	2.43306e+00	1.51131e+01
7.21258e-01	2.60372e+00	1.51131e+01
7.32552e-01	2.72068e+00	1.51131e+01
7.37727e-01	2.78020e+00	1.51131e+01
7.62248e-01	2.77322e+00	1.51131e+01
8.48749e-01	2.70729e+00	1.54296e+01
7.32118e-01	2.67836e+00	1.54296e+01
7.33635e-01	2.55989e+00	1.54296e+01
7.33377e-01	2.38585e+00	1.54296e+01
7.31747e-01	2.15969e+00	1.54296e+01
7.28937e-01	1.88596e+00	1.54296e+01
7.24268e-01	1.57044e+00	1.54296e+01
7.16791e-01	1.21983e+00	1.54296e+01
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3.27102e-01	-1.49354e+00	1.54296e+01
1.82495e-01	-1.80821e+00	1.54296e+01
2.59832e-02	-2.08391e+00	1.54296e+01
-1.34624e-01	-2.31626e+00	1.54296e+01
-2.92344e-01	-2.50149e+00	1.54296e+01
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-8.57437e-01	-2.64550e+00	1.54296e+01
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-1.07172e+00	-1.79608e+00	1.54296e+01
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-1.00951e+00	-1.14307e+00	1.54296e+01
-9.19639e-01	-7.85400e-01	1.54296e+01
-7.88091e-01	-4.17758e-01	1.54296e+01
-6.16462e-01	-4.79145e-02	1.54296e+01
-4.08796e-01	3.16746e-01	1.54296e+01
-1.79489e-01	6.71466e-01	1.54296e+01
5.22411e-02	1.01284e+00	1.54296e+01
2.62725e-01	1.33889e+00	1.54296e+01
4.32511e-01	1.64695e+00	1.54296e+01
5.53512e-01	1.93196e+00	1.54296e+01
6.30656e-01	2.18619e+00	1.54296e+01
6.72859e-01	2.40172e+00	1.54296e+01
6.93016e-01	2.57054e+00	1.54296e+01
7.02813e-01	2.68605e+00	1.54296e+01
7.10930e-01	2.74357e+00	1.54296e+01
7.30526e-01	2.73843e+00	1.54296e+01
7.60150e-01	2.69099e+00	1.57462e+01
7.08896e-01	2.64545e+00	1.57462e+01
7.10978e-01	2.52860e+00	1.57462e+01
7.11584e-01	2.35694e+00	1.57462e+01
7.11093e-01	2.13385e+00	1.57462e+01
7.09674e-01	1.86385e+00	1.57462e+01
7.06629e-01	1.55261e+00	1.57462e+01
7.00986e-01	1.20673e+00	1.57462e+01
6.91022e-01	8.33738e-01	1.57462e+01
6.74068e-01	4.41953e-01	1.57462e+01
6.46322e-01	4.04178e-02	1.57462e+01
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3.59285e-02	-2.05536e+00	1.57462e+01
-1.21340e-01	-2.28534e+00	1.57462e+01
-2.75994e-01	-2.46882e+00	1.57462e+01
-4.20803e-01	-2.60314e+00	1.57462e+01
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-1.04823e+00	-1.77689e+00	1.57462e+01
-1.03720e+00	-1.46770e+00	1.57462e+01
-9.90095e-01	-1.13247e+00	1.57462e+01
-9.03216e-01	-7.79234e-01	1.57462e+01
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-6.07818e-01	-5.03079e-02	1.57462e+01
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4.65228e-02	9.99284e-01	1.57462e+01
2.52525e-01	1.32193e+00	1.57462e+01
4.18472e-01	1.62662e+00	1.57462e+01
5.36415e-01	1.90835e+00	1.57462e+01
6.11250e-01	2.15950e+00	1.57462e+01
6.51813e-01	2.37229e+00	1.57462e+01
6.70860e-01	2.53891e+00	1.57462e+01

6.79953e-01	2.65289e+00	1.57462e+01
6.87769e-01	2.70964e+00	1.57462e+01
7.07029e-01	2.70469e+00	1.57462e+01
7.35183e-01	2.65789e+00	1.60628e+01
6.85674e-01	2.61255e+00	1.60628e+01
6.88321e-01	2.49732e+00	1.60628e+01
6.89790e-01	2.32802e+00	1.60628e+01
6.90440e-01	2.10801e+00	1.60628e+01
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6.88990e-01	1.53477e+00	1.60628e+01
6.85182e-01	1.19364e+00	1.60628e+01
6.77250e-01	8.25742e-01	1.60628e+01
6.62521e-01	4.39278e-01	1.60628e+01
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4.50068e-01	-1.10912e+00	1.60628e+01
3.32944e-01	-1.44957e+00	1.60628e+01
1.95391e-01	-1.75708e+00	1.60628e+01
4.58739e-02	-2.02681e+00	1.60628e+01
-1.08056e-01	-2.25441e+00	1.60628e+01
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-8.92625e-01	-2.44596e+00	1.60628e+01
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-1.00380e+00	-2.02924e+00	1.60628e+01
-1.02473e+00	-1.75769e+00	1.60628e+01
-1.01543e+00	-1.45271e+00	1.60628e+01
-9.70678e-01	-1.12187e+00	1.60628e+01
-8.86792e-01	-7.73068e-01	1.60628e+01
-7.62466e-01	-4.14154e-01	1.60628e+01
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-1.81231e-01	6.51377e-01	1.60628e+01
4.08044e-02	9.85730e-01	1.60628e+01
2.42324e-01	1.30496e+00	1.60628e+01
4.04434e-01	1.60630e+00	1.60628e+01
5.19317e-01	1.88474e+00	1.60628e+01
5.91844e-01	2.13280e+00	1.60628e+01
6.30766e-01	2.34286e+00	1.60628e+01
6.48704e-01	2.50728e+00	1.60628e+01
6.57092e-01	2.61973e+00	1.60628e+01
6.64609e-01	2.67572e+00	1.60628e+01
6.83531e-01	2.67096e+00	1.60628e+01
7.10216e-01	2.62478e+00	1.63794e+01
6.62452e-01	2.57964e+00	1.63794e+01
6.65664e-01	2.46603e+00	1.63794e+01
6.67997e-01	2.29911e+00	1.63794e+01
6.69786e-01	2.08218e+00	1.63794e+01
6.71150e-01	1.81961e+00	1.63794e+01
6.71350e-01	1.51693e+00	1.63794e+01
6.69378e-01	1.18055e+00	1.63794e+01
6.63478e-01	8.17747e-01	1.63794e+01
6.50973e-01	4.36603e-01	1.63794e+01
6.28072e-01	4.58688e-02	1.63794e+01
5.90513e-01	-3.45339e-01	1.63794e+01
5.33229e-01	-7.27623e-01	1.63794e+01
4.49575e-01	-1.09128e+00	1.63794e+01
3.35865e-01	-1.42759e+00	1.63794e+01
2.01839e-01	-1.73152e+00	1.63794e+01
5.58192e-02	-1.99827e+00	1.63794e+01
-9.47723e-02	-2.22349e+00	1.63794e+01
-2.43293e-01	-2.40348e+00	1.63794e+01
-3.82741e-01	-2.53556e+00	1.63794e+01
-5.06727e-01	-2.61791e+00	1.63794e+01
-6.48783e-01	-2.63999e+00	1.63794e+01
-7.83131e-01	-2.54998e+00	1.63794e+01
-8.67383e-01	-2.41646e+00	1.63794e+01
-9.33676e-01	-2.23382e+00	1.63794e+01
-9.79183e-01	-2.00614e+00	1.63794e+01
-1.00124e+00	-1.73849e+00	1.63794e+01
-9.93658e-01	-1.43772e+00	1.63794e+01
-9.51261e-01	-1.11127e+00	1.63794e+01
-8.70369e-01	-7.66903e-01	1.63794e+01
-7.49654e-01	-4.12351e-01	1.63794e+01
-5.90530e-01	-5.50947e-02	1.63794e+01
-3.96767e-01	2.97725e-01	1.63794e+01
-1.82102e-01	6.41333e-01	1.63794e+01
3.50860e-02	9.72175e-01	1.63794e+01
2.32124e-01	1.28800e+00	1.63794e+01
3.90396e-01	1.58598e+00	1.63794e+01
5.02220e-01	1.86113e+00	1.63794e+01
5.72439e-01	2.10610e+00	1.63794e+01
6.09720e-01	2.31343e+00	1.63794e+01
6.26549e-01	2.47564e+00	1.63794e+01
6.34232e-01	2.58657e+00	1.63794e+01
6.41449e-01	2.64180e+00	1.63794e+01
6.60034e-01	2.63722e+00	1.63794e+01
6.85248e-01	2.59168e+00	1.66960e+01
6.39230e-01	2.54673e+00	1.66960e+01
6.43006e-01	2.43475e+00	1.66960e+01
6.46203e-01	2.27020e+00	1.66960e+01
6.49133e-01	2.05634e+00	1.66960e+01

6.51887e-01	1.79749e+00	1.66960e+01
6.53711e-01	1.49909e+00	1.66960e+01
6.53574e-01	1.16745e+00	1.66960e+01
6.49706e-01	8.09751e-01	1.66960e+01
6.39425e-01	4.33927e-01	1.66960e+01
6.18946e-01	4.85943e-02	1.66960e+01
5.84020e-01	-3.37285e-01	1.66960e+01
5.29600e-01	-7.14473e-01	1.66960e+01
4.49082e-01	-1.07344e+00	1.66960e+01
3.38786e-01	-1.40560e+00	1.66960e+01
2.08288e-01	-1.70596e+00	1.66960e+01
6.57645e-02	-1.96972e+00	1.66960e+01
-8.14883e-02	-2.19257e+00	1.66960e+01
-2.26942e-01	-2.37081e+00	1.66960e+01
-3.63710e-01	-2.50177e+00	1.66960e+01
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2.93677e-02	9.58621e-01	1.66960e+01
2.21924e-01	1.27104e+00	1.66960e+01
3.76357e-01	1.56565e+00	1.66960e+01
4.85123e-01	1.83752e+00	1.66960e+01
5.53033e-01	2.07940e+00	1.66960e+01
5.88673e-01	2.28400e+00	1.66960e+01
6.04393e-01	2.44401e+00	1.66960e+01
6.11372e-01	2.55342e+00	1.66960e+01
6.18288e-01	2.60788e+00	1.66960e+01
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6.20349e-01	2.40346e+00	1.70126e+01
6.24410e-01	2.24129e+00	1.70126e+01
6.28479e-01	2.03051e+00	1.70126e+01
6.32625e-01	1.77538e+00	1.70126e+01
6.36072e-01	1.48126e+00	1.70126e+01
6.37769e-01	1.15436e+00	1.70126e+01
6.35934e-01	8.01755e-01	1.70126e+01
6.27878e-01	4.31252e-01	1.70126e+01
6.09821e-01	5.13198e-02	1.70126e+01
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5.25971e-01	-7.01322e-01	1.70126e+01
4.48589e-01	-1.05560e+00	1.70126e+01
3.41707e-01	-1.38362e+00	1.70126e+01
2.14736e-01	-1.68040e+00	1.70126e+01
7.57098e-02	-1.94117e+00	1.70126e+01
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-8.83279e-01	-2.18069e+00	1.70126e+01
-9.29945e-01	-1.95994e+00	1.70126e+01
-9.54255e-01	-1.70010e+00	1.70126e+01
-9.50118e-01	-1.40775e+00	1.70126e+01
-9.12427e-01	-1.09008e+00	1.70126e+01
-8.37522e-01	-7.54571e-01	1.70126e+01
-7.24029e-01	-4.08747e-01	1.70126e+01
-5.73242e-01	-5.98815e-02	1.70126e+01
-3.88748e-01	2.85045e-01	1.70126e+01
-1.83844e-01	6.21244e-01	1.70126e+01
2.36493e-02	9.45066e-01	1.70126e+01
2.11723e-01	1.25408e+00	1.70126e+01
3.62319e-01	1.54533e+00	1.70126e+01
4.68026e-01	1.81390e+00	1.70126e+01
5.33627e-01	2.05271e+00	1.70126e+01
5.67626e-01	2.25458e+00	1.70126e+01
5.82237e-01	2.41238e+00	1.70126e+01
5.88511e-01	2.52026e+00	1.70126e+01
5.95128e-01	2.57396e+00	1.70126e+01
6.13040e-01	2.56975e+00	1.70126e+01
6.35314e-01	2.52547e+00	1.73291e+01
5.92786e-01	2.48091e+00	1.73291e+01
5.97692e-01	2.37218e+00	1.73291e+01
6.02616e-01	2.21237e+00	1.73291e+01
6.07826e-01	2.00467e+00	1.73291e+01
6.13363e-01	1.75326e+00	1.73291e+01
6.18433e-01	1.46342e+00	1.73291e+01
6.21965e-01	1.14127e+00	1.73291e+01
6.22162e-01	7.93759e-01	1.73291e+01
6.16330e-01	4.28577e-01	1.73291e+01
6.00696e-01	5.40453e-02	1.73291e+01
5.71035e-01	-3.21177e-01	1.73291e+01
5.22342e-01	-6.88172e-01	1.73291e+01
4.48096e-01	-1.03776e+00	1.73291e+01

3.44627e-01	-1.36163e+00	1.73291e+01
2.21184e-01	-1.65484e+00	1.73291e+01
8.56551e-02	-1.91262e+00	1.73291e+01
-5.49203e-02	-2.13072e+00	1.73291e+01
-1.94242e-01	-2.30547e+00	1.73291e+01
-3.25647e-01	-2.43420e+00	1.73291e+01
-4.43048e-01	-2.51505e+00	1.73291e+01
-5.78723e-01	-2.53848e+00	1.73291e+01
-7.08825e-01	-2.45446e+00	1.73291e+01
-7.91656e-01	-2.32796e+00	1.73291e+01
-8.58081e-01	-2.15413e+00	1.73291e+01
-9.05326e-01	-1.93684e+00	1.73291e+01
-9.30763e-01	-1.68090e+00	1.73291e+01
-9.28348e-01	-1.39277e+00	1.73291e+01
-8.93011e-01	-1.07948e+00	1.73291e+01
-8.21098e-01	-7.48406e-01	1.73291e+01
-7.11217e-01	-4.06945e-01	1.73291e+01
-5.64598e-01	-6.22749e-02	1.73291e+01
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1.79309e-02	9.31512e-01	1.73291e+01
2.01523e-01	1.23712e+00	1.73291e+01
3.48280e-01	1.52501e+00	1.73291e+01
4.50928e-01	1.79029e+00	1.73291e+01
5.14221e-01	2.02601e+00	1.73291e+01
5.46580e-01	2.22515e+00	1.73291e+01
5.60081e-01	2.38075e+00	1.73291e+01
5.65651e-01	2.48710e+00	1.73291e+01
5.71968e-01	2.54003e+00	1.73291e+01
5.89542e-01	2.53602e+00	1.73291e+01
6.10347e-01	2.49237e+00	1.76457e+01
5.69564e-01	2.44801e+00	1.76457e+01
5.75035e-01	2.34089e+00	1.76457e+01
5.80823e-01	2.18346e+00	1.76457e+01
5.87172e-01	1.97883e+00	1.76457e+01
5.94101e-01	1.73114e+00	1.76457e+01
6.00794e-01	1.44558e+00	1.76457e+01
6.06161e-01	1.12818e+00	1.76457e+01
6.08390e-01	7.85763e-01	1.76457e+01
6.04782e-01	4.25902e-01	1.76457e+01
5.91571e-01	5.67708e-02	1.76457e+01
5.64542e-01	-3.13123e-01	1.76457e+01
5.18712e-01	-6.75022e-01	1.76457e+01
4.47602e-01	-1.01991e+00	1.76457e+01
3.47548e-01	-1.33965e+00	1.76457e+01
2.27632e-01	-1.62927e+00	1.76457e+01
9.56004e-02	-1.88407e+00	1.76457e+01
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-9.07270e-01	-1.66170e+00	1.76457e+01
-9.06578e-01	-1.37778e+00	1.76457e+01
-8.73594e-01	-1.06888e+00	1.76457e+01
-8.04675e-01	-7.42240e-01	1.76457e+01
-6.98404e-01	-4.05143e-01	1.76457e+01
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-3.80729e-01	2.72365e-01	1.76457e+01
-1.85585e-01	6.01155e-01	1.76457e+01
1.22126e-02	9.17957e-01	1.76457e+01
1.91323e-01	1.22016e+00	1.76457e+01
3.34242e-01	1.50469e+00	1.76457e+01
4.33831e-01	1.76668e+00	1.76457e+01
4.94815e-01	1.99931e+00	1.76457e+01
5.25533e-01	2.19572e+00	1.76457e+01
5.37925e-01	2.34912e+00	1.76457e+01
5.42790e-01	2.45394e+00	1.76457e+01
5.48807e-01	2.50611e+00	1.76457e+01
5.66045e-01	2.50228e+00	1.76457e+01
5.85380e-01	2.45927e+00	1.79623e+01
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5.66519e-01	1.95300e+00	1.79623e+01
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5.82446e-01	5.94963e-02	1.79623e+01
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4.47109e-01	-1.00207e+00	1.79623e+01
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1.05546e-01	-1.85553e+00	1.79623e+01
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1.81122e-01	1.20320e+00	1.79623e+01
3.20204e-01	1.48436e+00	1.79623e+01
4.16734e-01	1.74307e+00	1.79623e+01
4.75409e-01	1.97261e+00	1.79623e+01
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5.15769e-01	2.31749e+00	1.79623e+01
5.19930e-01	2.42078e+00	1.79623e+01
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5.60413e-01	2.42616e+00	1.82789e+01
5.23120e-01	2.38219e+00	1.82789e+01
5.29720e-01	2.27832e+00	1.82789e+01
5.37236e-01	2.12564e+00	1.82789e+01
5.45865e-01	1.92716e+00	1.82789e+01
5.55576e-01	1.68691e+00	1.82789e+01
5.65515e-01	1.40991e+00	1.82789e+01
5.74552e-01	1.10199e+00	1.82789e+01
5.80846e-01	7.69772e-01	1.82789e+01
5.81687e-01	4.20552e-01	1.82789e+01
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1.15491e-01	-1.82698e+00	1.82789e+01
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-7.71828e-01	-7.29908e-01	1.82789e+01
-6.72779e-01	-4.01539e-01	1.82789e+01
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-3.72710e-01	2.59685e-01	1.82789e+01
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7.75821e-04	8.90848e-01	1.82789e+01
1.70922e-01	1.18624e+00	1.82789e+01
3.06165e-01	1.46404e+00	1.82789e+01
3.99637e-01	1.71946e+00	1.82789e+01
4.56004e-01	1.94592e+00	1.82789e+01
4.83440e-01	2.13686e+00	1.82789e+01
4.93613e-01	2.28586e+00	1.82789e+01
4.97070e-01	2.38763e+00	1.82789e+01
5.02487e-01	2.43827e+00	1.82789e+01
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5.48967e-01	2.39481e+00	1.85955e+01
5.05496e-01	2.35266e+00	1.85955e+01
5.12472e-01	2.25021e+00	1.85955e+01
5.20568e-01	2.09960e+00	1.85955e+01
5.29967e-01	1.90382e+00	1.85955e+01
5.40620e-01	1.66683e+00	1.85955e+01
5.51662e-01	1.39357e+00	1.85955e+01
5.61952e-01	1.08982e+00	1.85955e+01
5.69643e-01	7.62067e-01	1.85955e+01
5.72031e-01	4.17523e-01	1.85955e+01
5.65376e-01	6.39472e-02	1.85955e+01
5.45505e-01	-2.90582e-01	1.85955e+01
5.07508e-01	-6.37763e-01	1.85955e+01
4.45034e-01	-9.69075e-01	1.85955e+01
3.54444e-01	-1.27677e+00	1.85955e+01
2.44352e-01	-1.55597e+00	1.85955e+01
1.22097e-01	-1.80203e+00	1.85955e+01
-5.77519e-03	-2.01076e+00	1.85955e+01
-1.33401e-01	-2.17859e+00	1.85955e+01
-2.54555e-01	-2.30283e+00	1.85955e+01
-3.63534e-01	-2.38162e+00	1.85955e+01
-4.90989e-01	-2.40665e+00	1.85955e+01
-6.15508e-01	-2.33022e+00	1.85955e+01
-6.96362e-01	-2.21270e+00	1.85955e+01
-7.62765e-01	-2.05019e+00	1.85955e+01
-8.12018e-01	-1.84628e+00	1.85955e+01
-8.41536e-01	-1.60543e+00	1.85955e+01
-8.45483e-01	-1.33363e+00	1.85955e+01
-8.18921e-01	-1.03739e+00	1.85955e+01
-7.58248e-01	-7.23589e-01	1.85955e+01
-6.61990e-01	-3.99170e-01	1.85955e+01
-5.31152e-01	-7.09293e-02	1.85955e+01

-3.68884e-01	2.54549e-01	1.85955e+01
-1.87414e-01	5.72459e-01	1.85955e+01
-3.20879e-03	8.78937e-01	1.85955e+01
1.63344e-01	1.17114e+00	1.85955e+01
2.95539e-01	1.44583e+00	1.85955e+01
3.86621e-01	1.69824e+00	1.85955e+01
4.41224e-01	1.92193e+00	1.85955e+01
4.67441e-01	2.11044e+00	1.85955e+01
4.76814e-01	2.25749e+00	1.85955e+01
4.79770e-01	2.35791e+00	1.85955e+01
4.84968e-01	2.40788e+00	1.85955e+01
5.01247e-01	2.40456e+00	1.85955e+01
5.25050e-01	2.36797e+00	1.89121e+01
4.90312e-01	2.32461e+00	1.89121e+01
4.97580e-01	2.22349e+00	1.89121e+01
5.06134e-01	2.07482e+00	1.89121e+01
5.16142e-01	1.88157e+00	1.89121e+01
5.27541e-01	1.64763e+00	1.89121e+01
5.39459e-01	1.37789e+00	1.89121e+01
5.50747e-01	1.07804e+00	1.89121e+01
5.59560e-01	7.54490e-01	1.89121e+01
5.63199e-01	4.14338e-01	1.89121e+01
5.57944e-01	6.52368e-02	1.89121e+01
5.39646e-01	-2.84855e-01	1.89121e+01
5.03423e-01	-6.27760e-01	1.89121e+01
4.42976e-01	-9.55089e-01	1.89121e+01
3.54684e-01	-1.25921e+00	1.89121e+01
2.47031e-01	-1.53526e+00	1.89121e+01
1.27247e-01	-1.77865e+00	1.89121e+01
1.77846e-03	-1.98520e+00	1.89121e+01
-1.23599e-01	-2.15137e+00	1.89121e+01
-2.42749e-01	-2.27448e+00	1.89121e+01
-3.50049e-01	-2.35267e+00	1.89121e+01
-4.75789e-01	-2.37786e+00	1.89121e+01
-5.99008e-01	-2.30286e+00	1.89121e+01
-6.79271e-01	-2.18714e+00	1.89121e+01
-7.45433e-01	-2.02694e+00	1.89121e+01
-7.94818e-01	-1.82581e+00	1.89121e+01
-8.24857e-01	-1.58814e+00	1.89121e+01
-8.29764e-01	-1.31981e+00	1.89121e+01
-8.04642e-01	-1.02724e+00	1.89121e+01
-7.45909e-01	-7.17204e-01	1.89121e+01
-6.52082e-01	-3.96554e-01	1.89121e+01
-5.24130e-01	-7.20027e-02	1.89121e+01
-3.65137e-01	2.49938e-01	1.89121e+01
-1.87159e-01	5.64479e-01	1.89121e+01
-6.43767e-03	8.67742e-01	1.89121e+01
1.56909e-01	1.15684e+00	1.89121e+01
2.86400e-01	1.42853e+00	1.89121e+01
3.75384e-01	1.67807e+00	1.89121e+01
4.28460e-01	1.89911e+00	1.89121e+01
4.53643e-01	2.08533e+00	1.89121e+01
4.62349e-01	2.23054e+00	1.89121e+01
4.64894e-01	2.32969e+00	1.89121e+01
4.69909e-01	2.37904e+00	1.89121e+01
4.85924e-01	2.37583e+00	1.89121e+01
5.08673e-01	2.33959e+00	1.92286e+01
4.75128e-01	2.29656e+00	1.92286e+01
4.82689e-01	2.19677e+00	1.92286e+01
4.91699e-01	2.05005e+00	1.92286e+01
5.02317e-01	1.85932e+00	1.92286e+01
5.14461e-01	1.62843e+00	1.92286e+01
5.27256e-01	1.36222e+00	1.92286e+01
5.39543e-01	1.06626e+00	1.92286e+01
5.49478e-01	7.46913e-01	1.92286e+01
5.54368e-01	4.11154e-01	1.92286e+01
5.50513e-01	6.65264e-02	1.92286e+01
5.33786e-01	-2.79129e-01	1.92286e+01
4.99338e-01	-6.17758e-01	1.92286e+01
4.40919e-01	-9.41104e-01	1.92286e+01
3.54924e-01	-1.24164e+00	1.92286e+01
2.49711e-01	-1.51456e+00	1.92286e+01
1.32397e-01	-1.75526e+00	1.92286e+01
9.33211e-03	-1.95964e+00	1.92286e+01
-1.13797e-01	-2.12414e+00	1.92286e+01
-2.30944e-01	-2.24613e+00	1.92286e+01
-3.36563e-01	-2.32372e+00	1.92286e+01
-4.60589e-01	-2.34906e+00	1.92286e+01
-5.82508e-01	-2.27550e+00	1.92286e+01
-6.62179e-01	-2.16157e+00	1.92286e+01
-7.28102e-01	-2.00369e+00	1.92286e+01
-7.77618e-01	-1.80534e+00	1.92286e+01
-8.08177e-01	-1.57084e+00	1.92286e+01
-8.14046e-01	-1.30598e+00	1.92286e+01
-7.90364e-01	-1.01708e+00	1.92286e+01
-7.33570e-01	-7.10818e-01	1.92286e+01
-6.42174e-01	-3.93937e-01	1.92286e+01
-5.17108e-01	-7.30762e-02	1.92286e+01
-3.61391e-01	2.45327e-01	1.92286e+01
-1.86904e-01	5.56498e-01	1.92286e+01
-9.66654e-03	8.56547e-01	1.92286e+01
1.50474e-01	1.14255e+00	1.92286e+01
2.77261e-01	1.41123e+00	1.92286e+01
3.64147e-01	1.65790e+00	1.92286e+01
4.15696e-01	1.87630e+00	1.92286e+01
4.39844e-01	2.06021e+00	1.92286e+01
4.47884e-01	2.20359e+00	1.92286e+01

4.50018e-01	2.30148e+00	1.92286e+01
4.54849e-01	2.35020e+00	1.92286e+01
4.70601e-01	2.34710e+00	1.92286e+01
4.92296e-01	2.31121e+00	1.95452e+01
4.59945e-01	2.26850e+00	1.95452e+01
4.67797e-01	2.17004e+00	1.95452e+01
4.77265e-01	2.02527e+00	1.95452e+01
4.88491e-01	1.83706e+00	1.95452e+01
5.01382e-01	1.60923e+00	1.95452e+01
5.15052e-01	1.34654e+00	1.95452e+01
5.28339e-01	1.05449e+00	1.95452e+01
5.39395e-01	7.39336e-01	1.95452e+01
5.45536e-01	4.07970e-01	1.95452e+01
5.43081e-01	6.78160e-02	1.95452e+01
5.27926e-01	-2.73402e-01	1.95452e+01
4.95253e-01	-6.07755e-01	1.95452e+01
4.38862e-01	-9.27118e-01	1.95452e+01
3.55164e-01	-1.22408e+00	1.95452e+01
2.52390e-01	-1.49385e+00	1.95452e+01
1.37547e-01	-1.73188e+00	1.95452e+01
1.68858e-02	-1.93407e+00	1.95452e+01
-1.03995e-01	-2.09692e+00	1.95452e+01
-2.19138e-01	-2.21777e+00	1.95452e+01
-3.23078e-01	-2.29477e+00	1.95452e+01
-4.45389e-01	-2.32027e+00	1.95452e+01
-5.66008e-01	-2.24814e+00	1.95452e+01
-6.45088e-01	-2.13601e+00	1.95452e+01
-7.10770e-01	-1.98045e+00	1.95452e+01
-7.60418e-01	-1.78487e+00	1.95452e+01
-7.91497e-01	-1.55355e+00	1.95452e+01
-7.98327e-01	-1.29216e+00	1.95452e+01
-7.76085e-01	-1.00693e+00	1.95452e+01
-7.21230e-01	-7.04432e-01	1.95452e+01
-6.32266e-01	-3.91321e-01	1.95452e+01
-5.10086e-01	-7.41496e-02	1.95452e+01
-3.57644e-01	2.40716e-01	1.95452e+01
-1.86649e-01	5.48517e-01	1.95452e+01
-1.28954e-02	8.45352e-01	1.95452e+01
1.44039e-01	1.12826e+00	1.95452e+01
2.68122e-01	1.39394e+00	1.95452e+01
3.52911e-01	1.63773e+00	1.95452e+01
4.02933e-01	1.85349e+00	1.95452e+01
4.26046e-01	2.03510e+00	1.95452e+01
4.33419e-01	2.17664e+00	1.95452e+01
4.35142e-01	2.27326e+00	1.95452e+01
4.39790e-01	2.32135e+00	1.95452e+01
4.55278e-01	2.31837e+00	1.95452e+01
4.75919e-01	2.28283e+00	1.98618e+01
4.44761e-01	2.24045e+00	1.98618e+01
4.52906e-01	2.14332e+00	1.98618e+01
4.62831e-01	2.00049e+00	1.98618e+01
4.74666e-01	1.81481e+00	1.98618e+01
4.88303e-01	1.59004e+00	1.98618e+01
5.02849e-01	1.33086e+00	1.98618e+01
5.17135e-01	1.04271e+00	1.98618e+01
5.29312e-01	7.31759e-01	1.98618e+01
5.36705e-01	4.04785e-01	1.98618e+01
5.35650e-01	6.91056e-02	1.98618e+01
5.22066e-01	-2.67676e-01	1.98618e+01
4.91169e-01	-5.97753e-01	1.98618e+01
4.36804e-01	-9.13133e-01	1.98618e+01
3.55404e-01	-1.20652e+00	1.98618e+01
2.55069e-01	-1.47315e+00	1.98618e+01
1.42697e-01	-1.70850e+00	1.98618e+01
2.44394e-02	-1.90851e+00	1.98618e+01
-9.41933e-02	-2.06970e+00	1.98618e+01
-2.07333e-01	-2.18942e+00	1.98618e+01
-3.09593e-01	-2.26582e+00	1.98618e+01
-4.30189e-01	-2.29148e+00	1.98618e+01
-5.49508e-01	-2.22078e+00	1.98618e+01
-6.27996e-01	-2.11045e+00	1.98618e+01
-6.93438e-01	-1.95720e+00	1.98618e+01
-7.43218e-01	-1.76440e+00	1.98618e+01
-7.74818e-01	-1.53625e+00	1.98618e+01
-7.82608e-01	-1.27834e+00	1.98618e+01
-7.61806e-01	-9.96779e-01	1.98618e+01
-7.08891e-01	-6.98046e-01	1.98618e+01
-6.22359e-01	-3.88705e-01	1.98618e+01
-5.03064e-01	-7.52231e-02	1.98618e+01
-3.53898e-01	2.36105e-01	1.98618e+01
-1.86394e-01	5.40536e-01	1.98618e+01
-1.61243e-02	8.34156e-01	1.98618e+01
1.37604e-01	1.11396e+00	1.98618e+01
2.58984e-01	1.37664e+00	1.98618e+01
3.41674e-01	1.61757e+00	1.98618e+01
3.90169e-01	1.83068e+00	1.98618e+01
4.12247e-01	2.00999e+00	1.98618e+01
4.18955e-01	2.14969e+00	1.98618e+01
4.20266e-01	2.24505e+00	1.98618e+01
4.24730e-01	2.29251e+00	1.98618e+01
4.39956e-01	2.28964e+00	1.98618e+01
4.59541e-01	2.25445e+00	2.01784e+01
4.29577e-01	2.21239e+00	2.01784e+01
4.38015e-01	2.11659e+00	2.01784e+01
4.48397e-01	1.97571e+00	2.01784e+01
4.60840e-01	1.79256e+00	2.01784e+01

4.75223e-01	1.57084e+00	2.01784e+01
4.90646e-01	1.31518e+00	2.01784e+01
5.05931e-01	1.03094e+00	2.01784e+01
5.19229e-01	7.24182e-01	2.01784e+01
5.27873e-01	4.01601e-01	2.01784e+01
5.28219e-01	7.03952e-02	2.01784e+01
5.16207e-01	-2.61949e-01	2.01784e+01
4.87084e-01	-5.87750e-01	2.01784e+01
4.34747e-01	-8.99147e-01	2.01784e+01
3.55643e-01	-1.18895e+00	2.01784e+01
2.57749e-01	-1.45244e+00	2.01784e+01
1.47848e-01	-1.68512e+00	2.01784e+01
3.19930e-02	-1.88295e+00	2.01784e+01
-8.43915e-02	-2.04247e+00	2.01784e+01
-1.95527e-01	-2.16107e+00	2.01784e+01
-2.96108e-01	-2.23687e+00	2.01784e+01
-4.14989e-01	-2.26268e+00	2.01784e+01
-5.33008e-01	-2.19342e+00	2.01784e+01
-6.10905e-01	-2.08488e+00	2.01784e+01
-6.76106e-01	-1.93395e+00	2.01784e+01
-7.26018e-01	-1.74393e+00	2.01784e+01
-7.58138e-01	-1.51895e+00	2.01784e+01
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-7.47528e-01	-9.86626e-01	2.01784e+01
-6.96551e-01	-6.91660e-01	2.01784e+01
-6.12451e-01	-3.86089e-01	2.01784e+01
-4.96042e-01	-7.62965e-02	2.01784e+01
-3.50151e-01	2.31494e-01	2.01784e+01
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-1.93532e-02	8.22961e-01	2.01784e+01
1.31169e-01	1.09967e+00	2.01784e+01
2.49845e-01	1.35934e+00	2.01784e+01
3.30437e-01	1.59740e+00	2.01784e+01
3.77406e-01	1.80786e+00	2.01784e+01
3.98449e-01	1.98487e+00	2.01784e+01
4.04490e-01	2.12274e+00	2.01784e+01
4.05390e-01	2.21683e+00	2.01784e+01
4.09671e-01	2.26366e+00	2.01784e+01
4.24633e-01	2.26091e+00	2.01784e+01
4.43164e-01	2.22607e+00	2.04950e+01
4.14393e-01	2.18434e+00	2.04950e+01
4.23123e-01	2.08987e+00	2.04950e+01
4.33963e-01	1.95094e+00	2.04950e+01
4.47015e-01	1.77031e+00	2.04950e+01
4.62144e-01	1.55164e+00	2.04950e+01
4.78442e-01	1.29950e+00	2.04950e+01
4.94727e-01	1.01916e+00	2.04950e+01
5.09146e-01	7.16605e-01	2.04950e+01
5.19042e-01	3.98416e-01	2.04950e+01
5.20787e-01	7.16848e-02	2.04950e+01
5.10347e-01	-2.56223e-01	2.04950e+01
4.82999e-01	-5.77748e-01	2.04950e+01
4.32689e-01	-8.85161e-01	2.04950e+01
3.55883e-01	-1.17139e+00	2.04950e+01
2.60428e-01	-1.43173e+00	2.04950e+01
1.52998e-01	-1.66173e+00	2.04950e+01
3.95467e-02	-1.85739e+00	2.04950e+01
-7.45897e-02	-2.01525e+00	2.04950e+01
-1.83722e-01	-2.13271e+00	2.04950e+01
-2.82623e-01	-2.20792e+00	2.04950e+01
-3.99789e-01	-2.23389e+00	2.04950e+01
-5.16508e-01	-2.16607e+00	2.04950e+01
-5.93813e-01	-2.05932e+00	2.04950e+01
-6.58774e-01	-1.91070e+00	2.04950e+01
-7.08818e-01	-1.72346e+00	2.04950e+01
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-7.51171e-01	-1.25069e+00	2.04950e+01
-7.33249e-01	-9.76473e-01	2.04950e+01
-6.84212e-01	-6.85274e-01	2.04950e+01
-6.02543e-01	-3.83473e-01	2.04950e+01
-4.89020e-01	-7.73700e-02	2.04950e+01
-3.46405e-01	2.26883e-01	2.04950e+01
-1.85884e-01	5.24574e-01	2.04950e+01
-2.25820e-02	8.11766e-01	2.04950e+01
1.24734e-01	1.08538e+00	2.04950e+01
2.40706e-01	1.34205e+00	2.04950e+01
3.19200e-01	1.57723e+00	2.04950e+01
3.64642e-01	1.78505e+00	2.04950e+01
3.84650e-01	1.95976e+00	2.04950e+01
3.90025e-01	2.09579e+00	2.04950e+01
3.90514e-01	2.18861e+00	2.04950e+01
3.94611e-01	2.23482e+00	2.04950e+01
4.09310e-01	2.23218e+00	2.04950e+01
4.26787e-01	2.19769e+00	2.08116e+01
3.99210e-01	2.15629e+00	2.08116e+01
4.08232e-01	2.06315e+00	2.08116e+01
4.19529e-01	1.92616e+00	2.08116e+01
4.33190e-01	1.74806e+00	2.08116e+01
4.49065e-01	1.53245e+00	2.08116e+01
4.66239e-01	1.28382e+00	2.08116e+01
4.83522e-01	1.00738e+00	2.08116e+01
4.99064e-01	7.09028e-01	2.08116e+01
5.10210e-01	3.95232e-01	2.08116e+01
5.13356e-01	7.29744e-02	2.08116e+01
5.04487e-01	-2.50496e-01	2.08116e+01
4.78914e-01	-5.67745e-01	2.08116e+01
4.30632e-01	-8.71176e-01	2.08116e+01

3.56123e-01	-1.15383e+00	2.08116e+01
2.63107e-01	-1.41103e+00	2.08116e+01
1.58148e-01	-1.63835e+00	2.08116e+01
4.71003e-02	-1.83182e+00	2.08116e+01
-6.47878e-02	-1.98803e+00	2.08116e+01
-1.71916e-01	-2.10436e+00	2.08116e+01
-2.69137e-01	-2.17897e+00	2.08116e+01
-3.84589e-01	-2.20509e+00	2.08116e+01
-5.00008e-01	-2.13871e+00	2.08116e+01
-5.76721e-01	-2.03376e+00	2.08116e+01
-6.41442e-01	-1.88745e+00	2.08116e+01
-6.91618e-01	-1.70299e+00	2.08116e+01
-7.24779e-01	-1.48436e+00	2.08116e+01
-7.35452e-01	-1.23687e+00	2.08116e+01
-7.18970e-01	-9.66320e-01	2.08116e+01
-6.71873e-01	-6.78888e-01	2.08116e+01
-5.92635e-01	-3.80856e-01	2.08116e+01
-4.81998e-01	-7.84434e-02	2.08116e+01
-3.42658e-01	2.22272e-01	2.08116e+01
-1.85629e-01	5.16593e-01	2.08116e+01
-2.58109e-02	8.00571e-01	2.08116e+01
1.18299e-01	1.07108e+00	2.08116e+01
2.31567e-01	1.32475e+00	2.08116e+01
3.07964e-01	1.55706e+00	2.08116e+01
3.51878e-01	1.76224e+00	2.08116e+01
3.70852e-01	1.93465e+00	2.08116e+01
3.75560e-01	2.06884e+00	2.08116e+01
3.75638e-01	2.16040e+00	2.08116e+01
3.79552e-01	2.20598e+00	2.08116e+01
3.93987e-01	2.20345e+00	2.08116e+01
4.10410e-01	2.16931e+00	2.11281e+01
3.84026e-01	2.12823e+00	2.11281e+01
3.93340e-01	2.03642e+00	2.11281e+01
4.05095e-01	1.90138e+00	2.11281e+01
4.19364e-01	1.72581e+00	2.11281e+01
4.35986e-01	1.51325e+00	2.11281e+01
4.54036e-01	1.26814e+00	2.11281e+01
4.72318e-01	9.95609e-01	2.11281e+01
4.88981e-01	7.01451e-01	2.11281e+01
5.01379e-01	3.92048e-01	2.11281e+01
5.05924e-01	7.42640e-02	2.11281e+01
4.98627e-01	-2.44770e-01	2.11281e+01
4.74829e-01	-5.57742e-01	2.11281e+01
4.28575e-01	-8.57190e-01	2.11281e+01
3.56363e-01	-1.13626e+00	2.11281e+01
2.65786e-01	-1.39032e+00	2.11281e+01
1.63298e-01	-1.61497e+00	2.11281e+01
5.46540e-02	-1.80626e+00	2.11281e+01
-5.49860e-02	-1.96081e+00	2.11281e+01
-1.60111e-01	-2.07601e+00	2.11281e+01
-2.55652e-01	-2.15002e+00	2.11281e+01
-3.69389e-01	-2.17630e+00	2.11281e+01
-4.83508e-01	-2.11135e+00	2.11281e+01
-5.59630e-01	-2.00820e+00	2.11281e+01
-6.24110e-01	-1.86421e+00	2.11281e+01
-6.74418e-01	-1.68251e+00	2.11281e+01
-7.08099e-01	-1.46706e+00	2.11281e+01
-7.19733e-01	-1.22304e+00	2.11281e+01
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-6.59533e-01	-6.72503e-01	2.11281e+01
-5.82727e-01	-3.78240e-01	2.11281e+01
-4.74976e-01	-7.95169e-02	2.11281e+01
-3.38911e-01	2.17660e-01	2.11281e+01
-1.85374e-01	5.08612e-01	2.11281e+01
-2.90398e-02	7.89376e-01	2.11281e+01
1.11864e-01	1.05679e+00	2.11281e+01
2.22428e-01	1.30745e+00	2.11281e+01
2.96727e-01	1.53689e+00	2.11281e+01
3.39115e-01	1.73943e+00	2.11281e+01
3.57053e-01	1.90953e+00	2.11281e+01
3.61096e-01	2.04189e+00	2.11281e+01
3.60762e-01	2.13218e+00	2.11281e+01
3.64492e-01	2.17713e+00	2.11281e+01
3.78665e-01	2.17472e+00	2.11281e+01
3.94032e-01	2.14093e+00	2.14447e+01
3.68842e-01	2.10018e+00	2.14447e+01
3.78449e-01	2.00970e+00	2.14447e+01
3.90661e-01	1.87660e+00	2.14447e+01
4.05539e-01	1.70355e+00	2.14447e+01
4.22906e-01	1.49405e+00	2.14447e+01
4.41833e-01	1.25246e+00	2.14447e+01
4.61114e-01	9.83833e-01	2.14447e+01
4.78898e-01	6.93874e-01	2.14447e+01
4.92548e-01	3.88863e-01	2.14447e+01
4.98493e-01	7.55536e-02	2.14447e+01
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4.70745e-01	-5.47740e-01	2.14447e+01
4.26517e-01	-8.43205e-01	2.14447e+01
3.56603e-01	-1.11870e+00	2.14447e+01
2.68466e-01	-1.36961e+00	2.14447e+01
1.68448e-01	-1.59159e+00	2.14447e+01
6.22076e-02	-1.78070e+00	2.14447e+01
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-1.48305e-01	-2.04765e+00	2.14447e+01
-2.42167e-01	-2.12107e+00	2.14447e+01
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-6.91420e-01	-1.44977e+00	2.14447e+01
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1.05429e-01	1.04250e+00	2.14447e+01
2.13289e-01	1.29016e+00	2.14447e+01
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3.26351e-01	1.71661e+00	2.14447e+01
3.43255e-01	1.88442e+00	2.14447e+01
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3.45886e-01	2.10397e+00	2.14447e+01
3.49433e-01	2.14829e+00	2.14447e+01
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3.88287e-01	2.11634e+00	2.17613e+01
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3.68303e-01	1.98741e+00	2.17613e+01
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4.13718e-01	1.47789e+00	2.17613e+01
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-5.30562e-01	-1.96113e+00	2.17613e+01
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1.01223e-01	1.03067e+00	2.17613e+01
2.07121e-01	1.27576e+00	2.17613e+01
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3.36840e-01	1.99250e+00	2.17613e+01
3.35850e-01	2.08049e+00	2.17613e+01
3.39278e-01	2.12430e+00	2.17613e+01
3.52982e-01	2.12208e+00	2.17613e+01
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3.48340e-01	2.05352e+00	2.20779e+01
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3.70923e-01	1.83526e+00	2.20779e+01
3.86440e-01	1.66631e+00	2.20779e+01
4.04595e-01	1.46177e+00	2.20779e+01
4.24455e-01	1.22590e+00	2.20779e+01
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4.82687e-01	-2.30692e-01	2.20779e+01
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7.08869e-02	-1.73901e+00	2.20779e+01
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-6.28383e-01	-6.54332e-01	2.20779e+01
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9.70546e-02	1.01889e+00	2.20779e+01
2.01003e-01	1.26142e+00	2.20779e+01
2.70240e-01	1.48313e+00	2.20779e+01
3.09016e-01	1.67862e+00	2.20779e+01
3.24573e-01	1.84263e+00	2.20779e+01
3.27129e-01	1.97014e+00	2.20779e+01
3.25896e-01	2.05709e+00	2.20779e+01
3.29206e-01	2.10039e+00	2.20779e+01
3.42707e-01	2.09824e+00	2.20779e+01
3.60863e-01	2.06994e+00	2.23945e+01
3.38130e-01	2.03023e+00	2.23945e+01
3.48173e-01	1.94298e+00	2.23945e+01
3.61092e-01	1.81462e+00	2.23945e+01
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3.95472e-01	1.44566e+00	2.23945e+01
4.15796e-01	1.21264e+00	2.23945e+01
4.36693e-01	9.53520e-01	2.23945e+01
4.56326e-01	6.73803e-01	2.23945e+01
4.72097e-01	3.79523e-01	2.23945e+01
4.80502e-01	7.71673e-02	2.23945e+01
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4.59025e-01	-5.24671e-01	2.23945e+01
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1.76133e-01	-1.53477e+00	2.23945e+01
7.51995e-02	-1.71820e+00	2.23945e+01
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-6.19001e-01	-6.48444e-01	2.23945e+01
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-3.25314e-01	2.02782e-01	2.23945e+01
-1.82938e-01	4.81634e-01	2.23945e+01
-3.77693e-02	7.50825e-01	2.23945e+01
9.28864e-02	1.00711e+00	2.23945e+01
1.94885e-01	1.24707e+00	2.23945e+01
2.62645e-01	1.46637e+00	2.23945e+01
3.00383e-01	1.65965e+00	2.23945e+01
3.15270e-01	1.82176e+00	2.23945e+01
3.17417e-01	1.94777e+00	2.23945e+01
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3.19134e-01	2.07648e+00	2.23945e+01
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3.49837e-01	2.04631e+00	2.27111e+01
3.27921e-01	2.00694e+00	2.27111e+01
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4.72620e-01	-2.22368e-01	2.27111e+01
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2.55050e-01	1.44960e+00	2.27111e+01
2.91750e-01	1.64069e+00	2.27111e+01
3.05967e-01	1.80090e+00	2.27111e+01
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3.38810e-01	2.02267e+00	2.30276e+01
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4.67587e-01	-2.18205e-01	2.30276e+01
4.51213e-01	-5.09318e-01	2.30276e+01
4.13735e-01	-7.88281e-01	2.30276e+01
3.51776e-01	-1.04880e+00	2.30276e+01
2.72303e-01	-1.28641e+00	2.30276e+01
1.81228e-01	-1.49693e+00	2.30276e+01
8.38247e-02	-1.67659e+00	2.30276e+01
-1.51759e-02	-1.82213e+00	2.30276e+01
-1.10703e-01	-1.93105e+00	2.30276e+01
-1.98083e-01	-2.00154e+00	2.30276e+01
-3.03237e-01	-2.02802e+00	2.30276e+01
-4.10416e-01	-1.96979e+00	2.30276e+01
-4.83003e-01	-1.87538e+00	2.30276e+01
-5.45525e-01	-1.74283e+00	2.30276e+01
-5.95575e-01	-1.57500e+00	2.30276e+01
-6.30801e-01	-1.37552e+00	2.30276e+01
-6.46073e-01	-1.14909e+00	2.30276e+01
-6.36988e-01	-9.00952e-01	2.30276e+01
-6.00239e-01	-6.36668e-01	2.30276e+01
-5.34292e-01	-3.61962e-01	2.30276e+01
-4.39685e-01	-8.25469e-02	2.30276e+01
-3.18753e-01	1.95951e-01	2.30276e+01
-1.81479e-01	4.68987e-01	2.30276e+01
-4.14208e-02	7.32611e-01	2.30276e+01
8.45500e-02	9.83554e-01	2.30276e+01
1.82650e-01	1.21839e+00	2.30276e+01
2.47456e-01	1.43284e+00	2.30276e+01
2.83116e-01	1.62172e+00	2.30276e+01
2.96663e-01	1.78004e+00	2.30276e+01
2.97995e-01	1.90304e+00	2.30276e+01
2.96032e-01	1.98689e+00	2.30276e+01
2.98991e-01	2.02866e+00	2.30276e+01
3.11881e-01	2.02673e+00	2.30276e+01
3.27783e-01	1.99904e+00	2.33442e+01
3.07501e-01	1.96036e+00	2.33442e+01
3.17977e-01	1.87634e+00	2.33442e+01
3.31601e-01	1.75271e+00	2.33442e+01
3.48385e-01	1.59195e+00	2.33442e+01
3.68104e-01	1.39732e+00	2.33442e+01
3.89819e-01	1.17285e+00	2.33442e+01
4.12323e-01	9.23235e-01	2.33442e+01
4.33797e-01	6.53747e-01	2.33442e+01
4.51680e-01	3.70184e-01	2.33442e+01
4.62535e-01	7.87684e-02	2.33442e+01
4.62554e-01	-2.14043e-01	2.33442e+01
4.47308e-01	-5.01642e-01	2.33442e+01
4.11177e-01	-7.77306e-01	2.33442e+01
3.50807e-01	-1.03483e+00	2.33442e+01
2.73064e-01	-1.26978e+00	2.33442e+01
1.83775e-01	-1.47802e+00	2.33442e+01
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-2.93064e-01	-2.00415e+00	2.33442e+01
-3.99115e-01	-1.94697e+00	2.33442e+01
-4.71113e-01	-1.85394e+00	2.33442e+01
-5.33292e-01	-1.72322e+00	2.33442e+01
-5.83263e-01	-1.55760e+00	2.33442e+01
-6.18693e-01	-1.36067e+00	2.33442e+01
-6.34499e-01	-1.13707e+00	2.33442e+01
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-3.15472e-01	1.92535e-01	2.33442e+01
-1.80749e-01	4.62663e-01	2.33442e+01
-4.32466e-02	7.23504e-01	2.33442e+01
8.03818e-02	9.71774e-01	2.33442e+01
1.76532e-01	1.20404e+00	2.33442e+01
2.39861e-01	1.41607e+00	2.33442e+01
2.74483e-01	1.60276e+00	2.33442e+01
2.87360e-01	1.75918e+00	2.33442e+01
2.88283e-01	1.88068e+00	2.33442e+01
2.86078e-01	1.96350e+00	2.33442e+01
2.88919e-01	2.00476e+00	2.33442e+01
3.01605e-01	2.00290e+00	2.33442e+01
3.16757e-01	1.97540e+00	2.36608e+01
2.97292e-01	1.93707e+00	2.36608e+01
3.07912e-01	1.85412e+00	2.36608e+01
3.21771e-01	1.73207e+00	2.36608e+01
3.38871e-01	1.57336e+00	2.36608e+01

3.58981e-01	1.38120e+00	2.36608e+01
3.81160e-01	1.15959e+00	2.36608e+01
4.04200e-01	9.13140e-01	2.36608e+01
4.26287e-01	6.47061e-01	2.36608e+01
4.44874e-01	3.67071e-01	2.36608e+01
4.56546e-01	7.93021e-02	2.36608e+01
4.57520e-01	-2.09881e-01	2.36608e+01
4.43402e-01	-4.93965e-01	2.36608e+01
4.08619e-01	-7.66332e-01	2.36608e+01
3.49838e-01	-1.02086e+00	2.36608e+01
2.73825e-01	-1.25316e+00	2.36608e+01
1.86322e-01	-1.45910e+00	2.36608e+01
9.24499e-02	-1.63498e+00	2.36608e+01
-3.19801e-03	-1.77758e+00	2.36608e+01
-9.56906e-02	-1.88444e+00	2.36608e+01
-1.80481e-01	-1.95376e+00	2.36608e+01
-2.82890e-01	-1.98027e+00	2.36608e+01
-3.87813e-01	-1.92414e+00	2.36608e+01
-4.59223e-01	-1.83251e+00	2.36608e+01
-5.21058e-01	-1.70360e+00	2.36608e+01
-5.70951e-01	-1.54021e+00	2.36608e+01
-6.06584e-01	-1.34583e+00	2.36608e+01
-6.22924e-01	-1.12505e+00	2.36608e+01
-6.15642e-01	-8.82935e-01	2.36608e+01
-5.81477e-01	-6.24892e-01	2.36608e+01
-5.18895e-01	-3.56497e-01	2.36608e+01
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-3.12191e-01	1.89120e-01	2.36608e+01
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7.62136e-02	9.59994e-01	2.36608e+01
1.70414e-01	1.18970e+00	2.36608e+01
2.32266e-01	1.39930e+00	2.36608e+01
2.65850e-01	1.58379e+00	2.36608e+01
2.78057e-01	1.73832e+00	2.36608e+01
2.78572e-01	1.85832e+00	2.36608e+01
2.76124e-01	1.94010e+00	2.36608e+01
2.78847e-01	1.98085e+00	2.36608e+01
2.91330e-01	1.97906e+00	2.36608e+01
3.05730e-01	1.95176e+00	2.39774e+01
2.87082e-01	1.91378e+00	2.39774e+01
2.97847e-01	1.83191e+00	2.39774e+01
3.11940e-01	1.71143e+00	2.39774e+01
3.29357e-01	1.55477e+00	2.39774e+01
3.49858e-01	1.36509e+00	2.39774e+01
3.72500e-01	1.14633e+00	2.39774e+01
3.96077e-01	9.03045e-01	2.39774e+01
4.18778e-01	6.40376e-01	2.39774e+01
4.38068e-01	3.63958e-01	2.39774e+01
4.50557e-01	7.98358e-02	2.39774e+01
4.52487e-01	-2.05718e-01	2.39774e+01
4.39496e-01	-4.86289e-01	2.39774e+01
4.06061e-01	-7.55357e-01	2.39774e+01
3.48868e-01	-1.00689e+00	2.39774e+01
2.74586e-01	-1.23653e+00	2.39774e+01
1.88869e-01	-1.44019e+00	2.39774e+01
9.67625e-02	-1.61417e+00	2.39774e+01
2.79094e-03	-1.75531e+00	2.39774e+01
-8.81845e-02	-1.86114e+00	2.39774e+01
-1.71680e-01	-1.92987e+00	2.39774e+01
-2.72716e-01	-1.95639e+00	2.39774e+01
-3.76512e-01	-1.90132e+00	2.39774e+01
-4.47333e-01	-1.81107e+00	2.39774e+01
-5.08825e-01	-1.68399e+00	2.39774e+01
-5.58638e-01	-1.52281e+00	2.39774e+01
-5.94476e-01	-1.33099e+00	2.39774e+01
-6.11350e-01	-1.11303e+00	2.39774e+01
-6.04969e-01	-8.73926e-01	2.39774e+01
-5.72096e-01	-6.19004e-01	2.39774e+01
-5.11197e-01	-3.53764e-01	2.39774e+01
-4.22737e-01	-8.37139e-02	2.39774e+01
-3.08910e-01	1.85704e-01	2.39774e+01
-1.79290e-01	4.50016e-01	2.39774e+01
-4.68981e-02	7.05290e-01	2.39774e+01
7.20455e-02	9.48214e-01	2.39774e+01
1.64296e-01	1.17535e+00	2.39774e+01
2.24671e-01	1.38254e+00	2.39774e+01
2.57217e-01	1.56482e+00	2.39774e+01
2.68753e-01	1.71745e+00	2.39774e+01
2.68860e-01	1.83595e+00	2.39774e+01
2.66169e-01	1.91670e+00	2.39774e+01
2.68775e-01	1.95694e+00	2.39774e+01
2.81054e-01	1.95522e+00	2.39774e+01
2.94703e-01	1.92813e+00	2.42940e+01
2.76872e-01	1.89049e+00	2.42940e+01
2.87782e-01	1.80970e+00	2.42940e+01
3.02110e-01	1.69080e+00	2.42940e+01
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3.40736e-01	1.34897e+00	2.42940e+01
3.63841e-01	1.13307e+00	2.42940e+01
3.87954e-01	8.92950e-01	2.42940e+01
4.11268e-01	6.33691e-01	2.42940e+01
4.31263e-01	3.60845e-01	2.42940e+01
4.44568e-01	8.03695e-02	2.42940e+01
4.47453e-01	-2.01556e-01	2.42940e+01
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3.47899e-01	-9.92920e-01	2.42940e+01
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-5.82367e-01	-1.31615e+00	2.42940e+01
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6.78773e-02	9.36434e-01	2.42940e+01
1.58178e-01	1.16101e+00	2.42940e+01
2.17077e-01	1.36577e+00	2.42940e+01
2.48584e-01	1.54586e+00	2.42940e+01
2.59450e-01	1.69659e+00	2.42940e+01
2.59149e-01	1.81359e+00	2.42940e+01
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2.58703e-01	1.93303e+00	2.42940e+01
2.70779e-01	1.93139e+00	2.42940e+01
2.88544e-01	1.90505e+00	2.46106e+01
2.67510e-01	1.86828e+00	2.46106e+01
2.78544e-01	1.78850e+00	2.46106e+01
2.93084e-01	1.67110e+00	2.46106e+01
3.11098e-01	1.51842e+00	2.46106e+01
3.32325e-01	1.33357e+00	2.46106e+01
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6.44725e-02	9.25169e-01	2.46106e+01
1.52532e-01	1.14733e+00	2.46106e+01
2.09838e-01	1.34981e+00	2.46106e+01
2.40335e-01	1.52780e+00	2.46106e+01
2.50666e-01	1.67672e+00	2.46106e+01
2.50130e-01	1.79227e+00	2.46106e+01
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2.76646e-01	1.88597e+00	2.49271e+01
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2.71540e-01	1.77006e+00	2.49271e+01
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4.19444e-01	3.54956e-01	2.49271e+01
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4.37846e-01	-1.94515e-01	2.49271e+01
4.26935e-01	-4.65244e-01	2.49271e+01
3.96085e-01	-7.25040e-01	2.49271e+01
3.42742e-01	-9.68184e-01	2.49271e+01
2.73092e-01	-1.19040e+00	2.49271e+01
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-2.88752e-01	1.75319e-01	2.49271e+01
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-4.67111e-02	6.79579e-01	2.49271e+01
6.31295e-02	9.15295e-01	2.49271e+01
1.48158e-01	1.13546e+00	2.49271e+01
2.03562e-01	1.33600e+00	2.49271e+01
2.33125e-01	1.51218e+00	2.49271e+01
2.43286e-01	1.65951e+00	2.49271e+01
2.42982e-01	1.77377e+00	2.49271e+01
2.40133e-01	1.85161e+00	2.49271e+01
2.42198e-01	1.89046e+00	2.49271e+01
2.54172e-01	1.88891e+00	2.49271e+01
2.68765e-01	1.86635e+00	2.52437e+01
2.53359e-01	1.82965e+00	2.52437e+01
2.64537e-01	1.75162e+00	2.52437e+01
2.79382e-01	1.63678e+00	2.52437e+01
2.97757e-01	1.48744e+00	2.52437e+01
3.19350e-01	1.30663e+00	2.52437e+01
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4.14049e-01	3.52108e-01	2.52437e+01
4.28854e-01	8.10898e-02	2.52437e+01
4.33175e-01	-1.91363e-01	2.52437e+01
4.22364e-01	-4.59130e-01	2.52437e+01
3.91715e-01	-7.16117e-01	2.52437e+01
3.39239e-01	-9.56736e-01	2.52437e+01
2.70880e-01	-1.17672e+00	2.52437e+01
1.91783e-01	-1.37207e+00	2.52437e+01
1.06569e-01	-1.53921e+00	2.52437e+01
1.94244e-02	-1.67504e+00	2.52437e+01
-6.51329e-02	-1.77715e+00	2.52437e+01
-1.43025e-01	-1.84376e+00	2.52437e+01
-2.39186e-01	-1.87016e+00	2.52437e+01
-3.32569e-01	-1.81940e+00	2.52437e+01
-3.96211e-01	-1.73457e+00	2.52437e+01
-4.51595e-01	-1.61445e+00	2.52437e+01
-4.96610e-01	-1.46161e+00	2.52437e+01
-5.29192e-01	-1.27932e+00	2.52437e+01
-5.44856e-01	-1.07179e+00	2.52437e+01
-5.39751e-01	-8.43681e-01	2.52437e+01
-5.10967e-01	-6.00040e-01	2.52437e+01
-4.57110e-01	-3.46080e-01	2.52437e+01
-3.78588e-01	-8.70598e-02	2.52437e+01
-2.77351e-01	1.71794e-01	2.52437e+01
-1.61964e-01	4.26052e-01	2.52437e+01
-4.40777e-02	6.71739e-01	2.52437e+01
6.17864e-02	9.05421e-01	2.52437e+01
1.43785e-01	1.12359e+00	2.52437e+01
1.97286e-01	1.32219e+00	2.52437e+01
2.25914e-01	1.49657e+00	2.52437e+01
2.35905e-01	1.64230e+00	2.52437e+01
2.35834e-01	1.75528e+00	2.52437e+01
2.33192e-01	1.83223e+00	2.52437e+01
2.34990e-01	1.87069e+00	2.52437e+01
2.47002e-01	1.86916e+00	2.52437e+01
2.60885e-01	1.84674e+00	2.55603e+01
2.46283e-01	1.81034e+00	2.55603e+01
2.57533e-01	1.73319e+00	2.55603e+01
2.72531e-01	1.61962e+00	2.55603e+01
2.91086e-01	1.47194e+00	2.55603e+01
3.12863e-01	1.29315e+00	2.55603e+01
3.37028e-01	1.08694e+00	2.55603e+01
3.62273e-01	8.57597e-01	2.55603e+01
3.86956e-01	6.09943e-01	2.55603e+01
4.08654e-01	3.49260e-01	2.55603e+01
4.23858e-01	8.12354e-02	2.55603e+01
4.28503e-01	-1.88212e-01	2.55603e+01
4.17794e-01	-4.53016e-01	2.55603e+01
3.87345e-01	-7.07195e-01	2.55603e+01
3.35736e-01	-9.45288e-01	2.55603e+01
2.68668e-01	-1.16304e+00	2.55603e+01
1.91126e-01	-1.35648e+00	2.55603e+01
1.07603e-01	-1.52204e+00	2.55603e+01
2.21879e-02	-1.65665e+00	2.55603e+01
-6.06982e-02	-1.75791e+00	2.55603e+01
-1.37108e-01	-1.82404e+00	2.55603e+01
-2.32168e-01	-1.85037e+00	2.55603e+01
-3.21823e-01	-1.80071e+00	2.55603e+01
-3.82752e-01	-1.71720e+00	2.55603e+01
-4.35708e-01	-1.59876e+00	2.55603e+01
-4.78669e-01	-1.44791e+00	2.55603e+01
-5.09661e-01	-1.26787e+00	2.55603e+01
-5.24386e-01	-1.06278e+00	2.55603e+01
-5.19155e-01	-8.37223e-01	2.55603e+01
-4.91188e-01	-5.96174e-01	2.55603e+01
-4.39152e-01	-3.44778e-01	2.55603e+01
-3.63445e-01	-8.82371e-02	2.55603e+01

-2.65949e-01	1.68270e-01	2.55603e+01
-1.54888e-01	4.20314e-01	2.55603e+01
-4.14442e-02	6.63899e-01	2.55603e+01
6.04434e-02	8.95546e-01	2.55603e+01
1.39411e-01	1.11172e+00	2.55603e+01
1.91010e-01	1.30838e+00	2.55603e+01
2.18704e-01	1.48096e+00	2.55603e+01
2.28525e-01	1.62509e+00	2.55603e+01
2.28686e-01	1.73678e+00	2.55603e+01
2.26250e-01	1.81285e+00	2.55603e+01
2.27782e-01	1.85091e+00	2.55603e+01
2.39831e-01	1.84941e+00	2.55603e+01
2.53004e-01	1.82713e+00	2.58769e+01
2.39208e-01	1.79103e+00	2.58769e+01
2.50530e-01	1.71475e+00	2.58769e+01
2.65680e-01	1.60246e+00	2.58769e+01
2.84415e-01	1.45645e+00	2.58769e+01
3.06375e-01	1.27968e+00	2.58769e+01
3.30761e-01	1.07579e+00	2.58769e+01
3.56233e-01	8.49039e-01	2.58769e+01
3.81197e-01	6.04174e-01	2.58769e+01
4.03259e-01	3.46413e-01	2.58769e+01
4.18861e-01	8.13811e-02	2.58769e+01
4.23832e-01	-1.85060e-01	2.58769e+01
4.13223e-01	-4.46902e-01	2.58769e+01
3.82976e-01	-6.98272e-01	2.58769e+01
3.32234e-01	-9.33840e-01	2.58769e+01
2.66455e-01	-1.14936e+00	2.58769e+01
1.90468e-01	-1.34089e+00	2.58769e+01
1.08637e-01	-1.50488e+00	2.58769e+01
2.49515e-02	-1.63827e+00	2.58769e+01
-5.62635e-02	-1.73867e+00	2.58769e+01
-1.31192e-01	-1.80431e+00	2.58769e+01
-2.25150e-01	-1.83059e+00	2.58769e+01
-3.11077e-01	-1.78202e+00	2.58769e+01
-3.69293e-01	-1.69983e+00	2.58769e+01
-4.19821e-01	-1.58307e+00	2.58769e+01
-4.60728e-01	-1.43421e+00	2.58769e+01
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-3.48303e-01	-8.94145e-02	2.58769e+01
-2.54548e-01	1.64745e-01	2.58769e+01
-1.47812e-01	4.14576e-01	2.58769e+01
-3.88108e-02	6.56060e-01	2.58769e+01
5.91003e-02	8.85672e-01	2.58769e+01
1.35037e-01	1.09985e+00	2.58769e+01
1.84734e-01	1.29458e+00	2.58769e+01
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2.21538e-01	1.71829e+00	2.58769e+01
2.19309e-01	1.79347e+00	2.58769e+01
2.20574e-01	1.83113e+00	2.58769e+01
2.32661e-01	1.82966e+00	2.58769e+01
2.45123e-01	1.80751e+00	2.61935e+01
2.32133e-01	1.77171e+00	2.61935e+01
2.43526e-01	1.69631e+00	2.61935e+01
2.58829e-01	1.58530e+00	2.61935e+01
2.77745e-01	1.44096e+00	2.61935e+01
2.99888e-01	1.26621e+00	2.61935e+01
3.24494e-01	1.06464e+00	2.61935e+01
3.50193e-01	8.40481e-01	2.61935e+01
3.75438e-01	5.98404e-01	2.61935e+01
3.97865e-01	3.43565e-01	2.61935e+01
4.13864e-01	8.15268e-02	2.61935e+01
4.19160e-01	-1.81908e-01	2.61935e+01
4.08652e-01	-4.40788e-01	2.61935e+01
3.78606e-01	-6.89350e-01	2.61935e+01
3.28731e-01	-9.22392e-01	2.61935e+01
2.64243e-01	-1.13569e+00	2.61935e+01
1.89811e-01	-1.32530e+00	2.61935e+01
1.09671e-01	-1.48771e+00	2.61935e+01
2.77151e-02	-1.61988e+00	2.61935e+01
-5.18288e-02	-1.71943e+00	2.61935e+01
-1.25275e-01	-1.78459e+00	2.61935e+01
-2.18132e-01	-1.81080e+00	2.61935e+01
-3.00332e-01	-1.76332e+00	2.61935e+01
-3.55834e-01	-1.68247e+00	2.61935e+01
-4.03933e-01	-1.56738e+00	2.61935e+01
-4.42787e-01	-1.42051e+00	2.61935e+01
-4.70600e-01	-1.24498e+00	2.61935e+01
-4.83445e-01	-1.04475e+00	2.61935e+01
-4.77964e-01	-8.24306e-01	2.61935e+01
-4.51630e-01	-5.88440e-01	2.61935e+01
-4.03235e-01	-3.42173e-01	2.61935e+01
-3.33160e-01	-9.05919e-02	2.61935e+01
-2.43147e-01	1.61221e-01	2.61935e+01
-1.40736e-01	4.08838e-01	2.61935e+01
-3.61774e-02	6.48220e-01	2.61935e+01
5.77573e-02	8.75798e-01	2.61935e+01
1.30664e-01	1.08798e+00	2.61935e+01
1.78458e-01	1.28077e+00	2.61935e+01
2.04283e-01	1.44973e+00	2.61935e+01
2.13764e-01	1.59067e+00	2.61935e+01
2.14389e-01	1.69979e+00	2.61935e+01

2.12368e-01	1.77409e+00	2.61935e+01
2.13367e-01	1.81136e+00	2.61935e+01
2.25491e-01	1.80991e+00	2.61935e+01
2.37242e-01	1.78790e+00	2.65101e+01
2.25057e-01	1.75240e+00	2.65101e+01
2.36523e-01	1.67787e+00	2.65101e+01
2.51978e-01	1.56815e+00	2.65101e+01
2.71074e-01	1.42547e+00	2.65101e+01
2.93400e-01	1.25273e+00	2.65101e+01
3.18228e-01	1.05350e+00	2.65101e+01
3.44153e-01	8.31923e-01	2.65101e+01
3.69680e-01	5.92634e-01	2.65101e+01
3.92470e-01	3.40717e-01	2.65101e+01
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4.14489e-01	-1.78756e-01	2.65101e+01
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2.62031e-01	-1.12201e+00	2.65101e+01
1.89154e-01	-1.30971e+00	2.65101e+01
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-4.73941e-02	-1.70019e+00	2.65101e+01
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-2.89586e-01	-1.74463e+00	2.65101e+01
-3.42375e-01	-1.66510e+00	2.65101e+01
-3.88046e-01	-1.55169e+00	2.65101e+01
-4.24846e-01	-1.40681e+00	2.65101e+01
-4.51069e-01	-1.23353e+00	2.65101e+01
-4.62975e-01	-1.03574e+00	2.65101e+01
-4.57369e-01	-8.17848e-01	2.65101e+01
-4.31851e-01	-5.84574e-01	2.65101e+01
-3.85276e-01	-3.40871e-01	2.65101e+01
-3.18018e-01	-9.17692e-02	2.65101e+01
-2.31745e-01	1.57696e-01	2.65101e+01
-1.33660e-01	4.03100e-01	2.65101e+01
-3.35440e-02	6.40380e-01	2.65101e+01
5.64142e-02	8.65924e-01	2.65101e+01
1.26290e-01	1.07611e+00	2.65101e+01
1.72182e-01	1.26696e+00	2.65101e+01
1.97073e-01	1.43411e+00	2.65101e+01
2.06383e-01	1.57346e+00	2.65101e+01
2.07241e-01	1.68130e+00	2.65101e+01
2.05426e-01	1.75471e+00	2.65101e+01
2.06159e-01	1.79158e+00	2.65101e+01
2.18320e-01	1.79016e+00	2.65101e+01
2.29362e-01	1.76829e+00	2.68266e+01
2.17982e-01	1.73309e+00	2.68266e+01
2.29519e-01	1.65943e+00	2.68266e+01
2.45127e-01	1.55099e+00	2.68266e+01
2.64403e-01	1.40997e+00	2.68266e+01
2.86913e-01	1.23926e+00	2.68266e+01
3.11961e-01	1.04235e+00	2.68266e+01
3.38113e-01	8.23365e-01	2.68266e+01
3.63921e-01	5.86864e-01	2.68266e+01
3.87075e-01	3.37869e-01	2.68266e+01
4.03871e-01	8.18182e-02	2.68266e+01
4.09817e-01	-1.75605e-01	2.68266e+01
3.99511e-01	-4.28560e-01	2.68266e+01
3.69866e-01	-6.71505e-01	2.68266e+01
3.21725e-01	-8.99496e-01	2.68266e+01
2.59819e-01	-1.10833e+00	2.68266e+01
1.88497e-01	-1.29412e+00	2.68266e+01
1.11739e-01	-1.45338e+00	2.68266e+01
3.32422e-02	-1.58311e+00	2.68266e+01
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-3.09105e-02	6.32540e-01	2.68266e+01
5.50711e-02	8.56049e-01	2.68266e+01
1.21917e-01	1.06424e+00	2.68266e+01
1.65906e-01	1.25315e+00	2.68266e+01
1.89863e-01	1.41850e+00	2.68266e+01
1.99003e-01	1.55626e+00	2.68266e+01
2.00093e-01	1.66280e+00	2.68266e+01
1.98485e-01	1.73533e+00	2.68266e+01
1.98951e-01	1.77181e+00	2.68266e+01
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2.21481e-01	1.74867e+00	2.71432e+01
2.10906e-01	1.71378e+00	2.71432e+01
2.22516e-01	1.64099e+00	2.71432e+01
2.38276e-01	1.53383e+00	2.71432e+01
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4.05146e-01	-1.72453e-01	2.71432e+01
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-2.87733e-01	-9.41240e-02	2.71432e+01
-2.08942e-01	1.50647e-01	2.71432e+01
-1.19508e-01	3.91624e-01	2.71432e+01
-2.82771e-02	6.24700e-01	2.71432e+01
5.37281e-02	8.46175e-01	2.71432e+01
1.17543e-01	1.05237e+00	2.71432e+01
1.59630e-01	1.23934e+00	2.71432e+01
1.82652e-01	1.40288e+00	2.71432e+01
1.91622e-01	1.53905e+00	2.71432e+01
1.92945e-01	1.64431e+00	2.71432e+01
1.91544e-01	1.71595e+00	2.71432e+01
1.91743e-01	1.75203e+00	2.71432e+01
2.03980e-01	1.75066e+00	2.71432e+01
2.13600e-01	1.72906e+00	2.74598e+01
2.03831e-01	1.69446e+00	2.74598e+01
2.15512e-01	1.62255e+00	2.74598e+01
2.31425e-01	1.51667e+00	2.74598e+01
2.51062e-01	1.37899e+00	2.74598e+01
2.73938e-01	1.21231e+00	2.74598e+01
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3.93878e-01	8.21096e-02	2.74598e+01
4.00474e-01	-1.69301e-01	2.74598e+01
3.90370e-01	-4.16332e-01	2.74598e+01
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3.14720e-01	-8.76600e-01	2.74598e+01
2.55394e-01	-1.08098e+00	2.74598e+01
1.87182e-01	-1.26294e+00	2.74598e+01
1.13806e-01	-1.41905e+00	2.74598e+01
3.87694e-02	-1.54634e+00	2.74598e+01
-3.40900e-02	-1.64247e+00	2.74598e+01
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-1.90060e-01	-1.73165e+00	2.74598e+01
-2.57349e-01	-1.68856e+00	2.74598e+01
-3.01997e-01	-1.61300e+00	2.74598e+01
-3.40384e-01	-1.50463e+00	2.74598e+01
-3.71022e-01	-1.36572e+00	2.74598e+01
-3.92477e-01	-1.19918e+00	2.74598e+01
-4.01564e-01	-1.00871e+00	2.74598e+01
-3.95582e-01	-7.98473e-01	2.74598e+01
-3.72514e-01	-5.72973e-01	2.74598e+01
-3.31400e-01	-3.36964e-01	2.74598e+01
-2.72590e-01	-9.53013e-02	2.74598e+01
-1.97541e-01	1.47123e-01	2.74598e+01
-1.12432e-01	3.85886e-01	2.74598e+01
-2.56437e-02	6.16861e-01	2.74598e+01
5.23850e-02	8.36301e-01	2.74598e+01
1.13169e-01	1.04050e+00	2.74598e+01
1.53354e-01	1.22553e+00	2.74598e+01
1.75442e-01	1.38727e+00	2.74598e+01
1.84242e-01	1.52184e+00	2.74598e+01
1.85797e-01	1.62581e+00	2.74598e+01
1.84602e-01	1.69657e+00	2.74598e+01
1.84535e-01	1.73225e+00	2.74598e+01
1.96809e-01	1.73091e+00	2.74598e+01
2.12133e-01	1.71070e+00	2.77764e+01
1.97890e-01	1.67695e+00	2.77764e+01
2.09612e-01	1.60583e+00	2.77764e+01
2.25613e-01	1.50110e+00	2.77764e+01
2.45362e-01	1.36491e+00	2.77764e+01
2.68363e-01	1.20006e+00	2.77764e+01
2.94005e-01	1.00990e+00	2.77764e+01
3.20783e-01	7.98425e-01	2.77764e+01
3.47368e-01	5.70011e-01	2.77764e+01
3.71526e-01	3.29494e-01	2.77764e+01
3.89452e-01	8.21259e-02	2.77764e+01
3.96410e-01	-1.66587e-01	2.77764e+01
3.86655e-01	-4.10980e-01	2.77764e+01
3.57899e-01	-6.45810e-01	2.77764e+01

3.12353e-01	-8.66472e-01	2.77764e+01
2.54125e-01	-1.06882e+00	2.77764e+01
1.87153e-01	-1.24902e+00	2.77764e+01
1.15079e-01	-1.40365e+00	2.77764e+01
4.13413e-02	-1.52979e+00	2.77764e+01
-3.02884e-02	-1.62510e+00	2.77764e+01
-9.67243e-02	-1.68782e+00	2.77764e+01
-1.84182e-01	-1.71371e+00	2.77764e+01
-2.49507e-01	-1.67140e+00	2.77764e+01
-2.92793e-01	-1.59689e+00	2.77764e+01
-3.30011e-01	-1.48988e+00	2.77764e+01
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-3.21346e-01	-3.34887e-01	2.77764e+01
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-1.91593e-01	1.44574e-01	2.77764e+01
-1.09094e-01	3.81149e-01	2.77764e+01
-2.49686e-02	6.10028e-01	2.77764e+01
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1.09574e-01	1.02973e+00	2.77764e+01
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1.69907e-01	1.37302e+00	2.77764e+01
1.78448e-01	1.50617e+00	2.77764e+01
1.79991e-01	1.60901e+00	2.77764e+01
1.78823e-01	1.67899e+00	2.77764e+01
1.78631e-01	1.71430e+00	2.77764e+01
1.90828e-01	1.71299e+00	2.77764e+01
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1.92851e-01	1.66086e+00	2.80930e+01
2.04590e-01	1.59046e+00	2.80930e+01
2.20627e-01	1.48678e+00	2.80930e+01
2.40434e-01	1.35197e+00	2.80930e+01
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3.67272e-01	3.26947e-01	2.80930e+01
3.85479e-01	8.20393e-02	2.80930e+01
3.92828e-01	-1.64220e-01	2.80930e+01
3.83620e-01	-4.06233e-01	2.80930e+01
3.55580e-01	-6.38812e-01	2.80930e+01
3.10888e-01	-8.57393e-01	2.80930e+01
2.53606e-01	-1.05786e+00	2.80930e+01
1.87623e-01	-1.23642e+00	2.80930e+01
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4.37610e-02	-1.51470e+00	2.80930e+01
-2.69898e-02	-1.60921e+00	2.80930e+01
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-2.86971e-01	-1.58176e+00	2.80930e+01
-3.24022e-01	-1.47587e+00	2.80930e+01
-3.53691e-01	-1.34002e+00	2.80930e+01
-3.74597e-01	-1.17705e+00	2.80930e+01
-3.83675e-01	-9.90568e-01	2.80930e+01
-3.78324e-01	-7.84619e-01	2.80930e+01
-3.56602e-01	-5.63613e-01	2.80930e+01
-3.17574e-01	-3.32194e-01	2.80930e+01
-2.61571e-01	-9.51227e-02	2.80930e+01
-1.89980e-01	1.42802e-01	2.80930e+01
-1.08728e-01	3.77207e-01	2.80930e+01
-2.58503e-02	6.03996e-01	2.80930e+01
4.86366e-02	8.19431e-01	2.80930e+01
1.06597e-01	1.01982e+00	2.80930e+01
1.44817e-01	1.20131e+00	2.80930e+01
1.65704e-01	1.35987e+00	2.80930e+01
1.73915e-01	1.49172e+00	2.80930e+01
1.75254e-01	1.59355e+00	2.80930e+01
1.73968e-01	1.66285e+00	2.80930e+01
1.73762e-01	1.69781e+00	2.80930e+01
1.85792e-01	1.69654e+00	2.80930e+01
1.96306e-01	1.67846e+00	2.84095e+01
1.87813e-01	1.64477e+00	2.84095e+01
1.99567e-01	1.57509e+00	2.84095e+01
2.15641e-01	1.47247e+00	2.84095e+01
2.35505e-01	1.33904e+00	2.84095e+01
2.58664e-01	1.17750e+00	2.84095e+01
2.84505e-01	9.91172e-01	2.84095e+01
3.11539e-01	7.83941e-01	2.84095e+01
3.38446e-01	5.60108e-01	2.84095e+01
3.63018e-01	3.24400e-01	2.84095e+01
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3.89246e-01	-1.61854e-01	2.84095e+01
3.80585e-01	-4.01486e-01	2.84095e+01
3.53261e-01	-6.31814e-01	2.84095e+01
3.09423e-01	-8.48314e-01	2.84095e+01
2.53086e-01	-1.04690e+00	2.84095e+01
1.88094e-01	-1.22381e+00	2.84095e+01
1.18006e-01	-1.37568e+00	2.84095e+01
4.61806e-02	-1.49961e+00	2.84095e+01
-2.36913e-02	-1.59332e+00	2.84095e+01
-8.85918e-02	-1.65505e+00	2.84095e+01
-1.74238e-01	-1.68076e+00	2.84095e+01
-2.38437e-01	-1.63952e+00	2.84095e+01

-2.81149e-01	-1.56663e+00	2.84095e+01
-3.18032e-01	-1.46186e+00	2.84095e+01
-3.47663e-01	-1.32741e+00	2.84095e+01
-3.68669e-01	-1.16610e+00	2.84095e+01
-3.78008e-01	-9.81478e-01	2.84095e+01
-3.73098e-01	-7.77560e-01	2.84095e+01
-3.52007e-01	-5.58702e-01	2.84095e+01
-3.13803e-01	-3.29502e-01	2.84095e+01
-2.58800e-01	-9.46731e-02	2.84095e+01
-1.88367e-01	1.41029e-01	2.84095e+01
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4.66113e-02	8.11405e-01	2.84095e+01
1.03620e-01	1.00992e+00	2.84095e+01
1.41120e-01	1.18969e+00	2.84095e+01
1.61501e-01	1.34671e+00	2.84095e+01
1.69382e-01	1.47727e+00	2.84095e+01
1.70516e-01	1.57810e+00	2.84095e+01
1.69112e-01	1.64671e+00	2.84095e+01
1.68893e-01	1.68131e+00	2.84095e+01
1.80756e-01	1.68008e+00	2.84095e+01
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1.82774e-01	1.62868e+00	2.87261e+01
1.94545e-01	1.55972e+00	2.87261e+01
2.10655e-01	1.45816e+00	2.87261e+01
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2.53814e-01	1.16622e+00	2.87261e+01
2.79755e-01	9.81805e-01	2.87261e+01
3.06916e-01	7.76699e-01	2.87261e+01
3.33984e-01	5.55156e-01	2.87261e+01
3.58764e-01	3.21853e-01	2.87261e+01
3.77532e-01	8.18662e-02	2.87261e+01
3.85664e-01	-1.59487e-01	2.87261e+01
3.77550e-01	-3.96740e-01	2.87261e+01
3.50942e-01	-6.24816e-01	2.87261e+01
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2.52566e-01	-1.03594e+00	2.87261e+01
1.88564e-01	-1.21121e+00	2.87261e+01
1.19469e-01	-1.36170e+00	2.87261e+01
4.86002e-02	-1.48453e+00	2.87261e+01
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-1.69266e-01	-1.66428e+00	2.87261e+01
-2.32902e-01	-1.62358e+00	2.87261e+01
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-3.72342e-01	-9.72389e-01	2.87261e+01
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-3.47413e-01	-5.53790e-01	2.87261e+01
-3.10031e-01	-3.26809e-01	2.87261e+01
-2.56030e-01	-9.42235e-02	2.87261e+01
-1.86754e-01	1.39257e-01	2.87261e+01
-1.07994e-01	3.69324e-01	2.87261e+01
-2.76135e-02	5.91932e-01	2.87261e+01
4.45860e-02	8.03380e-01	2.87261e+01
1.00643e-01	1.00002e+00	2.87261e+01
1.37423e-01	1.17806e+00	2.87261e+01
1.57299e-01	1.33355e+00	2.87261e+01
1.64850e-01	1.46282e+00	2.87261e+01
1.65778e-01	1.56264e+00	2.87261e+01
1.64257e-01	1.63056e+00	2.87261e+01
1.64024e-01	1.66482e+00	2.87261e+01
1.75720e-01	1.66363e+00	2.87261e+01
1.85509e-01	1.64568e+00	2.90427e+01
1.77735e-01	1.61259e+00	2.90427e+01
1.89522e-01	1.54435e+00	2.90427e+01
2.05669e-01	1.44385e+00	2.90427e+01
2.25649e-01	1.31316e+00	2.90427e+01
2.48965e-01	1.15494e+00	2.90427e+01
2.75005e-01	9.72439e-01	2.90427e+01
3.02294e-01	7.69457e-01	2.90427e+01
3.29523e-01	5.50205e-01	2.90427e+01
3.54510e-01	3.19306e-01	2.90427e+01
3.73559e-01	8.17796e-02	2.90427e+01
3.82082e-01	-1.57120e-01	2.90427e+01
3.74515e-01	-3.91993e-01	2.90427e+01
3.48623e-01	-6.17819e-01	2.90427e+01
3.06494e-01	-8.30156e-01	2.90427e+01
2.52047e-01	-1.02499e+00	2.90427e+01
1.89034e-01	-1.19861e+00	2.90427e+01
1.20932e-01	-1.34771e+00	2.90427e+01
5.10199e-02	-1.46944e+00	2.90427e+01
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-8.04594e-02	-1.62229e+00	2.90427e+01
-1.64293e-01	-1.64781e+00	2.90427e+01
-2.27367e-01	-1.60764e+00	2.90427e+01
-2.69504e-01	-1.53637e+00	2.90427e+01
-3.06052e-01	-1.43384e+00	2.90427e+01
-3.35608e-01	-1.30219e+00	2.90427e+01
-3.56814e-01	-1.14419e+00	2.90427e+01
-3.66675e-01	-9.63299e-01	2.90427e+01
-3.62647e-01	-7.63440e-01	2.90427e+01
-3.42819e-01	-5.48879e-01	2.90427e+01
-3.06260e-01	-3.24117e-01	2.90427e+01
-2.53260e-01	-9.37739e-02	2.90427e+01

-1.85141e-01	1.37484e-01	2.90427e+01
-1.07628e-01	3.65383e-01	2.90427e+01
-2.84952e-02	5.85901e-01	2.90427e+01
4.25607e-02	7.95354e-01	2.90427e+01
9.76662e-02	9.90120e-01	2.90427e+01
1.33725e-01	1.16643e+00	2.90427e+01
1.53096e-01	1.32039e+00	2.90427e+01
1.60317e-01	1.44837e+00	2.90427e+01
1.61041e-01	1.54719e+00	2.90427e+01
1.59401e-01	1.61442e+00	2.90427e+01
1.59155e-01	1.64832e+00	2.90427e+01
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1.80111e-01	1.62929e+00	2.93593e+01
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2.00683e-01	1.42953e+00	2.93593e+01
2.20720e-01	1.30022e+00	2.93593e+01
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3.78500e-01	-1.54753e-01	2.93593e+01
3.71480e-01	-3.87246e-01	2.93593e+01
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3.05029e-01	-8.21077e-01	2.93593e+01
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5.34395e-02	-1.45435e+00	2.93593e+01
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-3.02488e-01	-3.21424e-01	2.93593e+01
-2.50490e-01	-9.33244e-02	2.93593e+01
-1.83528e-01	1.35712e-01	2.93593e+01
-1.07261e-01	3.61442e-01	2.93593e+01
-2.93768e-02	5.79669e-01	2.93593e+01
4.05354e-02	7.87328e-01	2.93593e+01
9.46893e-02	9.80219e-01	2.93593e+01
1.30028e-01	1.15480e+00	2.93593e+01
1.48893e-01	1.30723e+00	2.93593e+01
1.55785e-01	1.43392e+00	2.93593e+01
1.56303e-01	1.53173e+00	2.93593e+01
1.54545e-01	1.59827e+00	2.93593e+01
1.54286e-01	1.63183e+00	2.93593e+01
1.65647e-01	1.63071e+00	2.93593e+01
1.74712e-01	1.61290e+00	2.96759e+01
1.67657e-01	1.58041e+00	2.96759e+01
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-2.16297e-01	-1.57576e+00	2.96759e+01
-2.57860e-01	-1.50612e+00	2.96759e+01
-2.94073e-01	-1.40582e+00	2.96759e+01
-3.23552e-01	-1.27698e+00	2.96759e+01
-3.44958e-01	-1.12228e+00	2.96759e+01
-3.55342e-01	-9.45119e-01	2.96759e+01
-3.52196e-01	-7.49320e-01	2.96759e+01
-3.33631e-01	-5.39056e-01	2.96759e+01
-2.98716e-01	-3.18732e-01	2.96759e+01
-2.47719e-01	-9.28748e-02	2.96759e+01
-1.81915e-01	1.33940e-01	2.96759e+01
-1.06895e-01	3.57500e-01	2.96759e+01
-3.02584e-02	5.73837e-01	2.96759e+01
3.85101e-02	7.79303e-01	2.96759e+01
9.17124e-02	9.70317e-01	2.96759e+01
1.26330e-01	1.14318e+00	2.96759e+01
1.44690e-01	1.29407e+00	2.96759e+01
1.51252e-01	1.41947e+00	2.96759e+01
1.51565e-01	1.51628e+00	2.96759e+01

1.49690e-01	1.58213e+00	2.96759e+01
1.49417e-01	1.61533e+00	2.96759e+01
1.60611e-01	1.61426e+00	2.96759e+01
1.69314e-01	1.59651e+00	2.99925e+01
1.62618e-01	1.56432e+00	2.99925e+01
1.74455e-01	1.49824e+00	2.99925e+01
1.90711e-01	1.40091e+00	2.99925e+01
2.10863e-01	1.27434e+00	2.99925e+01
2.34416e-01	1.12110e+00	2.99925e+01
2.60755e-01	9.44339e-01	2.99925e+01
2.88428e-01	7.47732e-01	2.99925e+01
3.16140e-01	5.35350e-01	2.99925e+01
3.41748e-01	3.11665e-01	2.99925e+01
3.61640e-01	8.15198e-02	2.99925e+01
3.71336e-01	-1.50020e-01	2.99925e+01
3.65410e-01	-3.77753e-01	2.99925e+01
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3.02100e-01	-8.02919e-01	2.99925e+01
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1.26785e-01	-1.29177e+00	3.03090e+01
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1.52541e-01	1.53214e+00	3.06256e+01
1.64410e-01	1.46750e+00	3.06256e+01
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3.07217e-01	5.25447e-01	3.06256e+01
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2.99171e-01	-7.84762e-01	3.06256e+01
2.49449e-01	-9.70204e-01	3.06256e+01
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-3.05469e-01	-1.23915e+00	3.06256e+01
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-3.38342e-01	-9.17849e-01	3.06256e+01
-3.36519e-01	-7.28141e-01	3.06256e+01
-3.19848e-01	-5.24322e-01	3.06256e+01
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-2.39408e-01	-9.15260e-02	3.06256e+01
-1.77076e-01	1.28622e-01	3.06256e+01
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-3.29033e-02	5.55741e-01	3.06256e+01
3.24342e-02	7.55226e-01	3.06256e+01
8.27816e-02	9.40612e-01	3.06256e+01
1.15238e-01	1.10829e+00	3.06256e+01
1.32081e-01	1.25460e+00	3.06256e+01
1.37654e-01	1.37612e+00	3.06256e+01
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1.34810e-01	1.56584e+00	3.06256e+01
1.45503e-01	1.56489e+00	3.06256e+01
1.56300e-01	1.54935e+00	3.09422e+01
1.48650e-01	1.51821e+00	3.09422e+01
1.60513e-01	1.45418e+00	3.09422e+01
1.76845e-01	1.35987e+00	3.09422e+01
1.97126e-01	1.23722e+00	3.09422e+01
2.20859e-01	1.08874e+00	3.09422e+01
2.47432e-01	9.17444e-01	3.09422e+01
2.75412e-01	7.26911e-01	3.09422e+01
3.03523e-01	5.21079e-01	3.09422e+01
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3.61035e-01	-1.43352e-01	3.09422e+01
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1.29128e-01	-1.26575e+00	3.09422e+01
6.47890e-02	-1.38100e+00	3.09422e+01
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-1.75645e-01	1.27145e-01	3.09422e+01
-1.05375e-01	3.42305e-01	3.09422e+01
-3.34978e-02	5.50543e-01	3.09422e+01
3.09121e-02	7.48286e-01	3.09422e+01
8.04935e-02	9.32038e-01	3.09422e+01
1.12378e-01	1.09822e+00	3.09422e+01
1.28828e-01	1.24319e+00	3.09422e+01
1.34153e-01	1.36361e+00	3.09422e+01
1.33702e-01	1.45652e+00	3.09422e+01
1.31391e-01	1.51971e+00	3.09422e+01
1.31069e-01	1.55156e+00	3.09422e+01
1.41624e-01	1.55064e+00	3.09422e+01
1.50417e-01	1.53572e+00	3.12588e+01
1.44971e-01	1.50467e+00	3.12588e+01
1.56825e-01	1.44124e+00	3.12588e+01
1.73154e-01	1.34781e+00	3.12588e+01
1.93439e-01	1.22631e+00	3.12588e+01
2.17185e-01	1.07920e+00	3.12588e+01
2.43779e-01	9.09504e-01	3.12588e+01
2.71799e-01	7.20742e-01	3.12588e+01
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3.57981e-01	-1.41498e-01	3.12588e+01
3.53935e-01	-3.60418e-01	3.12588e+01
3.32699e-01	-5.71143e-01	3.12588e+01

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2.47929e-01	-9.51723e-01	3.12588e+01
1.91442e-01	-1.11429e+00	3.12588e+01
1.29900e-01	-1.25407e+00	3.12588e+01
6.63216e-02	-1.36837e+00	3.12588e+01
4.03958e-03	-1.45505e+00	3.12588e+01
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-1.31998e-01	-1.53730e+00	3.12588e+01
-1.91274e-01	-1.50069e+00	3.12588e+01
-2.31444e-01	-1.43483e+00	3.12588e+01
-2.66805e-01	-1.33977e+00	3.12588e+01
-2.96024e-01	-1.21750e+00	3.12588e+01
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-2.34754e-01	-9.06041e-02	3.12588e+01
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2.94830e-02	7.41548e-01	3.12588e+01
7.83326e-02	9.23709e-01	3.12588e+01
1.09673e-01	1.08843e+00	3.12588e+01
1.25751e-01	1.23212e+00	3.12588e+01
1.30843e-01	1.35144e+00	3.12588e+01
1.30254e-01	1.44352e+00	3.12588e+01
1.27867e-01	1.50613e+00	3.12588e+01
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1.30672e-01	-1.24239e+00	3.15754e+01
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-1.72850e-01	1.24300e-01	3.15754e+01
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-3.45807e-02	5.40455e-01	3.15754e+01
2.80538e-02	7.34809e-01	3.15754e+01
7.61717e-02	9.15380e-01	3.15754e+01
1.06968e-01	1.07865e+00	3.15754e+01
1.22675e-01	1.22104e+00	3.15754e+01
1.27533e-01	1.33928e+00	3.15754e+01
1.26805e-01	1.43052e+00	3.15754e+01
1.24342e-01	1.49255e+00	3.15754e+01
1.24004e-01	1.52382e+00	3.15754e+01
1.34294e-01	1.52295e+00	3.15754e+01
1.42545e-01	1.50811e+00	3.18920e+01
1.37613e-01	1.47759e+00	3.18920e+01
1.49449e-01	1.41536e+00	3.18920e+01
1.65771e-01	1.32369e+00	3.18920e+01
1.86065e-01	1.20447e+00	3.18920e+01
2.09836e-01	1.06013e+00	3.18920e+01
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6.93869e-02	-1.34313e+00	3.18920e+01
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-3.08639e-01	-1.05197e+00	3.18920e+01
-3.20455e-01	-8.86682e-01	3.18920e+01
-3.19862e-01	-7.03833e-01	3.18920e+01
-3.05046e-01	-5.07294e-01	3.18920e+01
-2.75085e-01	-3.01166e-01	3.18920e+01
-2.30175e-01	-8.96802e-02	3.18920e+01
-1.71452e-01	1.22877e-01	3.18920e+01
-1.04085e-01	3.32507e-01	3.18920e+01
-3.51221e-02	5.35411e-01	3.18920e+01
2.66247e-02	7.28071e-01	3.18920e+01
7.40108e-02	9.07052e-01	3.18920e+01
1.04263e-01	1.06886e+00	3.18920e+01
1.19598e-01	1.20997e+00	3.18920e+01
1.24223e-01	1.32712e+00	3.18920e+01
1.23356e-01	1.41751e+00	3.18920e+01
1.20817e-01	1.47897e+00	3.18920e+01
1.20471e-01	1.50994e+00	3.18920e+01
1.30629e-01	1.50910e+00	3.18920e+01
1.38609e-01	1.49431e+00	3.22085e+01
1.33934e-01	1.46405e+00	3.22085e+01
1.45761e-01	1.40241e+00	3.22085e+01
1.62079e-01	1.31163e+00	3.22085e+01
1.82378e-01	1.19355e+00	3.22085e+01
2.06162e-01	1.05060e+00	3.22085e+01
2.32822e-01	8.85683e-01	3.22085e+01
2.60958e-01	7.02234e-01	3.22085e+01
2.89313e-01	5.04041e-01	3.22085e+01
3.15832e-01	2.95259e-01	3.22085e+01
3.37083e-01	8.03726e-02	3.22085e+01
3.48819e-01	-1.35935e-01	3.22085e+01
3.45911e-01	-3.48865e-01	3.22085e+01
3.26246e-01	-5.53906e-01	3.22085e+01
2.91644e-01	-7.46998e-01	3.22085e+01
2.45588e-01	-9.24437e-01	3.22085e+01
1.91414e-01	-1.08281e+00	3.22085e+01
1.32217e-01	-1.21904e+00	3.22085e+01
7.09195e-02	-1.33050e+00	3.22085e+01
1.07535e-02	-1.41509e+00	3.22085e+01
-4.56365e-02	-1.47120e+00	3.22085e+01
-1.21164e-01	-1.49573e+00	3.22085e+01
-1.78986e-01	-1.46040e+00	3.22085e+01
-2.18361e-01	-1.39653e+00	3.22085e+01
-2.53198e-01	-1.30425e+00	3.22085e+01
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-3.04057e-01	-1.04267e+00	3.22085e+01
-3.16032e-01	-8.78943e-01	3.22085e+01
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-3.01382e-01	-5.03064e-01	3.22085e+01
-2.72035e-01	-2.98807e-01	3.22085e+01
-2.27886e-01	-8.92183e-02	3.22085e+01
-1.70055e-01	1.21454e-01	3.22085e+01
-1.03655e-01	3.29240e-01	3.22085e+01
-3.56636e-02	5.30366e-01	3.22085e+01
2.51956e-02	7.21332e-01	3.22085e+01
7.18500e-02	8.98723e-01	3.22085e+01
1.01558e-01	1.05907e+00	3.22085e+01
1.16521e-01	1.19889e+00	3.22085e+01
1.20913e-01	1.31496e+00	3.22085e+01
1.19908e-01	1.40451e+00	3.22085e+01
1.17293e-01	1.46539e+00	3.22085e+01
1.16939e-01	1.49607e+00	3.22085e+01
1.26964e-01	1.49526e+00	3.22085e+01
1.34673e-01	1.48051e+00	3.25251e+01
1.30255e-01	1.45051e+00	3.25251e+01
1.42073e-01	1.38947e+00	3.25251e+01
1.58388e-01	1.29957e+00	3.25251e+01
1.78691e-01	1.18264e+00	3.25251e+01
2.02487e-01	1.04106e+00	3.25251e+01
2.29170e-01	8.77743e-01	3.25251e+01
2.57344e-01	6.96065e-01	3.25251e+01
2.85760e-01	4.99782e-01	3.25251e+01
3.12375e-01	2.93006e-01	3.25251e+01
3.33782e-01	8.01743e-02	3.25251e+01
3.45765e-01	-1.34081e-01	3.25251e+01
3.43236e-01	-3.45014e-01	3.25251e+01
3.24095e-01	-5.48160e-01	3.25251e+01
2.90138e-01	-7.39495e-01	3.25251e+01
2.44807e-01	-9.15342e-01	3.25251e+01
1.91405e-01	-1.07231e+00	3.25251e+01
1.32989e-01	-1.20736e+00	3.25251e+01
7.24521e-02	-1.31788e+00	3.25251e+01
1.29914e-02	-1.40177e+00	3.25251e+01
-4.27758e-02	-1.45744e+00	3.25251e+01
-1.17553e-01	-1.48187e+00	3.25251e+01
-1.74889e-01	-1.44696e+00	3.25251e+01
-2.14001e-01	-1.38377e+00	3.25251e+01
-2.48662e-01	-1.29241e+00	3.25251e+01
-2.77576e-01	-1.17479e+00	3.25251e+01
-2.99476e-01	-1.03337e+00	3.25251e+01
-3.11608e-01	-8.71203e-01	3.25251e+01
-3.11620e-01	-6.91759e-01	3.25251e+01
-2.97717e-01	-4.98833e-01	3.25251e+01
-2.68984e-01	-2.96448e-01	3.25251e+01
-2.25597e-01	-8.87564e-02	3.25251e+01

-1.68657e-01	1.20031e-01	3.25251e+01
-1.03225e-01	3.25974e-01	3.25251e+01
-3.62050e-02	5.25322e-01	3.25251e+01
2.37664e-02	7.14594e-01	3.25251e+01
6.96891e-02	8.90394e-01	3.25251e+01
9.88526e-02	1.04929e+00	3.25251e+01
1.13444e-01	1.18781e+00	3.25251e+01
1.17603e-01	1.30280e+00	3.25251e+01
1.16459e-01	1.39150e+00	3.25251e+01
1.13768e-01	1.45181e+00	3.25251e+01
1.13406e-01	1.48220e+00	3.25251e+01
1.23300e-01	1.48141e+00	3.25251e+01
1.30737e-01	1.46670e+00	3.28417e+01
1.26577e-01	1.43697e+00	3.28417e+01
1.38384e-01	1.37653e+00	3.28417e+01
1.54696e-01	1.28751e+00	3.28417e+01
1.75003e-01	1.17172e+00	3.28417e+01
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2.25517e-01	8.69803e-01	3.28417e+01
2.53731e-01	6.89895e-01	3.28417e+01
2.82208e-01	4.95522e-01	3.28417e+01
3.08919e-01	2.90753e-01	3.28417e+01
3.30480e-01	7.99760e-02	3.28417e+01
3.42711e-01	-1.32227e-01	3.28417e+01
3.40561e-01	-3.41163e-01	3.28417e+01
3.21944e-01	-5.42414e-01	3.28417e+01
2.88631e-01	-7.31991e-01	3.28417e+01
2.44027e-01	-9.06247e-01	3.28417e+01
1.91395e-01	-1.06182e+00	3.28417e+01
1.33761e-01	-1.19568e+00	3.28417e+01
7.39847e-02	-1.30526e+00	3.28417e+01
1.52294e-02	-1.38845e+00	3.28417e+01
-3.99150e-02	-1.44369e+00	3.28417e+01
-1.13942e-01	-1.46801e+00	3.28417e+01
-1.70793e-01	-1.43353e+00	3.28417e+01
-2.09640e-01	-1.37100e+00	3.28417e+01
-2.44127e-01	-1.28057e+00	3.28417e+01
-2.72965e-01	-1.16412e+00	3.28417e+01
-2.94894e-01	-1.02407e+00	3.28417e+01
-3.07185e-01	-8.63464e-01	3.28417e+01
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-2.94053e-01	-4.94603e-01	3.28417e+01
-2.65933e-01	-2.94089e-01	3.28417e+01
-2.23307e-01	-8.82944e-02	3.28417e+01
-1.67259e-01	1.18609e-01	3.28417e+01
-1.02795e-01	3.22708e-01	3.28417e+01
-3.67464e-02	5.20278e-01	3.28417e+01
2.23373e-02	7.07855e-01	3.28417e+01
6.75282e-02	8.82065e-01	3.28417e+01
9.61475e-02	1.03950e+00	3.28417e+01
1.10367e-01	1.17674e+00	3.28417e+01
1.14293e-01	1.29064e+00	3.28417e+01
1.13011e-01	1.37850e+00	3.28417e+01
1.10243e-01	1.43823e+00	3.28417e+01
1.09874e-01	1.46833e+00	3.28417e+01
1.19635e-01	1.46757e+00	3.28417e+01
1.26801e-01	1.45290e+00	3.31583e+01
1.22898e-01	1.42343e+00	3.31583e+01
1.34696e-01	1.36359e+00	3.31583e+01
1.51005e-01	1.27545e+00	3.31583e+01
1.71316e-01	1.16080e+00	3.31583e+01
1.95139e-01	1.02200e+00	3.31583e+01
2.21865e-01	8.61863e-01	3.31583e+01
2.50117e-01	6.83726e-01	3.31583e+01
2.78655e-01	4.91263e-01	3.31583e+01
3.05462e-01	2.88499e-01	3.31583e+01
3.27179e-01	7.97777e-02	3.31583e+01
3.39657e-01	-1.30372e-01	3.31583e+01
3.37886e-01	-3.37312e-01	3.31583e+01
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7.55173e-02	-1.29263e+00	3.31583e+01
1.74673e-02	-1.37514e+00	3.31583e+01
-3.70543e-02	-1.42994e+00	3.31583e+01
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-2.90389e-01	-4.90372e-01	3.31583e+01
-2.62882e-01	-2.91730e-01	3.31583e+01
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-1.65861e-01	1.17186e-01	3.31583e+01
-1.02364e-01	3.19442e-01	3.31583e+01
-3.72879e-02	5.15234e-01	3.31583e+01
-2.09082e-02	7.01117e-01	3.31583e+01
6.53674e-02	8.73736e-01	3.31583e+01
9.34423e-02	1.02971e+00	3.31583e+01
1.07290e-01	1.16566e+00	3.31583e+01
1.10983e-01	1.27848e+00	3.31583e+01
1.09562e-01	1.36550e+00	3.31583e+01

1.06719e-01	1.42465e+00	3.31583e+01
1.06341e-01	1.45446e+00	3.31583e+01
1.15970e-01	1.45372e+00	3.31583e+01
1.22865e-01	1.43909e+00	3.34749e+01
1.19219e-01	1.40989e+00	3.34749e+01
1.31008e-01	1.35065e+00	3.34749e+01
1.47313e-01	1.26339e+00	3.34749e+01
1.67629e-01	1.14988e+00	3.34749e+01
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2.18213e-01	8.53922e-01	3.34749e+01
2.46503e-01	6.77557e-01	3.34749e+01
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-2.59831e-01	-2.89371e-01	3.34749e+01
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-1.64464e-01	1.15763e-01	3.34749e+01
-1.01934e-01	3.16176e-01	3.34749e+01
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1.94790e-02	6.94378e-01	3.34749e+01
6.32065e-02	8.65408e-01	3.34749e+01
9.07371e-02	1.01992e+00	3.34749e+01
1.04213e-01	1.15458e+00	3.34749e+01
1.07673e-01	1.26632e+00	3.34749e+01
1.06113e-01	1.35249e+00	3.34749e+01
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1.02809e-01	1.44058e+00	3.34749e+01
1.12305e-01	1.43988e+00	3.34749e+01
1.20573e-01	1.42545e+00	3.37915e+01
1.15662e-01	1.39661e+00	3.37915e+01
1.27440e-01	1.33796e+00	3.37915e+01
1.43739e-01	1.25156e+00	3.37915e+01
1.64055e-01	1.13918e+00	3.37915e+01
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2.14662e-01	8.46132e-01	3.37915e+01
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2.71637e-01	4.82818e-01	3.37915e+01
2.98627e-01	2.84027e-01	3.37915e+01
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3.33605e-01	-1.26712e-01	3.37915e+01
3.32578e-01	-3.29698e-01	3.37915e+01
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-2.56851e-01	-2.87050e-01	3.37915e+01
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-1.63090e-01	1.14375e-01	3.37915e+01
-1.01502e-01	3.12979e-01	3.37915e+01
-3.83422e-02	5.05247e-01	3.37915e+01
1.81025e-02	6.87772e-01	3.37915e+01
6.11187e-02	8.57242e-01	3.37915e+01
8.81212e-02	1.01033e+00	3.37915e+01
1.01237e-01	1.14372e+00	3.37915e+01
1.04473e-01	1.25440e+00	3.37915e+01
1.02780e-01	1.33974e+00	3.37915e+01
9.97883e-02	1.39776e+00	3.37915e+01
9.93957e-02	1.42698e+00	3.37915e+01
1.08763e-01	1.42630e+00	3.37915e+01
1.16126e-01	1.41380e+00	3.41080e+01
1.12916e-01	1.38509e+00	3.41080e+01
1.24670e-01	1.32694e+00	3.41080e+01
1.40943e-01	1.24128e+00	3.41080e+01
1.61233e-01	1.12987e+00	3.41080e+01

1.85049e-01	9.94965e-01	3.41080e+01
2.11790e-01	8.39334e-01	3.41080e+01
2.40099e-01	6.66199e-01	3.41080e+01
2.68752e-01	4.79129e-01	3.41080e+01
2.95769e-01	2.82034e-01	3.41080e+01
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3.30978e-01	-1.25228e-01	3.41080e+01
3.30219e-01	-3.26519e-01	3.41080e+01
3.13548e-01	-5.20510e-01	3.41080e+01
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2.40771e-01	-8.71495e-01	3.41080e+01
1.91045e-01	-1.02169e+00	3.41080e+01
1.36364e-01	-1.15101e+00	3.41080e+01
7.94666e-02	-1.25695e+00	3.41080e+01
2.33874e-02	-1.33746e+00	3.41080e+01
-2.93888e-02	-1.39102e+00	3.41080e+01
-1.00549e-01	-1.41491e+00	3.41080e+01
-1.55533e-01	-1.38206e+00	3.41080e+01
-1.93348e-01	-1.32206e+00	3.41080e+01
-2.27139e-01	-1.23516e+00	3.41080e+01
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-2.90504e-01	-8.33740e-01	3.41080e+01
-2.91922e-01	-6.62516e-01	3.41080e+01
-2.80169e-01	-4.78318e-01	3.41080e+01
-2.54338e-01	-2.84978e-01	3.41080e+01
-2.14567e-01	-8.64597e-02	3.41080e+01
-1.61873e-01	1.13211e-01	3.41080e+01
-1.01052e-01	3.10234e-01	3.41080e+01
-3.86655e-02	5.00976e-01	3.41080e+01
1.70748e-02	6.82047e-01	3.41080e+01
5.95155e-02	8.50155e-01	3.41080e+01
8.60973e-02	1.00200e+00	3.41080e+01
9.89340e-02	1.13429e+00	3.41080e+01
1.02001e-01	1.24405e+00	3.41080e+01
1.00214e-01	1.32868e+00	3.41080e+01
9.71732e-02	1.38620e+00	3.41080e+01
9.67761e-02	1.41518e+00	3.41080e+01
1.06036e-01	1.41452e+00	3.41080e+01
1.13193e-01	1.40205e+00	3.44246e+01
1.10170e-01	1.37356e+00	3.44246e+01
1.21900e-01	1.31592e+00	3.44246e+01
1.38147e-01	1.23101e+00	3.44246e+01
1.58410e-01	1.12056e+00	3.44246e+01
1.82201e-01	9.86821e-01	3.44246e+01
2.08918e-01	8.32536e-01	3.44246e+01
2.37212e-01	6.60896e-01	3.44246e+01
2.65867e-01	4.75440e-01	3.44246e+01
2.92912e-01	2.80041e-01	3.44246e+01
3.15083e-01	7.88642e-02	3.44246e+01
3.28352e-01	-1.23744e-01	3.44246e+01
3.27859e-01	-3.23340e-01	3.44246e+01
3.11580e-01	-5.15718e-01	3.44246e+01
2.81176e-01	-6.97054e-01	3.44246e+01
2.39871e-01	-8.63839e-01	3.44246e+01
1.90759e-01	-1.01284e+00	3.44246e+01
1.36706e-01	-1.14114e+00	3.44246e+01
8.04258e-02	-1.24625e+00	3.44246e+01
2.49235e-02	-1.32616e+00	3.44246e+01
-2.73387e-02	-1.37934e+00	3.44246e+01
-9.78684e-02	-1.40312e+00	3.44246e+01
-1.52431e-01	-1.37061e+00	3.44246e+01
-1.90006e-01	-1.31117e+00	3.44246e+01
-2.23624e-01	-1.22504e+00	3.44246e+01
-2.52041e-01	-1.11402e+00	3.44246e+01
-2.74035e-01	-9.80416e-01	3.44246e+01
-2.86976e-01	-8.27092e-01	3.44246e+01
-2.88603e-01	-6.57312e-01	3.44246e+01
-2.77188e-01	-4.74651e-01	3.44246e+01
-2.51826e-01	-2.82906e-01	3.44246e+01
-2.12647e-01	-8.60092e-02	3.44246e+01
-1.60655e-01	1.12047e-01	3.44246e+01
-1.00603e-01	3.07489e-01	3.44246e+01
-3.89889e-02	4.96704e-01	3.44246e+01
1.60471e-02	6.76323e-01	3.44246e+01
5.79124e-02	8.43069e-01	3.44246e+01
8.40734e-02	9.93666e-01	3.44246e+01
9.66305e-02	1.12486e+00	3.44246e+01
9.95294e-02	1.23370e+00	3.44246e+01
9.76479e-02	1.31761e+00	3.44246e+01
9.45581e-02	1.37465e+00	3.44246e+01
9.41566e-02	1.40338e+00	3.44246e+01
1.03310e-01	1.40274e+00	3.44246e+01
1.10260e-01	1.39029e+00	3.47412e+01
1.07423e-01	1.36204e+00	3.47412e+01
1.19130e-01	1.30490e+00	3.47412e+01
1.35351e-01	1.22073e+00	3.47412e+01
1.55587e-01	1.11124e+00	3.47412e+01
1.79352e-01	9.78678e-01	3.47412e+01
2.06046e-01	8.25738e-01	3.47412e+01
2.34326e-01	6.55594e-01	3.47412e+01
2.62981e-01	4.71751e-01	3.47412e+01
2.90054e-01	2.78048e-01	3.47412e+01
3.12302e-01	7.86095e-02	3.47412e+01
3.25725e-01	-1.22260e-01	3.47412e+01
3.25500e-01	-3.20161e-01	3.47412e+01
3.09613e-01	-5.10925e-01	3.47412e+01

2.79706e-01	-6.90761e-01	3.47412e+01
2.38972e-01	-8.56184e-01	3.47412e+01
1.90473e-01	-1.00398e+00	3.47412e+01
1.37049e-01	-1.13126e+00	3.47412e+01
8.13850e-02	-1.23556e+00	3.47412e+01
2.64595e-02	-1.31486e+00	3.47412e+01
-2.52885e-02	-1.36766e+00	3.47412e+01
-9.51877e-02	-1.39133e+00	3.47412e+01
-1.49330e-01	-1.35917e+00	3.47412e+01
-1.86664e-01	-1.30028e+00	3.47412e+01
-2.20110e-01	-1.21493e+00	3.47412e+01
-2.48431e-01	-1.10489e+00	3.47412e+01
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-2.85285e-01	-6.52108e-01	3.47412e+01
-2.74208e-01	-4.70983e-01	3.47412e+01
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-1.59438e-01	1.10884e-01	3.47412e+01
-1.00153e-01	3.04744e-01	3.47412e+01
-3.93123e-02	4.92433e-01	3.47412e+01
1.50194e-02	6.70598e-01	3.47412e+01
5.63092e-02	8.35983e-01	3.47412e+01
8.20496e-02	9.85335e-01	3.47412e+01
9.43270e-02	1.11543e+00	3.47412e+01
9.70579e-02	1.22335e+00	3.47412e+01
9.50818e-02	1.30655e+00	3.47412e+01
9.19431e-02	1.36310e+00	3.47412e+01
9.15371e-02	1.39158e+00	3.47412e+01
1.00583e-01	1.39096e+00	3.47412e+01
1.07326e-01	1.37854e+00	3.50578e+01
1.04677e-01	1.35051e+00	3.50578e+01
1.16360e-01	1.29388e+00	3.50578e+01
1.32555e-01	1.21045e+00	3.50578e+01
1.52765e-01	1.10193e+00	3.50578e+01
1.76504e-01	9.70534e-01	3.50578e+01
2.03174e-01	8.18941e-01	3.50578e+01
2.31440e-01	6.50292e-01	3.50578e+01
2.60096e-01	4.68063e-01	3.50578e+01
2.87197e-01	2.76054e-01	3.50578e+01
3.09521e-01	7.83547e-02	3.50578e+01
3.23099e-01	-1.20776e-01	3.50578e+01
3.23141e-01	-3.16981e-01	3.50578e+01
3.07645e-01	-5.06133e-01	3.50578e+01
2.78236e-01	-6.84469e-01	3.50578e+01
2.38072e-01	-8.48529e-01	3.50578e+01
1.90188e-01	-9.95123e-01	3.50578e+01
1.37391e-01	-1.12139e+00	3.50578e+01
8.23442e-02	-1.22486e+00	3.50578e+01
2.79956e-02	-1.30356e+00	3.50578e+01
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-1.83321e-01	-1.28939e+00	3.50578e+01
-2.16595e-01	-1.20482e+00	3.50578e+01
-2.44822e-01	-1.09576e+00	3.50578e+01
-2.66794e-01	-9.64479e-01	3.50578e+01
-2.79919e-01	-8.13794e-01	3.50578e+01
-2.81967e-01	-6.46903e-01	3.50578e+01
-2.71227e-01	-4.67316e-01	3.50578e+01
-2.46801e-01	-2.78763e-01	3.50578e+01
-2.08807e-01	-8.51083e-02	3.50578e+01
-1.58221e-01	1.09720e-01	3.50578e+01
-9.97038e-02	3.01999e-01	3.50578e+01
-3.96357e-02	4.88162e-01	3.50578e+01
1.39917e-02	6.64873e-01	3.50578e+01
5.47061e-02	8.28896e-01	3.50578e+01
8.00257e-02	9.77003e-01	3.50578e+01
9.20235e-02	1.10601e+00	3.50578e+01
9.45863e-02	1.21300e+00	3.50578e+01
9.25156e-02	1.29548e+00	3.50578e+01
8.93280e-02	1.35154e+00	3.50578e+01
8.89175e-02	1.37978e+00	3.50578e+01
9.78568e-02	1.37918e+00	3.50578e+01
1.04393e-01	1.36678e+00	3.53744e+01
1.01931e-01	1.33899e+00	3.53744e+01
1.13590e-01	1.28286e+00	3.53744e+01
1.29759e-01	1.20018e+00	3.53744e+01
1.49942e-01	1.09262e+00	3.53744e+01
1.73655e-01	9.62390e-01	3.53744e+01
2.00301e-01	8.12143e-01	3.53744e+01
2.28553e-01	6.44989e-01	3.53744e+01
2.57210e-01	4.64374e-01	3.53744e+01
2.84339e-01	2.74061e-01	3.53744e+01
3.06740e-01	7.80999e-02	3.53744e+01
3.20472e-01	-1.19292e-01	3.53744e+01
3.20781e-01	-3.13802e-01	3.53744e+01
3.05678e-01	-5.01341e-01	3.53744e+01
2.76765e-01	-6.78176e-01	3.53744e+01
2.37173e-01	-8.40874e-01	3.53744e+01
1.89902e-01	-9.86266e-01	3.53744e+01
1.37734e-01	-1.11151e+00	3.53744e+01
8.33034e-02	-1.21417e+00	3.53744e+01
2.95317e-02	-1.29226e+00	3.53744e+01
-2.11882e-02	-1.34429e+00	3.53744e+01
-8.98262e-02	-1.36775e+00	3.53744e+01
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-1.79979e-01	-1.27850e+00	3.53744e+01
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-2.41212e-01	-1.08663e+00	3.53744e+01
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-1.57004e-01	1.08556e-01	3.53744e+01
-9.92544e-02	2.99254e-01	3.53744e+01
-3.99591e-02	4.83890e-01	3.53744e+01
1.29640e-02	6.59148e-01	3.53744e+01
5.31029e-02	8.21810e-01	3.53744e+01
7.80018e-02	9.68672e-01	3.53744e+01
8.97201e-02	1.09658e+00	3.53744e+01
9.21148e-02	1.20265e+00	3.53744e+01
8.99495e-02	1.28442e+00	3.53744e+01
8.67129e-02	1.33999e+00	3.53744e+01
8.62980e-02	1.36798e+00	3.53744e+01
9.51303e-02	1.36740e+00	3.53744e+01
1.01459e-01	1.35503e+00	3.56910e+01
9.91845e-02	1.32746e+00	3.56910e+01
1.10820e-01	1.27184e+00	3.56910e+01
1.26963e-01	1.18990e+00	3.56910e+01
1.47119e-01	1.08331e+00	3.56910e+01
1.70807e-01	9.54246e-01	3.56910e+01
1.97429e-01	8.05345e-01	3.56910e+01
2.25667e-01	6.39687e-01	3.56910e+01
2.54325e-01	4.60685e-01	3.56910e+01
2.81481e-01	2.72068e-01	3.56910e+01
3.03959e-01	7.78452e-02	3.56910e+01
3.17845e-01	-1.17808e-01	3.56910e+01
3.18422e-01	-3.10623e-01	3.56910e+01
3.03710e-01	-4.96549e-01	3.56910e+01
2.75295e-01	-6.71883e-01	3.56910e+01
2.36273e-01	-8.33218e-01	3.56910e+01
1.89616e-01	-9.77410e-01	3.56910e+01
1.38076e-01	-1.10164e+00	3.56910e+01
8.42626e-02	-1.20348e+00	3.56910e+01
3.10677e-02	-1.28096e+00	3.56910e+01
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-1.40026e-01	-1.32484e+00	3.56910e+01
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-2.09566e-01	-1.18459e+00	3.56910e+01
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-2.59554e-01	-9.48542e-01	3.56910e+01
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-2.75330e-01	-6.36495e-01	3.56910e+01
-2.65266e-01	-4.59982e-01	3.56910e+01
-2.41776e-01	-2.74620e-01	3.56910e+01
-2.04967e-01	-8.42073e-02	3.56910e+01
-1.55786e-01	1.07392e-01	3.56910e+01
-9.88049e-02	2.96509e-01	3.56910e+01
-4.02825e-02	4.79619e-01	3.56910e+01
1.19363e-02	6.53423e-01	3.56910e+01
5.14998e-02	8.14724e-01	3.56910e+01
7.59779e-02	9.60341e-01	3.56910e+01
8.74166e-02	1.08715e+00	3.56910e+01
8.96432e-02	1.19230e+00	3.56910e+01
8.73833e-02	1.27335e+00	3.56910e+01
8.40978e-02	1.32843e+00	3.56910e+01
8.36785e-02	1.35618e+00	3.56910e+01
9.24039e-02	1.35562e+00	3.56910e+01
9.85261e-02	1.34327e+00	3.60075e+01
9.64382e-02	1.31594e+00	3.60075e+01
1.08050e-01	1.26082e+00	3.60075e+01
1.24166e-01	1.17962e+00	3.60075e+01
1.44297e-01	1.07400e+00	3.60075e+01
1.67958e-01	9.46102e-01	3.60075e+01
1.94557e-01	7.98547e-01	3.60075e+01
2.22780e-01	6.34385e-01	3.60075e+01
2.51440e-01	4.56996e-01	3.60075e+01
2.78624e-01	2.70075e-01	3.60075e+01
3.01179e-01	7.75904e-02	3.60075e+01
3.15219e-01	-1.16324e-01	3.60075e+01
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3.01743e-01	-4.91757e-01	3.60075e+01
2.73825e-01	-6.65591e-01	3.60075e+01
2.35374e-01	-8.25563e-01	3.60075e+01
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3.26038e-02	-1.26966e+00	3.60075e+01
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-2.55933e-01	-9.40574e-01	3.60075e+01
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-2.03047e-01	-8.37568e-02	3.60075e+01

-1.54569e-01	1.06229e-01	3.60075e+01
-9.83555e-02	2.93764e-01	3.60075e+01
-4.06058e-02	4.75347e-01	3.60075e+01
1.09086e-02	6.47698e-01	3.60075e+01
4.98967e-02	8.07637e-01	3.60075e+01
7.39540e-02	9.52009e-01	3.60075e+01
8.51131e-02	1.07772e+00	3.60075e+01
8.71717e-02	1.18195e+00	3.60075e+01
8.48172e-02	1.26228e+00	3.60075e+01
8.14827e-02	1.31688e+00	3.60075e+01
8.10590e-02	1.34438e+00	3.60075e+01
8.96774e-02	1.34383e+00	3.60075e+01
9.55927e-02	1.33152e+00	3.63241e+01
9.36919e-02	1.30441e+00	3.63241e+01
1.05280e-01	1.24981e+00	3.63241e+01
1.21370e-01	1.16935e+00	3.63241e+01
1.41474e-01	1.06469e+00	3.63241e+01
1.65110e-01	9.37958e-01	3.63241e+01
1.91685e-01	7.91749e-01	3.63241e+01
2.19894e-01	6.29082e-01	3.63241e+01
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3.12592e-01	-1.14840e-01	3.63241e+01
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2.99775e-01	-4.86965e-01	3.63241e+01
2.72355e-01	-6.59298e-01	3.63241e+01
2.34475e-01	-8.17908e-01	3.63241e+01
1.89044e-01	-9.59697e-01	3.63241e+01
1.38761e-01	-1.08189e+00	3.63241e+01
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-2.30384e-01	-1.05923e+00	3.63241e+01
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-2.65807e-01	-7.87199e-01	3.63241e+01
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-2.59305e-01	-4.52647e-01	3.63241e+01
-2.36752e-01	-2.70476e-01	3.63241e+01
-2.01127e-01	-8.33064e-02	3.63241e+01
-1.53352e-01	1.05065e-01	3.63241e+01
-9.79060e-02	2.91019e-01	3.63241e+01
-4.09292e-02	4.71076e-01	3.63241e+01
9.88094e-03	6.41974e-01	3.63241e+01
4.82935e-02	8.00551e-01	3.63241e+01
7.19301e-02	9.43678e-01	3.63241e+01
8.28096e-02	1.06829e+00	3.63241e+01
8.47001e-02	1.17160e+00	3.63241e+01
8.22510e-02	1.25122e+00	3.63241e+01
7.88677e-02	1.30532e+00	3.63241e+01
7.84394e-02	1.33258e+00	3.63241e+01
8.69510e-02	1.33205e+00	3.63241e+01
9.26593e-02	1.31977e+00	3.66407e+01
9.09457e-02	1.29289e+00	3.66407e+01
1.02510e-01	1.23879e+00	3.66407e+01
1.18574e-01	1.15907e+00	3.66407e+01
1.38651e-01	1.05538e+00	3.66407e+01
1.62261e-01	9.29814e-01	3.66407e+01
1.88813e-01	7.84951e-01	3.66407e+01
2.17008e-01	6.23780e-01	3.66407e+01
2.45669e-01	4.49618e-01	3.66407e+01
2.72908e-01	2.66089e-01	3.66407e+01
2.95617e-01	7.70809e-02	3.66407e+01
3.09966e-01	-1.13356e-01	3.66407e+01
3.11344e-01	-3.01086e-01	3.66407e+01
2.97808e-01	-4.82173e-01	3.66407e+01
2.70885e-01	-6.53005e-01	3.66407e+01
2.33575e-01	-8.10253e-01	3.66407e+01
1.88758e-01	-9.50841e-01	3.66407e+01
1.39104e-01	-1.07201e+00	3.66407e+01
8.71402e-02	-1.17139e+00	3.66407e+01
3.56759e-02	-1.24706e+00	3.66407e+01
-1.29875e-02	-1.29756e+00	3.66407e+01
-7.91033e-02	-1.32058e+00	3.66407e+01
-1.30721e-01	-1.29051e+00	3.66407e+01
-1.66610e-01	-1.23495e+00	3.66407e+01
-1.99023e-01	-1.15425e+00	3.66407e+01
-2.26774e-01	-1.05010e+00	3.66407e+01
-2.48693e-01	-9.24637e-01	3.66407e+01
-2.62278e-01	-7.80551e-01	3.66407e+01
-2.65376e-01	-6.20883e-01	3.66407e+01
-2.56324e-01	-4.48980e-01	3.66407e+01
-2.34239e-01	-2.68405e-01	3.66407e+01
-1.99207e-01	-8.28559e-02	3.66407e+01
-1.52135e-01	1.03901e-01	3.66407e+01
-9.74565e-02	2.88274e-01	3.66407e+01
-4.12526e-02	4.66805e-01	3.66407e+01
8.85324e-03	6.36249e-01	3.66407e+01
4.66904e-02	7.93465e-01	3.66407e+01
6.99063e-02	9.35347e-01	3.66407e+01
8.05061e-02	1.05886e+00	3.66407e+01
8.22286e-02	1.16125e+00	3.66407e+01
7.96848e-02	1.24015e+00	3.66407e+01

7.62526e-02	1.29377e+00	3.66407e+01
7.58199e-02	1.32077e+00	3.66407e+01
8.42245e-02	1.32027e+00	3.66407e+01
9.11938e-02	1.30864e+00	3.69573e+01
8.84753e-02	1.28205e+00	3.69573e+01
1.00013e-01	1.22843e+00	3.69573e+01
1.16046e-01	1.14941e+00	3.69573e+01
1.36089e-01	1.04662e+00	3.69573e+01
1.59664e-01	9.22150e-01	3.69573e+01
1.86181e-01	7.78549e-01	3.69573e+01
2.14349e-01	6.18781e-01	3.69573e+01
2.42996e-01	4.46132e-01	3.69573e+01
2.70246e-01	2.64193e-01	3.69573e+01
2.93011e-01	7.68165e-02	3.69573e+01
3.07489e-01	-1.11988e-01	3.69573e+01
3.09102e-01	-2.98125e-01	3.69573e+01
2.95919e-01	-4.77696e-01	3.69573e+01
2.69448e-01	-6.47116e-01	3.69573e+01
2.32661e-01	-8.03081e-01	3.69573e+01
1.88409e-01	-9.42537e-01	3.69573e+01
1.39335e-01	-1.06275e+00	3.69573e+01
8.79421e-02	-1.16135e+00	3.69573e+01
3.70133e-02	-1.23645e+00	3.69573e+01
-1.11715e-02	-1.28658e+00	3.69573e+01
-7.66969e-02	-1.30949e+00	3.69573e+01
-1.27917e-01	-1.27974e+00	3.69573e+01
-1.63574e-01	-1.22470e+00	3.69573e+01
-1.95819e-01	-1.14473e+00	3.69573e+01
-2.23472e-01	-1.04150e+00	3.69573e+01
-2.45370e-01	-9.17130e-01	3.69573e+01
-2.59030e-01	-7.74282e-01	3.69573e+01
-2.62311e-01	-6.15971e-01	3.69573e+01
-2.53563e-01	-4.45513e-01	3.69573e+01
-2.31902e-01	-2.66439e-01	3.69573e+01
-1.97410e-01	-8.24158e-02	3.69573e+01
-1.50982e-01	1.02821e-01	3.69573e+01
-9.70099e-02	2.85703e-01	3.69573e+01
-4.15168e-02	4.62796e-01	3.69573e+01
7.94171e-03	6.30870e-01	3.69573e+01
4.52516e-02	7.86803e-01	3.69573e+01
6.80842e-02	9.27514e-01	3.69573e+01
7.84319e-02	1.04999e+00	3.69573e+01
8.00051e-02	1.15152e+00	3.69573e+01
7.73792e-02	1.22975e+00	3.69573e+01
7.39055e-02	1.28291e+00	3.69573e+01
7.34692e-02	1.30968e+00	3.69573e+01
8.17751e-02	1.30920e+00	3.69573e+01
8.77991e-02	1.29864e+00	3.72739e+01
8.63893e-02	1.27218e+00	3.72739e+01
9.78960e-02	1.21898e+00	3.72739e+01
1.13890e-01	1.14060e+00	3.72739e+01
1.33889e-01	1.03863e+00	3.72739e+01
1.57417e-01	9.15154e-01	3.72739e+01
1.83884e-01	7.72698e-01	3.72739e+01
2.12008e-01	6.14203e-01	3.72739e+01
2.40620e-01	4.42928e-01	3.72739e+01
2.67857e-01	2.62433e-01	3.72739e+01
2.90649e-01	7.65387e-02	3.72739e+01
3.05220e-01	-1.10782e-01	3.72739e+01
3.07024e-01	-2.95470e-01	3.72739e+01
2.94138e-01	-4.73659e-01	3.72739e+01
2.68059e-01	-6.41791e-01	3.72739e+01
2.31728e-01	-7.96582e-01	3.72739e+01
1.87972e-01	-9.35002e-01	3.72739e+01
1.39410e-01	-1.05433e+00	3.72739e+01
8.85246e-02	-1.15223e+00	3.72739e+01
3.80737e-02	-1.22680e+00	3.72739e+01
-9.68165e-03	-1.27659e+00	3.72739e+01
-7.46725e-02	-1.29939e+00	3.72739e+01
-1.25527e-01	-1.26993e+00	3.72739e+01
-1.60967e-01	-1.21536e+00	3.72739e+01
-1.93048e-01	-1.13605e+00	3.72739e+01
-2.20599e-01	-1.03364e+00	3.72739e+01
-2.42462e-01	-9.10266e-01	3.72739e+01
-2.56171e-01	-7.68544e-01	3.72739e+01
-2.59600e-01	-6.11468e-01	3.72739e+01
-2.51105e-01	-4.42325e-01	3.72739e+01
-2.29808e-01	-2.64620e-01	3.72739e+01
-1.95785e-01	-8.19902e-02	3.72739e+01
-1.49920e-01	1.01857e-01	3.72739e+01
-9.65671e-02	2.83376e-01	3.72739e+01
-4.16987e-02	4.59152e-01	3.72739e+01
7.19202e-03	6.25973e-01	3.72739e+01
4.40417e-02	7.80735e-01	3.72739e+01
6.65434e-02	9.20376e-01	3.72739e+01
7.66770e-02	1.04191e+00	3.72739e+01
7.81273e-02	1.14265e+00	3.72739e+01
7.54365e-02	1.22027e+00	3.72739e+01
7.19319e-02	1.27301e+00	3.72739e+01
7.14931e-02	1.29958e+00	3.72739e+01
7.97116e-02	1.29911e+00	3.72739e+01
8.55756e-02	1.28857e+00	3.75905e+01
8.43033e-02	1.26231e+00	3.75905e+01
9.57788e-02	1.20954e+00	3.75905e+01
1.11734e-01	1.13178e+00	3.75905e+01
1.31690e-01	1.03064e+00	3.75905e+01

1.55170e-01	9.08157e-01	3.75905e+01
1.81588e-01	7.66847e-01	3.75905e+01
2.09666e-01	6.09625e-01	3.75905e+01
2.38244e-01	4.39723e-01	3.75905e+01
2.65467e-01	2.60674e-01	3.75905e+01
2.88287e-01	7.62609e-02	3.75905e+01
3.02951e-01	-1.09575e-01	3.75905e+01
3.04945e-01	-2.92815e-01	3.75905e+01
2.92358e-01	-4.69621e-01	3.75905e+01
2.66669e-01	-6.36465e-01	3.75905e+01
2.30795e-01	-7.90084e-01	3.75905e+01
1.87535e-01	-9.27468e-01	3.75905e+01
1.39486e-01	-1.04592e+00	3.75905e+01
8.91072e-02	-1.14310e+00	3.75905e+01
3.91342e-02	-1.21714e+00	3.75905e+01
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-1.58359e-01	-1.20601e+00	3.75905e+01
-1.90277e-01	-1.12736e+00	3.75905e+01
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-2.39554e-01	-9.03402e-01	3.75905e+01
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-2.27714e-01	-2.62801e-01	3.75905e+01
-1.94161e-01	-8.15646e-02	3.75905e+01
-1.48858e-01	1.00893e-01	3.75905e+01
-9.61243e-02	2.81048e-01	3.75905e+01
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6.44233e-03	6.21076e-01	3.75905e+01
4.28319e-02	7.74666e-01	3.75905e+01
6.50026e-02	9.13238e-01	3.75905e+01
7.49222e-02	1.03383e+00	3.75905e+01
7.62496e-02	1.13378e+00	3.75905e+01
7.34938e-02	1.21080e+00	3.75905e+01
6.99583e-02	1.26312e+00	3.75905e+01
6.95170e-02	1.28947e+00	3.75905e+01
7.76481e-02	1.28901e+00	3.75905e+01
8.33521e-02	1.27849e+00	3.79070e+01
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9.36615e-02	1.20010e+00	3.79070e+01
1.09579e-01	1.12297e+00	3.79070e+01
1.29490e-01	1.02265e+00	3.79070e+01
1.52923e-01	9.01161e-01	3.79070e+01
1.79291e-01	7.60997e-01	3.79070e+01
2.07325e-01	6.05047e-01	3.79070e+01
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3.00682e-01	-1.08369e-01	3.79070e+01
3.02867e-01	-2.90160e-01	3.79070e+01
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2.65280e-01	-6.31139e-01	3.79070e+01
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1.87098e-01	-9.19933e-01	3.79070e+01
1.39561e-01	-1.03750e+00	3.79070e+01
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-2.25621e-01	-2.60982e-01	3.79070e+01
-1.92536e-01	-8.11391e-02	3.79070e+01
-1.47797e-01	9.99286e-02	3.79070e+01
-9.56815e-02	2.78721e-01	3.79070e+01
-4.20623e-02	4.51864e-01	3.79070e+01
5.69264e-03	6.16179e-01	3.79070e+01
4.16220e-02	7.68597e-01	3.79070e+01
6.34617e-02	9.06100e-01	3.79070e+01
7.31673e-02	1.02576e+00	3.79070e+01
7.43718e-02	1.12492e+00	3.79070e+01
7.15511e-02	1.20132e+00	3.79070e+01
6.79846e-02	1.25322e+00	3.79070e+01
6.75409e-02	1.27937e+00	3.79070e+01
7.55846e-02	1.27892e+00	3.79070e+01
8.11285e-02	1.26842e+00	3.82236e+01
8.01313e-02	1.24256e+00	3.82236e+01
9.15443e-02	1.19065e+00	3.82236e+01
1.07423e-01	1.11416e+00	3.82236e+01
1.27291e-01	1.01466e+00	3.82236e+01
1.50676e-01	8.94164e-01	3.82236e+01
1.76994e-01	7.55146e-01	3.82236e+01
2.04983e-01	6.00469e-01	3.82236e+01
2.33492e-01	4.33315e-01	3.82236e+01
2.60688e-01	2.57154e-01	3.82236e+01
2.83562e-01	7.57053e-02	3.82236e+01
2.98413e-01	-1.07163e-01	3.82236e+01
3.00788e-01	-2.87504e-01	3.82236e+01
2.88798e-01	-4.61547e-01	3.82236e+01

2.63890e-01	-6.25814e-01	3.82236e+01
2.28928e-01	-7.77087e-01	3.82236e+01
1.86660e-01	-9.12399e-01	3.82236e+01
1.39637e-01	-1.02908e+00	3.82236e+01
9.02724e-02	-1.12485e+00	3.82236e+01
4.12552e-02	-1.19783e+00	3.82236e+01
-5.21214e-03	-1.24661e+00	3.82236e+01
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-1.18356e-01	-1.24050e+00	3.82236e+01
-1.53145e-01	-1.18732e+00	3.82236e+01
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-2.23527e-01	-2.59163e-01	3.82236e+01
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-9.52387e-02	2.76394e-01	3.82236e+01
-4.22442e-02	4.48220e-01	3.82236e+01
4.94294e-03	6.11282e-01	3.82236e+01
4.04122e-02	7.62528e-01	3.82236e+01
6.19209e-02	8.98962e-01	3.82236e+01
7.14125e-02	1.01768e+00	3.82236e+01
7.24940e-02	1.11605e+00	3.82236e+01
6.96085e-02	1.19184e+00	3.82236e+01
6.60110e-02	1.24333e+00	3.82236e+01
6.55649e-02	1.26926e+00	3.82236e+01
7.35211e-02	1.26883e+00	3.82236e+01
7.89050e-02	1.25834e+00	3.85402e+01
7.80453e-02	1.23269e+00	3.85402e+01
8.94271e-02	1.18121e+00	3.85402e+01
1.05267e-01	1.10535e+00	3.85402e+01
1.25091e-01	1.00667e+00	3.85402e+01
1.48428e-01	8.87168e-01	3.85402e+01
1.74697e-01	7.49295e-01	3.85402e+01
2.02642e-01	5.95891e-01	3.85402e+01
2.31116e-01	4.30111e-01	3.85402e+01
2.58299e-01	2.55395e-01	3.85402e+01
2.81200e-01	7.54275e-02	3.85402e+01
2.96144e-01	-1.05957e-01	3.85402e+01
2.98710e-01	-2.84849e-01	3.85402e+01
2.87018e-01	-4.57510e-01	3.85402e+01
2.62500e-01	-6.20488e-01	3.85402e+01
2.27994e-01	-7.70589e-01	3.85402e+01
1.86223e-01	-9.04864e-01	3.85402e+01
1.39712e-01	-1.02067e+00	3.85402e+01
9.08550e-02	-1.11572e+00	3.85402e+01
4.23157e-02	-1.18818e+00	3.85402e+01
-3.72231e-03	-1.23662e+00	3.85402e+01
-6.65752e-02	-1.25900e+00	3.85402e+01
-1.15966e-01	-1.23069e+00	3.85402e+01
-1.50537e-01	-1.17798e+00	3.85402e+01
-1.81964e-01	-1.10130e+00	3.85402e+01
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-2.41275e-01	-4.29574e-01	3.85402e+01
-2.21433e-01	-2.57344e-01	3.85402e+01
-1.89286e-01	-8.02879e-02	3.85402e+01
-1.45673e-01	9.80003e-02	3.85402e+01
-9.47959e-02	2.74066e-01	3.85402e+01
-4.24260e-02	4.44576e-01	3.85402e+01
4.19325e-03	6.06385e-01	3.85402e+01
3.92023e-02	7.56460e-01	3.85402e+01
6.03801e-02	8.91824e-01	3.85402e+01
6.96576e-02	1.00960e+00	3.85402e+01
7.06162e-02	1.10718e+00	3.85402e+01
6.76658e-02	1.18236e+00	3.85402e+01
6.40373e-02	1.23343e+00	3.85402e+01
6.35888e-02	1.25916e+00	3.85402e+01
7.14576e-02	1.25874e+00	3.85402e+01
7.66815e-02	1.24827e+00	3.88568e+01
7.59593e-02	1.22282e+00	3.88568e+01
8.73098e-02	1.17177e+00	3.88568e+01
1.03111e-01	1.09654e+00	3.88568e+01
1.22892e-01	9.98676e-01	3.88568e+01
1.46181e-01	8.80172e-01	3.88568e+01
1.72401e-01	7.43444e-01	3.88568e+01
2.00300e-01	5.91314e-01	3.88568e+01
2.28740e-01	4.26907e-01	3.88568e+01
2.55909e-01	2.53635e-01	3.88568e+01
2.78838e-01	7.51497e-02	3.88568e+01
2.93875e-01	-1.04751e-01	3.88568e+01
2.96632e-01	-2.82194e-01	3.88568e+01
2.85238e-01	-4.53473e-01	3.88568e+01
2.61111e-01	-6.15162e-01	3.88568e+01
2.27061e-01	-7.64091e-01	3.88568e+01
1.85786e-01	-8.97330e-01	3.88568e+01
1.39787e-01	-1.01225e+00	3.88568e+01
9.14376e-02	-1.10660e+00	3.88568e+01
4.33762e-02	-1.17852e+00	3.88568e+01
-2.23247e-03	-1.22663e+00	3.88568e+01
-6.45509e-02	-1.24890e+00	3.88568e+01
-1.13576e-01	-1.22087e+00	3.88568e+01

-1.47930e-01	-1.16863e+00	3.88568e+01
-1.79194e-01	-1.09261e+00	3.88568e+01
-2.06232e-01	-9.94369e-01	3.88568e+01
-2.27922e-01	-8.75947e-01	3.88568e+01
-2.41879e-01	-7.39853e-01	3.88568e+01
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-2.19339e-01	-2.55525e-01	3.88568e+01
-1.87661e-01	-7.98623e-02	3.88568e+01
-1.44611e-01	9.70361e-02	3.88568e+01
-9.43531e-02	2.71739e-01	3.88568e+01
-4.26078e-02	4.40932e-01	3.88568e+01
3.44356e-03	6.01488e-01	3.88568e+01
3.79925e-02	7.50391e-01	3.88568e+01
5.88392e-02	8.84686e-01	3.88568e+01
6.79028e-02	1.00152e+00	3.88568e+01
6.87384e-02	1.09832e+00	3.88568e+01
6.57231e-02	1.17288e+00	3.88568e+01
6.20637e-02	1.22354e+00	3.88568e+01
6.16127e-02	1.24905e+00	3.88568e+01
6.93941e-02	1.24865e+00	3.88568e+01
7.44580e-02	1.23820e+00	3.91734e+01
7.38733e-02	1.21295e+00	3.91734e+01
8.51926e-02	1.16232e+00	3.91734e+01
1.00956e-01	1.08773e+00	3.91734e+01
1.20692e-01	9.90686e-01	3.91734e+01
1.43934e-01	8.73175e-01	3.91734e+01
1.70104e-01	7.37593e-01	3.91734e+01
1.97958e-01	5.86736e-01	3.91734e+01
2.26364e-01	4.23703e-01	3.91734e+01
2.53520e-01	2.51875e-01	3.91734e+01
2.76476e-01	7.48719e-02	3.91734e+01
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2.94553e-01	-2.79539e-01	3.91734e+01
2.83458e-01	-4.49436e-01	3.91734e+01
2.59721e-01	-6.09836e-01	3.91734e+01
2.26128e-01	-7.57592e-01	3.91734e+01
1.85349e-01	-8.89795e-01	3.91734e+01
1.39863e-01	-1.00384e+00	3.91734e+01
9.20202e-02	-1.09747e+00	3.91734e+01
4.44367e-02	-1.16887e+00	3.91734e+01
-7.42633e-04	-1.21664e+00	3.91734e+01
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-1.11186e-01	-1.21106e+00	3.91734e+01
-1.45322e-01	-1.15929e+00	3.91734e+01
-1.76423e-01	-1.08392e+00	3.91734e+01
-2.03358e-01	-9.86514e-01	3.91734e+01
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-2.43331e-01	-5.84446e-01	3.91734e+01
-2.36360e-01	-4.23199e-01	3.91734e+01
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-9.39103e-02	2.69411e-01	3.91734e+01
-4.27897e-02	4.37288e-01	3.91734e+01
2.69386e-03	5.96592e-01	3.91734e+01
3.67826e-02	7.44322e-01	3.91734e+01
5.72984e-02	8.77548e-01	3.91734e+01
6.61479e-02	9.93440e-01	3.91734e+01
6.68606e-02	1.08945e+00	3.91734e+01
6.37804e-02	1.16341e+00	3.91734e+01
6.00901e-02	1.21364e+00	3.91734e+01
5.96366e-02	1.23895e+00	3.91734e+01
6.73306e-02	1.23856e+00	3.91734e+01
7.22345e-02	1.22812e+00	3.94899e+01
7.17873e-02	1.20307e+00	3.94899e+01
8.30753e-02	1.15288e+00	3.94899e+01
9.87999e-02	1.07892e+00	3.94899e+01
1.18493e-01	9.82697e-01	3.94899e+01
1.41687e-01	8.66179e-01	3.94899e+01
1.67807e-01	7.31742e-01	3.94899e+01
1.95617e-01	5.82158e-01	3.94899e+01
2.23988e-01	4.20499e-01	3.94899e+01
2.51130e-01	2.50116e-01	3.94899e+01
2.74114e-01	7.45941e-02	3.94899e+01
2.89338e-01	-1.02339e-01	3.94899e+01
2.92475e-01	-2.76883e-01	3.94899e+01
2.81678e-01	-4.45398e-01	3.94899e+01
2.58332e-01	-6.04511e-01	3.94899e+01
2.25194e-01	-7.51094e-01	3.94899e+01
1.84911e-01	-8.82261e-01	3.94899e+01
1.39938e-01	-9.95422e-01	3.94899e+01
9.26028e-02	-1.08834e+00	3.94899e+01
4.54972e-02	-1.15921e+00	3.94899e+01
7.47204e-04	-1.20665e+00	3.94899e+01
-6.05022e-02	-1.22870e+00	3.94899e+01
-1.08796e-01	-1.20125e+00	3.94899e+01
-1.42715e-01	-1.14994e+00	3.94899e+01
-1.73652e-01	-1.07523e+00	3.94899e+01
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-2.22106e-01	-8.62219e-01	3.94899e+01
-2.36162e-01	-7.28376e-01	3.94899e+01
-2.40620e-01	-5.79942e-01	3.94899e+01
-2.33903e-01	-4.20011e-01	3.94899e+01
-2.15152e-01	-2.51887e-01	3.94899e+01
-1.84411e-01	-7.90111e-02	3.94899e+01

-1.42487e-01	9.51078e-02	3.94899e+01
-9.34676e-02	2.67084e-01	3.94899e+01
-4.29715e-02	4.33644e-01	3.94899e+01
1.94417e-03	5.91695e-01	3.94899e+01
3.55728e-02	7.38254e-01	3.94899e+01
5.57576e-02	8.70410e-01	3.94899e+01
6.43931e-02	9.85360e-01	3.94899e+01
6.49828e-02	1.08058e+00	3.94899e+01
6.18377e-02	1.15393e+00	3.94899e+01
5.81164e-02	1.20375e+00	3.94899e+01
5.76605e-02	1.22884e+00	3.94899e+01
6.52671e-02	1.22847e+00	3.94899e+01
7.00110e-02	1.21805e+00	3.98065e+01
6.97012e-02	1.19320e+00	3.98065e+01
8.09581e-02	1.14344e+00	3.98065e+01
9.66441e-02	1.07011e+00	3.98065e+01
1.16293e-01	9.74707e-01	3.98065e+01
1.39440e-01	8.59182e-01	3.98065e+01
1.65511e-01	7.25891e-01	3.98065e+01
1.93275e-01	5.77580e-01	3.98065e+01
2.21612e-01	4.17295e-01	3.98065e+01
2.48741e-01	2.48356e-01	3.98065e+01
2.71751e-01	7.43163e-02	3.98065e+01
2.87069e-01	-1.01133e-01	3.98065e+01
2.90396e-01	-2.74228e-01	3.98065e+01
2.79897e-01	-4.41361e-01	3.98065e+01
2.56942e-01	-5.99185e-01	3.98065e+01
2.24261e-01	-7.44596e-01	3.98065e+01
1.84474e-01	-8.74726e-01	3.98065e+01
1.40014e-01	-9.87006e-01	3.98065e+01
9.31853e-02	-1.07922e+00	3.98065e+01
4.65577e-02	-1.14956e+00	3.98065e+01
2.23704e-03	-1.19665e+00	3.98065e+01
-5.84778e-02	-1.21860e+00	3.98065e+01
-1.06405e-01	-1.19144e+00	3.98065e+01
-1.40108e-01	-1.14060e+00	3.98065e+01
-1.70881e-01	-1.06654e+00	3.98065e+01
-1.97612e-01	-9.70804e-01	3.98065e+01
-2.19198e-01	-8.55355e-01	3.98065e+01
-2.33303e-01	-7.22638e-01	3.98065e+01
-2.37908e-01	-5.75438e-01	3.98065e+01
-2.31445e-01	-4.16823e-01	3.98065e+01
-2.13058e-01	-2.50069e-01	3.98065e+01
-1.82786e-01	-7.85855e-02	3.98065e+01
-1.41425e-01	9.41437e-02	3.98065e+01
-9.30248e-02	2.64757e-01	3.98065e+01
-4.31533e-02	4.30000e-01	3.98065e+01
1.19448e-03	5.86798e-01	3.98065e+01
3.43629e-02	7.32185e-01	3.98065e+01
5.42167e-02	8.63272e-01	3.98065e+01
6.26382e-02	9.77281e-01	3.98065e+01
6.31050e-02	1.07172e+00	3.98065e+01
5.98950e-02	1.14445e+00	3.98065e+01
5.61428e-02	1.19385e+00	3.98065e+01
5.56844e-02	1.21874e+00	3.98065e+01
6.32036e-02	1.21838e+00	3.98065e+01
6.90667e-02	1.20888e+00	4.01231e+01
6.79496e-02	1.18426e+00	4.01231e+01
7.91725e-02	1.13489e+00	4.01231e+01
9.48154e-02	1.06212e+00	4.01231e+01
1.14414e-01	9.67464e-01	4.01231e+01
1.37505e-01	8.52836e-01	4.01231e+01
1.63515e-01	7.20579e-01	4.01231e+01
1.91223e-01	5.73416e-01	4.01231e+01
2.19510e-01	4.14371e-01	4.01231e+01
2.46607e-01	2.46736e-01	4.01231e+01
2.69623e-01	7.40341e-02	4.01231e+01
2.85004e-01	-1.00074e-01	4.01231e+01
2.88484e-01	-2.71859e-01	4.01231e+01
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2.55617e-01	-5.94396e-01	4.01231e+01
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1.83979e-01	-8.67930e-01	4.01231e+01
1.39970e-01	-9.79408e-01	4.01231e+01
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4.73864e-02	-1.14083e+00	4.01231e+01
3.44889e-03	-1.18761e+00	4.01231e+01
-5.67849e-02	-1.20946e+00	4.01231e+01
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-2.30799e-01	-7.17420e-01	4.01231e+01
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-1.81325e-01	-7.81797e-02	4.01231e+01
-1.40454e-01	9.32874e-02	4.01231e+01
-9.25940e-02	2.62662e-01	4.01231e+01
-4.32689e-02	4.26708e-01	4.01231e+01
5.82921e-04	5.82367e-01	4.01231e+01
3.33516e-02	7.26690e-01	4.01231e+01
5.29208e-02	8.56808e-01	4.01231e+01
6.11616e-02	9.69964e-01	4.01231e+01
6.15279e-02	1.06369e+00	4.01231e+01
5.82675e-02	1.13587e+00	4.01231e+01

5.44929e-02	1.18489e+00	4.01231e+01
5.40329e-02	1.20959e+00	4.01231e+01
6.14752e-02	1.20924e+00	4.01231e+01
6.64328e-02	1.20020e+00	4.04397e+01
6.63380e-02	1.17571e+00	4.04397e+01
7.75259e-02	1.12670e+00	4.04397e+01
9.31236e-02	1.05449e+00	4.04397e+01
1.12669e-01	9.60535e-01	4.04397e+01
1.35700e-01	8.46762e-01	4.04397e+01
1.61646e-01	7.15491e-01	4.04397e+01
1.89291e-01	5.69426e-01	4.04397e+01
2.17522e-01	4.11564e-01	4.04397e+01
2.44581e-01	2.45174e-01	4.04397e+01
2.67591e-01	7.37499e-02	4.04397e+01
2.83026e-01	-9.90773e-02	4.04397e+01
2.86643e-01	-2.69609e-01	4.04397e+01
2.76626e-01	-4.34294e-01	4.04397e+01
2.54318e-01	-5.89831e-01	4.04397e+01
2.22404e-01	-7.33157e-01	4.04397e+01
1.83460e-01	-8.61443e-01	4.04397e+01
1.39876e-01	-9.72151e-01	4.04397e+01
9.39193e-02	-1.06309e+00	4.04397e+01
4.81181e-02	-1.13248e+00	4.04397e+01
4.54429e-03	-1.17897e+00	4.04397e+01
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-1.79932e-01	-7.77821e-02	4.04397e+01
-1.39520e-01	9.24763e-02	4.04397e+01
-9.21683e-02	2.60664e-01	4.04397e+01
-4.33568e-02	4.23564e-01	4.04397e+01
2.92329e-05	5.78131e-01	4.04397e+01
3.24234e-02	7.21435e-01	4.04397e+01
5.17275e-02	8.50625e-01	4.04397e+01
5.98016e-02	9.62966e-01	4.04397e+01
6.00768e-02	1.05601e+00	4.04397e+01
5.67720e-02	1.12766e+00	4.04397e+01
5.29786e-02	1.17633e+00	4.04397e+01
5.25175e-02	1.20084e+00	4.04397e+01
5.98871e-02	1.20050e+00	4.04397e+01
6.47189e-02	1.19147e+00	4.07563e+01
6.47264e-02	1.16716e+00	4.07563e+01
7.58792e-02	1.11852e+00	4.07563e+01
9.14318e-02	1.04685e+00	4.07563e+01
1.10924e-01	9.53605e-01	4.07563e+01
1.33895e-01	8.40689e-01	4.07563e+01
1.59777e-01	7.10404e-01	4.07563e+01
1.87359e-01	5.65435e-01	4.07563e+01
2.15535e-01	4.08757e-01	4.07563e+01
2.42554e-01	2.43612e-01	4.07563e+01
2.65560e-01	7.34658e-02	4.07563e+01
2.81047e-01	-9.80804e-02	4.07563e+01
2.84801e-01	-2.67359e-01	4.07563e+01
2.75015e-01	-4.30847e-01	4.07563e+01
2.53020e-01	-5.85266e-01	4.07563e+01
2.21476e-01	-7.27573e-01	4.07563e+01
1.82940e-01	-8.54956e-01	4.07563e+01
1.39782e-01	-9.64895e-01	4.07563e+01
9.42490e-02	-1.05521e+00	4.07563e+01
4.88497e-02	-1.12414e+00	4.07563e+01
5.63968e-03	-1.17033e+00	4.07563e+01
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-2.07651e-01	-2.45183e-01	4.07563e+01
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3.14951e-02	7.16181e-01	4.07563e+01
5.05342e-02	8.44443e-01	4.07563e+01
5.84416e-02	9.55969e-01	4.07563e+01
5.86257e-02	1.04833e+00	4.07563e+01
5.52765e-02	1.11945e+00	4.07563e+01
5.14643e-02	1.16776e+00	4.07563e+01
5.10020e-02	1.19209e+00	4.07563e+01
5.82991e-02	1.19176e+00	4.07563e+01
6.30050e-02	1.18275e+00	4.10729e+01
6.31148e-02	1.15861e+00	4.10729e+01
7.42326e-02	1.11034e+00	4.10729e+01
8.97400e-02	1.03921e+00	4.10729e+01
1.09179e-01	9.46676e-01	4.10729e+01

1.32090e-01	8.34615e-01	4.10729e+01
1.57907e-01	7.05317e-01	4.10729e+01
1.85427e-01	5.61444e-01	4.10729e+01
2.13547e-01	4.05950e-01	4.10729e+01
2.40527e-01	2.42050e-01	4.10729e+01
2.63529e-01	7.31816e-02	4.10729e+01
2.79068e-01	-9.70836e-02	4.10729e+01
2.82959e-01	-2.65110e-01	4.10729e+01
2.73404e-01	-4.27400e-01	4.10729e+01
2.51721e-01	-5.80701e-01	4.10729e+01
2.20549e-01	-7.21988e-01	4.10729e+01
1.82421e-01	-8.48469e-01	4.10729e+01
1.39687e-01	-9.57639e-01	4.10729e+01
9.45787e-02	-1.04733e+00	4.10729e+01
4.95814e-02	-1.11580e+00	4.10729e+01
6.73508e-03	-1.16169e+00	4.10729e+01
-5.21224e-02	-1.18323e+00	4.10729e+01
-9.87535e-02	-1.15704e+00	4.10729e+01
-1.31665e-01	-1.10781e+00	4.10729e+01
-1.61822e-01	-1.03604e+00	4.10729e+01
-1.88137e-01	-9.43198e-01	4.10729e+01
-2.09532e-01	-8.31203e-01	4.10729e+01
-2.23732e-01	-7.02417e-01	4.10729e+01
-2.28764e-01	-5.59535e-01	4.10729e+01
-2.23094e-01	-4.05529e-01	4.10729e+01
-2.05880e-01	-2.43576e-01	4.10729e+01
-1.77147e-01	-7.69869e-02	4.10729e+01
-1.37653e-01	9.08541e-02	4.10729e+01
-9.13169e-02	2.56668e-01	4.10729e+01
-4.35325e-02	4.17274e-01	4.10729e+01
-1.07814e-03	5.69660e-01	4.10729e+01
3.05669e-02	7.10926e-01	4.10729e+01
4.93408e-02	8.38260e-01	4.10729e+01
5.70815e-02	9.48971e-01	4.10729e+01
5.71746e-02	1.04065e+00	4.10729e+01
5.37810e-02	1.11124e+00	4.10729e+01
4.99500e-02	1.15919e+00	4.10729e+01
4.94865e-02	1.18334e+00	4.10729e+01
5.67110e-02	1.18303e+00	4.10729e+01
6.12911e-02	1.17402e+00	4.13894e+01
6.15032e-02	1.15006e+00	4.13894e+01
7.25860e-02	1.10216e+00	4.13894e+01
8.80483e-02	1.03158e+00	4.13894e+01
1.07434e-01	9.39746e-01	4.13894e+01
1.30285e-01	8.28541e-01	4.13894e+01
1.56038e-01	7.00230e-01	4.13894e+01
1.83495e-01	5.57454e-01	4.13894e+01
2.11559e-01	4.03143e-01	4.13894e+01
2.38501e-01	2.40487e-01	4.13894e+01
2.61498e-01	7.28975e-02	4.13894e+01
2.77090e-01	-9.60868e-02	4.13894e+01
2.81117e-01	-2.62860e-01	4.13894e+01
2.71793e-01	-4.23954e-01	4.13894e+01
2.50423e-01	-5.76136e-01	4.13894e+01
2.19621e-01	-7.16404e-01	4.13894e+01
1.81901e-01	-8.41982e-01	4.13894e+01
1.39593e-01	-9.50382e-01	4.13894e+01
9.49083e-02	-1.03946e+00	4.13894e+01
5.03130e-02	-1.10745e+00	4.13894e+01
7.83047e-03	-1.15305e+00	4.13894e+01
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-2.21376e-01	-6.97417e-01	4.13894e+01
-2.26512e-01	-5.55602e-01	4.13894e+01
-2.21036e-01	-4.02734e-01	4.13894e+01
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-1.75754e-01	-7.65893e-02	4.13894e+01
-1.36719e-01	9.00430e-02	4.13894e+01
-9.08911e-02	2.54670e-01	4.13894e+01
-4.36203e-02	4.14129e-01	4.13894e+01
-1.63183e-03	5.65424e-01	4.13894e+01
2.96387e-02	7.05672e-01	4.13894e+01
4.81475e-02	8.32077e-01	4.13894e+01
5.57215e-02	9.41973e-01	4.13894e+01
5.57234e-02	1.03297e+00	4.13894e+01
5.22855e-02	1.10304e+00	4.13894e+01
4.84357e-02	1.15062e+00	4.13894e+01
4.79711e-02	1.17460e+00	4.13894e+01
5.51229e-02	1.17429e+00	4.13894e+01
5.95772e-02	1.16529e+00	4.17060e+01
5.98917e-02	1.14151e+00	4.17060e+01
7.09393e-02	1.09398e+00	4.17060e+01
8.63565e-02	1.02394e+00	4.17060e+01
1.05689e-01	9.32817e-01	4.17060e+01
1.28480e-01	8.22467e-01	4.17060e+01
1.54169e-01	6.95142e-01	4.17060e+01
1.81563e-01	5.53463e-01	4.17060e+01
2.09572e-01	4.00335e-01	4.17060e+01
2.36474e-01	2.38925e-01	4.17060e+01
2.59467e-01	7.26133e-02	4.17060e+01
2.75111e-01	-9.50899e-02	4.17060e+01
2.79275e-01	-2.60610e-01	4.17060e+01
2.70182e-01	-4.20507e-01	4.17060e+01

2.49124e-01	-5.71572e-01	4.17060e+01
2.18694e-01	-7.10820e-01	4.17060e+01
1.81382e-01	-8.35495e-01	4.17060e+01
1.39499e-01	-9.43126e-01	4.17060e+01
9.52380e-02	-1.03158e+00	4.17060e+01
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8.92587e-03	-1.14440e+00	4.17060e+01
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-2.18978e-01	-3.99940e-01	4.17060e+01
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5.78634e-02	1.15657e+00	4.20226e+01
5.82801e-02	1.13296e+00	4.20226e+01
6.92927e-02	1.08580e+00	4.20226e+01
8.46647e-02	1.01631e+00	4.20226e+01
1.03944e-01	9.25888e-01	4.20226e+01
1.26676e-01	8.16393e-01	4.20226e+01
1.52300e-01	6.90055e-01	4.20226e+01
1.79631e-01	5.49472e-01	4.20226e+01
2.07584e-01	3.97528e-01	4.20226e+01
2.34447e-01	2.37363e-01	4.20226e+01
2.57436e-01	7.23292e-02	4.20226e+01
2.73132e-01	-9.40931e-02	4.20226e+01
2.77433e-01	-2.58361e-01	4.20226e+01
2.68571e-01	-4.17060e-01	4.20226e+01
2.47826e-01	-5.67007e-01	4.20226e+01
2.17766e-01	-7.05235e-01	4.20226e+01
1.80863e-01	-8.29009e-01	4.20226e+01
1.39405e-01	-9.35870e-01	4.20226e+01
9.55676e-02	-1.02370e+00	4.20226e+01
5.17763e-02	-1.09077e+00	4.20226e+01
1.00213e-02	-1.13576e+00	4.20226e+01
-4.74600e-02	-1.15700e+00	4.20226e+01
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-1.25452e-01	-1.08350e+00	4.20226e+01
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-1.34852e-01	8.84208e-02	4.20226e+01
-9.00397e-02	2.50674e-01	4.20226e+01
-4.37960e-02	4.07840e-01	4.20226e+01
-2.73921e-03	5.56953e-01	4.20226e+01
2.77823e-02	6.95163e-01	4.20226e+01
4.57609e-02	8.19712e-01	4.20226e+01
5.30015e-02	9.27977e-01	4.20226e+01
5.28212e-02	1.01761e+00	4.20226e+01
4.92945e-02	1.08662e+00	4.20226e+01
4.54071e-02	1.13349e+00	4.20226e+01
4.49402e-02	1.15710e+00	4.20226e+01
5.19468e-02	1.15681e+00	4.20226e+01
5.61495e-02	1.14784e+00	4.23392e+01
5.66685e-02	1.12441e+00	4.23392e+01
6.76460e-02	1.07762e+00	4.23392e+01
8.29729e-02	1.00867e+00	4.23392e+01
1.02199e-01	9.18958e-01	4.23392e+01
1.24871e-01	8.10320e-01	4.23392e+01
1.50430e-01	6.84968e-01	4.23392e+01
1.77699e-01	5.45482e-01	4.23392e+01
2.05596e-01	3.94721e-01	4.23392e+01
2.32421e-01	2.35801e-01	4.23392e+01
2.55405e-01	7.20450e-02	4.23392e+01
2.71153e-01	-9.30963e-02	4.23392e+01
2.75591e-01	-2.56111e-01	4.23392e+01
2.66960e-01	-4.13614e-01	4.23392e+01
2.46527e-01	-5.62442e-01	4.23392e+01
2.16839e-01	-6.99651e-01	4.23392e+01
1.80343e-01	-8.22522e-01	4.23392e+01
1.39311e-01	-9.28613e-01	4.23392e+01
9.58973e-02	-1.01582e+00	4.23392e+01
5.25080e-02	-1.08243e+00	4.23392e+01
1.11167e-02	-1.12712e+00	4.23392e+01
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-1.71576e-01	-7.53964e-02	4.23392e+01
-1.33918e-01	8.76097e-02	4.23392e+01
-8.96140e-02	2.48676e-01	4.23392e+01
-4.38839e-02	4.04695e-01	4.23392e+01
-3.29290e-03	5.52718e-01	4.23392e+01
2.68541e-02	6.89908e-01	4.23392e+01
4.45675e-02	8.13530e-01	4.23392e+01
5.16414e-02	9.20979e-01	4.23392e+01
5.13701e-02	1.00993e+00	4.23392e+01
4.77990e-02	1.07842e+00	4.23392e+01
4.38928e-02	1.12492e+00	4.23392e+01
4.34247e-02	1.14835e+00	4.23392e+01
5.03588e-02	1.14807e+00	4.23392e+01
5.44356e-02	1.13911e+00	4.26558e+01
5.50569e-02	1.11586e+00	4.26558e+01
6.59994e-02	1.06944e+00	4.26558e+01
8.12811e-02	1.00103e+00	4.26558e+01
1.00454e-01	9.12029e-01	4.26558e+01
1.23066e-01	8.04246e-01	4.26558e+01
1.48561e-01	6.79881e-01	4.26558e+01
1.75767e-01	5.41491e-01	4.26558e+01
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2.30394e-01	2.34239e-01	4.26558e+01
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1.79824e-01	-8.16035e-01	4.26558e+01
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9.62269e-02	-1.00794e+00	4.26558e+01
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2.59259e-02	6.84654e-01	4.26558e+01
4.33742e-02	8.07347e-01	4.26558e+01
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4.87707e-02	1.13934e+00	4.26558e+01
5.27217e-02	1.13039e+00	4.29724e+01
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6.43528e-02	1.06126e+00	4.29724e+01
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1.73835e-01	5.37501e-01	4.29724e+01
2.01621e-01	3.89107e-01	4.29724e+01
2.28368e-01	2.32677e-01	4.29724e+01
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2.63738e-01	-4.06720e-01	4.29724e+01
2.43930e-01	-5.53312e-01	4.29724e+01
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1.79304e-01	-8.09548e-01	4.29724e+01
1.39123e-01	-9.14101e-01	4.29724e+01
9.65566e-02	-1.00006e+00	4.29724e+01
5.39713e-02	-1.06574e+00	4.29724e+01
1.33074e-02	-1.10984e+00	4.29724e+01
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4.48080e-02	1.06200e+00	4.29724e+01
4.08642e-02	1.10778e+00	4.29724e+01
4.03938e-02	1.13085e+00	4.29724e+01
4.71826e-02	1.13060e+00	4.29724e+01
5.21136e-02	1.12268e+00	4.32889e+01
5.21785e-02	1.09978e+00	4.32889e+01
6.30489e-02	1.05406e+00	4.32889e+01
7.82366e-02	9.86671e-01	4.32889e+01
9.72973e-02	8.98993e-01	4.32889e+01
1.19782e-01	7.92815e-01	4.32889e+01
1.45140e-01	6.70301e-01	4.32889e+01
1.72209e-01	5.33968e-01	4.32889e+01
1.99926e-01	3.86612e-01	4.32889e+01
2.26617e-01	2.31273e-01	4.32889e+01
2.49567e-01	7.11928e-02	4.32889e+01
2.65444e-01	-9.02624e-02	4.32889e+01
2.70253e-01	-2.49670e-01	4.32889e+01
2.62267e-01	-4.03726e-01	4.32889e+01
2.42713e-01	-5.49332e-01	4.32889e+01
2.14075e-01	-6.83602e-01	4.32889e+01
1.78738e-01	-8.03868e-01	4.32889e+01
1.38916e-01	-9.07740e-01	4.32889e+01
9.67108e-02	-9.93148e-01	4.32889e+01
5.44701e-02	-1.05841e+00	4.32889e+01
1.41201e-02	-1.10224e+00	4.32889e+01
-4.15844e-02	-1.12307e+00	4.32889e+01
-8.60052e-02	-1.09853e+00	4.32889e+01
-1.17561e-01	-1.05204e+00	4.32889e+01
-1.46653e-01	-9.84137e-01	4.32889e+01
-1.72240e-01	-8.96218e-01	4.32889e+01
-1.93287e-01	-7.90093e-01	4.32889e+01
-2.07618e-01	-6.67987e-01	4.32889e+01
-2.13343e-01	-5.32445e-01	4.32889e+01
-2.08988e-01	-3.86276e-01	4.32889e+01
-1.93732e-01	-2.32490e-01	4.32889e+01
-1.67576e-01	-7.42287e-02	4.32889e+01
-1.31218e-01	8.52923e-02	4.32889e+01
-8.83551e-02	2.42935e-01	4.32889e+01
-4.40828e-02	3.95647e-01	4.32889e+01
-4.81330e-03	5.40523e-01	4.32889e+01
2.42737e-02	6.74776e-01	4.32889e+01
4.12403e-02	7.95723e-01	4.32889e+01
4.78486e-02	9.00824e-01	4.32889e+01
4.73268e-02	9.87815e-01	4.32889e+01
4.36371e-02	1.05478e+00	4.32889e+01
3.96830e-02	1.10025e+00	4.32889e+01
3.92123e-02	1.12316e+00	4.32889e+01
4.59397e-02	1.12291e+00	4.32889e+01
5.00367e-02	1.11503e+00	4.36055e+01
5.09146e-02	1.09227e+00	4.36055e+01
6.17479e-02	1.04687e+00	4.36055e+01
7.68867e-02	9.79953e-01	4.36055e+01
9.58886e-02	8.92894e-01	4.36055e+01
1.18306e-01	7.87464e-01	4.36055e+01
1.43590e-01	6.65813e-01	4.36055e+01
1.70586e-01	5.30440e-01	4.36055e+01
1.98234e-01	3.84119e-01	4.36055e+01
2.24869e-01	2.29869e-01	4.36055e+01
2.47793e-01	7.09089e-02	4.36055e+01
2.63694e-01	-8.94234e-02	4.36055e+01
2.68601e-01	-2.47732e-01	4.36055e+01
2.60797e-01	-4.00735e-01	4.36055e+01
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2.13166e-01	-6.78727e-01	4.36055e+01
1.78171e-01	-7.98196e-01	4.36055e+01
1.38709e-01	-9.01386e-01	4.36055e+01
9.68636e-02	-9.86242e-01	4.36055e+01
5.49669e-02	-1.05109e+00	4.36055e+01
1.49303e-02	-1.09465e+00	4.36055e+01
-4.03741e-02	-1.11538e+00	4.36055e+01
-8.45098e-02	-1.09105e+00	4.36055e+01
-1.15887e-01	-1.04491e+00	4.36055e+01
-1.44835e-01	-9.77494e-01	4.36055e+01
-1.70319e-01	-8.90200e-01	4.36055e+01
-1.91308e-01	-7.84822e-01	4.36055e+01
-2.05641e-01	-6.63567e-01	4.36055e+01
-2.11438e-01	-5.28961e-01	4.36055e+01
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-1.92209e-01	-2.31052e-01	4.36055e+01
-1.66363e-01	-7.38565e-02	4.36055e+01
-1.30387e-01	8.45981e-02	4.36055e+01
-8.79477e-02	2.41193e-01	4.36055e+01
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-5.22515e-03	5.36804e-01	4.36055e+01
2.35514e-02	6.70158e-01	4.36055e+01
4.03019e-02	7.90288e-01	4.36055e+01
4.67782e-02	8.94672e-01	4.36055e+01
4.61884e-02	9.81064e-01	4.36055e+01
4.24689e-02	1.04756e+00	4.36055e+01

3.85045e-02	1.09272e+00	4.36055e+01
3.80337e-02	1.11547e+00	4.36055e+01
4.46996e-02	1.11523e+00	4.36055e+01
4.86957e-02	1.10736e+00	4.39221e+01
4.96506e-02	1.08475e+00	4.39221e+01
6.04469e-02	1.03967e+00	4.39221e+01
7.55367e-02	9.73236e-01	4.39221e+01
9.44798e-02	8.86795e-01	4.39221e+01
1.16831e-01	7.82113e-01	4.39221e+01
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1.68963e-01	5.26911e-01	4.39221e+01
1.96541e-01	3.81626e-01	4.39221e+01
2.23121e-01	2.28466e-01	4.39221e+01
2.46019e-01	7.06250e-02	4.39221e+01
2.61943e-01	-8.85845e-02	4.39221e+01
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2.40281e-01	-5.41381e-01	4.39221e+01
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1.77604e-01	-7.92524e-01	4.39221e+01
1.38501e-01	-8.95032e-01	4.39221e+01
9.70164e-02	-9.79336e-01	4.39221e+01
5.54637e-02	-1.04377e+00	4.39221e+01
1.57406e-02	-1.08706e+00	4.39221e+01
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-2.03664e-01	-6.59146e-01	4.39221e+01
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-2.05479e-01	-3.81310e-01	4.39221e+01
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-1.29555e-01	8.39038e-02	4.39221e+01
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2.28291e-02	6.65540e-01	4.39221e+01
3.93634e-02	7.84852e-01	4.39221e+01
4.57078e-02	8.88519e-01	4.39221e+01
4.50500e-02	9.74314e-01	4.39221e+01
4.13008e-02	1.04035e+00	4.39221e+01
3.73261e-02	1.08519e+00	4.39221e+01
3.68551e-02	1.10778e+00	4.39221e+01
4.34595e-02	1.10755e+00	4.39221e+01
4.73547e-02	1.09969e+00	4.42387e+01
4.83867e-02	1.07723e+00	4.42387e+01
5.91459e-02	1.03248e+00	4.42387e+01
7.41868e-02	9.66518e-01	4.42387e+01
9.30710e-02	8.80696e-01	4.42387e+01
1.15355e-01	7.76763e-01	4.42387e+01
1.40492e-01	6.56837e-01	4.42387e+01
1.67340e-01	5.23383e-01	4.42387e+01
1.94849e-01	3.79133e-01	4.42387e+01
2.21373e-01	2.27063e-01	4.42387e+01
2.44245e-01	7.03411e-02	4.42387e+01
2.60193e-01	-8.77456e-02	4.42387e+01
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2.57856e-01	-3.94753e-01	4.42387e+01
2.39065e-01	-5.37405e-01	4.42387e+01
2.11349e-01	-6.68977e-01	4.42387e+01
1.77037e-01	-7.86851e-01	4.42387e+01
1.38293e-01	-8.88679e-01	4.42387e+01
9.71691e-02	-9.72430e-01	4.42387e+01
5.59606e-02	-1.03645e+00	4.42387e+01
1.65508e-02	-1.07948e+00	4.42387e+01
-3.79536e-02	-1.10001e+00	4.42387e+01
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-1.41199e-01	-9.64206e-01	4.42387e+01
-1.66476e-01	-8.78163e-01	4.42387e+01
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-2.01687e-01	-6.54726e-01	4.42387e+01
-2.07629e-01	-5.21993e-01	4.42387e+01
-2.03725e-01	-3.78827e-01	4.42387e+01
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2.21068e-02	6.60922e-01	4.42387e+01
3.84249e-02	7.79417e-01	4.42387e+01
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4.39116e-02	9.67563e-01	4.42387e+01
4.01326e-02	1.03314e+00	4.42387e+01
3.61477e-02	1.07766e+00	4.42387e+01
3.56765e-02	1.10009e+00	4.42387e+01
4.22194e-02	1.09987e+00	4.42387e+01
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1.13879e-01	7.71412e-01	4.45553e+01
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1.65716e-01	5.19854e-01	4.45553e+01
1.93156e-01	3.76641e-01	4.45553e+01
2.19625e-01	2.25660e-01	4.45553e+01
2.42471e-01	7.00572e-02	4.45553e+01
2.58443e-01	-8.69067e-02	4.45553e+01
2.63645e-01	-2.41916e-01	4.45553e+01
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2.37849e-01	-5.33429e-01	4.45553e+01
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1.76471e-01	-7.81179e-01	4.45553e+01
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9.73219e-02	-9.65524e-01	4.45553e+01
5.64574e-02	-1.02913e+00	4.45553e+01
1.73611e-02	-1.07189e+00	4.45553e+01
-3.67433e-02	-1.09233e+00	4.45553e+01
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-1.10865e-01	-1.02350e+00	4.45553e+01
-1.39381e-01	-9.57562e-01	4.45553e+01
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-1.85372e-01	-7.69007e-01	4.45553e+01
-1.99710e-01	-6.50305e-01	4.45553e+01
-2.05724e-01	-5.18509e-01	4.45553e+01
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-1.87640e-01	-2.26736e-01	4.45553e+01
-1.62724e-01	-7.27397e-02	4.45553e+01
-1.27891e-01	8.25153e-02	4.45553e+01
-8.67257e-02	2.35965e-01	4.45553e+01
-4.41735e-02	3.84624e-01	4.45553e+01
-6.46068e-03	5.25646e-01	4.45553e+01
2.13845e-02	6.56304e-01	4.45553e+01
3.74865e-02	7.73982e-01	4.45553e+01
4.35670e-02	8.76215e-01	4.45553e+01
4.27732e-02	9.60812e-01	4.45553e+01
3.89644e-02	1.02592e+00	4.45553e+01
3.49693e-02	1.07013e+00	4.45553e+01
3.44979e-02	1.09240e+00	4.45553e+01
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4.46728e-02	1.08434e+00	4.48719e+01
4.58589e-02	1.06220e+00	4.48719e+01
5.65438e-02	1.01809e+00	4.48719e+01
7.14870e-02	9.53083e-01	4.48719e+01
9.02535e-02	8.68498e-01	4.48719e+01
1.12403e-01	7.66061e-01	4.48719e+01
1.37393e-01	6.47862e-01	4.48719e+01
1.64093e-01	5.16326e-01	4.48719e+01
1.91464e-01	3.74148e-01	4.48719e+01
2.17877e-01	2.24257e-01	4.48719e+01
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2.61993e-01	-2.39978e-01	4.48719e+01
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2.36633e-01	-5.29454e-01	4.48719e+01
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5.69542e-02	-1.02181e+00	4.48719e+01
1.81713e-02	-1.06430e+00	4.48719e+01
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2.06622e-02	6.51686e-01	4.48719e+01
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4.16348e-02	9.54061e-01	4.48719e+01
3.77962e-02	1.01871e+00	4.48719e+01
3.37909e-02	1.06260e+00	4.48719e+01
3.33192e-02	1.08472e+00	4.48719e+01
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4.33318e-02	1.07667e+00	4.51884e+01
4.45949e-02	1.05469e+00	4.51884e+01
5.52428e-02	1.01090e+00	4.51884e+01
7.01370e-02	9.46366e-01	4.51884e+01
8.88447e-02	8.62399e-01	4.51884e+01
1.10927e-01	7.60710e-01	4.51884e+01
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1.62470e-01	5.12797e-01	4.51884e+01
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2.38923e-01	6.94894e-02	4.51884e+01
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2.60341e-01	-2.38039e-01	4.51884e+01
2.53445e-01	-3.85780e-01	4.51884e+01

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1.75337e-01	-7.69834e-01	4.51884e+01
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3.66281e-02	1.01149e+00	4.51884e+01
3.26124e-02	1.05507e+00	4.51884e+01
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4.19909e-02	1.06900e+00	4.55050e+01
4.33310e-02	1.04717e+00	4.55050e+01
5.39418e-02	1.00371e+00	4.55050e+01
6.87871e-02	9.39648e-01	4.55050e+01
8.74359e-02	8.56300e-01	4.55050e+01
1.09451e-01	7.55360e-01	4.55050e+01
1.34294e-01	6.38886e-01	4.55050e+01
1.60847e-01	5.09269e-01	4.55050e+01
1.88079e-01	3.69162e-01	4.55050e+01
2.14381e-01	2.21451e-01	4.55050e+01
2.37149e-01	6.92055e-02	4.55050e+01
2.53192e-01	-8.43900e-02	4.55050e+01
2.58689e-01	-2.36100e-01	4.55050e+01
2.51974e-01	-3.82790e-01	4.55050e+01
2.34200e-01	-5.21503e-01	4.55050e+01
2.07714e-01	-6.49477e-01	4.55050e+01
1.74770e-01	-7.64161e-01	4.55050e+01
1.37463e-01	-8.63264e-01	4.55050e+01
9.77802e-02	-9.44807e-01	4.55050e+01
5.79479e-02	-1.00717e+00	4.55050e+01
1.97918e-02	-1.04912e+00	4.55050e+01
-3.31125e-02	-1.06927e+00	4.55050e+01
-7.55374e-02	-1.04617e+00	4.55050e+01
-1.05843e-01	-1.00210e+00	4.55050e+01
-1.33927e-01	-9.37631e-01	4.55050e+01
-1.58789e-01	-8.54090e-01	4.55050e+01
-1.79435e-01	-7.53192e-01	4.55050e+01
-1.93780e-01	-6.37044e-01	4.55050e+01
-2.00010e-01	-5.08056e-01	4.55050e+01
-1.96707e-01	-3.68895e-01	4.55050e+01
-1.83071e-01	-2.22420e-01	4.55050e+01
-1.59085e-01	-7.16229e-02	4.55050e+01
-1.25396e-01	8.04325e-02	4.55050e+01
-8.55037e-02	2.30737e-01	4.55050e+01
-4.42415e-02	3.76357e-01	4.55050e+01
-7.69621e-03	5.14488e-01	4.55050e+01
1.92177e-02	6.42450e-01	4.55050e+01
3.46711e-02	7.57675e-01	4.55050e+01
4.03558e-02	8.57757e-01	4.55050e+01
3.93579e-02	9.40560e-01	4.55050e+01
3.54599e-02	1.00428e+00	4.55050e+01
3.14340e-02	1.04754e+00	4.55050e+01
3.09620e-02	1.06934e+00	4.55050e+01
3.72591e-02	1.06915e+00	4.55050e+01
4.06499e-02	1.06133e+00	4.58216e+01
4.20671e-02	1.03965e+00	4.58216e+01
5.26408e-02	9.96514e-01	4.58216e+01
6.74372e-02	9.32931e-01	4.58216e+01
8.60272e-02	8.50200e-01	4.58216e+01
1.07975e-01	7.50009e-01	4.58216e+01
1.32745e-01	6.34398e-01	4.58216e+01
1.59224e-01	5.05741e-01	4.58216e+01
1.86387e-01	3.66670e-01	4.58216e+01
2.12633e-01	2.20048e-01	4.58216e+01
2.35375e-01	6.89215e-02	4.58216e+01
2.51442e-01	-8.35511e-02	4.58216e+01
2.57037e-01	-2.34162e-01	4.58216e+01
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2.06805e-01	-6.44602e-01	4.58216e+01
1.74203e-01	-7.58489e-01	4.58216e+01
1.37255e-01	-8.56910e-01	4.58216e+01
9.79330e-02	-9.37901e-01	4.58216e+01
5.84447e-02	-9.99853e-01	4.58216e+01
2.06020e-02	-1.04154e+00	4.58216e+01
-3.19022e-02	-1.06159e+00	4.58216e+01
-7.40420e-02	-1.03869e+00	4.58216e+01

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-1.56868e-01	-8.48072e-01	4.58216e+01
-1.77456e-01	-7.47921e-01	4.58216e+01
-1.91803e-01	-6.32623e-01	4.58216e+01
-1.98106e-01	-5.04572e-01	4.58216e+01
-1.94953e-01	-3.66412e-01	4.58216e+01
-1.81548e-01	-2.20982e-01	4.58216e+01
-1.57873e-01	-7.12506e-02	4.58216e+01
-1.24565e-01	7.97382e-02	4.58216e+01
-8.50964e-02	2.28995e-01	4.58216e+01
-4.42642e-02	3.73601e-01	4.58216e+01
-8.10805e-03	5.10769e-01	4.58216e+01
1.84954e-02	6.37832e-01	4.58216e+01
3.37327e-02	7.52240e-01	4.58216e+01
3.92854e-02	8.51605e-01	4.58216e+01
3.82195e-02	9.33809e-01	4.58216e+01
3.42917e-02	9.97066e-01	4.58216e+01
3.02556e-02	1.04001e+00	4.58216e+01
2.97834e-02	1.06165e+00	4.58216e+01
3.60190e-02	1.06146e+00	4.58216e+01
3.99984e-02	1.05390e+00	4.61382e+01
4.07818e-02	1.03240e+00	4.61382e+01
5.13091e-02	9.89574e-01	4.61382e+01
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1.84719e-01	3.64260e-01	4.61382e+01
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1.36642e-01	-8.50775e-01	4.61382e+01
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2.11523e-02	-1.03421e+00	4.61382e+01
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1.74889e-02	6.33382e-01	4.61382e+01
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3.71424e-02	9.27294e-01	4.61382e+01
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2.92836e-02	1.03274e+00	4.61382e+01
2.89452e-02	1.05423e+00	4.61382e+01
3.47415e-02	1.05406e+00	4.61382e+01
3.73252e-02	1.04723e+00	4.64548e+01
3.94414e-02	1.02583e+00	4.64548e+01
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6.45372e-02	9.20580e-01	4.64548e+01
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1.56092e-01	4.99236e-01	4.64548e+01
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2.09039e-01	2.17449e-01	4.64548e+01
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2.52397e-01	-2.30595e-01	4.64548e+01
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2.28467e-01	-5.10196e-01	4.64548e+01
2.02848e-01	-6.35614e-01	4.64548e+01
1.71019e-01	-7.48033e-01	4.64548e+01
1.34978e-01	-8.45203e-01	4.64548e+01
9.66208e-02	-9.25180e-01	4.64548e+01
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2.10293e-02	-1.02756e+00	4.64548e+01
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-9.97221e-02	-9.81865e-01	4.64548e+01
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-1.49253e-01	-8.37075e-01	4.64548e+01
-1.68542e-01	-7.38321e-01	4.64548e+01
-1.82060e-01	-6.24605e-01	4.64548e+01
-1.88163e-01	-4.98282e-01	4.64548e+01
-1.85521e-01	-3.61958e-01	4.64548e+01
-1.73249e-01	-2.18435e-01	4.64548e+01
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-4.38575e-02	3.68530e-01	4.64548e+01
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1.57464e-02	6.29369e-01	4.64548e+01
3.07421e-02	7.42269e-01	4.64548e+01
3.66827e-02	8.40302e-01	4.64548e+01
3.62238e-02	9.21389e-01	4.64548e+01
3.27414e-02	9.83784e-01	4.64548e+01
2.88460e-02	1.02614e+00	4.64548e+01
2.89886e-02	1.04748e+00	4.64548e+01
3.33671e-02	1.04735e+00	4.64548e+01
3.54814e-02	1.04053e+00	4.67714e+01
3.81009e-02	1.01927e+00	4.67714e+01
4.84865e-02	9.77004e-01	4.67714e+01
6.30308e-02	9.14711e-01	4.67714e+01
8.14400e-02	8.33654e-01	4.67714e+01
1.03327e-01	7.35484e-01	4.67714e+01
1.28099e-01	6.22205e-01	4.67714e+01
1.54558e-01	4.96139e-01	4.67714e+01
1.81511e-01	3.59872e-01	4.67714e+01
2.07214e-01	2.16209e-01	4.67714e+01
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1.68848e-01	-7.43057e-01	4.67714e+01
1.33315e-01	-8.39632e-01	4.67714e+01
9.55083e-02	-9.19127e-01	4.67714e+01
5.74605e-02	-9.79959e-01	4.67714e+01
2.09064e-02	-1.02092e+00	4.67714e+01
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-1.81461e-01	-4.95329e-01	4.67714e+01
-1.79134e-01	-3.59876e-01	4.67714e+01
-1.67618e-01	-2.17254e-01	4.67714e+01
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-1.15911e-01	7.77424e-02	4.67714e+01
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1.40039e-02	6.25356e-01	4.67714e+01
2.89327e-02	7.37538e-01	4.67714e+01
3.52510e-02	8.34932e-01	4.67714e+01
3.53052e-02	9.15485e-01	4.67714e+01
3.21880e-02	9.77466e-01	4.67714e+01
2.84085e-02	1.01955e+00	4.67714e+01
2.90320e-02	1.04074e+00	4.67714e+01
3.19926e-02	1.04064e+00	4.67714e+01
3.36377e-02	1.03384e+00	4.70879e+01
3.67605e-02	1.01270e+00	4.70879e+01
4.70751e-02	9.70719e-01	4.70879e+01
6.15243e-02	9.08841e-01	4.70879e+01
7.98729e-02	8.28323e-01	4.70879e+01
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1.53025e-01	4.93043e-01	4.70879e+01
1.79907e-01	3.57678e-01	4.70879e+01
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2.42005e-01	-8.05544e-02	4.70879e+01
2.47004e-01	-2.27202e-01	4.70879e+01
2.39955e-01	-3.69037e-01	4.70879e+01
2.22774e-01	-5.03216e-01	4.70879e+01
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1.66677e-01	-7.38080e-01	4.70879e+01
1.31651e-01	-8.34061e-01	4.70879e+01
9.43958e-02	-9.13075e-01	4.70879e+01
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2.07834e-02	-1.01427e+00	4.70879e+01
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-1.72747e-01	-3.57793e-01	4.70879e+01
-1.61987e-01	-2.16074e-01	4.70879e+01
-1.41366e-01	-7.01315e-02	4.70879e+01
-1.12377e-01	7.70863e-02	4.70879e+01
-7.83278e-02	2.22656e-01	4.70879e+01
-4.31958e-02	3.63707e-01	4.70879e+01
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1.22614e-02	6.21343e-01	4.70879e+01
2.71233e-02	7.32806e-01	4.70879e+01
3.38193e-02	8.29563e-01	4.70879e+01
3.43866e-02	9.09580e-01	4.70879e+01
3.16346e-02	9.71148e-01	4.70879e+01

2.79710e-02	1.01295e+00	4.70879e+01
2.90754e-02	1.03399e+00	4.70879e+01
3.06181e-02	1.03393e+00	4.70879e+01
3.17940e-02	1.02714e+00	4.74045e+01
3.54200e-02	1.00613e+00	4.74045e+01
4.56638e-02	9.64434e-01	4.74045e+01
6.00179e-02	9.02971e-01	4.74045e+01
7.83058e-02	8.22992e-01	4.74045e+01
1.00183e-01	7.26124e-01	4.74045e+01
1.25002e-01	6.14343e-01	4.74045e+01
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1.78303e-01	3.55484e-01	4.74045e+01
2.03564e-01	2.13727e-01	4.74045e+01
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2.39512e-01	-7.98249e-02	4.74045e+01
2.44308e-01	-2.25506e-01	4.74045e+01
2.37060e-01	-3.66413e-01	4.74045e+01
2.19928e-01	-4.99725e-01	4.74045e+01
1.95102e-01	-6.22778e-01	4.74045e+01
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1.29988e-01	-8.28490e-01	4.74045e+01
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5.62820e-02	-9.67133e-01	4.74045e+01
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2.53138e-02	7.28075e-01	4.74045e+01
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2.75334e-02	1.00635e+00	4.74045e+01
2.91189e-02	1.02725e+00	4.74045e+01
2.92437e-02	1.02722e+00	4.74045e+01
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4.42525e-02	9.58150e-01	4.77211e+01
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9.86103e-02	7.21443e-01	4.77211e+01
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1.76699e-01	3.53290e-01	4.77211e+01
2.01739e-01	2.12487e-01	4.77211e+01
2.22723e-01	6.73544e-02	4.77211e+01
2.37019e-01	-7.90955e-02	4.77211e+01
2.41611e-01	-2.23810e-01	4.77211e+01
2.34166e-01	-3.63789e-01	4.77211e+01
2.17081e-01	-4.96235e-01	4.77211e+01
1.92520e-01	-6.18499e-01	4.77211e+01
1.62336e-01	-7.28127e-01	4.77211e+01
1.28324e-01	-8.22919e-01	4.77211e+01
9.21708e-02	-9.00970e-01	4.77211e+01
5.56927e-02	-9.60720e-01	4.77211e+01
2.05375e-02	-1.00098e+00	4.77211e+01
-2.59995e-02	-1.02057e+00	4.77211e+01
-6.52322e-02	-9.98772e-01	4.77211e+01
-8.95886e-02	-9.56980e-01	4.77211e+01
-1.11096e-01	-8.95729e-01	4.77211e+01
-1.29769e-01	-8.16257e-01	4.77211e+01
-1.45119e-01	-7.20188e-01	4.77211e+01
-1.56040e-01	-6.09498e-01	4.77211e+01
-1.61356e-01	-4.86468e-01	4.77211e+01
-1.59973e-01	-3.53629e-01	4.77211e+01
-1.50724e-01	-2.13713e-01	4.77211e+01
-1.31831e-01	-6.96126e-02	4.77211e+01
-1.05309e-01	7.57742e-02	4.77211e+01
-7.43781e-02	2.19557e-01	4.77211e+01
-4.25341e-02	3.58884e-01	4.77211e+01
-1.37785e-02	4.91011e-01	4.77211e+01
8.77637e-03	6.13317e-01	4.77211e+01
2.35044e-02	7.23344e-01	4.77211e+01
3.09560e-02	8.18824e-01	4.77211e+01
3.25495e-02	8.97771e-01	4.77211e+01
3.05278e-02	9.58513e-01	4.77211e+01
2.70959e-02	9.99758e-01	4.77211e+01
2.91623e-02	1.02050e+00	4.77211e+01
2.78692e-02	1.02051e+00	4.77211e+01
2.81065e-02	1.01375e+00	4.80377e+01
3.27391e-02	9.92997e-01	4.80377e+01
4.28412e-02	9.51865e-01	4.80377e+01
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9.70379e-02	7.16763e-01	4.80377e+01
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1.48425e-01	4.83753e-01	4.80377e+01
1.75095e-01	3.51096e-01	4.80377e+01
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2.14235e-01	-4.92745e-01	4.80377e+01
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1.60165e-01	-7.23151e-01	4.80377e+01
1.26661e-01	-8.17347e-01	4.80377e+01
9.10584e-02	-8.94917e-01	4.80377e+01
5.51035e-02	-9.54306e-01	4.80377e+01
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-6.37676e-02	-9.92232e-01	4.80377e+01
-8.70552e-02	-9.50759e-01	4.80377e+01
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7.03387e-03	6.09304e-01	4.80377e+01
2.16950e-02	7.18612e-01	4.80377e+01
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2.99744e-02	9.52195e-01	4.80377e+01
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2.92057e-02	1.01376e+00	4.80377e+01
2.64948e-02	1.01380e+00	4.80377e+01
2.62628e-02	1.00706e+00	4.83543e+01
3.13987e-02	9.86430e-01	4.83543e+01
4.14299e-02	9.45580e-01	4.83543e+01
5.54986e-02	8.85363e-01	4.83543e+01
7.36044e-02	8.06999e-01	4.83543e+01
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1.46891e-01	4.80656e-01	4.83543e+01
1.73491e-01	3.48902e-01	4.83543e+01
1.98090e-01	2.10006e-01	4.83543e+01
2.18414e-01	6.68383e-02	4.83543e+01
2.32032e-01	-7.76367e-02	4.83543e+01
2.36218e-01	-2.20418e-01	4.83543e+01
2.28378e-01	-3.58540e-01	4.83543e+01
2.11388e-01	-4.89255e-01	4.83543e+01
1.87357e-01	-6.09941e-01	4.83543e+01
1.57994e-01	-7.18174e-01	4.83543e+01
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8.99459e-02	-8.88865e-01	4.83543e+01
5.45142e-02	-9.47893e-01	4.83543e+01
2.02915e-02	-9.87682e-01	4.83543e+01
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-8.45218e-02	-9.44537e-01	4.83543e+01
-1.03631e-01	-8.84185e-01	4.83543e+01
-1.20027e-01	-8.05848e-01	4.83543e+01
-1.33408e-01	-7.11121e-01	4.83543e+01
-1.43031e-01	-6.01944e-01	4.83543e+01
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-1.47199e-01	-3.49464e-01	4.83543e+01
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-9.82400e-02	7.44620e-02	4.83543e+01
-7.04284e-02	2.16457e-01	4.83543e+01
-4.18724e-02	3.54061e-01	4.83543e+01
-1.58144e-02	4.84541e-01	4.83543e+01
5.29136e-03	6.05291e-01	4.83543e+01
1.98855e-02	7.13881e-01	4.83543e+01
2.80926e-02	8.08086e-01	4.83543e+01
3.07123e-02	8.85961e-01	4.83543e+01
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2.62209e-02	9.86566e-01	4.83543e+01
2.92491e-02	1.00701e+00	4.83543e+01
2.51203e-02	1.00709e+00	4.83543e+01
2.44191e-02	1.00036e+00	4.86709e+01
3.00583e-02	9.79863e-01	4.86709e+01
4.00186e-02	9.39295e-01	4.86709e+01
5.39921e-02	8.79493e-01	4.86709e+01
7.20373e-02	8.01668e-01	4.86709e+01
9.38932e-02	7.07402e-01	4.86709e+01
1.18808e-01	5.98621e-01	4.86709e+01
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1.71887e-01	3.46708e-01	4.86709e+01
1.96265e-01	2.08765e-01	4.86709e+01
2.16259e-01	6.65802e-02	4.86709e+01
2.29539e-01	-7.69072e-02	4.86709e+01
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1.55824e-01	-7.13198e-01	4.86709e+01
1.23334e-01	-8.06205e-01	4.86709e+01
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2.01685e-02	-9.81035e-01	4.86709e+01
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-1.41251e-01	-4.77608e-01	4.86709e+01
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-6.84536e-02	2.14907e-01	4.86709e+01
-4.15415e-02	3.51649e-01	4.86709e+01
-1.68323e-02	4.81306e-01	4.86709e+01
3.54885e-03	6.01279e-01	4.86709e+01
1.80761e-02	7.09149e-01	4.86709e+01
2.66609e-02	8.02717e-01	4.86709e+01
2.97937e-02	8.80057e-01	4.86709e+01
2.88677e-02	9.39560e-01	4.86709e+01
2.57833e-02	9.79970e-01	4.86709e+01
2.92926e-02	1.00027e+00	4.86709e+01
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2.25753e-02	9.93670e-01	4.89874e+01
2.87178e-02	9.73296e-01	4.89874e+01
3.86072e-02	9.33010e-01	4.89874e+01
5.24857e-02	8.73624e-01	4.89874e+01
7.04701e-02	7.96338e-01	4.89874e+01
9.23208e-02	7.02722e-01	4.89874e+01
1.17259e-01	5.94691e-01	4.89874e+01
1.43824e-01	4.74463e-01	4.89874e+01
1.70283e-01	3.44514e-01	4.89874e+01
1.94440e-01	2.07525e-01	4.89874e+01
2.14105e-01	6.63221e-02	4.89874e+01
2.27046e-01	-7.61778e-02	4.89874e+01
2.30825e-01	-2.17025e-01	4.89874e+01
2.22590e-01	-3.53292e-01	4.89874e+01
2.05695e-01	-4.82274e-01	4.89874e+01
1.82193e-01	-6.01384e-01	4.89874e+01
1.53653e-01	-7.08221e-01	4.89874e+01
1.21670e-01	-8.00634e-01	4.89874e+01
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-4.12106e-02	3.49238e-01	4.89874e+01
-1.78503e-02	4.78072e-01	4.89874e+01
1.80634e-03	5.97266e-01	4.89874e+01
1.62667e-02	7.04418e-01	4.89874e+01
2.52292e-02	7.97347e-01	4.89874e+01
2.88752e-02	8.74152e-01	4.89874e+01
2.83143e-02	9.33242e-01	4.89874e+01
2.53458e-02	9.73374e-01	4.89874e+01
2.93360e-02	9.93525e-01	4.89874e+01
2.23714e-02	9.93676e-01	4.89874e+01
2.89923e-02	9.87224e-01	4.93040e+01
2.76932e-02	9.67145e-01	4.93040e+01
3.75326e-02	9.27121e-01	4.93040e+01
5.13436e-02	8.68123e-01	4.93040e+01
6.92680e-02	7.91340e-01	4.93040e+01
9.10752e-02	6.98333e-01	4.93040e+01
1.15977e-01	5.91003e-01	4.93040e+01
1.42501e-01	4.71556e-01	4.93040e+01
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2.25219e-01	-7.55217e-02	4.93040e+01
2.28948e-01	-2.15469e-01	4.93040e+01
2.20703e-01	-3.50872e-01	4.93040e+01
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1.80668e-01	-5.97416e-01	4.93040e+01
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1.46405e-02	6.95750e-01	4.96206e+01
2.35896e-02	7.87537e-01	4.96206e+01
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6.74246e-02	7.81856e-01	4.99372e+01
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1.90317e-01	2.04098e-01	4.99372e+01
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2.18479e-01	-3.46348e-01	4.99372e+01
2.02075e-01	-4.72992e-01	4.99372e+01
1.79241e-01	-5.89960e-01	4.99372e+01
1.51491e-01	-6.94893e-01	4.99372e+01
1.20369e-01	-7.85674e-01	4.99372e+01
8.73036e-02	-8.60471e-01	4.99372e+01
5.37681e-02	-9.17778e-01	4.99372e+01
2.12557e-02	-9.56445e-01	4.99372e+01
-1.98336e-02	-9.75475e-01	4.99372e+01
-5.62592e-02	-9.54869e-01	4.99372e+01
-7.56381e-02	-9.15147e-01	4.99372e+01
-9.16745e-02	-8.56832e-01	4.99372e+01
-1.05212e-01	-7.81087e-01	4.99372e+01
-1.16177e-01	-6.89451e-01	4.99372e+01
-1.24268e-01	-5.83789e-01	4.99372e+01
-1.28835e-01	-4.66264e-01	4.99372e+01
-1.29057e-01	-3.39282e-01	4.99372e+01
-1.23470e-01	-2.05457e-01	4.99372e+01
-1.08866e-01	-6.75827e-02	4.99372e+01
-8.83676e-02	7.15929e-02	4.99372e+01
-6.49502e-02	2.09288e-01	4.99372e+01

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-1.87814e-02	4.69265e-01	4.99372e+01
6.89080e-05	5.86315e-01	4.99372e+01
1.41040e-02	6.91526e-01	4.99372e+01
2.29396e-02	7.82760e-01	4.99372e+01
2.66247e-02	8.58155e-01	4.99372e+01
2.61334e-02	9.16153e-01	4.99372e+01
2.31956e-02	9.55541e-01	4.99372e+01
2.71870e-02	9.75325e-01	4.99372e+01
1.90772e-02	9.69517e-01	5.02538e+01
2.53474e-02	9.49649e-01	5.02538e+01
3.50845e-02	9.10371e-01	5.02538e+01
4.87574e-02	8.52469e-01	5.02538e+01
6.65029e-02	7.77114e-01	5.02538e+01
8.80917e-02	6.85836e-01	5.02538e+01
1.12746e-01	5.80501e-01	5.02538e+01
1.39017e-01	4.63272e-01	5.02538e+01
1.65169e-01	3.36558e-01	5.02538e+01
1.89020e-01	2.02973e-01	5.02538e+01
2.08439e-01	6.52666e-02	5.02538e+01
2.21277e-01	-7.37221e-02	5.02538e+01
2.25205e-01	-2.11123e-01	5.02538e+01
2.17368e-01	-3.44086e-01	5.02538e+01
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1.78528e-01	-5.86232e-01	5.02538e+01
1.51016e-01	-6.90541e-01	5.02538e+01
1.20144e-01	-7.80786e-01	5.02538e+01
8.73293e-02	-8.55147e-01	5.02538e+01
5.40364e-02	-9.12125e-01	5.02538e+01
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-1.90772e-02	-9.69517e-01	5.02538e+01
-5.52932e-02	-9.49060e-01	5.02538e+01
-7.45793e-02	-9.09594e-01	5.02538e+01
-9.05556e-02	-8.51647e-01	5.02538e+01
-1.04059e-01	-7.76375e-01	5.02538e+01
-1.15017e-01	-6.85306e-01	5.02538e+01
-1.23127e-01	-5.80297e-01	5.02538e+01
-1.27743e-01	-4.63494e-01	5.02538e+01
-1.28046e-01	-3.37289e-01	5.02538e+01
-1.22581e-01	-2.04280e-01	5.02538e+01
-1.08155e-01	-6.72399e-02	5.02538e+01
-8.78733e-02	7.10972e-02	5.02538e+01
-6.46889e-02	2.07965e-01	5.02538e+01
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-1.89719e-02	4.66381e-01	5.02538e+01
-3.13194e-04	5.82726e-01	5.02538e+01
1.35675e-02	6.87302e-01	5.02538e+01
2.22896e-02	7.77984e-01	5.02538e+01
2.59031e-02	8.52919e-01	5.02538e+01
2.53770e-02	9.10562e-01	5.02538e+01
2.24316e-02	9.49707e-01	5.02538e+01
2.63420e-02	9.69373e-01	5.02538e+01
1.83208e-02	9.63559e-01	5.05704e+01
2.45654e-02	9.43818e-01	5.05704e+01
3.42685e-02	9.04788e-01	5.05704e+01
4.78953e-02	8.47251e-01	5.05704e+01
6.55812e-02	7.72372e-01	5.05704e+01
8.70972e-02	6.81670e-01	5.05704e+01
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1.87722e-01	2.01847e-01	5.05704e+01
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1.77815e-01	-5.82505e-01	5.05704e+01
1.50541e-01	-6.86189e-01	5.05704e+01
1.19918e-01	-7.75899e-01	5.05704e+01
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2.22407e-02	-9.44708e-01	5.05704e+01
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-7.35206e-02	-9.04042e-01	5.05704e+01
-8.94366e-02	-8.46462e-01	5.05704e+01
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-1.27035e-01	-3.35296e-01	5.05704e+01
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-1.07445e-01	-6.68971e-02	5.05704e+01
-8.73790e-02	7.06014e-02	5.05704e+01
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-1.91624e-02	4.63496e-01	5.05704e+01
-6.95296e-04	5.79137e-01	5.05704e+01
1.30310e-02	6.83079e-01	5.05704e+01
2.16396e-02	7.73207e-01	5.05704e+01
2.51815e-02	8.47683e-01	5.05704e+01
2.46206e-02	9.04971e-01	5.05704e+01
2.16675e-02	9.43873e-01	5.05704e+01
2.54969e-02	9.63422e-01	5.05704e+01
1.75643e-02	9.57601e-01	5.08869e+01

2.37835e-02	9.37986e-01	5.08869e+01
3.34524e-02	8.99205e-01	5.08869e+01
4.70333e-02	8.42034e-01	5.08869e+01
6.46595e-02	7.67630e-01	5.08869e+01
8.61027e-02	6.77505e-01	5.08869e+01
1.10593e-01	5.73500e-01	5.08869e+01
1.36694e-01	4.57749e-01	5.08869e+01
1.62691e-01	3.32631e-01	5.08869e+01
1.86424e-01	2.00722e-01	5.08869e+01
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2.18648e-01	-7.25224e-02	5.08869e+01
2.22709e-01	-2.08226e-01	5.08869e+01
2.15144e-01	-3.39561e-01	5.08869e+01
1.99288e-01	-4.63912e-01	5.08869e+01
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1.50066e-01	-6.81836e-01	5.08869e+01
1.19693e-01	-7.71011e-01	5.08869e+01
8.73808e-02	-8.44499e-01	5.08869e+01
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2.27332e-02	-9.38839e-01	5.08869e+01
-1.75643e-02	-9.57601e-01	5.08869e+01
-5.33612e-02	-9.37444e-01	5.08869e+01
-7.24618e-02	-8.98489e-01	5.08869e+01
-8.83177e-02	-8.41276e-01	5.08869e+01
-1.01754e-01	-7.66949e-01	5.08869e+01
-1.12697e-01	-6.77017e-01	5.08869e+01
-1.20846e-01	-5.73312e-01	5.08869e+01
-1.25559e-01	-4.57953e-01	5.08869e+01
-1.26024e-01	-3.33303e-01	5.08869e+01
-1.20802e-01	-2.01926e-01	5.08869e+01
-1.06734e-01	-6.65543e-02	5.08869e+01
-8.68847e-02	7.01056e-02	5.08869e+01
-6.41662e-02	2.05318e-01	5.08869e+01
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-1.93529e-02	4.60611e-01	5.08869e+01
-1.07740e-03	5.75548e-01	5.08869e+01
1.24945e-02	6.78855e-01	5.08869e+01
2.09896e-02	7.68431e-01	5.08869e+01
2.44599e-02	8.42448e-01	5.08869e+01
2.38643e-02	8.99380e-01	5.08869e+01
2.09035e-02	9.38039e-01	5.08869e+01
2.46519e-02	9.57470e-01	5.08869e+01
1.68079e-02	9.51643e-01	5.12035e+01
2.30015e-02	9.32154e-01	5.12035e+01
3.26364e-02	8.93621e-01	5.12035e+01
4.61712e-02	8.36816e-01	5.12035e+01
6.37378e-02	7.62888e-01	5.12035e+01
8.51082e-02	6.73339e-01	5.12035e+01
1.09516e-01	5.69999e-01	5.12035e+01
1.35533e-01	4.54988e-01	5.12035e+01
1.61452e-01	3.30667e-01	5.12035e+01
1.85126e-01	1.99596e-01	5.12035e+01
2.04459e-01	6.44729e-02	5.12035e+01
2.17334e-01	-7.19226e-02	5.12035e+01
2.21461e-01	-2.06778e-01	5.12035e+01
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1.98359e-01	-4.60885e-01	5.12035e+01
1.76389e-01	-5.75049e-01	5.12035e+01
1.49591e-01	-6.77484e-01	5.12035e+01
1.19468e-01	-7.66123e-01	5.12035e+01
8.74066e-02	-8.39175e-01	5.12035e+01
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-6.39049e-02	2.03995e-01	5.12035e+01
-4.08842e-02	3.34241e-01	5.12035e+01
-1.95435e-02	4.57727e-01	5.12035e+01
-1.45950e-03	5.71959e-01	5.12035e+01
1.19580e-02	6.74631e-01	5.12035e+01
2.03397e-02	7.63654e-01	5.12035e+01
2.37383e-02	8.37212e-01	5.12035e+01
2.31079e-02	8.93790e-01	5.12035e+01
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2.38069e-02	9.51518e-01	5.12035e+01
1.60515e-02	9.45684e-01	5.15201e+01
2.22196e-02	9.26323e-01	5.15201e+01
3.18204e-02	8.88038e-01	5.15201e+01
4.53091e-02	8.31598e-01	5.15201e+01
6.28161e-02	7.58146e-01	5.15201e+01
8.41137e-02	6.69173e-01	5.15201e+01
1.08439e-01	5.66498e-01	5.15201e+01
1.34371e-01	4.52226e-01	5.15201e+01
1.60213e-01	3.28703e-01	5.15201e+01
1.83828e-01	1.98471e-01	5.15201e+01
2.03132e-01	6.42083e-02	5.15201e+01
2.16020e-01	-7.13228e-02	5.15201e+01

2.20214e-01	-2.05329e-01	5.15201e+01
2.12920e-01	-3.35036e-01	5.15201e+01
1.97430e-01	-4.57858e-01	5.15201e+01
1.75676e-01	-5.71321e-01	5.15201e+01
1.49116e-01	-6.73132e-01	5.15201e+01
1.19242e-01	-7.61236e-01	5.15201e+01
8.74323e-02	-8.33851e-01	5.15201e+01
5.51097e-02	-8.89513e-01	5.15201e+01
2.37183e-02	-9.27102e-01	5.15201e+01
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-6.36435e-02	2.02671e-01	5.15201e+01
-4.08564e-02	3.32116e-01	5.15201e+01
-1.97340e-02	4.54842e-01	5.15201e+01
-1.84160e-03	5.68370e-01	5.15201e+01
1.14215e-02	6.70407e-01	5.15201e+01
1.96897e-02	7.58878e-01	5.15201e+01
2.30166e-02	8.31977e-01	5.15201e+01
2.23515e-02	8.88199e-01	5.15201e+01
1.93755e-02	9.26371e-01	5.15201e+01
2.29619e-02	9.45567e-01	5.15201e+01
1.52951e-02	9.39726e-01	5.18367e+01
2.14376e-02	9.20491e-01	5.18367e+01
3.10044e-02	8.82454e-01	5.18367e+01
4.44471e-02	8.26380e-01	5.18367e+01
6.18944e-02	7.53404e-01	5.18367e+01
8.31192e-02	6.65008e-01	5.18367e+01
1.07362e-01	5.62998e-01	5.18367e+01
1.33210e-01	4.49465e-01	5.18367e+01
1.58974e-01	3.26739e-01	5.18367e+01
1.82530e-01	1.97346e-01	5.18367e+01
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2.18966e-01	-2.03880e-01	5.18367e+01
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1.08850e-02	6.66183e-01	5.18367e+01
1.90397e-02	7.54101e-01	5.18367e+01
2.22950e-02	8.26741e-01	5.18367e+01
2.15952e-02	8.82608e-01	5.18367e+01
1.86114e-02	9.20537e-01	5.18367e+01
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1.45386e-02	9.33768e-01	5.21533e+01
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4.35850e-02	8.21163e-01	5.21533e+01
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8.74838e-02	-8.23203e-01	5.21533e+01
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-1.08057e-01	-6.60438e-01	5.21533e+01
-1.16283e-01	-5.59341e-01	5.21533e+01
-1.21190e-01	-4.46872e-01	5.21533e+01
-1.21981e-01	-3.25332e-01	5.21533e+01
-1.17244e-01	-1.97216e-01	5.21533e+01
-1.03893e-01	-6.51830e-02	5.21533e+01
-8.49074e-02	6.81226e-02	5.21533e+01
-6.31209e-02	2.00025e-01	5.21533e+01
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-2.60581e-03	5.61192e-01	5.21533e+01
1.03485e-02	6.61960e-01	5.21533e+01
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2.15734e-02	8.21505e-01	5.21533e+01
2.08388e-02	8.77017e-01	5.21533e+01
1.78474e-02	9.14703e-01	5.21533e+01
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1.39089e-02	9.28375e-01	5.24698e+01
2.00017e-02	9.09381e-01	5.24698e+01
2.95013e-02	8.71817e-01	5.24698e+01
4.28530e-02	8.16438e-01	5.24698e+01
6.01826e-02	7.44367e-01	5.24698e+01
8.12635e-02	6.57067e-01	5.24698e+01
1.05343e-01	5.56323e-01	5.24698e+01
1.31023e-01	4.44197e-01	5.24698e+01
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2.12199e-01	-6.95928e-02	5.24698e+01
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1.18544e-01	-7.47044e-01	5.24698e+01
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1.67936e-02	7.36511e-01	5.31030e+01
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1.20854e-02	9.12491e-01	5.34196e+01
1.81063e-02	8.93832e-01	5.34196e+01
2.75072e-02	8.56928e-01	5.34196e+01
4.07247e-02	8.02521e-01	5.34196e+01
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1.28020e-01	4.36809e-01	5.34196e+01
1.53397e-01	3.17722e-01	5.34196e+01
1.76651e-01	1.92154e-01	5.34196e+01
1.95759e-01	6.26813e-02	5.34196e+01
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2.06639e-01	-3.22490e-01	5.34196e+01
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1.14776e-02	9.07196e-01	5.37362e+01

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2.07509e-01	-6.75197e-02	5.37362e+01
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7.31505e-03	6.35615e-01	5.43693e+01
1.46899e-02	7.19530e-01	5.43693e+01
1.74608e-02	7.88846e-01	5.43693e+01
1.65357e-02	8.42142e-01	5.43693e+01
1.35144e-02	8.78314e-01	5.43693e+01
1.64735e-02	8.96534e-01	5.43693e+01
9.65417e-03	8.91311e-01	5.46859e+01
1.55791e-02	8.73100e-01	5.46859e+01
2.48485e-02	8.37076e-01	5.46859e+01
3.78870e-02	7.83965e-01	5.46859e+01
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9.89149e-02	5.34499e-01	5.46859e+01
1.24015e-01	4.26958e-01	5.46859e+01
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2.03991e-01	-6.59648e-02	5.46859e+01
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1.16780e-01	-7.16703e-01	5.46859e+01
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2.77175e-02	-8.73569e-01	5.46859e+01
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-6.11575e-02	-8.36683e-01	5.46859e+01
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1.38187e-02	8.58121e-01	5.56357e+01
2.29904e-02	8.22731e-01	5.56357e+01
3.58956e-02	7.70556e-01	5.56357e+01
5.26466e-02	7.02652e-01	5.56357e+01
7.30218e-02	6.20401e-01	5.56357e+01
9.63010e-02	5.25481e-01	5.56357e+01
1.21154e-01	4.19832e-01	5.56357e+01
1.45989e-01	3.05614e-01	5.56357e+01
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7.47411e-03	8.71224e-01	5.59523e+01
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1.17656e-02	8.39388e-01	5.69020e+01
2.08090e-02	8.04791e-01	5.69020e+01
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5.00619e-02	6.87397e-01	5.69020e+01
7.01595e-02	6.06986e-01	5.69020e+01
9.31234e-02	5.14189e-01	5.69020e+01
1.17648e-01	4.10900e-01	5.69020e+01
1.42174e-01	2.99230e-01	5.69020e+01
1.64744e-01	1.81461e-01	5.69020e+01
1.83444e-01	6.00026e-02	5.69020e+01
1.96344e-01	-6.26633e-02	5.69020e+01
2.01353e-01	-1.84026e-01	5.69020e+01
1.95917e-01	-3.01598e-01	5.69020e+01
1.82992e-01	-4.13006e-01	5.69020e+01
1.64309e-01	-5.15993e-01	5.69020e+01
1.41167e-01	-6.08467e-01	5.69020e+01
1.14899e-01	-6.88553e-01	5.69020e+01
8.67368e-02	-7.54625e-01	5.69020e+01
5.79573e-02	-8.05343e-01	5.69020e+01
2.98581e-02	-8.39679e-01	5.69020e+01
-6.00292e-03	-8.56873e-01	5.69020e+01
-3.82322e-02	-8.39202e-01	5.69020e+01
-5.57151e-02	-8.04546e-01	5.69020e+01
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-8.32547e-02	-6.87165e-01	5.69020e+01
-9.39566e-02	-6.06820e-01	5.69020e+01
-1.02298e-01	-5.14125e-01	5.69020e+01
-1.07684e-01	-4.10970e-01	5.69020e+01
-1.09365e-01	-2.99460e-01	5.69020e+01
-1.06025e-01	-1.81873e-01	5.69020e+01
-9.48125e-02	-6.06235e-02	5.69020e+01
-7.84402e-02	6.18373e-02	5.69020e+01
-5.94868e-02	1.83032e-01	5.69020e+01
-4.00119e-02	3.00506e-01	5.69020e+01
-2.19845e-02	4.11878e-01	5.69020e+01
-6.80023e-03	5.14889e-01	5.69020e+01
4.29396e-03	6.07448e-01	5.69020e+01
1.09856e-02	6.87671e-01	5.69020e+01
1.33396e-02	7.53924e-01	5.69020e+01
1.22294e-02	8.04851e-01	5.69020e+01
9.18861e-03	8.39406e-01	5.69020e+01
1.16761e-02	8.56832e-01	5.69020e+01
5.51253e-03	8.52089e-01	5.72186e+01
1.12524e-02	8.34705e-01	5.72186e+01
2.02637e-02	8.00306e-01	5.72186e+01
3.29493e-02	7.49589e-01	5.72186e+01
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9.23290e-02	5.11366e-01	5.72186e+01
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1.07391e-02	8.30021e-01	5.75352e+01
1.97183e-02	7.95821e-01	5.75352e+01
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1.15896e-01	4.06434e-01	5.75352e+01
1.40266e-01	2.96038e-01	5.75352e+01
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1.81331e-01	5.95278e-02	5.75352e+01
1.94217e-01	-6.17537e-02	5.75352e+01
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1.81388e-01	-4.08197e-01	5.75352e+01
1.63023e-01	-5.10052e-01	5.75352e+01
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-7.76297e-02	6.10627e-02	5.75352e+01
-5.90173e-02	1.80924e-01	5.75352e+01
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3.59823e-03	6.00665e-01	5.75352e+01
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1.12345e-02	7.95871e-01	5.75352e+01
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1.05685e-02	8.47271e-01	5.75352e+01
4.53174e-03	8.42521e-01	5.78518e+01
1.02258e-02	8.25338e-01	5.78518e+01
1.91730e-02	7.91336e-01	5.78518e+01
3.17707e-02	7.41202e-01	5.78518e+01
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9.07402e-02	5.05720e-01	5.78518e+01
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1.98272e-01	-1.80620e-01	5.78518e+01
1.93115e-01	-2.96230e-01	5.78518e+01
1.80586e-01	-4.05792e-01	5.78518e+01
1.62381e-01	-5.07082e-01	5.78518e+01
1.39775e-01	-5.98043e-01	5.78518e+01
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3.07007e-02	-8.25558e-01	5.78518e+01
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-3.62491e-02	-8.25198e-01	5.78518e+01

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-9.31264e-02	-5.97591e-02	5.78518e+01
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-3.98219e-02	2.95406e-01	5.78518e+01
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-7.50983e-03	5.06249e-01	5.78518e+01
3.25037e-03	5.97274e-01	5.78518e+01
9.70153e-03	6.76163e-01	5.78518e+01
1.19101e-02	7.41309e-01	5.78518e+01
1.07371e-02	7.91381e-01	5.78518e+01
7.69197e-03	8.25352e-01	5.78518e+01
1.00147e-02	8.42491e-01	5.78518e+01
4.04134e-03	8.37737e-01	5.81683e+01
9.71255e-03	8.20655e-01	5.81683e+01
1.86276e-02	7.86851e-01	5.81683e+01
3.11815e-02	7.37009e-01	5.81683e+01
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1.60676e-01	1.77758e-01	5.81683e+01
1.79217e-01	5.90529e-02	5.81683e+01
1.92090e-01	-6.08442e-02	5.81683e+01
1.97246e-01	-1.79484e-01	5.81683e+01
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2.90251e-03	5.93882e-01	5.81683e+01
9.27351e-03	6.72327e-01	5.81683e+01
1.14336e-02	7.37104e-01	5.81683e+01
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3.58951e-03	8.33149e-01	5.84849e+01
8.75990e-03	8.16164e-01	5.84849e+01
1.74985e-02	7.82549e-01	5.84849e+01
2.99013e-02	7.32987e-01	5.84849e+01
4.60940e-02	6.68484e-01	5.84849e+01
6.58697e-02	5.90354e-01	5.84849e+01
8.85327e-02	5.00188e-01	5.84849e+01
1.12795e-01	3.99825e-01	5.84849e+01
1.37096e-01	2.91313e-01	5.84849e+01
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1.77968e-01	5.88213e-02	5.84849e+01
1.90785e-01	-6.04123e-02	5.84849e+01
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1.90906e-01	-2.92729e-01	5.84849e+01
1.78633e-01	-4.01084e-01	5.84849e+01
1.60749e-01	-5.01266e-01	5.84849e+01
1.38508e-01	-5.91239e-01	5.84849e+01
1.13206e-01	-6.69175e-01	5.84849e+01
8.60422e-02	-7.33491e-01	5.84849e+01
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9.27195e-03	7.82585e-01	5.84849e+01
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1.45356e-03	8.28826e-01	5.88015e+01
7.21567e-03	8.11931e-01	5.88015e+01
1.55834e-02	7.78495e-01	5.88015e+01
2.76909e-02	7.29196e-01	5.88015e+01
4.37185e-02	6.65035e-01	5.88015e+01
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1.76461e-01	5.85974e-02	5.88015e+01
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8.76468e-03	7.29271e-01	5.88015e+01
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1.36683e-02	7.74441e-01	5.91181e+01
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1.87527e-01	-5.96105e-02	5.91181e+01
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1.57838e-01	-4.95915e-01	5.91181e+01
1.35992e-01	-5.84974e-01	5.91181e+01
1.11160e-01	-6.62124e-01	5.91181e+01
8.45478e-02	-7.25797e-01	5.91181e+01
5.72559e-02	-7.74698e-01	5.91181e+01
3.06917e-02	-8.07835e-01	5.91181e+01
-2.78965e-03	-8.24503e-01	5.91181e+01
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-6.07808e-02	-7.25284e-01	5.91181e+01
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-1.78077e-03	8.20179e-01	5.94347e+01

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2.32700e-02	7.21613e-01	5.94347e+01
3.89675e-02	6.58138e-01	5.94347e+01
5.87124e-02	5.81251e-01	5.94347e+01
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1.06850e-01	3.93754e-01	5.94347e+01
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3.72575e-03	7.17869e-01	5.97513e+01
2.86902e-03	7.66353e-01	5.97513e+01
7.39368e-04	7.99239e-01	5.97513e+01
2.18237e-04	8.15842e-01	5.97513e+01
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1.03875e-03	7.95001e-01	6.00678e+01
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1.88492e-02	7.14031e-01	6.00678e+01
3.42164e-02	6.51240e-01	6.00678e+01
5.39408e-02	5.75183e-01	6.00678e+01
7.73035e-02	4.87409e-01	6.00678e+01
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1.52147e-01	1.72641e-01	6.00678e+01
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1.70530e-01	-3.90266e-01	6.00678e+01
1.53472e-01	-4.87888e-01	6.00678e+01
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1.08091e-01	-6.51548e-01	6.00678e+01
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-7.75218e-02	-1.72747e-01	6.00678e+01
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2.04610e-03	7.14069e-01	6.00678e+01
1.26829e-03	7.62295e-01	6.00678e+01
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6.00790e-03	7.58224e-01	6.03844e+01
1.66388e-02	7.10240e-01	6.03844e+01
3.18409e-02	6.47791e-01	6.03844e+01
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7.50576e-02	4.84853e-01	6.03844e+01
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3.66455e-04	7.10268e-01	6.03844e+01
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-3.81450e-03	8.07201e-01	6.03844e+01
-8.24945e-03	8.02885e-01	6.07010e+01
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4.09281e-03	7.54170e-01	6.07010e+01
1.44283e-02	7.06449e-01	6.07010e+01
2.94654e-02	6.44342e-01	6.07010e+01
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-5.02201e-02	-6.37444e-01	6.13342e+01
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-3.93145e-02	5.56236e-02	6.26005e+01
-2.94577e-02	1.65889e-01	6.26005e+01
-2.20462e-02	2.72772e-01	6.26005e+01
-1.62011e-02	3.74099e-01	6.26005e+01
-1.16015e-02	4.67808e-01	6.26005e+01
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-6.64639e-03	6.24934e-01	6.26005e+01
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4.20617e-02	5.52077e-01	6.26005e+01
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9.20993e-02	3.74285e-01	6.26005e+01
1.18894e-01	2.73014e-01	6.26005e+01
1.42973e-01	1.66185e-01	6.26005e+01
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1.72568e-01	-5.53948e-02	6.26005e+01
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1.41413e-01	1.64660e-01	6.32337e+01
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2.14846e-02	6.15716e-01	6.35503e+01
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